

Beyond the Tide

Integrating the management of New Zealand's coasts

Raewyn Peart

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To Charles
for his intelligence and humanity



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Acronyms

AEE	Assessment of environmental effects
AMA	Aquaculture management area
ARC	Auckland Regional Council
BNZ	Biosecurity New Zealand
B_{MSY}	Biomass level that can support the maximum sustainable yield
CMA	Coastal marine area
CPUE	Catch per unit effort
DoC	Department of Conservation
EDS	Environmental Defence Society
EIA	Environmental impact assessment
EW	Environment Waikato
GIS	Geographic information system
ICM	Integrated coastal management
LAMS	Local area management strategies
ITQ	Individual transferable quota
LTCCP	Long term council community plan
MFE	Ministry for the Environment
MFish	Ministry of Fisheries
MHWS	Mean high water springs
MPA	Marine protected area
MSY	Maximum sustainable yield
NAFMAC	National Fisheries Management Advisory Committee
NIWA	National Institute of Water and Atmospheric Research
NGO	Non-governmental organisation
NRC	Northland Regional Council
NZCPS	New Zealand Coastal Policy Statement
QMA	Quota management area

QMS	Quota management system
RMA	Resource Management Act 1991
TA	Territorial authority
TAC	Total allowable catch
TACC	Total allowable commercial catch
TCDC	Thames-Coromandel District Council
UNCED	United Nations Conference on Environment and Development
USA	United States of America

Glossary

Bach	Small, simply furnished weekend or holiday cottage
Benthic	Of flora and fauna found at the bottom of the sea
Boatie	Person who owns a small boat
Hapū	A Māori sub-tribe
Hui	Meeting
Iwi	A Māori tribe
Kaimoana	Seafood
Kaitiaki	A Māori guardian/manager
Kaitiakitanga	The exercise of guardianship by the tangata whenua of an area in accordance with tikanga Māori in relation to natural and physical resources
Kina	Edible green New Zealand sea urchin
Mataitai	Reserves declared under the Kaimoana Customary Fishing Regulations 1998 where tangata whenua manage non-commercial fishing activity
Pā	Fortified Māori settlement
Rohe	Territory
Rohe moana	Marine areas within the authority of a Māori tribe
Taiapure	Local fishery areas established under the Fisheries Act 1996 where a management committee, representative of the local Māori community, is established
Tangata whenua	Māori people of a particular locality
Tikanga Māori	Māori custom
Wāhi tapu	Sacred place
Waka	Māori canoe
Whanau	Extended family

Foreword

The coast is important to many New Zealanders and is the focus of much passionate debate. The Hauraki Gulf and Kaipara Harbour are particularly special parts of our coast, treasured by Aucklanders and visitors alike. They are quite different in character and management regimes, and are excellent case studies for this report's examination of integrated coastal management.

The Hauraki Gulf is a popular area for fishing, sailing and other recreation; over one million people live in its catchment and it is within day-trip distance for over half of New Zealand's population. Significant for tangata whenua and with a rich history of settlement and use, the Gulf is also of ecological and conservation importance, being home to rare and endangered species. It also supports many commercial activities such as fishing, tourism and marine transport. The national significance of the Gulf was recognised in 2000 with the passing of the Hauraki Gulf Marine Park Act.

The Kaipara Harbour is further from the main population centres than parts of the Hauraki Gulf but this contributes to its qualities. Nationally significant for several bird species, it is also visited by the endangered Maui's dolphin. Dramatic landscapes and numerous cultural heritage sites of importance to tangata whenua and Pākehā fall within its catchment which spreads over much of the Northland and Auckland regions. The harbour has an active fishing industry and has also been the focus of several controversial development proposals such as sand mining, energy generation and marine farms.

Both areas are under pressure from increasing population and development. This places a strain on resources and leads to tension between conservation and development. This has generated various coastal management issues involving a wide range of agencies, legislation and activities.

The Auckland Regional Council has coastal management responsibilities in these two areas under the Resource Management Act, Biosecurity Act, Local Government Act, and in managing our regional parks. Through our work we actively recognise the need for integrated coastal management but find that it can be challenging to achieve. As Raewyn notes in this report, while integrated management is a widely accepted approach for managing the coast, it has yet to be clearly defined.

Integrated management needs to take many forms. It should occur across the land and water boundary, and between the different functions of local authorities and government departments. Government agencies cannot address coastal management issues alone and need to work with communities and tangata whenua. We also need to recognise that changes in the coast may impact

on ecology, recreation and cultural heritage amongst other things. Strong links between research, policy development, implementation and monitoring are vital. There are many barriers to integration and we must look for better ways to align our activities and make the most of the coast's diverse values.

The Auckland Regional Council has supported the development of this report in order to learn more about the issues around integrated coastal management. It's crucial that we use it to review our progress and identify opportunities for improvement. These are beautiful places with complex issues and management requirements. We need to ensure our actions are integrated as much as possible so that management is effective and these special areas are passed on in a good state.

I congratulate Raewyn on the completion of the report. It is timely in terms of current work such as the oceans policy and New Zealand Coastal Policy review as well as many local projects. I hope that other councils and the many other parties and individuals involved in coastal management in New Zealand will find it a useful document that assists them in their work.

Cr Paul Walbran
Chair, Regional Strategy and Planning Committee
Auckland Regional Council

Executive Summary

Introduction

This report presents the results of a detailed study into the management of the Hauraki Gulf and Kaipara Harbour. The study includes an international literature review and in-depth interviews with sixty people involved in coastal management. It focuses on providing an understanding of how environmental managers and other stakeholders see environmental management issues in these marine systems, how they have approached their tasks, what they have learned from what they have done, and how they see future challenges. As well as describing how these areas are currently managed, the report explores the extent to which management is integrated, the presence of factors which are likely to promote integrated coastal management, and how these could be more strongly supported.

Both the Hauraki Gulf and Kaipara Harbour have considerable ecological, social, cultural and economic importance. The Hauraki Gulf, on the east coast of the North Island, supports a rich diversity of marine species and is a particularly important spawning and nursery area for fish. It was one of the earliest places to be settled in New Zealand, it contains the country's busiest port and is adjacent to New Zealand's largest urban area. It provides important recreational opportunities for a burgeoning population. There are enormous pressures on the Gulf's marine environment from sedimentation, pollution, the overharvesting of marine life and exotic invasive species.

The Kaipara Harbour, located on the west coast, is New Zealand's largest enclosed harbour and estuarine system. It is ecologically very important, containing extensive mangrove forests, eelgrass beds, salt marshes and habitats for coastal and estuarine wading birds. The catchments of the harbour are largely rural, with the main impacts on the ecology of the harbour being from agriculture and fishing activity. Rural-residential subdivision is starting to increase on coastal land accessible to Auckland.

The evolution of integrated coastal management

Within New Zealand, coastal management is largely split between eight types of statutory management agencies with overlapping responsibilities: the Department of Conservation (DoC), the Ministry of Fisheries (MFish), Biosecurity New Zealand, Maritime New Zealand, the Ministry of Energy, regional councils, territorial authorities, and iwi and hapū.

Coastal management in New Zealand has evolved considerably over the past forty years. A concern for better management of the coast emerged during the 1970s with a range of legislation passed which sought to promote economic exploitation of marine resources as well as to provide some protection. The 1980s saw the beginnings of a more sophisticated approach to coastal management leading to concepts of sustainability, integrated management and ecosystems-based management being incorporated into coastal legislation during the 1990s. The 2000s has been a period of consolidation with important policy initiatives such as the oceans policy, the marine protected areas policy and the review of the New Zealand Coastal Policy Statement providing the opportunity to embed and better apply such concepts to coastal management going forward.

Although integrated management is widely accepted as the appropriate approach to apply to the management of the coast, it is a broad concept which has no prescribed meaning. Key elements which emerge from the literature are the importance of participatory processes, the need for coastal planning, and the advantages of a strategic approach. International experience has shown that the application of integrated coastal management has not been easy in practice. It is likely to require coastal managers who are facilitative, proactive and entrepreneurial rather than hierarchical and bureaucratic. It also requires a strong focus on applying consensus-building techniques.

Conflicts underlying New Zealand's coastal management system

In practice, integrating coastal management has been difficult in New Zealand because of underlying conflicts embedded within the management system itself. These include conflicting values and management approaches adopted by the various management agencies, conflicts over information and conflicts between exclusive and non-exclusive rights to use the marine area.

Coastal management within the Hauraki Gulf and Kaipara Harbour

The results of the sixty stakeholder interviews undertaken for the project have been distilled and written up into nine 'case studies'. These studies indicate that the environmental requirements placed on different economic activities can be highly variable and do not appear to be linked to their potential to cause significant environmental harm to the marine environment. The current uneven playing field in environmental regulation effectively favours old-style environmentally-damaging economic activities and can act as a disincentive for the establishment of newer and 'cleaner' resource-based activities. The key findings of these case studies are summarized below.

Case Study 1: Planning for the coastal marine area

This case study describes attempts at undertaking regional coastal planning under the RMA, fisheries planning under the Fisheries Act 1996 and marine protection planning, which is currently non-statutory, but which is being undertaken within the framework of the Marine Protected Areas Policy Statement and Implementation Plan. The case study identifies significant innovation in coastal marine planning, with regional councils exploring ways to more effectively span the land/water divide through sub-regional plans such as community compartment plans. The Department of Conservation is moving away from a conflict-ridden and ad hoc approach to establishing marine reserves and MFish is trialling Ministry-led multi-stakeholder planning for fisheries.

Many of the initiatives are very new and it is not entirely clear how successful they will be. They all face the difficulties of planning in a poorly understood and dynamic environment and where there has been a long history of conflictual relationships between stakeholders. There are good prospects for further integration between the different planning processes, although there is currently a poor interface between regional coastal planning and that for fisheries.

Case Study 2: Managing coastal development

This case study focuses on the management of coastal development on the Coromandel Peninsula, the Hauraki Gulf Islands, the Hibiscus Coast and the land surrounding the Kaipara Harbour. There is broad agreement on the overall approach to be adopted towards managing coastal development, which is to consolidate urban development in existing developed areas and avoid urban development in rural coastal areas. The strategy has primarily been implemented through district plan rules which provide for subdivision opportunities within townships and restrict subdivision in rural areas. Different councils, however, vary in the level of sophistication of controls which are applied.

All of the four territorial authorities studied have incorporated conservation lot provisions in their district plans as an incentive for landowners to provide environmental benefits. These provisions increase the number of lots which landowners can create through subdivision if they permanently protect other areas of land which have environmental importance. This has created an unresolved tension between the overall strategy of restricting rural subdivision, and the conservation lot approach which facilitates it.

The prospects of integration between territorial authorities planning for common coastal/marine environments do not look strong. Councils have approached planning from different perspectives, the plans are structured in different ways and all the councils are at different stages of plan development.

Case Study 3: Reducing sedimentation

This case study examines how regional councils are approaching the management of activities which result in human-induced sedimentation of the marine area. It identifies significant management problems, particularly in effectively reducing sedimentation from rural activities such as agriculture and forestry. In these areas regulation has proved politically problematic, and it is difficult to scale up voluntary mechanisms to achieve catchment-wide improvements. Although large earthworks are more strongly regulated, they also contribute to marine sedimentation due to the lack of efficiency of sediment ponds. Small earthworks are not well-regulated and, cumulatively, can also have a significant impact.

Councils are piloting innovative ways of more closely linking land-management decisions to the health and carrying capacity of affected marine systems, such as estuaries and harbours. These include the Mahurangi Action Plan, the Okura Catchment Study and the Muddy Feet project in the Firth of Thames. These show some promise, but are resource-intensive and are only likely to be applied to a small number of priority areas.

Case Study 4: Marine biosecurity

Marine biosecurity is a recent management activity primarily undertaken by a national agency, Biosecurity New Zealand. It is yet unclear what role regional councils will undertake and they have proved reluctant to enter a potentially costly new management area outside their current area of expertise.

Effective marine biosecurity is extremely difficult to achieve because of problems in regulating international vessels without international agreements, difficulties in identifying new incursions in a timely fashion and the lack of effective tools to eradicate or manage new species once they have arrived. There is potential to better integrate biosecurity considerations into regional councils' coastal planning activities. One way to achieve this could be to incorporate marine biosecurity guidance into the New Zealand Coastal Policy Statement.

Case Study 5: Hauraki Gulf amateur fishers

This case study examines issues concerning the amateur fishing sector and how their non-resolution has resulted in escalating levels of conflict spilling over into multiple areas of marine management in the Hauraki Gulf. Much of the conflict stems from the uncertain status of amateur fishing rights, where fish stocks are shared with other sectors with better-defined rights. The conflict is exacerbated by depleted fish stocks which are being managed to a target size which generates a hypothetical maximum sustainable yield (B_{MSY}) which primarily benefits commercial fishers. The battle over the establishment of additional marine reserves in the Gulf has added to this underlying conflict and

provided the recently mobilised amateur fishing sector with a targeted outlet for its frustration.

Case Study 6: Kaipara Harbour commercial fisheries

This case study examines how local conflicts between commercial fishers in the Kaipara Harbour have arisen and how they have been responded to. The conflict has revolved around the depletion of fish stocks within the Kaipara Harbour, poor fishing practices and competition for the harbour's fish between locally-based commercial fishers and mobile 'outsiders' who can trailer their boats to the harbour.

The case study highlights the significant difficulties experienced in attempting to address local fisheries-management issues within the current quota management system, which provides little room for local management or local solutions. It also indicates that unresolved conflicts can absorb considerable energy whilst more insidious impacts on fisheries (such as sedimentation) which would require a more integrated and collaborative management approach go largely unaddressed.

Case Study 7: Kaipara Harbour aquaculture

How the ARC and the Northland Regional Council (NRC) approached planning for aquaculture in the Kaipara Harbour, and the challenges that they faced, is the subject of this case study. Ultimately both councils abandoned attempts to incorporate aquaculture management areas (AMAs) into their regional coastal plans through council-led plan changes. This was due to a combination of factors including public opposition, a perception that the councils were promoting the aquaculture industry, a lack of ecological information, council political changes and negative decisions on resource consent applications.

The case study provides interesting insights into the difficulties of planning spatially for individual marine uses in the absence of a broader strategic framework. It also highlights the pitfalls which can be encountered when assessing potential activities on a case-by-case basis in the absence of a broader understanding of the marine system. It identifies the need for a strategic framework and a long-term research and monitoring programme to assist and guide future decision making.

Case Study 8: Integrating management of the Hauraki Gulf

Several initiatives which have been undertaken to better integrate the management of all or part of the Hauraki Gulf are reviewed in this case study, including the Hauraki Gulf Forum, local area management strategies, coastal compartment plans and Blueprint Coromandel. The Hauraki Gulf Forum represents a move

away from attempting to achieve integration through the establishment of a single management body, towards developing stronger relationships between a multiplicity of existing management bodies. Although this is a less well-defined and messier approach, it appears to be slowly advancing integrated management through improved information flows and the development of collaborative projects.

The case study illustrates the dilemma over whether to seek to integrate planning at a broad strategic level or at a more local level. Local planning, such as through the development of coastal compartment plans, can address specific issues of direct concern to residents and achieve an observable difference in the short term. Such plans are weak, however, in addressing the larger, longer-term issues and are difficult to scale up to a regional level. Broader strategic planning, as being attempted in the Blueprint Coromandel project, faces the multiple challenges of demonstrating relevance to local communities, of achieving change on the ground and of maintaining political interest and support over the longer time frame required to complete the planning process.

Case Study 9: Integrating management of the Kaipara Harbour

The need for integration in relation to the Kaipara Harbour has only recently reached the political agenda. This case study examines two recent initiatives to promote more integrated management of the Kaipara Harbour, one led by a regional council and one led by iwi. They are both in the early stages so it is difficult to evaluate their impact.

The case study illustrates the tensions between the need for integration and the difficulties of getting fully-integrated projects off the ground because of their large size and complexity and the resultant need for significant funding. It also highlights the difficulties that iwi face in exercising leadership in coastal management when the bulk of the management resources are held by statutory bodies which may be reluctant to give up control.

Approaches to achieving better integration

Efforts to achieve better integration within the Hauraki Gulf and Kaipara Harbour have focused around improving communication and information sharing, developing joint policies and plans and engaging in joint projects to address shared problems. A major issue for integrative initiatives is the lack of a statutory framework within which to ground them. Experiences with these initiatives highlight the tension between the increased complexity which results from integration and the need to ensure that an initiative is 'doable'. A doable project is one where the information needs are modest, where the number of

stakeholders is manageable, and where the time period required is short enough to enable political and stakeholder interest to be maintained.

In the future, integrative efforts are likely to focus on localised hotspots, where interagency collaboration is required to address issues of immediate mutual concern, rather than on large-scale integrated planning efforts where the outcomes are uncertain and politically more risky.

Supporting integrated coastal management

The report identified eight elements which may help support integrated coastal management. The extent to which these elements are currently in place and how they might be supported is explored in the following sections.

Broad agency alignment

Different coastal agencies are focused on managing different things, utilising different tools to achieve different objectives. At times these objectives and/or methods can be in direct conflict, such as when the establishment of marine reserves by the Minister of Conservation reduces the ability to exercise rights to extract fish issued by the Minister of Fisheries. At other times it can result in an organisational cultural divide, such as that between the relatively narrow private-property-right orientation of fisheries management and the more holistic environmental impact assessment (EIA) orientation of RMA marine management. Other organisational cultural differences are evident in the way management efforts are undertaken, with a participatory culture now well embedded in local councils' coastal planning efforts but not yet applied to nearly the same extent in marine protection and fisheries management. Such conflicts and divides can make integrated management efforts difficult.

More recently the divergence between the different agencies appears to have narrowed somewhat. In conjunction with MFish, DoC is embarking on a process to identify marine protected areas which will take into account, upfront, other users of the marine area. The Ministry of Fisheries is starting to incorporate some of the environmental effects of fishing activity into its decision-making processes, although still shying away from a full EIA approach. Both DoC and MFish are now seeking greater stakeholder involvement in decision making through their marine-protection and fisheries-planning exercises.

Inter-agency dependency

The extent to which a management agency can achieve its goals independently, rather than being reliant on other agencies to succeed, can have a major impact on the degree of collaboration which occurs between agencies. Collaboration can bring with it a loss of control and therefore increased risk. Managers of

coastal agencies are therefore unlikely to embark on collaborative efforts unless they have no choice, or if the countervailing benefits are seen as substantive. The current relationships between the different coastal management agencies can be summarised as follows:

- Relationships between regional councils and territorial authorities have in the past been problematic, but there is evidence of more positive relationships now developing.
- There is a strong overlap between the roles of regional councils and DoC in coastal marine areas under the RMA. This statutory interdependency has not, however, always resulted in positive relationships developing between agency staff, largely due to different philosophical approaches.
- There is a strong ecological linkage between the roles of regional councils and MFish in marine management, but little practical or statutory interdependency between the agencies.
- Despite the marked philosophical differences between MFish and DoC, the two agencies are now working much more closely together on marine issues.
- There is a very unequal relationship between iwi and other coastal management agencies. Iwi often lack the resources to undertake significant resource management activities and feel unable to influence important coastal management decision making. There have been some recent positive initiatives which are improving this situation.

Climate of trust

The interviews identified strong levels of mistrust in some areas, with growing trust relationships being developed in others. The highest levels of mistrust were expressed by stakeholders involved in the marine reserve activities undertaken by DoC and in fisheries management undertaken by MFish. This may be the result of poorer stakeholder participation and conflict-resolution processes in these two areas than provided for in RMA processes.

On the other hand, as already discussed, positive relationships are being developed between regional councils and some district councils, as well as between regional councils themselves. This should help facilitate moving towards a more integrated management approach for RMA functions. Positive relationships are also being developed between some councils and tangata whenua.

Compatible scales

In order for integration to occur, it is important that management agencies are operating at spatial scales which are able to link together. Spatial planning enables locally-specific issues to be addressed and provides an interface where different coastal management regimes can interact and can complement each other.

Spatial planning is an important part of coastal management under the RMA, through the preparation of regional and district plans. Although these plans have largely divided planning for the land and marine areas, there is the potential for integration across MHWS as well as across coastal systems.

Fisheries management has moved away from the spatial planning approach evident in the fisheries management plans of the 1980s, in which all fisheries management issues within a spatial region were considered, to a single species management model. The spatial scale of the fisheries areas managed by MFish is much larger than the spatial scale of coastal management under the RMA or marine protection management undertaken by DoC. This makes it difficult for fisheries management to interface with these other management regimes.

Decentralised decision making

Decentralised decision making enables a degree of local autonomy and this in turn enables local partnership building and innovation. Decision making under the RMA is heavily decentralized, with councils having autonomy within the broad framework of the Act. As a result, one might expect to see high levels of innovation and this was evident in some of the councils involved in the study.

The central government departments involved in coastal management have different degrees of decentralisation. Although DoC is a very large organisation, it is relatively decentralised and organised around semi-autonomous conservancies. The Ministry of Fisheries is a much smaller and more highly centralised government agency. The Ministry's decision making on some important regional issues has been based in Wellington. This large spatial separation, between MFish decision makers who are based in the capital, and the decision makers in other coastal management agencies which are based regionally or locally, makes effective integration less likely. It can also mean that local innovations are unwelcomed by the Ministry which takes a national rather than regional view because of the perceived precedent effect on other areas. In contrast, other coastal management agencies appeared more comfortable with trialling different approaches for different areas and issues.

Strategic approach

Integrated coastal management requires a strategic focus so that the myriad activities that impact on a single coastal system can be guided towards a common long-term goal. Conversely, a strategic approach requires integration because, for example, proactive forward-looking planning of marine areas requires consideration of what is happening in other sectors and on the adjacent land.

Several of the council planners interviewed identified the lack of a strategic focus in current planning efforts as a major difficulty, because of the uncertainty this created both for applicants wanting to carry out activities and for communities, in terms of the impacts on their places and resources. It also made it difficult to plan for infrastructure provision and to address the cumulative effects of activities. Most of the agencies involved in the study are moving towards a more strategic approach. The Ministry of Fisheries is developing fisheries plans with the intention of placing more emphasis on objectives-based management. The Department of Conservation is moving away from the ad hoc establishment of marine reserves to the development of marine protection plans. District councils are developing coastal and growth management strategies.

Good information base

Information plays an essential role in coastal management. However, it can be expensive to obtain and there are large gaps in the current knowledge of the state of the coastal environment and coastal processes. Very little monitoring is undertaken of the state of the marine environment. Both the ARC and EW are developing monitoring regimes for benthic ecology but these currently have limited coverage. The monitoring which does occur differs between councils and there is no common approach or methodology. Most of the information gathered by MFish relates to fish stocks and not the marine environment within which they are located. The monitoring of marine reserves by DoC is not well-funded and has not been undertaken in a consistent or comparable manner, although the Department is developing a new nature heritage management system to address this.

Adequate resourcing

Coastal users do not currently pay the full cost of their use of public resources and the public is reluctant to pay for coastal management. As a result, coastal management is poorly funded. The Ministry of Fisheries adopts a user-pays approach to the costs of managing fish stocks. It has resulted in relatively low management costs but may have resulted in commercial fishing interests having too strong a say in management issues.

Conclusions and recommendations

The coast is extraordinarily important in terms of New Zealand's natural and cultural heritage, quality of life and economy. At the same time, the coast is a high conflict zone, where there are many competing uses and where sensitive ecological systems are under growing pressure from human impacts. This study has demonstrated that effective management of the coastal environment at a regional level has proved difficult to achieve.

New Zealand's current coastal management system is fragmented and suffers from inconsistent objectives and management approaches. The legislative framework has developed in an ad hoc manner and without a clear vision of how integrated coastal management might be delivered at a regional level. There is a wide disparity in the extent to which key elements of a successful integrated coastal management system are being implemented in the various management spheres. Such elements include strategic planning, a spatial orientation, decentralised decision making, participative processes and effective dispute resolution procedures. The levels of information and resourcing to support effective coastal management are insufficient. There is no legislative framework to support integrative initiatives when they do occur.

There is clearly a need to step through the complexities of the current administration arrangements to develop a more strategic and integrated approach. This does not require a grand national oceans plan to overlay the other multiplicity of planning activity already underway. What it does require is a review and reform of the key pieces of legislation impacting on coastal management including the RMA, the Fisheries Act, the Marine Reserves Act and the Biosecurity Act to ensure that a fully integrated and effective coastal management system can be practically delivered at the regional level.

Much can be also done to support greater integration between New Zealand's coastal management efforts prior to such a legislative reform program. The recommendations below identify some ways in which this could be achieved.

Summary of recommendations

This study has identified some serious shortcomings in the current governance framework and the extent to which it can deliver effective and integrated coastal management at a regional level. It is therefore recommended that:

- The Minister in charge of oceans policy, as a matter of urgency, initiates a comprehensive review of legislation impacting on coastal management, with the objective of implementing a programme of legislative reform to ensure that effective and integrated coastal management can be delivered at a regional level. (**Recommendation 1**)

Pending the implementation of the results of such a review, a number of recommendations have been developed which identify ways in which integrated coastal management could be strengthened at a regional level. It is recommended that:

- Broad alignment between coastal management agencies be encouraged and supported through the identification of core common elements which are required to underpin all coastal management efforts. These could include commonalities in the objectives sought to be achieved, such as integrated ecosystems-based management, as well as the approaches and processes which are used to reach the objectives. The core common elements could be incorporated into a formal memorandum of understanding between the agencies in the short-term but would need to be embedded into legislation in the longer term. **(Recommendation 2)**
- A clearer statutory relationship between MFish and regional councils be created so that they are jointly charged with managing the ecological health of the marine environment and have defined roles in each other's decision-making processes. Pending any necessary legislative changes, formal and regular consultative engagements should take place. **(Recommendation 3)**
- A clearer statutory basis be developed for the role of tangata whenua in coastal management. **(Recommendation 4)**
- Strengthened statutory procedural requirements be developed for stakeholder engagement and broader public participation in decisions affecting the coastal environment, which are to be applied to all areas of coastal management as a minimum standard, accompanied by a best-practice guideline. **(Recommendation 5)**
- Additional stakeholder capacity-building funding be made available, particularly in relation to stakeholder engagement in fisheries management and marine protection. **(Recommendation 6)**
- Effective conflict-resolution procedures, including mediation and an independent review of the merits of the decision, be incorporated into the Fisheries Act 1996 and the Marine Reserves Act 1971. **(Recommendation 7)**
- Statutory regional spatial planning be explicitly provided for in marine protection and fisheries management, and that a stronger interface between plans prepared under the different marine management systems be defined. **(Recommendation 8)**
- Those elements of MFish decision making which are of regional significance to regional offices be decentralised, to foster greater regional

diversity in approaches to fisheries management, to enable greater integration with other coastal management agencies, to enable regional and local circumstances to be addressed, and to promote innovation and richer learning opportunities. **(Recommendation 9)**

- The move towards strategic planning for the coast be encouraged through the provision of legislative support. This could include providing for a strategic element in RMA Plans, providing a more detailed strategic framework for fisheries plans in the Fisheries Act and providing a legislative framework for marine-protection plans in new marine-protection legislation. **(Recommendation 10)**
- A comprehensive monitoring framework for the coastal environment, which promotes better consistency and integration between existing programmes and helps progressively fill key knowledge gaps, be developed and resourced. **(Recommendation 11)**
- Additional funding sources be developed to support coastal management, including an effective and implementable coastal user-charging system, while ensuring that those paying user charges do not directly influence how they are spent. **(Recommendation 12)**



one

Introduction

My vision is real clear. I want to see my grandchildren and their children swimming, fishing, living and enjoying a Hauraki Gulf of at least the quality that I do, if not better. That's why I do this stuff.
(respondent 48)

Why am I here? So we can feed all our children. That's what I'm here for, the future generations ...
(Wiremu Heihei, Hokianga Accord, 2006, 56)

The Beginnings

Some of my earliest childhood memories are of rowing my father out to sea in a small wooden dinghy off Omana Beach, just as dawn was breaking, and putting a line overboard to catch a fresh snapper for breakfast. The snapper would be sizzled over a small stove in the kitchen of our family bach, a simple fibrolite structure built in the late 1950s by my father and grandfather. There were few modern conveniences. Rather brownish water came from the roof and was stored in a rusty corrugated iron tank. The bathroom consisted of a small and often smelly long-drop that had been dug out the back of the property and was inhabited by aggressive blowflies.

This bach experience was followed in my later childhood by memorable summers exploring the Hauraki Gulf and beyond in small boats. Frequently we would row ashore laden with food, cooking equipment and bedding, build a roaring camp fire to cook dinner and then snuggle into our sleeping bags to watch the stars as the fire died down and we drifted off to sleep. It never occurred to me at the time that we might be trespassing on someone else's land. The coast, including the extensive farmland that bordered the sea, was a place to be experienced and explored, there to be enjoyed by all.

I remember a sea which was full of life. We would sail through schools of splashing fish being dive-bombed by screaming gulls, drop a lure off the back of the boat and catch dinner. Playful dolphins surfed in our bow wave. Sunfish lazily cruised across our path. Snorkelling off rocky reefs would reveal forests of kelp, crayfish hiding under rocks, octopuses lying in wait and shoals of small reef fish.

In those days, during the late 1960s and early 1970s, overnight boating was undertaken by hardy souls, those who set out to sea in small, homebuilt craft with very rudimentary equipment. 'Boaties' were kindred spirits, with a common bond which transcended disparities between their lives back on dry land. As those long hot summer days drew to a close, and boats nestled into safe anchorages, stories of triumphs and narrow escapes, words of advice and caution, and many bottles of beer were shared over communal beach camp fires at 'home base'.

But life moves on. I finished school, left home and sought to carve out an adult life for myself. Rogernomics came, and the safe but over-prescribed and somewhat dull society I grew up in changed for ever. Private interests and private enterprise became ascendant, public interest and communality beat a retreat. As with all substantive changes there are winners and losers. I was one of the lucky ones. As a young professional I was able to benefit from rising incomes and

property values and enjoy the increased diversity of our more outward-looking nation. But what did all this mean for the coast and the coastal experiences that I had so loved and enjoyed as a child?

Like many others of my baby boomer generation I spent some years overseas. Now married, I returned with a young daughter and an eagerness to introduce her to the kind of coastal outdoor experiences that I had enjoyed as a child. We looked for a bach to buy within one hour's driving distance of Auckland. Omana Beach, the location of my family bach now long sold, had been swallowed up by the fast-expanding Auckland metropolitan area. The old baches had been replaced by large modern homes, cafes had sprung up along the beachfront, and there was now a service station. Luckily, we did find what we were looking for at Orere Point. It was only slightly further away and still a traditional bach settlement with a camping ground, a small dairy and no cafes.

We were fortunate that my father still owned a yacht so we were able to revisit some of my childhood haunts in the Hauraki Gulf and further afield. But somehow it was not quite the same. The sea seemed strangely opaque, empty and silent – bereft of once abundant fish and birdlife. Snorkelling along the rocky headlands revealed barren rocks stripped of vegetation and dotted with armies of kina (sea eggs). Many coastal farms had been subdivided and were dotted with large 'in your face' houses perched on prominent ridges. Anchorages were crammed full with large, gleaming 'gin palaces' and yachts bristling with the latest in electronic gadgetry and boating equipment. The groups of convivial boaties communing around campfires had given way to private onboard cocktails and socialising. It was no longer possible to dive over the side for a refreshing swim due to the emptying of numerous boat toilets into the bay.

Even a visit to our bach at Orere Point proved to be a disquieting experience. When we drove through the small bustling rural township of Clevedon, there were protest signs opposing a canal development on the nearby estuary. As the road hit the coast and meandered along the tidal embayment which forms Kawakawa Bay, signs erected on the foreshore announced 'Warning! Polluted water. Do not swim. Do not take shellfish.' After every downpour, we could see a wide plume of brown sediment-laden water pouring out of the Orere Stream into the Firth of Thames. Back in Auckland my father's berth at Westpark Marina was rapidly filling up with silt, necessitating dredging work of such expense that many berthholders were simply walking away from their leases.

So what was happening and why? It was the desire to answer this question, and in doing so to contribute to the preservation of our coastal heritage, that led me to undertake this research project on coastal management.

Those that helped

I would like to acknowledge all the people who generously agreed to be interviewed for this project and therefore made it possible. Although I fear that I have not done justice to the wealth of wisdom which was imparted to me, I hope that I have captured sufficient to make your time worthwhile. Keir Volkerling helped facilitate interviews with iwi representatives and provided comments on the draft report. Charles Crothers and Gary Taylor also provided valuable comments on the first draft. Margaret Peart helped improve my written expression. I would like to thank the many others, including interviewees and anonymous reviewers, who also provided feedback on the draft report.

I also wish to acknowledge the Auckland Regional Council (ARC), Environment Waikato (EW) and the Thames-Coromandel District Council for generously providing financial assistance for the project.

Background

This study had its beginnings in 2003, when the Environmental Defence Society (EDS) formed a partnership with USA-based Conservation International to review the New Zealand Government's then attempt to develop an oceans policy. EDS saw the government's initiative as an excellent opportunity to establish a world-leading oceans governance structure designed for the twenty-first century. The government project was unfortunately stalled in June 2003 by a fracas over the ownership of the seabed and foreshore. Ministerial responsibility for the policy was subsequently shifted and, although work on developing the policy has recommenced, it has yet to re-establish significant momentum.

The EDS-Conservation International study reviewed the international thinking behind national oceans policies and their development in Australia, Canada, the United States and New Zealand. It then focused specifically on the New Zealand situation, assessing the adequacy of our current ocean governance structure within the context of international best practice and local environmental issues. EDS published the results of this work in 2005 in a report titled *Looking out to sea: New Zealand as a model for ocean governance*.

When the EDS-Conservation International project was initially conceptualised, it was always envisaged that a subsequent stage of the study would examine how a national oceans policy could best be implemented at a sub-national (regional) level. Financial support provided by the ARC and the Thames-Coromandel District Council enabled case studies of the management of the Hauraki Gulf and the Kaipara Harbour to be undertaken. Financial

support from EW enabled them to be published. This report presents the findings of these studies.

The case study areas

The selection of the Hauraki Gulf and the Kaipara Harbour as case study areas was initially based on the complexity of their governance structures. Both are partially enclosed marine systems bisected by regional council boundaries as well as being managed by multiple territorial authorities and conservancy offices of the Department of Conservation (DoC). The two areas provide an interesting contrast in terms of environmental pressures, with the Hauraki Gulf being adjacent to the country's largest metropolitan area and the Kaipara Harbour catchments being largely rural. As several management bodies have jurisdiction over parts of both the Hauraki Gulf and the Kaipara Harbour, the study has also provided an interesting contrast between how each organisation has approached the management of the two different areas. The Hauraki Gulf and the Kaipara Harbour both have high marine ecological values which are being negatively impacted upon by a range of human activities.

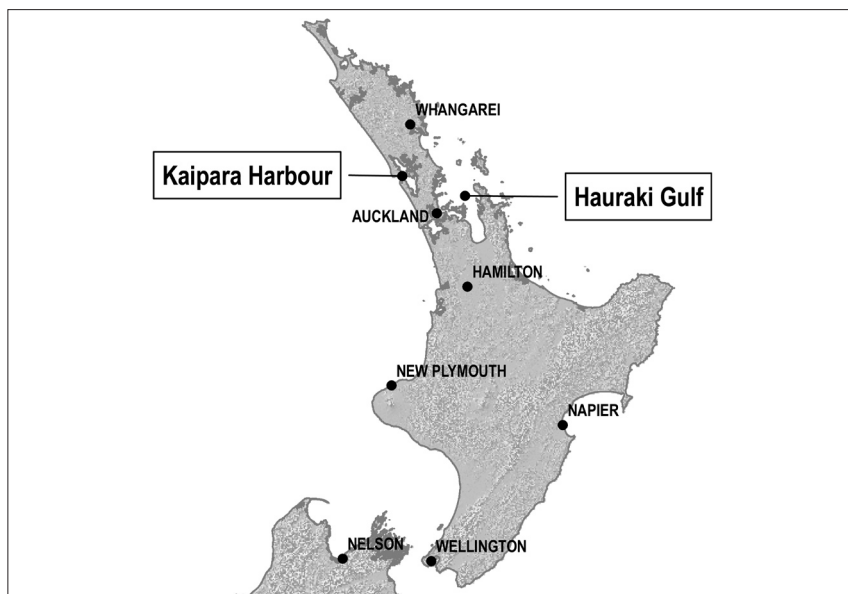


Figure 1.1: Location of the Hauraki Gulf and the Kaipara Harbour Source: ARC



Figure 1.2: Home Bay, Motutapu Island, Hauraki Gulf

There are fewer and fewer places in the Auckland region where you can go and feel that you've escaped. (respondent 1)

The Hauraki Gulf is located on the east coast of the northern part of the North Island. As defined under the Hauraki Gulf Marine Park Act 2001, it incorporates the eastern catchments of the Auckland and Waikato regions and extends over the marine area to the edge of the territorial sea (see Figure 1.3)¹. The Gulf covers approximately 13,900 square kilometres and has a coastline which is about 2,550 kilometres long.

The Hauraki Gulf has a wide variety of marine habitats including rocky reefs, sandy beaches, dune systems, enclosed harbours and estuaries. These support a rich diversity of marine species. The Gulf is a particularly important spawning and nursery area for some species of fish including snapper. Schools of common dolphins and several species of whales frequent the Gulf. The Gulf supports a wide range of birds and includes a wetland of international significance on the edge of the Firth of Thames where 132 species of birds have

¹ The material in this section was largely sourced from the Hauraki Gulf State of the Environment Report prepared by the Hauraki Gulf Forum in 2004.

been recorded. Several islands in the Gulf are pest free and provide important habitat for threatened indigenous terrestrial species.

The Hauraki Gulf was one of the earliest places to be settled in New Zealand and many well-known waka passed through the Gulf. The area supported extensive Māori settlements, which is reflected in the large number of pā sites still visible today. Various iwi have occupied the Gulf at different times due to ongoing warfare and migration.

European settlement resulted in significant landscape changes through the felling of most of the remaining forests and the draining of extensive wetlands. Rapid urbanisation followed, with the country's largest urban area developing within the catchments of the Gulf. Extensive areas of the catchment were also converted to farmland or planted in exotic forests. More recently the Gulf's catchments have experienced high levels of population growth resulting in urban intensification and increasing sprawl, with rural-residential development locating on the urban fringes. The Gulf contains the country's busiest port, a large number of recreational boats and boating facilities and many marine farms.

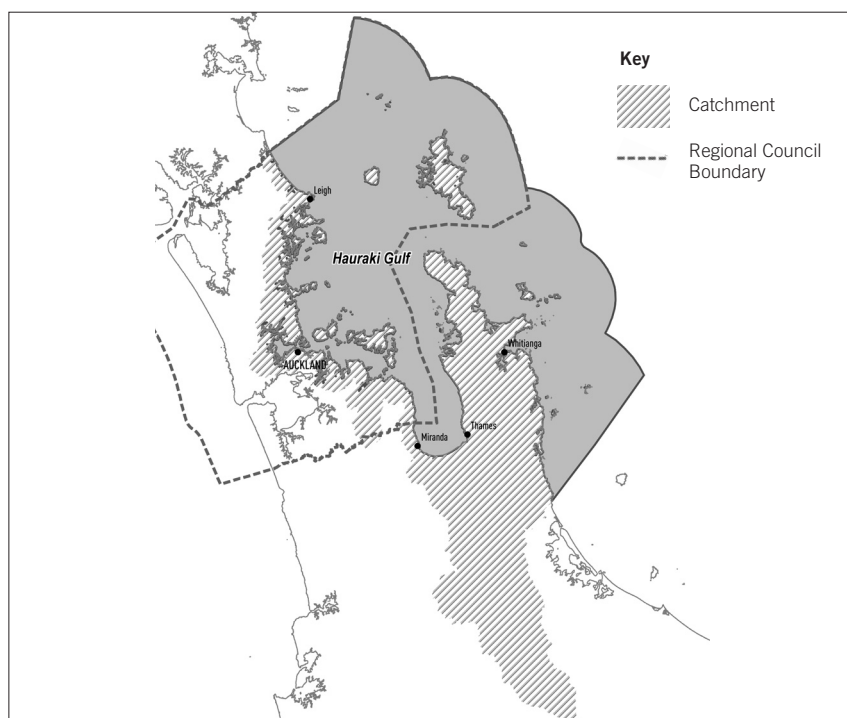


Figure 1.3: Boundaries of the Hauraki Gulf Source: ARC

The marine area of the Hauraki Gulf has been heavily impacted on by urban and rural development as well as marine-based activities. Key environmental issues for the marine area include accelerated rates of sedimentation, the accumulation of contaminants in seabed sediments, the overharvesting of wild seafood, the impacts of invasive marine species and the loss of natural character.

High levels of sedimentation in the estuaries and harbours of the Gulf are resulting in degraded and less diverse ecological systems. Activities which are contributing to the problem include earthworks, forestry and grazing activities. The contamination of harbour sediments is increasing in many urbanised catchments, with cars and unpainted iron roofs being major sources of zinc.

Several habitats of particular importance to fish species have declined in the Gulf. These include extensive mussel beds which were destroyed by dredging activities, intertidal seagrass beds which have probably declined due to increased sedimentation and sponge gardens for which the reason for decline is not well understood. 'Urchin barrens', where sea urchins have demolished kelp forests have appeared on rocky reefs, due to the heavy fishing of predator species such as snapper and crayfish. Much damage was caused in the early days of trawling:

The Hauraki Gulf out here, now it's sort of like a bowling green whereas before trawling there would have been little pockets of what we call oasis communities where over thousands of years probably there's built up populations of sponges and black coral and all sorts of high relief stuff like that which is very important for fish to live in ... In the early days of trawling they were pulling up black coral trees all the time, sponges and just flattening the bottom ... one of the things they used to do back in the 1950s, if they were working a new area, they would stretch a chain between two boats and just drag it across the bottom and deliberately flatten everything in the way so they could then bring the trawlers along to have a clean sweep of the bottom.

(respondent 55)

Wild seafood within the Gulf has been heavily exploited. Many of the popularly targeted fish species such as snapper, kahawai and flounder were heavily fished during the 1970s and 1980s which led to stock levels dropping to low levels. Populations are now thought to be slowly rebuilding. Shellfish such as cockle and pipi appear to still be in decline which may be due to a combination of sedimentation, contamination and overharvesting.

Being the location of New Zealand's busiest port, the Gulf is a high-risk area for the arrival of invasive marine species and many exotic species have established in the area. The Asian date mussel, the Pacific oyster and an introduced parchment worm and file shell are of particular risk to indigenous species.

The Hauraki Gulf is managed by a large number of agencies including the Ministry of Fisheries (MFish), Biosecurity New Zealand, three conservancies of DoC, two regional councils and ten territorial authorities.

The Kaipara Harbour



Figure 1.4: Arapaoa River, Kaipara Harbour

The Kaipara has been a forgotten harbour and that's been part of its saving. But now, because of the low cost of land or because of opportunities in terms of our culture, they think they might be able to do things in it. It just seems to be a great area to exploit and not to have to worry about costs to any great extent. (respondent 16)

The Kaipara Harbour, which is located on the west coast of the northern part of the North Island, is New Zealand's largest enclosed harbour and estuarine system. It consists of a system of drowned river valleys enclosed by two long sand barriers (North and South Heads) at the harbour mouth. The harbour extends over 947 square kilometres, contains many peninsulas, and is fringed by 612 kilometres of coastal edge. It is very tidal, with around 409 square kilometres of the harbour exposed at low tide. The harbour mouth is seven kilometres across and an enormous two billion cubic metres of water is exchanged during each tidal flow.

This harbour is ecologically very important. It contains extensive mangrove forests, eelgrass beds and salt marshes. These provide important rearing grounds for juvenile fish, including many commercially harvested species. Many areas of the sea floor, particularly in the southern part of the harbour, support high levels of biodiversity. The harbour contains a number of species, including sponges

and bryozoans, that are commonly associated with pristine environments. Several species of invasive shellfish are present in the harbour including the Pacific Oyster and Asian Date Mussel, but not in as large numbers as in the Hauraki Gulf.

The harbour is internationally and nationally important as a major feeding, roosting and breeding area for coastal and estuarine wading birds, including godwits, the critically endangered New Zealand fairy tern and the endangered New Zealand dotterel. It is used by common dolphins, the critically endangered Maui's dolphin and orca whales. South Head is a fur seal haul-out area (Kaipara District Council 2005, 6-6).

The land surrounding the harbour is relatively low lying, with hills on the South Head Peninsula rising to about 100 metres. Although most of the original forest cover around the harbour has been removed, many of the remnants are unique to the area. They contain an unusual mix of canopy species such as kowhai, kauri, mangero and matai often without pohutukawa. They support significant populations of tui and pigeon along with fantail, kingfisher, morepork and silvereye. North Kaipara Head contains rare indigenous dune field vegetation.

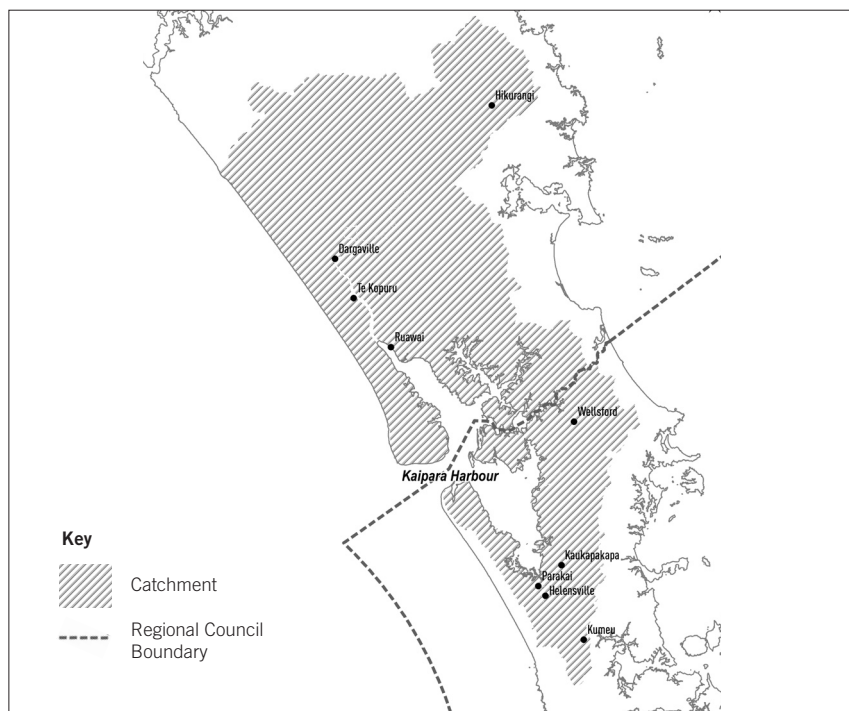


Figure 1.5: Map of the Kaipara Harbour Source: ARC

The Kaipara Harbour provided a very attractive environment for Māori settlement, with its abundant fish and shellfish, numerous streams and swamps providing eels and wild fowl, extensive forests and well-drained fertile flats which were particularly suited to growing kumara. It contains numerous pā, terraces, pits and middens (few of which are recorded), indicating that the area was long occupied and frequently fought over (Waitangi Tribunal, 2006, 13).

During Māori occupation significant forest areas, around the harbour shoreline and along important routes between Ōtamatea and Mangawhai, and the Kaipara River and the upper Waitemata Harbour, were cleared. At the beginning of the nineteenth century the northern part of the harbour was largely occupied by Te Uri o Hau and the southern part by people of Te Taoū, Ngāti Whātua Tūturu, Ngāti Rongo and other related groups, often referred to collectively as Ngāti Whātua (Waitangi Tribunal, 2006, 13).

The difficult harbour entrance slowed European settlement and it was not until the late 1830s that any Pākeha communities were established. This was mainly in the north, around the Wairoa River, because of the accessible kauri timber resources. By the 1890s the present pattern of towns and rural settlements had been established with a stable community of Pākeha farmers and small service towns. From 1880 to 1910 the Kaipara Harbour was the largest single timber export port in New Zealand but, with the demise of the timber industry, the port was closed in 1947 (Ryburn, 1999; Waitangi Tribunal, 2006, 22, 26, 35 - 40). During the twentieth century the Kaipara district developed rapidly as a dairy farming region.



Figure 1.6: Land for sale signs along Oneriri Road, Kaipara Harbour

There are major settlements to the north of the harbour (Dargaville), to the east (Wellsford) and to the south (Helensville). A tourist boat trip operates between Helensville and Dargaville from time to time. Most land is used for primary production activities including exotic forests in the south, intensive dairying on

the Dargaville-Ruawai flats and sheep and cattle grazing as well as horticulture elsewhere.

Residential subdivision and development are occurring outside urban areas in places along South Head, on the eastern shore of the harbour, and on the Pahi Peninsula.

Farming has probably had the biggest impact on the ecology of the harbour through increased sedimentation and nutrient inflow. Much of the coastal margin is in private ownership including 70 per cent of the coastline within the Kaipara District (Kaipara District Council, 2005, 6-9). Livestock graze to the waters' edge in many places.

Increasing urban development is resulting in increased sedimentation from earthworks and vegetation clearance, increased discharges of contaminated runoff and the loss of natural landscapes.

Marine-based activities are also having an impact. The harbour fisheries are becoming depleted, with popular commercial- and recreational-targeted species such as flounder, mullet and rig appearing to be in decline. Sand extraction may have resulted in increased coastal erosion and the decline of some tuatua beds.

Although fewer than is the case with the Hauraki Gulf, the Kaipara Harbour is managed by numerous statutory agencies. These include MFish, Biosecurity New Zealand, two conservancies of DoC, two regional councils and two territorial authorities. Two main iwi, Te Uri o Hau and Ngati Whatua, exercise kaitiakitanga over the harbour.

The Methodology

The focus of the study is on the marine area and the management of activities, both land- and water-based, which impact on the marine environment. Rather than comprising an encyclopaedic review of all environmental and management information pertaining to the two areas, or a stocktake of the state of the environment, this study has focused on providing an understanding of how environmental managers and other stakeholders see the environmental management issues, how they have approached their tasks, what they have learned from what they have done, and how they see future challenges.

The material for the study has been gathered through undertaking face-to-face interviews with sixty environmental managers and stakeholders. Potential interviewees were identified through the researcher's knowledge of people working in the field and through recommendations of key contacts. When selecting the interviewees, endeavours were made to cover a range of managers

and stakeholders in both the Hauraki Gulf and the Kaipara Harbour. A breakdown of the categories of people interviewed is set out in Figure 1.7. This indicates that the largest number of interviewees came from regional councils. This was primarily because regional councils have the broadest mandate for coastal management and are involved in a wide range of activities related to catchment and marine management.

The interviews followed a semi-structured pre-prepared questionnaire which was formulated around a series of key issues:

- The interviewee's background and experience in marine issues
- The role of the interviewee's organisation in marine management
- What the organisation was trying to achieve in respect of marine management in the case study areas
- To what extent the organisation's goals had been achieved
- What the interviewee saw as the key issues in managing the case study areas
- How effective management efforts have been in addressing these issues
- What the main barriers have been to effectively addressing the issues
- Examples of where management efforts had worked particularly well or poorly
- The nature of the organisation's relationship with other organisations and individuals involved in marine environmental issues in the case study areas

In practice, however, the interviews followed areas of the interviewee's expertise and insight rather than being confined to specific questions. Each interviewee was given an information sheet describing the project and asked to complete a consent form before the interview commenced. Interviews typically took around one and a half hours. They were recorded and subsequently transcribed. The transcriptions were then reviewed and relevant material from each interview combined into a master document ordered by key topics areas. This master document was used as a basis for writing the report. Direct quotes which were selected were subsequently sent back to the individual interviewees who were given the opportunity of editing their quotes to more accurately reflect their viewpoints.

As a result of the methodology adopted, the material presented in this report is largely based on interviewees' perceptions and recollection of events rather than verified factual information. However, where information could be readily verified or supplemented by written documentation, such as planning documents or reviews, this was been undertaken. The study is therefore founded on the premise that the perceptions of environmental managers and stakeholders

are a valid basis for analysis, because their subsequent actions are significantly influenced by their perceptions of issues and of other stakeholders.

Figure 1.7: Sectoral breakdown of interviewees

Sector	No. people interviewed
Regional councillors	4
Regional council staff	19
Territorial authority staff	9
Department of Conservation staff	5
Ministry of Fisheries staff	4
Biosecurity New Zealand staff	1
Iwi	4
Environmental/community NGOs	5
Recreational fishing NGOs	2
Scientists/consultants	4
Private sector	3
TOTAL	60

Structure of Report

Part One of the report focuses on coastal management in practice. It describes how the concept of integrated coastal management has evolved, both internationally and within New Zealand (chapter two). It investigates the conflicts which underlie New Zealand's coastal management system and which make integration problematic (chapter three). Part One then goes on to summarise the findings of the research into management within the Hauraki Gulf and the Kaipara Harbour (chapter four). This part of the report concludes with an analysis of how integrated coastal management could be more effectively supported within New Zealand (chapter five).

Part Two of the report presents in more detail the material gathered through the interviews in the form of nine case studies focused on specific issues relevant to the management of the Hauraki Gulf and Kaipara Harbour. These case studies cover a broad range of management issues including:

- Coastal planning (chapter six)

- Coastal development (chapter seven)
- Sedimentation (chapter eight)
- Marine biosecurity (chapter nine)
- Fisheries (chapters ten and eleven)
- Aquaculture (chapter twelve)
- Integrating management (chapters thirteen and fourteen)

Some of the case studies (including coastal planning, coastal development and sedimentation) have included management within both the Hauraki Gulf and Kaipara Harbour to provide a broader basis for comparison of different approaches. Other case studies have focused on specific management issues within one area because they provided a particularly interesting insight into management challenges (including Hauraki Gulf amateur fishers, Kaipara Harbour commercial fishers and Kaipara Harbour aquaculture). Two case studies focus directly on the integration challenge for each of the case study areas and efforts specifically directed at integrating their management. Not all significant management issues within the case study areas are covered by the case studies because of the need to keep the study to a manageable size. In particular, significant issues which are not addressed include sand mining, power generation, marine pollution and coastal erosion.

The report concludes with a series of recommendations on how government policy could more effectively support integrated coastal management (chapter fifteen) .

PART ONE

COASTAL MANAGEMENT IN PRACTICE



two

The Evolution of Integrated Coastal Management

Coastal zone management at the beginning of the new millennium is a maelstrom of issues, conflicts, values, interests, multiple stakeholders, and institutions. (Kay et al., 2003, 225)

The concept of integrated coastal management has been evolving over the past thirty or so years, both internationally and within New Zealand. Although there is still a diverse range of views on what it might be and how one might achieve it, a series of common themes are emerging. These are summarised in the sections below.

New Zealand's institutional framework

Coastal management is largely split between eight types of statutory management agencies with overlapping responsibilities (see Figure 2.1). The Minister of Conservation has a broad role, through its Resource Management Act 1991 (RMA) responsibilities in the coastal environment, its management of marine and terrestrial reserves and its advocacy role for the conservation of natural and cultural resources more generally. Regional councils also play a broad role through their management under the RMA of both catchments and activities in the marine area. Territorial authorities have a more limited role for coastal management under the RMA: being primarily focused on land use and development. The role is significant, however, as land development can have major impacts on the coastal environment.

The Ministry of Fisheries (and the Minister) has a narrower focus on the management of fisheries, but plays a significant role in marine management because of the extent of fishing activity and the significance of the agency's impacts on the marine environment.

Biosecurity New Zealand, a recently-established division of the Ministry of Agriculture and Forestry, has a narrow focus on biosecurity issues. Marine biosecurity is a small part of its activities, which are more terrestrially focused, but capacity is gradually being built up.

Maritime New Zealand has a role in managing shipping and marine pollution and the Ministry of Energy is involved in regulating mining activities in the marine area. These two agencies were not interviewed for the study so as to keep the project within a manageable scope.

Iwi and hapū are increasingly seeking to re-establish a strong kaitiakitanga role in the coastal environment. Although such a role is not specifically enshrined in statute, there is some recognition of it in legislation such as the Fisheries Act 1996 and the RMA.

Figure 2.1: Key New Zealand coastal management agencies

Management Agency	Areas of coastal management focus	Key legislation
Minister and/or Department of Conservation	Conservation of natural and historic resources (land and marine reserves, protected indigenous species, advocacy for conservation) Coastal management under the RMA (preparation of NZCPS, approval of regional coastal plans, consent authority for restricted coastal activities); advocacy for conservation	Conservation Act 1987 Marine Reserves Act 1971 Reserves Act 1977 Marine Mammals Protection Act 1978 Wildlife Act 1952 Resource Management Act 1991
Minister and/or Ministry of Fisheries	Fisheries resources	Fisheries Act 1996
Biosecurity New Zealand	Invasive non-indigenous species	Biosecurity Act 1993
Maritime New Zealand	Shipping and marine pollution	Maritime Transport Act 1994
Ministry of Energy	Mining	Crown Minerals Act 1991 Continental Shelf Act 1964
Regional Councils	Catchment and marine management (excluding fisheries)	Resource Management Act 1991 Local Government Act 2002
Territorial authorities	Land use	Resource Management Act 1991 Local Government Act 2002
Iwi and hapū	All coastal resources	Resource Management Act 1991 Fisheries Act 1996

Evolution of international approaches to coastal management

The evolution of thinking about coastal management and its application has been conceptualised as broadly consisting of a series of stages which commenced during the late 1960s (see Figure 2.2).

By the late 1960s, coastal management was primarily focused on addressing a few specific issues of public concern, such as coastal erosion and loss of public access, arising from inappropriate development on the coastal edge (Cicin-Sain and Knecht, 1998, 32-33; Vallega 2001, 122).

The thinking around how to approach coastal management expanded considerably with the establishment in the USA of the Commission on Marine Science, Engineering, and Resources, commonly known as the ‘Stratton

Commission'. In its influential report *Our Nation and the Sea*, which was released in 1969, the Commission recommended that ocean and coastal areas be managed as 'systems' recognising the interaction and interdependencies between different parts.

Figure 2.2: Stages in the international evolution of coastal management.

(Adapted from Vallega 2001, 122)

Stage	Objective	Management focus	Geographical coverage
Late 1960s Pre-take-off	Address specific environmental issues	A few coastal activities	Coastal edge
1970s Take-off	Manage uses and protect the environment	A few coastal activities	Coastal edge and arbitrarily defined coastal marine area
1980s Drive to Maturity	Manage uses and protect the environment	All coastal activities	Coastal edge and arbitrarily defined coastal marine area
1990s Maturity	Integrate management	Coastal ecosystems	Expanded to include water catchments and the entire EEZ
2000s Consolidation	Integrate management	Coastal ecosystems	Water catchments and entire EEZ

The report led directly to the passing of the Coastal Zone Management Act in 1972, which was probably the first formal effort by a country to develop a management framework for the coast. Under the legislation, the federal government provided incentives to state governments to prepare and implement coastal management programmes. In practice, however, these programmes focused on the management of shore-based activities rather than coastal water-related issues (Cicin-Sain and Knecht 1998, 33).

Following the lead set by the USA, many other countries undertook coastal management efforts during the 1970s. These early initiatives, however, generally focused on a particular coastal region, a specific development project or an important coastal problem such as erosion. Management programmes tended to target a limited number of coastal activities, without taking into account the interaction between them or the bigger picture. The spatial focus of such efforts also tended to be limited, encompassing the intertidal area, an area delineated in an arbitrary way or one coinciding with the jurisdiction of administering local authorities (Cicin-Sain and Knecht 1998, 33, Vallega 2001, 122).

During the 1980s, the integrated pursuit of both environmental protection and economic development came increasingly to be seen as the true goal of

coastal management. There was also greater focus on the living elements of ecosystems, with growing interest in the establishment of marine parks and reserves and the protection of habitats and endangered species. The seaward extent of coastal management efforts began to expand to coincide with the 200 nautical mile national jurisdiction under the United Nations Convention on the Law of the Sea (Vallega 2001, 123).

The defining point in the evolution of coastal management approaches during the 1990s was the 1992 United National Conference on Environment and Development (UNCED) and Agenda 21 which was adopted at the conference and endorsed by the New Zealand Government. Chapter 17 of Agenda 21 addressed coastal management. The document urged countries to commit themselves to the integrated management and sustainable development of coastal areas and the marine environment under their jurisdiction. It called on all coastal nations to formulate and implement coastal management programmes by the end of the decade (Coastal Resources Centre 1991, 6).

During the 1990s and early 2000s, greater emphasis has been placed on the development of oceans policies encompassing entire marine areas within national jurisdiction. In 1996 Canada passed an Oceans Act and Australia commenced the development of an oceans policy. A key implementation tool adopted by those countries has been the development of regional marine plans which encompass large marine ecosystems crossing multi-jurisdictional boundaries. The large scale and complexity of the plans, however, has meant that they have been slow to develop and difficult to implement.

So, by the new millennium, the concept of applying an integrated and systems-based management approach to the coast was well entrenched internationally, as was acceptance of the need to include water catchments and extensive marine areas within the scope of coastal management.

Evolution of coastal management in New Zealand

The evolution of coastal management in New Zealand can be mapped onto the series of stages which have been identified internationally, as shown in Figure 2.3. This indicates a concern for better management of the coast emerging during the 1970s with the passage of legislation for the creation of marine reserves, the protection of marine mammals, the management of marine pollution, the development of the country's first coastal policy and the introduction of maritime planning through the Town and Country Planning Act 1977.

Figure 2.3: Stages in the evolution of coastal management in New Zealand.

Stage	New legislation/policy	Objective	Management focus	Geographical coverage
Late 1960s Pre-take-off	Continental Shelf Act 1964	Facilitate resource exploitation	Minerals exploitation	Continental Shelf
1970s Take-off	Marine Reserves Act 1971	Promote scientific study	Activities impacting on the marine area	Territorial sea
	Marine Farming Act 1971	Facilitate marine farming	Aquaculture	Territorial sea
	Coastal Policy 1973	Provide for active and passive recreational opportunities; preserve coastal scenery	Coastal subdivision	Coastal land
	Marine Pollution Act 1974	Prevent and manage marine pollution	Shipping and oil exploration	Territorial sea, continental shelf
	Town and Country Planning Act 1977	Conserve, protect and enhance the environment; use and manage resources wisely; contain urban development; protect natural character; maintain the relationship between Māori and their ancestral land	All land activities; some marine	Land; some territorial sea where maritime planning scheme and/or regional planning scheme
	Territorial Sea and Exclusive Economic Zone Act 1977	Exert NZ jurisdiction over marine areas Provide for exploitation and conservation of resources	Foreign fishing vessels	EEZ
	Reserves Act 1977	Preserve and manage public reserves (including ensuring public access to the coast and the preservation of the natural character of the coastal environment)	Activities impacting on reserve land	Land
	Marine Mammals Protection Act 1978	Protect marine mammals	Marine mammals	Territorial sea and EEZ
1980s Drive to maturity	Fisheries Act 1983	Manage and conserve fisheries	Fishing	Territorial sea and EEZ

1990s Maturity	Resource Management Act 1991	Promote sustainable management Integrate management	Most coastal and marine activities (excluding fishing)	Land and territorial sea
	New Zealand Coastal Policy Statement 1994	Promote sustainable management	Most coastal and marine activities (excluding fishing)	Coastal land and territorial sea
	Maritime Transport Act 1994	Manage maritime transport Protect the marine environment	Maritime transport	Territorial sea and EEZ
	Fisheries Act 1996	Provide for sustainable utilisation	Fishing	Territorial sea and EEZ
2000s: Consolidation	Hauraki Gulf Marine Park Act 2000	Integrate management Protect natural, historic and physical resources	All activities	Hauraki Gulf Marine Park (land catchments and territorial sea)
	Oceans policy initiative 2000	Ensure healthy oceans	All activities	Land, territorial sea and EEZ
	Foreshore and Seabed Act 2004	Protect the foreshore and seabed	Public access	Territorial sea
	Marine Protected Areas Policy and Implementation Plan 2005	Establish a comprehensive and representative network of marine protected areas	All marine activities	Territorial sea and EEZ
	Review of NZCPS 2006	Promote sustainable management	Most coastal and marine activities (excluding fishing)	Coastal land and territorial sea

Between the late 1920s and early 1960s the government took a fairly conservationist stance towards the management of fisheries, controlling fishing effort through a system of licensing, area closures and gear restrictions. This changed significantly in 1963 when the government aggressively sought to expand the commercial fishing industry. It removed licensing and opened up commercial fishing to all New Zealand residents and companies with more than fifty per cent New Zealand ownership. It also provided over 50 million dollars

worth of loans nationwide over a twenty year period to finance the purchase of new fishing vessels (Hersoug 2002, 18).

These ill-thought out policies rapidly resulted in the over-capitalisation of the fishing fleet, with vessel numbers throughout the country increasing from 1,727 in 1963 to 5,178 ten years later. There was also further depletion of stocks of inshore fish such as snapper. Subsequent management efforts to control the massively-expanded fishing effort had limited effect on reducing pressure on fish stocks (Hersoug, 2002, 19).

The 1980s saw the beginnings of a more sophisticated approach to coastal management. By the early 1980s there was a crisis in the state of inshore fisheries. The (then) National Fisheries Management Advisory Committee (NAFMAC) stated that 'failure to take action will not only result in a biological and commercial fishing disaster in the very near future but will also ensure the consequent economic collapse of a large part of the inshore fishing industry.' (NAFMAC, 1983 in Hersoug, 2002, 25-26). A revised Fisheries Act was developed which sought to apply an integrated management approach to fisheries through the development of comprehensive fisheries management plans. However, this was not immediate enough to stem the crisis in inshore fishery stocks which were severely depleted, and the move towards integration was overtaken by the introduction of the quota management system in 1986.

In 1987 the Minister of Conservation initiated a major review of coastal legislation and a year later this was integrated into the Resource Management Law reform process.

During the 1990s a resource management framework embracing integrated management with a focus on ecosystems was put into place. The RMA came into force in 1991 and had at its core the concept of sustainable management. The RMA and associated reforms to local government resulted in the establishment of regional councils whose jurisdiction extended from water catchments to the extent of the territorial sea (twelve nautical miles seawards). Their focus was to achieve the 'integrated management of the natural and physical resources of the region' (section 30(1)(a) RMA).

The RMA mandated the preparation of a New Zealand Coastal Policy Statement (NZCPS) which was first drafted in 1990 and became operative in 1994. The policy statement focused primarily on the coastal edge and avoided directly addressing broader catchment management issues such as sedimentation.

During the 1990s the fisheries legislation was revised, resulting in a new Fisheries Act in 1996. This introduced the concept of sustainability (while also providing for utilisation). Environmental principles contained in the Act incorporated the concepts of protecting habitats of particular significance to

fisheries, maintaining biological diversity and addressing impacts on associated or dependent species (incorporating the concept of ecosystem linkages).

In practice, however, fisheries management has been strongly focused around the concept of maximum sustainable yield (MSY) which is designed to maximise the total biomass which can be extracted from each stock of fish over time. It fails to incorporate consideration of the effects of such extraction on marine ecosystems. Implementation of the environmental aspects of the new Act has been very slow.

The preparation of a national oceans policy was initiated in 2000. Its purpose was to provide a more coherent and integrated policy for the entire coastal and marine area. This initiative, however, was partially derailed in 2003 by dispute over potential Māori ownership of the foreshore and seabed and has failed to regain momentum. It provides an opportunity to support more effective integration in coastal management but it is not yet clear whether this potential will be realised.

Also during 2000 the Hauraki Gulf Marine Park Act passed into law. This legislation sought to promote the integrated management of the Gulf and its catchments and the protection of its natural, historic and physical resources. The legislation established a set of objectives to guide the management of the Gulf, set up the Hauraki Gulf Forum as a co-ordinating body and created a Hauraki Gulf Marine Park.

The development of a marine protected areas policy and implementation plan, which was finalised in 2005, represents an attempt to integrate the fragmented system for spatial marine protection. Efforts to update the Marine Reserves Act 1971, which is dated and fails to reflect a more ecologically based and integrated view of the marine area, are stalled. A Marine Reserves Bill was introduced into Parliament in 2002 but has yet to be passed into law.

The current review of the New Zealand Coastal Policy Statement provides an opportunity to update the policy to reflect a broader and more integrated view of coastal management and to better realise the potential of regional councils as integrative agencies.

In summary, New Zealand has come a long way over the past forty years in developing approaches to coastal management, but has yet to achieve a truly integrated ecosystems-based approach. The oceans policy initiative and the review of the New Zealand Coastal Policy Statement provide excellent opportunities to further strengthen an integrated approach to coastal management.

Achieving integrated coastal management

Nothing falls through the cracks, nothing gets done twice, [and] nothing suffers from an uncoordinated work stream. That to me is getting integrated management. (respondent 1)

Although integrated coastal management is widely accepted as the appropriate approach to apply to the management of the coast, it is a broad concept which has no prescribed meaning and has been defined in different ways. A common approach is to define it as a process rather than an outcome:

Integrated coastal management can be defined as a continuous and dynamic process by which decisions are made for the sustainable use, development, and protection of coastal and marine areas and resources

(Cicin-Sain and Knecht 1998, 39).

ICM [integrated coastal management] is a sustained participatory process of rational decision making based on the application of the best available knowledge and technology ... it is a sustained and iterative process, not an end result in itself (Coastal Resources Centre, 2001, 3).

Some commentators see planning as being an important element of the enterprise:

A continuous and dynamic process that unites government and the community, science and management, sectoral and public interests in preparing and implementing an integrated plan for the protection and development of coastal ecosystems and resources (Olsen et al 1997, 157).

Others emphasise the need for a strategic approach to management. For example, the Council of the European Union has urged member states to be strategic in their management of their coastal zones and to develop national strategic plans to promote integrated management (European Parliament and European Council 2002).

The integration of coastal management can be sought across of range of dimensions including (Sorenson in Westcott, 2004, 96):

- *Institutional integration* - integrating the activities of all agencies and stakeholders at any particular level of coastal management (horizontal integration) and across various levels of government (vertical integration)

- *Ecological integration* - planning for and managing the catchment, coast and marine area as an interlinked and interdependent system
- *Inter-disciplinary integration* - integrating disciplines such as natural science, social science, kaitiakitanga, economics and politics that study specific aspects of the coastal environment

Another key area where integration has been sought is between traditional ecological knowledge applicable to coastal management and Western science. The latter currently dominates coastal management efforts, but there is increasing recognition of the contribution that traditional knowledge can make. This knowledge has been developed over long periods of time and is based on close observation and dynamic interactions between people and the environment. Western science, on the other hand, is often based on a small number of observations undertaken over a limited time period (Volkerling, 2006, 7).

A key element of integrated management is co-ordination between different activities and collaboration between different management bodies involved in coastal management. This is to ensure that managers are working towards common overriding objectives and that the impacts of a management decision on other management spheres are considered and addressed. It is also to ensure that there are no significant gaps in the overall management system.

Coastal management can be undertaken at a range of scales and can be conceived as a nested management system operating simultaneously from the local level to the global, with different levels of governance playing different roles in the system (Coastal Resources Centre, 2001, 3).

The European Commission has proposed a delineation of the appropriate role of each level of governance, based on the assumption that integrated solutions to concrete problems can only be found and implemented at the local and regional level. It also pointed out that the integration of policies at these levels is only possible if the higher levels of administration provide an integrated legal and institutional context, as well as take measures to enable local and regional action. The Commission proposes the following roles as being appropriate at the national, regional and local levels, which broadly coincide with central government, regional councils and territorial authorities in the New Zealand context (Commission of the European Communities, 2000, 9-10):

- *National level* – this must provide a legal and statutory framework to adequately enable implementation of integrated coastal management at the lower levels. It needs to ensure coherence of national legislation and programmes which affect the coastal zone. It needs to promote a national

vision and to give guidance and support to promote coherent activities at regional and local levels.

- *Regional/catchment level* – this level performs a key role in integrated planning and management of the coastal zone. This level is still closely aware of the specific context on the ground but has a broad enough remit to take a strategic outlook. It can promote coordination between local councils and ensure that local initiatives have a larger holistic, regional context for their activities. Guidance from this level can be a counterbalance to the powerful short-term political and economic interests that may operate at a local level to promote unsustainable decisions.
- *Local level* – concrete integration actions occur in the context of detailed planning, problem solving and territorial management. This is the best level to collect information about local conditions, to involve local stakeholders, to develop consensus and to ensure the optimal routine application of integration.

Although it looks good in theory, the application of integrated coastal management has not been easy in practice.

It can be difficult to shift the attitudes of decision makers and stakeholders towards supporting more integrated and sustainable outcomes. Treby and Clark (2004) argue that the underlying attitudes of formal decision-makers and stakeholders towards the environment often mean that sustainable policy options are not considered acceptable. Therefore, unless these attitudes are addressed first, coastal management can not advance substantially. They suggest that the passive absorption of information by stakeholders is not likely to be helpful and that stakeholders need to be actively engaged in a process which enhances mutual awareness of interests and preferences.

There is some evidence that projects with higher levels of participation by the intended beneficiaries perform better. Tobey and Volk (2002) suggest that this is because, in ideal situations, it may provide the following benefits:

- Opening the management process to significant stakeholders enhances problem-solving capacity due to the increase in the number of pathways along which information can flow. Participants bring more information and more points of view to bear. This enables more existing local knowledge and experience to be integrated into the planning and management process, producing better policy targeting.
- Active participation helps resolve conflicts and build consensus. A group of people meeting and working together is the most effective means of solving complex problems.

- The participatory process can unleash people's energy and knowledge, thus stimulating their creativity and problem-solving capacities.
- Participation builds and preserves trust and instills the ideals of cooperation, partnership and public-spiritedness. Trust and civic cooperation are core features of social capital which is linked to better performance of government institutions and inter-sectoral problem solving.

Treby & Clark (2004) argue that sustainable development and consensus building require stakeholders to sacrifice part of their interests to other stakeholders and to future generations and that they are more likely to make this sacrifice if they are involved in reaching the decision.

Integrated coastal management also requires more sophisticated management of information. Relevant information needs to be available, accessible, appropriate and credible (Doody, 2003, 165).

So what is the best institutional structure through which to promote integrated coastal management? Public sector reforms during the 1980s and 1990s in New Zealand and elsewhere have been characterised by the decentralisation of authority, devolution to lower levels of government, the downsizing of the public service and contracting out. This has resulted in a move from a strong hierarchical model of governance towards an emerging network and systems model which may be better suited to integrated management approaches (Kay et al., 2003).

Kay et al. (2003) argue that there is no 'best' institutional arrangement and that complex stakeholder networks, consensus-building approaches, power elites and market forces are not easily accommodated by simple vertical and horizontal institutional models. They suggest that successful intergovernmental management requires mutual adjustments which are more likely to be achieved where there is a network of relations based on reciprocity, interdependence and strong trust.

Thomas (2003) argues that a necessary condition for cooperation to occur between agencies is interdependence, which he defines as the extent to which each has the resources or performs the functions desired by other. He argues that top-down approaches to cooperation are doomed to failure, as they are imposed and give insufficient attention to individual incentives to cooperate. He suggests that, rather than imposing coordination, one should provide incentives which lead individuals to work cooperatively together. Such cooperation involves sharing resources, informing one another of their planned behaviours and coordinating their activities to pursue mutually consistent strategies. This

enables them to work together to achieve things they are unable to achieve individually.

Thomas suggests that, within organisations, there are three categories of individuals who have different incentives and disincentives to cooperate. Line managers are the least likely to seek cooperation with other agencies as their strategy is to seek autonomy and certainty for their organisational units. Professionals obtain rewards and esteem from their professional communities rather than their agency and are therefore more likely to develop interagency networks. Field staff who live and work in a community are more likely to identify with local concerns than the agency's interests and develop local networks. Thomas's research into biodiversity management found that inter-agency cooperation was more likely to occur if there was a level of interdependence between agencies and if there was staff stability, delegated authority within the organisation and a large network of local offices. Conversely he found that a high level of staff rotations or turnover, centrally located staff and centralised authority within the organisation impeded cooperation.

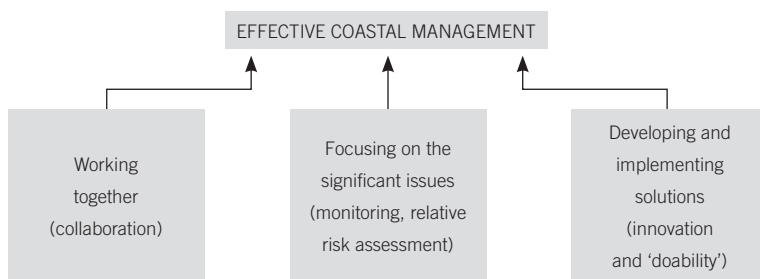
And how is coastal management to be funded? Kay et al. (2003) observe that there is currently an expectation of increased management of the coast by government, but at the same time there are decreasing resources within the government to manage and therefore to meet the expected levels of management. This is because much of the coast is used for public purposes and coastal industries are generally not prepared to subsidise public use while the public refuses to pay the real cost of management. In addition, coastal users often do not pay the real cost of using the resource. Because of this funding shortfall, government has been forced to look at how to manage the coast in a more cost-effective way through such mechanisms as devolving power to local councils and community groups while at the same time setting performance measures.

So what are the ways forward? Several commentators suggest that coastal managers need to adopt new roles which are facilitative, proactive and entrepreneurial rather than bureaucratic, reactive and regulative. Institutions need to be less hierarchical and coastal managers need to focus on adopting consensus-building techniques. Effective integration requires managers who can act as 'boundary spanners', to connect participants, knowledge and strategies, reinforce shared activities and maintain people's willingness to remain engaged (Jentoft & Buanes, 2005, 158; Kay et al., 2003; Cowell, 2006). Coastal managers therefore need to have strong people skills as well as strong technical skills.

So, effective coastal management is dependent on coastal managers working together, working on the issues that matter, and implementing solutions

which effectively address those issues (see Figure 2.4). This requires effective collaboration, methods to identify significant issues such as monitoring and relative risk assessment technologies, and the ability to innovate in order to develop effective and ‘doable’ solutions.

Figure 2.4: Key elements of effective coastal management



A key requirement for integrated coastal management is that coastal managers work together on interrelated or shared issues. The literature review has identified a range of factors which potentially support inter-agency collaboration. These include:

- *Broad agency alignment* – where coastal managers are seeking to achieve compatible outcomes. Incompatible goals can result in conflict and therefore less willingness to cooperate.
- *Inter-agency dependency* – where coastal managers are dependent on other coastal managers to achieve their desired outcomes. This provides a strong incentive to work cooperatively.
- *Climate of trust* – where coastal managers feel comfortable working together and in trialling new approaches. Mutual trust can reduce the perceived risks of collaboration. The development of trust can be facilitated by engagement in networking opportunities, undertaking joint projects, adopting a participatory approach to decision making and applying effective conflict resolution procedures when conflict arises.
- *Compatible scales* – where coastal managers are working, at least some of the time, at similar spatial scales, which provides a common level at which interaction can take place.

- *Decentralised decision making* – where coastal managers have the autonomy to address local issues, to forge local partnerships and to experiment with new solutions.
- *Strategic approach* – where coastal managers proactively plan ahead rather than reacting to issues on an ad hoc basis. This provides the opportunity to consider the wider impacts of decisions on other coastal management spheres and vice versa. The strategic planning process can also provide a useful forum where collaboration can take place.
- *Good information base* – which alerts coastal managers to key management issues in a timely manner and provides an understanding of the linkages between different management spheres. Lack of information can result in poor or inconsistent decision making or inaction.
- *Adequate resourcing* – which helps make available the information and staff time required to engage in integrative initiatives.

These eight factors have been used to structure the analysis of the case study material in chapter five and to assist in the development of recommendations. The extent to which these factors exert a causal influence over coastal management, and their relative importance, is a subject which requires further research.

Figure 2.5: Summary of factors supporting inter-agency collaboration

Broad agency alignment

Inter-agency dependency

Climate of trust

Compatible scales

Decentralised decision making

Strategic approach

Good information base

Adequate resourcing



three

Conflicts Underlying New Zealand's Coastal Management System

In practice, integrating coastal management has proved difficult in New Zealand because of underlying conflicts embedded within the management system itself. These include conflicting values and management approaches, conflicts over information and conflicts between exclusive and non-exclusive rights to use the marine area. Although there will inevitably be some discord in coastal management, and this is not always a bad thing, it can serve to seriously undermine integrated coastal management efforts when disputes become magnified through poor legislative design and through the lack of effective conflict-resolution procedures.

Conflicting values

Integrated coastal management requires the management efforts of coastal agencies to be broadly aligned so as to achieve common outcomes. In the case study areas, although most coastal management agencies were seeking to manage values to some extent, they were doing so in very different ways. Each agency differs in its approach, as do different offices, departments and staff, but it is possible to discern underlying broad orientations (see Figure 3.1). Incompatibility in these underlying philosophies can make it difficult to integrate the different management approaches, and can result in head-on conflicts.

Regional council planners interviewed talked about identifying the different ‘values’ of the coastal environment and seeking to maintain these while enabling activities to take place, or maintaining a balance between them. This enables people to co-exist with nature. The approach assumes that there are multiple differing values, all of which have legitimacy, and that somehow a delicate balance is negotiated between them. Discretionary decision making within a regulatory planning framework is seen as the prime tool to adjudicate conflict between values.

Because of their focus on land use, district council planners are more concerned with managing the coastal land development process, which includes the availability of land for development and the provision of supporting infrastructure such as roads, water and waste water treatment services. This can result in the management process focusing more on meeting the needs of people rather than those of nature. This is explicitly stated on the home page of the Kaipara District Council’s website¹ which indicates that the council values ‘People First – The council realises that its first responsibility is to the people of the district’.

¹ www.kaipara.govt.nz

Figure 3.1: Underlying orientation of different coastal management agencies

Management agency	Underlying approach	Assumption about people-nature relationship
Territorial authorities	Managing development	People are more important than nature
Ministry of Fisheries	Maximising value	People utilise nature
Regional councils	Balancing values	People co-exist with nature
Tangata whenua	Maintaining inter-relationships between people and nature	People are part of nature
Department of Conservation	Maximising conservation values	People are in conflict with nature

The Ministry of Fisheries has a utilitarian approach to the management of natural resources. Its corporate goal is:

‘ ... to maximise the value New Zealanders obtain from the sustainable use of fisheries resources and protection of the aquatic environment.’

(Ministry of Fisheries, 2005a, 1)

This goal refers to ‘maximising value’, in a singular sense. The approach is aligned with a more typical economic valuation approach, based on the concept of maximising utility. It implies that different values can be converted into the same measurement framework (often monetary) and then added together. The management response is to allocate the resource to the combination of uses which derive the highest additive value as determined by the measurement framework. The market mechanism can be potentially harnessed to achieve the highest value allocation of the resource. This approach in its simplest form does not address equitable allocation issues.

This is a similar underlying philosophy to that set out in a recent Treasury paper proposing principles for royalties on non-mineral natural resources (Guerin 2006, 5). An overall design principle proposed in the paper is that ‘Any royalty regime should aim to ensure that any given resource is allocated to its highest value use (in the widest sense of value) and allow for that allocation to adjust over time as appropriate.’

The Department of Conservation is more focused on maximising conservation values, which is only one category of the values councils and MFish are addressing. The Department’s corporate goal is ‘to conserve New Zealand’s natural and historic heritage for all to enjoy now and in the future’ (DoC, 2006).

Although this goal includes public enjoyment of nature, the underlying concept behind conservation is that people are separate from nature and have a negative impact on it. The logical management response to this approach is to protect nature by excluding people, or at least excluding them from undertaking extractive and damaging activities, such as through the establishment of marine protected areas.

Tangata whenua are seeking to apply a whole different set of values to resource management. These are founded on the spiritual connection between people and nature, and between past, present and future generations. The approach incorporates concepts of preservation, use and equitable sharing of resources. It eschews conservation for its own sake, which values nature apart from people, potentially putting it in direct conflict with the underlying DoC approach.

Conflicts over issues such as the establishment of marine reserves can be understood in terms of different underlying orientations. The fisheries sector can oppose marine reserves because this management mechanism excludes utilisation of marine life which is the rationale of the sector. Tangata whenua can oppose such reserves on the basis that they extinguish customary fishing rights within the area and permanently exclude people from nature, rather than maintaining an ongoing relationship where seafood may be harvested in the future when stocks have recovered.

Conflicting marine management approaches

Each management regime within the marine area focuses on a different core approach, utilises different mechanisms to allocate access to the resource, adopts different measures of success and has different approaches to conflict resolution (see Figure 3.2).

Resource Management Act

Under the RMA, the focus is on minimising the adverse environmental effects of activities. The broad approach behind the RMA is that the market should be left to allocate the use of resources, as this would involve their allocation to uses generating the highest value, and that regulators should focus on controlling the effects of activities. As a result, the RMA does not satisfactorily deal with allocation issues regarding publicly-owned resources where the market does not operate. The default position has been that allocation has been on a first come first served basis, where marine space and access to marine resources have

been allocated to the first person to lodge a resource consent application and demonstrate acceptable environmental effects.

Figure 3.2 Comparison of marine management regimes

Management regime	Resource management	Marine protection	Fisheries	Marine Biosecurity
Core approach	Minimise the effects of activities	Allocate space for conservation	Manipulate fish extraction levels	Exclude new organisms
Mechanism to allocate use of public resource	First come first served	Central government regulation	Market through tradeable rights (commercial) Restricted open access (recreational) Collective management (customary)	Not applicable
Core management objective	Enable development while maintaining environmental quality	10 per cent of representative marine space in MPAs	Fish stock at B_{MSY}	No damaging invasions
Prime management tool	Environmental impact assessment	Marine reserve establishment	Setting the TACC and TAC	Regulating international vessel practices
Measure of success	No clear measures	Percentage of marine space in MPAs	Fish stock biomass	Number and severity of invasions
Prime beneficiaries	Users of the marine area	Environment	Commercial fish stock quota holders	Users of marine ecosystems (e.g. aquaculture, fishers, divers)
Formal conflict resolution process	Mediation; Environment Court	None	Fisheries Dispute Commissioner	None

The difficulties in proactively allocating resources under the RMA to specific uses have been demonstrated by the Kaipara Harbour aquaculture case study. Both the regional councils managing the harbour experienced difficulties in proactively allocating marine areas to this use. They backed off in favour of an invited private plan change procedure which reinstates a more reactive approach to aquaculture planning, although within a broad framework which identifies

areas which may not be suitable for aquaculture. It is not yet clear who will get priority if several parties apply for a private plan change applying to overlapping areas.

The prime purpose of resource management under the RMA is stated as being 'sustainable management'. In practice this has meant enabling development to occur while seeking to maintain environmental quality. For example, in aquaculture, this translated into regional councils considering it appropriate to accommodate the use in their marine areas rather than to exclude it. They sought to locate marine farms where they would have the least environmental impact and create the least conflict with other uses of the marine area. The approach ran into problems because some members of the community challenged the initial assumption that provision should be made for aquaculture at all, at least in the part of the marine area they valued.

There are no clear measures of success under the RMA, which has made the monitoring of progress and providing clear accountability for results, problematic. Monitoring has generally been patchy, although the ARC and EW have been gradually implementing a well-designed monitoring regime for marine ecology. It is therefore difficult to assess how successful marine management under the RMA has been.

There has been less outright conflict in RMA processes affecting the marine area than in the marine reserve and fisheries arenas. This may be due to the broader scope of enquiry, the ability to balance various interests in the decision-making process and the availability of an accessible formal conflict-resolution process through council hearings and appeals to the Environment Court.

Marine protection

Marine protection has focused on the allocation of space for conservation through the creation of marine reserves, although it also includes many other activities such as protecting marine mammals and sea birds. Although the Marine Reserves Act establishes, as the sole purpose of marine reserves, the preservation of marine areas for the scientific study of marine life, they have in practice been used as a key marine conservation tool. The allocation of space is largely achieved through central government regulation under the Marine Reserves Act 1971. A target of ten per cent of New Zealand's entire marine area in marine protected areas by 2010 has been formally established in the New Zealand Biodiversity Strategy (DoC & MFE, 2000, 67). This target includes not only marine reserves but other less stringent forms of protection such as areas closed to fishing activity under the Fisheries Act 1996. More detailed policy on how this is to be achieved is contained in the Marine Protected Areas Policy and Implementation Plan (DoC & MFish, 2005).

In terms of the study areas, the current marine reserves in the Hauraki Gulf cover less than 0.3 per cent of the marine area (Hauraki Gulf Forum, 2005, 118) and there are no marine reserves in the Kaipara Harbour. A range of fishing restrictions apply in both harbours and there is a submarine cable and pipeline protection area running through the Hauraki Gulf.

The focus of marine reserves to date has been on excluding fishing activity from marine areas in order to derive conservation benefits within the reserves. The potential of marine reserves to increase the ecological health of the broader marine area, and therefore to increase fish stocks, has not been a major goal to date. Such benefits are hard to demonstrate scientifically, because of the small size of current reserves in comparison to the balance of the marine area, and the dynamic nature of marine systems. In addition, monitoring of marine reserves had not been strategically designed or well-funded and there has been little research into broader effects. Some research has, however, provided evidence of improved fishing along the boundaries of the reserves (Taylor and Buckenham, 2003, 31).

Marine reserves, therefore, can be seen by some stakeholders to benefit the narrow section of conservation interests rather than all marine users. In addition, because of their spatial allocation nature, they can be seen as being in direct conflict with, and restricting the operation of, non-spatial fishing rights (see below). The proposed establishment of marine reserves has in some cases, including the Great Barrier Island proposal, been characterised by high levels of conflict. There are no formal conflict resolution procedures under the Marine Reserves Act 1971, apart from judicial review proceedings in the High Court. This means that conflicts are left unresolved and potentially spill over into other marine management issues and future marine reserve proposals.

Fisheries management

Fisheries management is primarily focused on manipulating the levels of fish extraction. The allocation of fisheries rights is split between a market mechanism consisting of tradeable rights for commercial fishers, restricted open access for amateur fishers and collective management for customary fishers. Commercial fishers must acquire or lease individual transferable quotas in order to fish. Quotas are initially allocated by the Crown and then can be bought off existing quota holders. A quota is a right in perpetuity to extract a proportion of the total allowable commercial catch (TACC) within a specified (usually large) quota management area. Changing the level of the total allowable catch (TAC), which is the amount which can be taken by both commercial and non-commercial fishers, and the TACC are the prime mechanisms used to manage fisheries.

Amateur fishers have open access to fish, to the extent that anyone can go fishing without authorisation, with the main restriction being on the size and amount each fisher takes (a bag limit). This limit, however, is hard to enforce and information on the total fish extracted by amateur fishers is very poor. Recreational take is therefore primarily limited by the availability of fish which can be caught by recreational equipment and fishing methods.

In terms of customary fishing, the Treaty of Waitangi included the guarantee to Māori of the full, exclusive and undisturbed possession of their fisheries. In response to the creation of a property right in the form of individual transferable quotas, Māori fisheries claimants in the late 1980s sought redress for the alienation of customary fishing rights. The Treaty of Waitangi (Fisheries Claims) Settlement Act 1992 gave effect to the settlement of these claims by allocating quotas and fishing assets to iwi to address commercial rights, and by determining that policies for non-commercial Māori fishing rights would be developed. While those non-commercial rights could in principle include a wide range of usages, in practice only allowance for some hui and tangi are provided for by regulation.

Under the Fisheries (Amateur Fishing) Regulations 1986, customary fishing can be managed by statutorily defined kaitiaki, who authorise customary harvest. In practice, implementation of the regulations has been problematic, mechanisms for mediation and resourcing for the process are limited, and in many areas kaitiaki are yet to be appointed. The Ministry of Fisheries has been encouraging greater uptake of the Fisheries (Kaimoana Customary Fishing) Regulations 1998 where the first step is for tangata whenua to define the rohe moana (marine areas of authority) for each kaitiaki. While this may aid the resolution of conflicts in time, responsibilities are often still ambiguous, and the amount of take uncertain. In many areas it is probable that far lower customary take is utilised than that allowed for pursuant to the Fisheries Act.

The core management objective of fisheries, which is to keep fish stocks at the biomass level that can support their maximum sustainable yield (B_{MSY}) is the source of considerable conflict. The maximum sustainable yield is defined in section two of the Fisheries Act as ‘the greatest yield that can be achieved over time while maintaining the stock’s productive capacity...’. Although it is hard to precisely define the B_{MSY} for a particular fish stock, it is typically between twenty or thirty per cent of the virgin biomass (original size of the stock before human exploitation).

Fishing a stock down to such a small proportion of its original size benefits commercial quota holders and fishers who theoretically maximise the amount of fish they can extract. However, it has negative effects for both amateur fishers and the environment. For amateur fishers, the fish become harder to catch

and are generally smaller. For the environment, the removal of up to 80 per cent of a dominant species, such as snapper, can result in profound changes to the ecosystem. In the Hauraki Gulf, for example, it has resulted in extensive sea urchin barrens, because the snapper and crayfish which prey on the sea urchins are substantially reduced in numbers, and the increased population of sea urchins browsing on the kelp has completely removed kelp forests and their associated habitat in some areas.

Some amateur fishers feel threatened because fisheries management does not appear to incorporate their interests, and neither do the resource management or marine protection regimes. Commercial fishers can also feel threatened by the less readily controlled nature of recreational fishing as well as other incursions on their non-spatial rights (see below). This can result in high levels of conflict both within the fishing commercial sector and with other sectors such as recreational and environmental interests.

Although there are conflict resolution provisions in the Fisheries Act these are limited, cumbersome and not often utilised. They only apply to disputes amongst people with fishing interests and exclude disputes between stakeholders and the Minister or MFish and disputes about sustainability or the environmental effects of fishing (section 114 (b)). Aggrieved parties can initiate judicial review proceedings in the High Court to challenge the Minister's decisions, but this is a cumbersome and expensive way to solve disputes, and the court's enquiry is limited to the legality of the decision and does not include its merits.

Marine biosecurity

Marine biosecurity focuses on excluding new organisms from New Zealand waters. It can only achieve this through controlling practices on international vessels entering New Zealand waters and this is extremely difficult without international agreements. Such an agreement has been reached for ballast water and New Zealand has introduced regulations to control the exchange of such water in the country's marine areas. No such agreement has been reached for hull fouling, which is another major way in which invasive species are transported, and there is currently no regulatory control on this within New Zealand. Instituting an effective marine biosecurity system is problematic because it is a sphere in which international pressures, which can only effectively be controlled at an international level, impact directly at a local level. Regional councils, which have no border control powers, are reluctant to take the job on, and Biosecurity New Zealand has little regional or local capacity on the ground.

Conflicts over information

Information is vitally important in coastal management, because much less is known and understood about the marine environment compared to the terrestrial one, and many of the impacts of activities on the marine environment cannot be readily identified without scientific investigation. Information collection can be much more difficult and expensive because of the dynamic nature of the marine environment and the need to use boats and specialist equipment to measure it. Conflicts can arise over the sufficiency of information to support decision making, who has access to information, and the relative importance given to different types of information.

A lack of high-quality information and an inadequate understanding of the marine system can lead to poor planning and inconsistent decision making. This is because decisions which are based on weak information are likely to be politically, rather than technically, based. This is illustrated by the experience of planning for aquaculture in the southern Kaipara Harbour, where a decision to delineate aquaculture management areas was overturned by a subsequently elected, differently constituted, council.

Because information is expensive to collect, councils can be reluctant to require resource consent applicants to provide comprehensive information as part of their assessment of environmental effects. What information is provided can be difficult to properly evaluate in the absence of a broader understanding of the marine system. Such understanding can require more spatially extensive and longer-term research projects which councils can find difficult to fund. This is illustrated by the Biomarine Limited application for an oyster farm in the Kaipara Harbour, where the ecological significance of the affected area of seabed was not revealed in the assessment of effects but only through a much more extensive council monitoring study.

Information can have considerable commercial and political value. Battles are fought over what information is made available to whom and at what stage of the decision-making process. The inability to access information can make it difficult for opponents to proposals to mount successful cases. It can also undermine the credibility of scientific findings if they are not seen to be open to peer review. Such disputes in the case study areas were identified in relation to sand mining and aquaculture.

There is an uneasy relationship between the respective roles of scientific knowledge and indigenous knowledge in informing coastal management. Tangata whenua representatives interviewed all expressed dissatisfaction with the resource consent process under the RMA and their ability to have any

impact on it. This was because indigenous knowledge was considered to be inferior to scientifically derived knowledge. For example:

In 1996 our elders asked that no sand be taken and gave reasons for it. At the very same meeting that was ignored. So if you're asking me if we have any meaningful input, the answer has to be no. What they did, after we got our people up speaking, is they trotted out their scientists. The regional council was there and as the science did not support what we were saying our comments appeared to be ignored. (respondent 23b)

Because scientific information has a major impact on many coastal management decisions, it can be hotly contested. This is particularly the case in fisheries management where decisions on setting the total allowable catch for fish stocks have major commercial implications and are largely based on scientific assessments. The Ministry of Fisheries has established technical working groups which include stakeholder representatives as well as MFish staff and the scientists undertaking MFish-commissioned research. Commercial fishing representatives put considerable effort into seeking to influence these decisions and have from time to time engaged overseas scientists to critique the work undertaken by local researchers.

The only way commercial guys can get their hands on catch rights is to argue the TACC, that's the only thing that matters ... If the TACC is going to be set by numerical modelling of fish stocks then you make sure you've got better numerical modellers on your side than anybody else. If it's going to be a legal amendment [to the Act] make sure you've got better lawyers working for you and lobbying than the Crown has. That's how it's managed. Whatever it takes, whatever form the debate is being held in, make sure you are powered up more because you've got to win it ... (respondent 58)

As a result, the methodology employed in the research is put under intense scrutiny. Although this probably results in more robust science, it is resource-intensive, making the science more expensive, and acts as a deterrent to innovation and seeking answers to more complex questions:

... if every little step is going to be hotly contested then there's the cost of that process ... the whole system is set up to favour short-term programmes most of the answer of which we already know. We just want better answers. It doesn't favour risky work ... (respondent 35)

Conflict between exclusive and non-exclusive rights

There is an inherent conflict between users of the marine environment that have an exclusive right to a marine area, such as aquaculture, and those whose rights are non-exclusive. This is because exclusive rights necessarily constrain non-exclusive rights where they extend over the same area. This situation is leading to conflict, particularly in commercial fisheries.

Holders of individual transferable quotas have a perpetual right to harvest a proportion of a TACC within a large quota management area (QMA). This is in effect an extraction right over a moving resource bounded by the QMA. When parts of the QMA are allocated for a competing exclusive spatial use, this restricts the rights of the quota holder. There is, therefore, an incentive for a quota holder to oppose any grant of exclusive spatial rights in the marine area. The conflict is more acute the larger the spatial allocations are in proportion to the size of the overall QMA, and also in proportion to the fishable areas within the QMA. Large marine reserves, areas closed to fishing activity or AMAs can therefore be a threat, as can reducing the size of the QMA through partitioning, as proposed by local fishers in the Kaipara Harbour. It is not therefore surprising that this proposal was not supported by large quota holders.

Although the interests of fishers are taken into account when marine reserves and AMAs are established, the assessment is only undertaken for individual proposals and therefore fails to address progressive and cumulative encroachment of quota holder rights. A similar ‘squeezing out’ effect also has comparable impacts recreational fishers who also have a non-exclusive right, and who, although not restricted to QMAs, are in practice restricted to marine areas accessible to where they live or holiday.

Conflicts over resolving conflicts

The approach which should be adopted to resolve these conflicts is itself contested, particularly the extent to which private property rights should be created in the public marine commons as a way of addressing allocation and environmental management issues. The formulation of an oceans policy provides a forum for this debate.

One view that is not officially held by MFish, but which was presented to a recent conference by one of its senior policy analysts, incorporates a vision of extending the private property rights approach to all other oceans users, not just

commercial fishers, and extending the 'effects-based approach' applied in the RMA to managing the impacts of the exercise of those rights:

Where possible, the right to occupy certain spatial areas should be defined, and allow the occupation of such areas able to be transferred between different uses (eg marine farm to floating restaurant to tidal energy generation site), subject to compliance with the standards. With community input, particular areas of the marine environment would be zoned as suitable for occupation, and the maximum effects permitted in those areas defined (eg high, medium and low impact areas). Where the current rights holder agreed, the right to occupy that space could be transferred between different users undertaking the same activity, or between different activities, provided that the limits on effects were not breached. (Taylor, 2006)

This approach, which is more in line with the creation of private property rights approach currently applied in fisheries management, can be contrasted to how coastal and marine management is evolving at a local government level. It perhaps illustrates the underlying chasm between management approaches applied in the fisheries and RMA management regimes.

Councils, prompted by the Local Government Act 2002, are increasingly focusing on achieving community outcomes and proactively planning to do so. The case studies indicate a recent move away from a strong reliance on an effects-based approach by RMA marine planners, because of the difficulties of applying this approach in the marine area. This is because there is a lack of information to provide the context within which to assess the effects of individual applications, uncertainty of what might be appropriate where, and difficulties in addressing cumulative impacts. Their experience has indicated a need for a broader strategic framework within which effects of individual activities can be assessed.

Overcoming the divide between the management approaches of fisheries and the RMA management approach is the biggest challenge in achieving integrated coastal management.



four

Coastal management within the Hauraki Gulf and Kaipara Harbour

This chapter provides a summary of the findings of the case studies prepared on the Hauraki Gulf and the Kaipara Harbour with a particular focus on achieving integrated coastal management. The full case studies are presented in Section Three of this report. The first section of this chapter provides an overview of the case-study findings. The second section discusses the approaches which have been developed to better integrate management of these areas.

Overview of case study findings

The results of the sixty stakeholder interviews undertaken for this project have been distilled and written up into nine ‘case studies’ which seek to describe and illustrate the challenges that are currently being faced in the management of different aspects of the coastal environment, how coastal managers have sought to address those challenges, and what the prospects for greater integration in the future might be. The key findings from each of these case studies are described in the sections below and summarised in Figure 4.1.

Case Study 1: Planning for the coastal marine area

This case study investigates how coastal managers have approached planning for the coastal marine area. It describes attempts at undertaking regional coastal planning under the RMA, fisheries planning under the Fisheries Act and marine protection planning, which is currently non-statutory, but which is being undertaken within the framework of the Marine Protected Areas Policy Statement and Implementation Plan.

The case study identifies significant innovation in coastal marine planning, with regional councils exploring ways to more effectively span the land/water divide through sub-regional plans such as community compartment plans. The Department of Conservation is moving away from a conflict ridden and ad hoc approach to establishing marine reserves, and MFish is trialling a Ministry-led multi-stakeholder planning for fisheries.

Many of the initiatives are very new and it is not entirely clear how successful they will be. They all face the difficulties of planning in a poorly understood and dynamic environment and where there has been a long history of conflictual relationships between stakeholders. There are good prospects for further integration between the different planning processes, although there is currently a poor interface between regional coastal planning and that for fisheries.

Case Study 2: Managing coastal development

The approach of territorial authority planners to the management of coastal development is the subject of this case study. It focuses on management of the Coromandel Peninsula, the Hauraki Gulf Islands, the Hibiscus Coast and the Kaipara Harbour. The case study identifies broad agreement on the overall approach to be adopted towards managing coastal development. This is to consolidate urban development in existing developed areas and avoid urban development in rural coastal areas. The strategy has primarily been implemented through district plan rules which provide for subdivision opportunities within townships and restrict subdivision in rural areas. Different councils, however, vary in the level of sophistication of controls which are applied.

All of the four territorial authorities studied have incorporated conservation lot provisions in their district plans, as an incentive for landowners to provide environmental benefits. These provisions increase the number of lots which landowners can create through subdivision if they permanently protect other areas of land which have environmental importance. This has created an unresolved tension between the overall strategy of restricting rural subdivision, and the conservation lot approach which facilitates rural subdivision.

In the Hauraki Gulf, all the councils studied applied additional controls over the location and design of buildings within the coastal environment. Such additional controls on buildings have not, however, been applied to the Kaipara Harbour. This reflects the lower landscape values which have been attributed to the low-lying estuarine environment, the lower level of development pressure, and the harbour surroundings being largely a working environment dominated by rural production activities. This leaves the Kaipara Harbour vulnerable to significant negative landscape impacts of poorly-designed and located buildings and structures.

Within a broad common framework, councils have approached planning in different ways and have experienced varying levels of success. They have grappled with a range of challenges including those related to the RMA itself, those related to councils' internal operations and those related to external factors linked to development pressure.

The prospects of integration between territorial authorities planning for common coastal/marine environments do not look strong. Councils have approached planning from different perspectives, the plans are structured in different ways and all the councils are at different stages of plan development. Within the Kaipara Harbour, progress is unlikely to be made on integrated land management until the harbour is identified as a distinct planning unit in both district plans. It currently has some recognition in the Kaipara District Plan in

the form of an overlay policy area, but nothing similar in the Rodney District Plan.

Case Study 3: Reducing sedimentation

This case study examines how regional councils are approaching the management of activities which result in human-induced sedimentation of the marine area. It identifies significant management problems, particularly in effectively reducing sedimentation from rural activities such as agriculture and forestry. In these areas regulation has proved politically problematic and it is difficult to scale up voluntary mechanisms to achieve catchment-wide improvements. Although large earthworks are more strongly regulated, they also contribute to marine sedimentation due to the lack of efficiency of sediment ponds. Small earthworks are not well-regulated and, cumulatively, can also have a significant impact.

Councils are piloting innovative ways of more closely linking land-management decisions to the health and carrying capacity of affected marine systems, such as estuaries and harbours. These include the Mahurangi Action Plan, the Okura Catchment Study and the Muddy Feet project in the Firth of Thames. These show some promise, but are resource-intensive and are only likely to be applied to a small number of priority areas.

Case Study 4: Marine biosecurity

How marine biosecurity is currently managed in the case study areas was the focus of this case study. Marine biosecurity is a recent management activity primarily undertaken by a national agency, Biosecurity New Zealand. It is yet unclear what role regional councils will undertake and they have proved reluctant to enter a potentially costly new management area outside their current area of expertise.

Effective marine biosecurity is extremely difficult to achieve because of problems in regulating international vessels without international agreements, difficulties in identifying new incursions in a timely fashion and the lack of effective tools to eradicate or manage new species once they have arrived.

There is potential to better integrate biosecurity considerations into regional councils coastal planning activities. One way to achieve this could be to incorporate marine biosecurity guidance into the New Zealand Coastal Policy Statement.

Case Study 5: Hauraki Gulf amateur fishers

This case study examines issues concerning the amateur fishing sector and how their non-resolution has resulted in escalating levels of conflict spilling over

into multiple areas of marine management in the Hauraki Gulf. Much of the conflict stems from the uncertain status of amateur fishing rights, where fish stocks are shared with other sectors with better-defined rights. The conflict is exacerbated by depleted fish stocks which are being managed to a target (B_{MSY}) which primarily benefits commercial fishers. The battle over the establishment of additional marine reserves in the Gulf has added to this underlying conflict and provided the recently mobilised amateur fishing sector with a targeted outlet for its frustration.

As a result of these unresolved conflicts, possibly the largest user group of the Hauraki Gulf's marine area (excluding beaches) has withdrawn from a positive stewardship role into a negative combative and defensive position which is likely to make future collaborative marine management difficult.

Case Study 6: Kaipara Harbour commercial fisheries

This case study examines how local conflicts between commercial fishers in the Kaipara Harbour have arisen and how they have been responded to. The conflict has revolved around the depletion of fish stocks within the Kaipara Harbour, poor fishing practices and competition for the harbour's fish between locally-based commercial fishers and mobile 'outsiders' who can trailer their boats to the harbour.

The case study highlights the significant difficulties experienced in attempting to address local fisheries-management issues within the current quota management system, which provides little room for local management or local solutions. It also indicates that unresolved conflicts can absorb considerable energy, whilst more insidious impacts on fisheries (such as sedimentation) which would require a more integrated and collaborative management approach go largely unaddressed.

Case Study 7: Kaipara Harbour aquaculture

This case study investigates how the ARC and the Northland Regional Council (NRC) approached planning for aquaculture in the Kaipara Harbour and the challenges that they faced. Ultimately both councils abandoned attempts to incorporate aquaculture management areas (AMAs) into their regional coastal plans through council-led plan changes. This was due to a combination of factors including public opposition, a lack of ecological information, council political changes and negative decisions on resource consent applications.

The case study provides interesting insights into the difficulties of planning spatially for individual marine uses in the absence of a broader strategic framework. It also highlights the pitfalls which can be encountered when

assessing potential activities on a case-by-case basis in the absence of a broader understanding of the marine system. It identifies the need for a strategic framework and a long-term research and monitoring programme to assist and guide future decision making.

Case Study 8: Integrating management of the Hauraki Gulf

Several initiatives which have been undertaken to better integrate the management of all or part of the Hauraki Gulf are reviewed in this case study, including the Hauraki Gulf Forum, local area management strategies, coastal compartment plans and Blueprint Coromandel.

There has been a long history of integrative management efforts in the Gulf, commencing in the 1960s. During that time, there has been a move away from attempts to achieve integration through establishing a single management body, towards developing stronger relationships and collaboration between the multiplicity of existing management bodies operating within the Gulf. Although the latter is a less well-defined and potentially ‘messier’ approach, it appears to be slowly advancing integrated management through improved information flow and the development of collaborative projects.

There has also been a more recent shift towards incorporating tangata whenua as partners in the information gathering, planning and management processes. This is still in the early stages, but the platform and resources made available by the Hauraki Gulf Forum are providing the opportunity for improved cross-cultural communication, relationship building and tool development.

The case study illustrates the dilemma over whether to seek to integrate planning at a broad strategic level or at a more local level. Local planning, such as through the development of coastal compartment plans, can address specific issues of direct concern to residents and achieve an observable difference in the short term. Such plans are weak, however, in addressing the larger, longer-term issues and are difficult to scale up to a regional level. Broader strategic planning, as being attempted in the Blueprint Coromandel project, faces the multiple challenges of demonstrating relevance to local communities, of achieving change on the ground and of maintaining political interest and support over the longer time frame required to complete the planning process.

Case Study 9: Integrating management of the Kaipara Harbour

Unlike the Hauraki Gulf, which has a rich history of integrated management initiatives, the need for integration in relation to the Kaipara Harbour has only recently reached the political agenda. This case study examines two recent initiatives to promote more integrated management of the Kaipara Harbour,

one led by a regional council and one led by iwi. They are both in the early stages, so it is difficult to evaluate their impact.

The regional council project is quite modest in its aims, being confined to regional council management responsibilities within the coastal environment. It should initially result in a better understanding of the management issues within the harbour and also provide greater insight into the extent to which management agencies can justify devoting additional resources to the area. The project is being undertaken jointly by the two regional councils managing the harbour so is likely, to result at least in better integration between these agencies. The iwi-led project is much broader in scope and ambitious in its aims, seeking to include the entire catchment of the harbour and economic and cultural issues. It has the support of a wide group of management bodies and stakeholders but, crucially, lacks funding.

The case study illustrates the tensions between the need for integration and the difficulties of getting fully integrated projects off the ground because of their large size and complexity and the resultant need for significant funding. It also highlights the difficulties that iwi face in exercising leadership in coastal management when the bulk of the management resources are held by statutory bodies which may be reluctant to give up control.

Uneven ‘playing field’ for economic activities

The case studies indicate that the environmental requirements placed on different economic activities can be highly variable and do not appear to be linked to their potential to cause significant environmental harm to the marine environment.

For example, proposals to establish marine farms are required to undergo a detailed and costly environmental impact assessment process in order to obtain the necessary resource consent to proceed. Although marine farms can have negative visual and ecological impacts, these are often less than those of other major activities which impact on the marine environment. And on the positive side, filter-feeding farmed species can improve marine water quality.

In contrast, commercial fishing activity, which has a much greater negative impact on the marine environment through removing large amounts of fish and other species from the marine ecosystem and disturbing habitats through trawling and dredging and which has no obvious positive environmental impact, is not currently required to undergo any environmental assessment process. Similarly, on land, farming and forestry activities are generally permitted and are not

required to undergo any environmental impact assessment process, despite the significant negative effects such activities can have through producing sediment which ends up smothering seabed organisms.

In this respect, it was interesting that a representative from the Federated Farmers made submissions to the Environment Court when it was considering the Biomarine Limited oyster farm application for the Kaipara Harbour. The submissions were based on concerns that the marine farm might lead to constraints on farming operations, or in effect it might restrict the ability of farmers to pollute the harbour (Biomarine Limited, 2006, 40).

The current uneven playing field in environmental regulation effectively favours old-style environmentally-damaging economic activities and acts as a disincentive for the establishment of newer and cleaner resource-based ones.

Figure 4.1 Summary of key information from case studies

Marine planning	Coastal development	Sedimentation management	Biosecurity management	Hauraki Gulf amateur fishing	Kaipara Harbour commercial fishing	Kaipara Harbour aquaculture
Key issues						
Difficulties zoning the marine environment due to lack of information	Lack of a strategic perspective to address long-term development	Lack of cost-effective technologies to effectively remove sediment from run-off	Difficulties of regulating international vessels without international agreements	Uncertain status of amateur fishing rights	Failure to address local fisheries management issues within quota	Failure to address the impact of sedimentation on fish stocks
Difficulties managing outcomes without zoning	Strong focus on site-specific issues	Little control over the cumulative impacts of small earthworks	Problems identifying incursions in a timely fashion	Fisheries management goal which primarily benefits commercial fishers	Lack of an effective conflict resolution process to address high levels of conflict	Lack of sufficient ecological information to zone for aquaculture
Lack of planning integration across MHWS	Conservation lot provisions undermining strategy to concentrate development	Historical reluctance to regulate rural activities resulting in weak controls over forestry harvesting and grazing activities	Lack of cost-effective tools to eradicate or manage invasive species	High levels of conflict generated by the marine reserve establishment process	Poor information on the state of fish stocks	Difficulty in zoning for one marine use without a broader strategic spatial plan addressing multiple uses
Time and cost implications of undertaking detailed planning on a region-wide level	Poor linkage between district plan provisions and infrastructure provision	Lack of a catchment-level focus when considering resource consent applications	Uncertain role of regional councils in marine biosecurity vis a vis Biosecurity New Zealand			
High levels of conflict in some areas	Cumbersome plan change procedures	Poor linkages between catchment management and the ecological carrying capacity of the recipient marine area				
Lack of a statutory framework for integrated, strategic and marine protection planning	Lack of resources and information in smaller councils	Cumbersome plan change procedures				
	Lack of priority given to the Kaipara Harbour resulting in weak development controls					Difficulty in implementing complex legislative provisions

Marine planning	Coastal development	Sedimentation management	Biosecurity management	Hauraki Gulf amateur fishing	Kaipara Harbour commercial fishing	Kaipara Harbour aquaculture
Recent Developments						
Move towards sub-regional planning (focused on the land/water coastal edge)	Community-based planning Move towards strategic growth management and planning	Greater regulation of the agricultural sector Piloting of catchment-based management approaches in small areas	Marine biosecurity monitoring programme Work towards expanding the biosecurity toolkit	A more strategic and stakeholder driven approach to fisheries management (fisheries plans) A more strategic and stakeholder driven approach to the establishment of marine reserves (marine protection plans) Attempts to better define amateur fishing rights	Voluntary code of conduct	Invited private plan change mechanism Application of adaptive management approaches
Move towards strategic planning		Development of more sophisticated monitoring programmes to identify marine areas at high risk from sedimentation				
Greater stakeholder involvement in fisheries and marine protection planning						
Prospects for better integration						
Further integration between marine plans likely	Integration between district plans unlikely due to different approaches and timeframes	Integration between the management of a marine area and its catchments only likely in targeted hot spot areas	Better integration likely through the New Zealand Coastal Policy Statement	Unlikely until high levels of conflict around fishing and marine reserves addressed	Integration unlikely until an effective mechanism is created for local management in inshore areas	Only likely if strategic spatial planning is undertaken for the Kaipara Harbour
Integration between regional coastal planning and fisheries planning unlikely due to poor interface						

Approaches to Achieving Better Integration

Efforts to achieve better integration within the Hauraki Gulf and Kaipara Harbour have focused around improving communication and information sharing, developing joint policies and plans and engaging in joint projects to address shared problems. These are described more fully in the following sections.

Improving communication

Good communication between different management agencies can help ensure that actions taken by one agency do not have unintended consequences in another management regime. It can help to reduce duplication in effort and identify management gaps. It also can assist in the development of relationships and build-up of trust, both precursors to successful integration.

Interviewees referred to a myriad of networks and cross-agency committees and forums, particularly those attended by local government councillors and planners. Often these were not solely focused on coastal management, although they included coastal issues on their agenda. Short-term cross-agency forums have been set up from time to time to deal with specific coastal issues such as aquaculture and mangrove management. The responses of the interviewees indicated that some of the strongest communication links were between councils and some of the weakest between councils and MFish, although this area requires more detailed investigation.

The most holistic body established to focus on coastal management is the Hauraki Gulf Forum. Unlike other more informal arrangements, this forum has a statutory basis under the Hauraki Gulf Marine Park Act 2000. The Forum has political representatives from two regional councils and ten territorial authorities; one appointee each by the Ministers of Conservation, Fisheries and Māori Affairs; and six tangata whenua representatives, amongst others. The Forum meets four times a year with the supporting Technical Officers Group meeting at least monthly. For each meeting of the Forum each constituent party is required to report on coastal activities and this usefully provides an update on activities within the Gulf.

Although interviewees reported that the Forum had had little direct impact on the coastal management activities of individual agencies, it appears to have facilitated inter-agency communication and relationship building which indirectly may have had an impact through supporting a greater readiness to participate in inter-agency projects.

Integrating planning

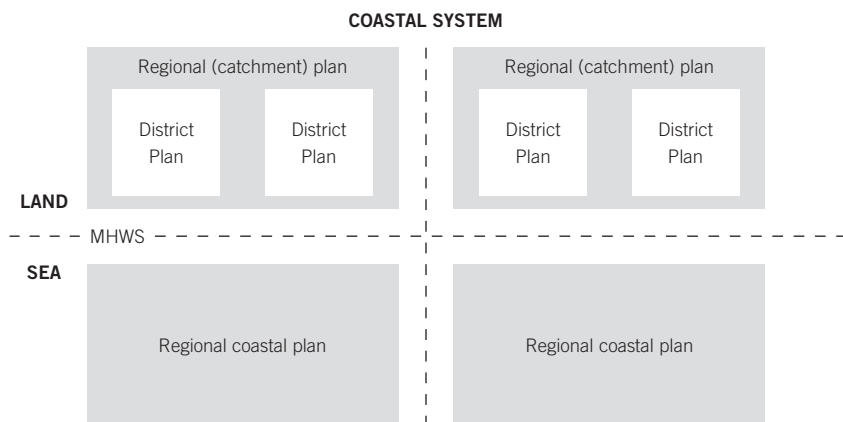
Different approaches have been adopted to integrate coastal planning. They include endeavours to integrate local government coastal planning, incorporating planning under the RMA and the Local Government Act 2002. Approaches are also being developed to integrate marine protection planning across both local government and central government jurisdictions.

Local government coastal planning

Coastal planning under the RMA is fragmented between different plans and different management bodies. It is split ‘horizontally’ between planning for catchments (through regional plans) and planning for marine areas (through regional coastal plans). It is also split ‘vertically’ between planning for different parts of the same marine system, different catchments draining into the same marine system, and different districts within catchments (district plans) (see Figure 4.2).

In the case study areas, the Hauraki Gulf marine area is managed by two regional councils, the ARC and EW. The catchments draining into the Gulf are also managed by these two regional councils as well as ten territorial authorities. The management of the Kaipara Harbour is also split between two regional councils, the NRC and the ARC. As well as being managed by these regional councils, the harbour’s catchments are also managed by two territorial authorities, Rodney District Council and Kaipara District Council.

Figure 4.2: Fragmentation of RMA coastal planning system



There are several strategies which could potentially be used to integrate this planning system. These include:

Combining the two regional coastal plans for the marine area.

This has not yet been attempted in either of the study areas. It seems more likely in the Kaipara Harbour where there are fewer pressures on the marine environment and therefore management is less complex. It may result from the current joint initiative for the harbour that is being undertaken by the ARC and the NRC.

Combining the regional (catchment) plan with the regional coastal plan

This has not yet been attempted in the study areas because of the increased complexity of planning which would result. Coastal managers interviewed indicated that they seek to reduce undue complexity, to make planning tasks manageable. Preparing a regional coastal or catchment plan on its own is a difficult task, let alone combining them. The plans also have somewhat different focuses, with coastal plans more focused on spatial planning for marine uses, whereas catchment planning is more focused on non-spatial minimisation of pollution and run-off.

Some regional councils have prepared coastal environment plans which include objectives and policies for the entire coastal environment, but only rules for the coastal marine area. These plans can help bridge the divide at mean high water springs, particularly if they effectively influence land management by territorial authorities. However, they fall short of encompassing broader catchment management.

Preparing one integrated RMA plan for the entire system

This could incorporate the regional, regional coastal and district plans. This would multiply the complexity of planning through increasing the range of issues to be addressed and the number of management agencies involved. It has not yet been attempted. An alternative softer approach is to prepare a combined non-statutory plan for the entire system, the provisions of which can then be implemented through the various statutory instruments through making the required changes to regional and district plans. This is the broad approach being adopted by Te Uri o Hau for the Kaipara Harbour, although the scope of that planning initiative is very wide, also incorporating fisheries and marine conservation issues outside the RMA.

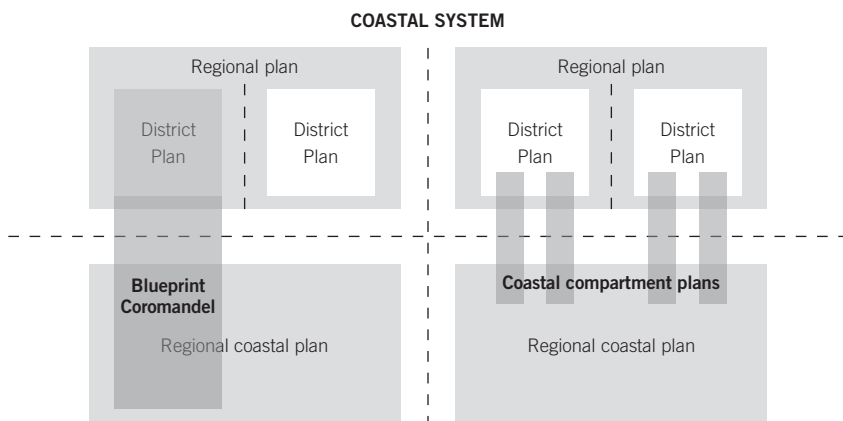
Instead of adopting these global approaches, regional councils have been approaching the integration challenge on a smaller spatial scale, so as to render the task more manageable. The two main approaches identified in the research, which are both being applied to the Hauraki Gulf, are (see Figure 4.3):

Stitching the regional coastal plan and district planning together across mean high water springs through preparing a series of coastal compartment plans. The

plans are largely confined to the coastal edge rather than addressing catchment issues. This approach has been adopted by the ARC in the Hauraki Gulf as well as other areas, and the plans are non-statutory. This has raised some challenges, including how to scale the planning exercise up to cover the bulk of the coastal interface, and how to incorporate the provisions in non-statutory plans into the statutory framework.

Slicing the system into chunks and undertaking combined regional and district strategic planning for the catchment and marine area. This is the approach being undertaken by EW and the Thames-Coromandel District Council for the Coromandel Peninsula. It is reliant on both councils being ready to undertake strategic planning for the same area at the same time. In this case, the Thames-Coromandel District Council was planning to prepare a growth strategy and EW was looking at preparing harbour plans. The two initiatives were compatible so could be combined. Like coastal compartment plans, the plans are non-statutory.

Figure 4.3: Integrating RMA coastal planning on a smaller scale



Marine protection planning

Marine protection planning is an attempt to integrate one aspect of marine management potentially across the marine conservation, fisheries and RMA management systems (see Figure 4.4). The integration will be achieved through the joint preparation of a non-statutory spatial plan, focusing on a common approach applied in different ways by different management agencies – excluding specified activities from spatial areas to achieve conservation objectives. Implementation relies on adoption through separate processes under

at least three separate pieces of legislation which have different purposes, the Marine Reserves Act 1971, the Fisheries Act 1996 and the RMA.

Figure 4.4: Integrating spatial marine protection

MARINE SYSTEM		
Marine Conservation	Fisheries management	RMA marine management
MARINE PROTECTION PLANS		
Marine reserves	Areas excluded from fishing activity	Conservation zones
Species marine protection	Setting TAC and TACC Other 'sustainability' measures	Other marine zones and rules Resource consent processing

Undertaking joint projects

Where agencies share common or overlapping issues they may undertake joint projects, often in association with other stakeholders, to address issues of mutual concern. The research identified a myriad of joint projects including joint research projects, joint planning projects and joint implementation projects. Joint research projects have been undertaken in diverse areas such as determining the carrying capacity of the marine area for aquaculture (Firth of Thames), the impacts of sedimentation likely to be generated by urban rezoning on the marine area (Okura and Whitford) and the relative risk factors affecting the marine area (Muddy Feet II - Firth of Thames).

Many joint planning projects have been undertaken as described in the sections above. Numerous implementation projects have also been undertaken jointly including the Mahurangi Action Plan and the Peninsula Project on the Coromandel Peninsula. Undertaking joint projects helps to strengthen the relationship between staff in different agencies and can result in the development of joint information all of which helps build a solid foundation for future collaboration.

Conclusions

The initiatives which have been developed to achieve better integration highlight the tension between the increased complexity which results from integration and the need to ensure that an initiative is 'doable'. A doable project is likely to be one where the information needs are modest, where the number of stakeholders is manageable and where the time period required to complete the project is short enough to enable political and stakeholder interest to be maintained. As a result, integration is currently happening on a relatively small spatial scale which is more manageable. This suggests that future integrative efforts are likely to focus on localised hotspots, where interagency collaboration is required to address issues of immediate mutual concern, rather than on large-scale integrated planning efforts where the outcomes are uncertain and politically more risky.

A major issue for integrated initiatives is the lack of a statutory framework within which to ground them. All the integrated planning initiatives identified are being undertaken as non-statutory planning processes. They all therefore face the major hurdle of achieving implementation through the myriad statutory processes required to be undertaken by different management agencies under different pieces of legislation. This provides no certainty that the actual outcome will closely mirror the intention in the non-statutory planning documents. It indicates a need for the development of a statutory framework to support integrative initiatives and to provide the outcomes with some legal standing.



five

Supporting Integrated Coastal Management

Chapter two reviewed a range of literature on the concept of integrated coastal management and identified eight elements which may help support this. These were:

- Broad agency alignment
- Inter-agency dependency
- Climate of trust
- Compatible scales
- Decentralised decision making
- Strategic approach
- Good information base
- Adequate resourcing

The extent to which these elements are currently in place and how they might be supported is explored in the following sections. The conclusions drawn in this chapter are based on the findings of the research undertaken into the management of the Hauraki Gulf and Kaipara Harbour. In this section, these findings have been applied to coastal management throughout New Zealand more generally. This is on the basis that experiences in managing these two study areas are likely to be replicated, to a greater or lesser extent, in other marine areas which are managed under the same legislative and policy framework.

Broad agency alignment

The discussion in chapter three explored the nature of conflicting values and approaches underpinning the activities of different management agencies. Different agencies are focused on managing different things and utilising different tools to achieve different objectives. At times these objectives and/or methods can be in direct conflict, such as when the establishment of marine reserves by the Minister of Conservation reduces the ability to exercise rights to extract fish issued by the Minister of Fisheries. At other times it can result in an organisational cultural divide, such as that between the relatively narrow private property right orientation of fisheries management and the more holistic environmental impact assessment (EIA) orientation of RMA marine management. Other organisational cultural differences are evident in the way management efforts are undertaken, with a participatory culture now well embedded in local councils' coastal planning efforts but not yet applied to

nearly the same extent in marine protection and fisheries management. Such conflicts and divides can make integrated management efforts difficult.

More recently the divergence between the different agencies appears to have narrowed somewhat. In conjunction with MFish, DoC is embarking on a process to identify marine protected areas which will take into account, upfront, other users of the marine area. The Ministry of Fisheries is starting to incorporate some of the environmental effects of fishing activity into its decision-making processes, although still shying away from a full EIA approach. Both DoC and MFish are now seeking greater stakeholder involvement in decision-making through their marine protection and fisheries planning exercises. Such convergences when they occur should be encouraged.

Inter-agency dependency

The extent to which a management agency can achieve its goals independently, rather than being dependent on other agencies to succeed, can have a major impact on the degree of collaboration which occurs between agencies. Collaboration can bring with it a loss of control and therefore increased risk. Managers of coastal agencies are therefore unlikely to embark on collaborative efforts unless they have no choice, or if the countervailing benefits are seen as substantive. The sections below explore the degree of interdependence between the various coastal management agencies interviewed for the case studies.

Regional councils and territorial authorities

Regional councils and territorial authorities are joint managers of catchments under the RMA. The relationship between the two levels of local government has in the past been problematic, but there is evidence of more positive relationships developing.

The establishment of regional councils experienced a shaky start. When local government reorganization was undertaken in the late 1980s, the Local Government Commission which was in charge of delineating administrative boundaries was not sympathetic to the concept of regional councils and favoured the establishment of large unitary authorities instead. Although the Commission was required by law to establish regional councils, it sought to limit their power by establishing the councils, not as a second tier of government, but as a complementary part of a single system. As a result, the relationship between regional and district councils lacked clarity with areas of potential overlap but no clear hierarchy (Memon, 1993, 81).

When the central government changed shortly after the new local government entities were established, the new Minister for Local Government recommended that regional councils be abolished. Although he was not successful in achieving this countrywide, he did achieve the abolition of the Nelson-Marlborough Regional Council and the designation of the three territorial authorities as unitary authorities undertaking the functions of both regional and district councils. Many regional councils have struggled to establish an electoral constituency when faced with this uncertain central government political support, with large and powerful district councils and with having few service delivery functions to perform for ratepayers themselves (Memon, 1993, 82).

This chequered history has made it difficult for many regional councils to assert a strong integrating role over functions exercised by territorial authorities. A low-risk approach adopted by many regional councils was to define their role narrowly, to include those functions that are solely undertaken by regional councils, such as the management of water, air, soil and the coastal marine area, but to avoid getting involved in subdivision and urban land-use issues. This enabled regional councils to reduce potential conflict with territorial authorities and to avoid the perception by their ratepayers that they were needlessly funding a duplication of tasks. A notable exception to this approach has been the ARC which, formed out of the formally powerful Auckland Regional Authority, had a stronger institutional basis on which to intervene in land-use planning decisions.

Greater clarity about the relationship between regional council and territorial authority planning has recently been achieved with the 2005 amendments to the RMA. These have created a stronger hierarchy of planning documents, with district plans now required to 'give effect' to regional policy statements rather than merely having to be 'not inconsistent with' them (section 75(3)).

One direct area of conflict between the different levels of local government has been when regional councils have prosecuted territorial authorities for breaching the provisions of the RMA for service provision such as with sewerage overflows from council-operated works. Regional councils also are the consent authority for the discharge permits often required by territorial authorities for stormwater and wastewater discharge outlets.

The research identified a recent positive shift in relationships between regional councils and territorial authorities. For example, the relationship between EW and the Thames-Coromandel District Council has shifted from one of conflict and suspicion to a more positive association. Officers from both councils are working cooperatively together on a number of projects including

Blueprint Coromandel. The ARC is also working positively with several territorial authorities on the development of coastal compartment plans.

So what is encouraging regional councils and territorial authorities to cooperate? From a territorial authority perspective the research pointed to a couple of important factors. As councils consulted their coastal communities on community outcomes, as required under the Local Government Act, it became increasingly apparent that many of those outcomes were not within the control of the council to deliver, particularly those which crossed the land-marine boundary. Examples of cross-boundary local issues included mangrove expansion, coastal erosion and the provision of boat launching facilities. To deliver on community expectations, territorial authorities needed to engage with regional councils.

In addition, small territorial authorities perceived regional councils as having the technical and informational resources which they lacked and could ill afford. Regional councils were therefore being seen as a source of much-needed technical expertise which could be brought to bear on local issues.

Conversely regional councils are also dependent on territorial authorities, in particular, to achieve their resource management goals. This is because land use has a major impact on water quality and the health of the marine area, amongst other things. As they have completed their first generation RMA plans, regional councils have been able to refocus their attention and resources on how to negotiate more cooperative relationships with territorial authorities to achieve more integrated management.

Regional councils and the Department of Conservation

There is a strong overlap between the roles of regional councils and DoC in coastal marine areas, with the Minister of Conservation developing the New Zealand Coastal Policy Statement and approving regional coastal plans and restricted coastal activities. This statutory interdependency has not, however, always resulted in positive relationships. The research identified a shifting relationship between the agencies, seemingly largely dependent on the personal relationships between DoC and regional council officers.

Ministry of Fisheries and regional councils

There is a strong ecological linkage between the roles of regional councils and MFish. Regional councils are managing catchments and marine developments and uses which can have a major impact on the health of fish stocks. Conversely, MFish is charged with managing fishing activity, which can have a major

impact on the environmental quality of marine areas, which regional councils are seeking to maintain.

Despite this ecological linkage, there is little statutory or practical interdependency between the activities of these agencies. The Ministry of Fisheries has defined its role narrowly, in practice, as being focused on managing fishing activity. It does not, for example, engage as an advocate for important fisheries habitats in RMA processes. This has minimised the Ministry's dependence on regional councils in order to achieve its management goals. Regional councils, in turn, have defined their role as excluding the management of the environmental effects of fishing. This is despite the provisions of the RMA which, through section 30(3), preclude councils from controlling the harvesting of aquatic organisms in order to manage fish stocks, but do not preclude controlling fishing activity to protect marine ecosystems. The interviews identified a weak relationship between regional councils and MFish, with each agency generally undertaking its activities independently of the other.

Ministry of Fisheries and the Department of Conservation

There is a high degree of statutory and policy interdependence between the roles of MFish and DoC. Under the Marine Reserves Act 1971, the Minister of Fisheries and Minister of Transport are required to concur with the decision of the Minister of Conservation before a marine reserve can be designated. Despite this joint role, historically there has been little coordination between DoC and MFish regarding the process of marine reserve applications, with MFish only becoming involved once the Minister of Conservation had reached a decision. The two agencies have now negotiated a marine reserves protocol to facilitate the processing of applications and MFish is becoming involved earlier on in the process.

The two agencies have also worked together over several years to reduce the impact of fishing activity (managed by MFish) on protected marine mammals and seabirds (managed by DoC). More recently the two agencies have been jointly charged by government with implementing the Marine Protected Areas Policy and Implementation Plan. Conflict still arises, however, due to the different cultures of the two organisations with MFish having a history of being centralised, having stronger industry links and being less consultative than DoC. Philosophically the two agencies are poles apart, with MFish focused on utilisation and DoC focused on preservation.

Tangata whenua and other agencies

Tangata whenua are very dependent on other statutory agencies in order to exercise a kaitiakitanga role in respect of the coastal environment. This is because the role of tangata whenua as coastal managers largely lacks a statutory basis. Such a role can include both managing coastal resources of significance to iwi and contributing to coastal management more broadly. An exception to this is in regards to customary fishing activity under the Fisheries Act 1996, where kaitiaki have a defined role in managing the customary harvest. There is also provision under the Act for a kaitiakitanga role to be exercised where mataitai reserves and taiapure-local fisheries areas are established.

Tangata whenua also often lack the resources to undertake significant resource management activities, with management resources largely residing in government agencies. This issue has been particularly problematic in the Kaipara Harbour where Te Uri o Hau is seeking to lead an integrated planning project but has struggled to secure the necessary resources.

On the other hand, statutory coastal management agencies are not generally dependent on tangata whenua in order to carry out their activities. This can create a very unequal relationship and iwi representatives interviewed for the case studies expressed much frustration at their inability to influence coastal management decision making, particularly with respect to RMA processes.

There have been some positive developments which have strengthened the relationship between tangata whenua and other coastal managers. The Ministry of Fisheries, the Department of Conservation and many councils have employed iwi liaison officers to facilitate relationship building with iwi. The Hauraki Gulf Forum provides the opportunity for tangata whenua representatives to sit around the table as equal partners with other management agencies, with resourcing provided for technical support officers. In addition, the Blueprint Coromandel project has incorporated the Hauraki Maori Trust Board as a partner at the early planning stages.

Climate of trust

The literature review on integrated coastal management identified high levels of trust as an important precursor to effective integrated management approaches. This is because integration is based on individuals within management agencies and stakeholder groups cooperating with each other and they are only likely to do so if they trust each other.

The interviews identified strong levels of mistrust in some areas, with growing trust relationships being developed in others. The highest levels of mistrust were expressed by stakeholders involved in the marine reserve activities undertaken by DoC and in fisheries management undertaken by MFish. This

may be the result of poorer stakeholder participation and conflict resolution processes in these two areas than is provided for in RMA processes. For example, in fisheries management, a recent study into information transfer concluded that litigation remains the only channel through which stakeholders can communicate effectively (Massey, 2005, 235). The mistrust has built up over several years and is likely to make moving towards an integrated approach in the areas difficult.

On the other hand, as already discussed, positive relationships are being developed between regional councils and some district councils, as well as between regional councils themselves. This should help facilitate a move towards a more integrated management approach for RMA functions. Positive relationships are also being developed between some councils and tangata whenua.

The development of ongoing positive relationships can be hindered by both frequent staff changes in councils and the physical distance between agencies. Several stakeholders expressed frustration at the high staff turnover in councils and therefore the inability to form ongoing relationships. Proximity can also be important. The Hauraki Māori Trust Board has been able to form a close relationship with EW as the Board's headquarters are physically located relatively close to EW's Hamilton headquarters. The Board has a poorer relationship with the ARC whose Auckland headquarters are much further away.

Stakeholder participation in coastal management is important because it can help to improve the information available to managers and, if facilitated well, can also help to resolve conflicts, develop trust and build consensus around solutions.

The RMA incorporates extensive participation processes both in plan making and the processing of significant resource consents. This has helped to embed a public participation culture within many local authorities. This has been supported by the provisions of the Local Government Act 2002 which require councils to consult their communities on desired community outcomes and on significant policy decisions. Most of the councils studied also included extensive public consultation in non-statutory processes such as the preparation of community plans and strategic planning documents.

The case studies indicated that high levels of consultation can be time-consuming and resource-hungry. They can also both raise community expectations and lead to consultation exhaustion. On the other hand, consultation can help mobilise communities to support coastal management, improve policy targeting and acceptance and help build positive relationships.

Stakeholder consultation is not so embedded in the legislation or cultures of the fisheries management or marine protection regimes, although the

level of consultation is increasing. Within fisheries management, MFish has been establishing customary and recreational forums to improve stakeholder relationships. Stakeholder working groups have also been established to provide input into research projects and stock assessment exercises. These are very technical processes and stakeholders require a sophisticated understanding of fisheries science to meaningfully participate. The debate within the group sessions focuses on the robustness of the scientific research and complex stock assessment mathematical models. In practice, the main participants are fisheries scientists engaged by various parties and specialist fisheries representatives of a few national environmental organisations.

Similar working groups have not been established to develop or consider management approaches or decisions on the management of specific fish stocks. These decisions are made through a process of Ministry staff developing draft advice to the Minister in-house and stakeholders having the opportunity to make written submissions on this advice. Ministry staff then finalise the advice and the Minister makes a final decision. The Ministry's initiative, commencing during 2006 to facilitate the development of fisheries plans by stakeholders, should increase the level of participation in developing strategies to guide fisheries management decisions.

The process to establish marine reserves under the Marine Reserves Act 1971 does not include extensive public participation provisions, but only makes provision for written submissions to be lodged on a proposed marine reserve. Although in practice DoC does consult more widely in respect of marine protection, this has been on the basis of specific marine reserve proposals rather than more broadly defining potential areas for marine protection. The initiative to develop marine protection plans should result in wider stakeholder engagement in marine protection decision making.

Effective stakeholder participation may require management agencies to develop the capacity of stakeholders to participate, as well as provide financial support to cover expenses such as travel to attend meetings. The Ministry for the Environment's RMA Education and Advisory Fund helps to build capacity within the RMA system. Similar funding schemes could assist with building capacity under the fisheries and marine protection systems.

Effective conflict resolution processes can also be important in reducing conflict and increasing trust. The RMA has well-developed conflict resolution procedures with submitters having the right to be heard at a council hearing and the right to appeal to the Environment Court and receive either court-assisted mediation or an independent hearing where the merits of the case are reviewed afresh. There are no similar provisions in the Marine Reserves Act,

with the only legal recourse for aggrieved stakeholders being judicial review in the High Court where the merits of the decision can not be considered.

Although there are dispute resolution provisions in the Fisheries Act, which include the ability of the Minister to appoint a Fisheries Dispute Commissioner when there are conflicts between people with fishing interests, these are limited, do not appear to be well understood or supported by MFish staff or stakeholders and are rarely used. More commonly, parties initiate judicial review proceedings in the High Court to challenge decisions reached by the Minister of Fisheries. Such proceedings can only challenge such decisions on the basis that the Minister failed to act in accordance with his or her relevant statutory obligations and the Court is not able to review the merits of decisions.

Parties are, however, attempting to use the High Court as an independent reviewer of the merits of fisheries decisions. This indicates a need for an effective appeal mechanism where decisions made under the Fisheries Act can be independently reviewed. A recent decision by Judge Harrison expresses the frustration felt by the Court over this issue:

It should have been unnecessary to recite that it is not my function to determine whether the Minister's decisions were objectively right or correct. It is not within the High Court's power to second-guess the merits of his decisions ... Nevertheless both challengers filed voluminous amounts of written material, much of it of an opinion nature and amounting to submissions camouflaged as evidence ... I am not the first to voice these frustrations. In judicial review proceedings between commercial interests and the Minister in 1997 the Court of Appeal expressed its dissatisfaction with 'the forest of paper' with which it was presented and the 'improper' treatment of the proceedings 'as if they were an appeal on matters of fact.

(The NZ Recreational Fishing Council, 2007, paras 5-7)

Compatible scales

In order for integration to be able to occur, it is important that management agencies are operating at spatial scales which are able to link together. Spatial planning enables locally-specific issues to be addressed and provides an interface where different coastal management regimes can interact and complement each other.

Spatial planning is an important part of coastal management under the RMA through the preparation of regional and district plans. Although these plans have largely divided planning for the land and marine areas, there is the potential for integration across MHWS as well as across coastal systems. Such integration is being attempted through various initiatives such as the coastal compartment plans and Blueprint Coromandel.

Integration is easier where plans are comparable and this is why, in the Kaipara Harbour, it is likely to be easier to integrate the regional coastal plans which have a broadly similar approach, than the district plans which adopt a very different approach to coastal subdivision and development.

Fisheries management has moved away from a broader spatial planning approach, which was evident in the fisheries management plans of the 1980s in which all fisheries management issues within a spatial region were considered, to a single species management model. Although the Ministry is moving towards a more strategic management model with the development of the latest generation fisheries plans, these are currently focused on single species. Subsequent plans may adopt a multi-species focus, such as for the west coast harbours of the North Island.

This focus on single-species management seems likely to continue as quota rights are species-specific and commercial stakeholders consequently see the primary goal of marine management as being to ensure the sustainable utilisation of individual fish stocks, not marine systems (Massey, 2005, 250). It provides little interface with other forms of coastal management, and is likely to be a significant impediment to incorporating fisheries management into integrated coastal management. Figure 5.1 illustrates the very large management areas adopted for fisheries management, in this case for snapper.

Another example of spatial mis-match is that between fisheries management and marine reserves. Under the Marine Reserves Act 1971, a marine reserve cannot be declared where it would 'interfere unduly with commercial fishing' (section 6(c)). Decisions on this issue largely favour marine reserve proposals, as they are comparatively only a very small proportion of very large quota management areas.

The Department of Conservation, in association with MFish, is moving towards spatial planning for marine protected areas. Such plans, however, lack regulatory teeth in that they provide a menu of possible options for marine protection which may or may not be implemented through various legislative mechanisms. These plans will overlap the areas covered by regional coastal plans and the relationship between the two is not entirely clear. There must, however, be considerable potential to integrate the two plans through regional coastal plans, reflecting potential marine protection areas, and marine protection planning,

reflecting uses provided for in regional plans. Indeed it is difficult to see how either plan could have much credibility if it is not integrated with the other.

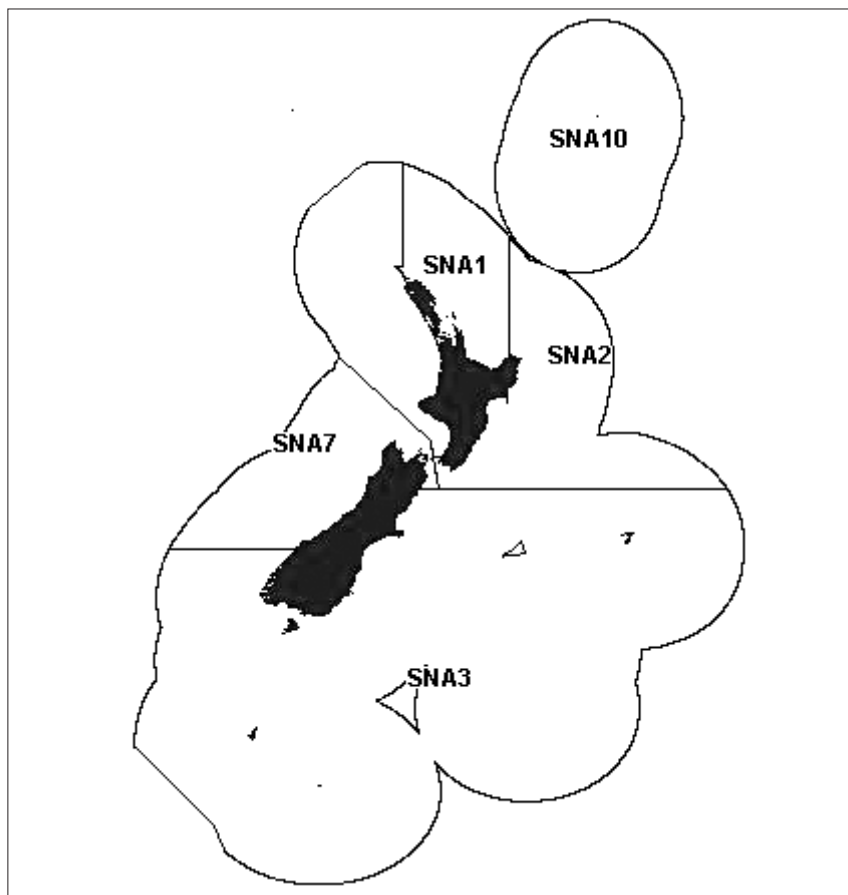


Figure 5.1: Quota management areas for snapper Source: www.fish.govt.nz

Decentralised decision making

Decentralised decision making enables a degree of local autonomy and this in turn enables local partnership building and innovation. For example, a recent study of local government in 40 European towns and cities found that, when

local governments were granted higher levels of autonomy and independence, they respond by being more proactive and adventurous in their policy making and implementation (Evans et al, 2005, 109).

Decision making under the RMA is heavily decentralized, with councils having autonomy within the broad framework of the Act. As a result, one might expect to see high levels of innovation and this was evident in some of the councils involved in the study.

Several of the staff interviewed at EW described a culture where mistakes were regarded, not as a bad thing, but as a learning opportunity. The council has evolved a process of piloting different approaches to integration and, based on lessons learnt, is trialling new approaches. This can be seen in the move from a focus on Integrated Area Management to Integrated Catchment Management and the more strategic Blueprint approach. Environment Waikato has also been developing adaptive management and risk-based methodologies to apply to the marine area.

The ARC has led the way in the development of a marine monitoring programme which has already demonstrated its value as an early warning system for marine degradation. The ARC has also piloted a range of different approaches to integrated management, including the development of a coastal environment regional plan, catchment modelling for greenfields urban development and the development of joint regional and district coastal compartment plans.

The Auckland City Council trialled an ecologically-based, land-unit approach when planning for the Hauraki Gulf Islands and has recently modified the approach based on implementation experience.

The central government departments involved in coastal management have different degrees of decentralisation. Although DoC is a very large organisation with around 1600 staff members, it is relatively decentralised into fourteen conservancies whose boundaries broadly match those of regional councils. Each conservancy develops its own conservation management strategy and has its own independent conservation board to provide advice on priorities for action. This has enabled a degree of autonomy at a local level and has facilitated, for example, the development of the approach behind the Hauraki Gulf marine protection plan to be initially developed by the Auckland conservancy office rather than centrally.

The Ministry of Fisheries is a much smaller and more highly centralised government agency with around 400 staff members. Policy development is largely undertaken in Wellington and Nelson with other regional offices engaged in operational and compliance issues, including the review of fish stocks and the development of fisheries plans. This has meant that the Ministry's decision making on some important regional issues, such as the Great Barrier marine

reserve proposal, and the application of the marine protected areas policy and implementation plan in the Hauraki Gulf, has been based in Wellington. The recent initiation of the fisheries plan process has helped to regionalise some decision-making processes.

This large spatial separation, between MFish decision makers who are based in Wellington and the decision makers in other coastal management agencies which are based regionally or locally, makes effective integration less likely. It can also mean that local innovations, such as those developed by the Sustainable Fisheries Management Study Group for the Kaipara Harbour, are unwelcomed by the Ministry which takes a national rather than regional view because of the perceived precedent effect on other areas. In contrast, other coastal management agencies studies appear more comfortable with trialling different approaches for different areas and issues.

Strategic approach

Integrated coastal management requires a strategic focus so that the myriad activities that impact on a single coastal system can be guided towards a common long-term goal. Conversely a strategic approach requires integration because, for example, proactive forward-looking planning of marine areas requires consideration of what is happening in other sectors and on the adjacent land.

Several of the council planners interviewed identified the lack of a strategic focus in current planning efforts as a major difficulty, because of the uncertainty this created both for applicants wanting to carry out activities and for communities in terms of the impacts on their places and resources. It also made it difficult to plan for infrastructure provision and to address the cumulative effects of activities.

Most of the agencies involved in the study are moving towards a more strategic approach. The Ministry of Fisheries is developing fisheries plans with the intention of placing more emphasis on objectives-based management. The Department of Conservation is moving away from the ad hoc establishment of marine reserves to the development of marine protection plans. District councils are developing coastal and growth management strategies.

Good information base

Information plays an essential role in coastal management. However, it can be expensive to obtain and there are large gaps in the current knowledge of the state of the coastal environment and coastal processes. Very little monitoring is undertaken of the state of the marine environment. The monitoring which does occur differs between councils and there is no common approach or methodology. The Auckland Regional Council and EW are putting in place more rigorous marine monitoring systems but their coverage is still patchy. Most of the information gathered by MFish relates to fish stocks and not the marine environment within which they are located. The monitoring of marine reserves by DoC is not well-funded and has not been undertaken in a consistent or comparable manner.

Adequate resourcing

Coastal users do not currently pay the cost of their use of public resources and the public is reluctant to pay for coastal management. As a result, coastal management is poorly-funded. For example, in the year ended 30 June 2006, the ARC spent only 1.7 per cent of its total operating expenditure on the management of coasts and seas. Of the three million dollars the council spent in this area, \$1.2 million was spent on maritime operations and only \$1.8 million on coastal management (see Figure 5.2). Environment Waikato also devoted a small proportion (2.8 per cent) of its spending to coastal management, with expenditure on the coast totalling just over two million dollars (see Figure 5.3).

Under the RMA (section 64A), regional councils are empowered to impose coastal occupation charges on persons occupying the crown-owned coastal marine area. The money raised from such charges is to be only used for the sustainable management of the coastal marine area. In reality, it has been too practically and politically difficult for councils to impose the charges and attempts to revise the regime to make it more workable have been stalled at the central government level (Palmer et al, 2005).

The Ministry of Fisheries adopts a user-pays approach to the costs of managing fish stocks. This is a very complex system which endeavours to allocate costs incurred to specific fish stocks or groups of stocks and the related quota holders. It has resulted in relatively low management costs, as those that are paying a substantive proportion of the costs, the quota holders, have an incentive to minimise expenditure. Environmental and amateur fishing stakeholders have concerns, however, that this approach has resulted in

commercial fishing interests having too strong a say in management questions on the basis of ‘user-pays user says’ (Hersoug, 2002, 12).

Figure 5.2: ARC operating expenditure for year ended 30 June 2006 Source: ARC, 2006a

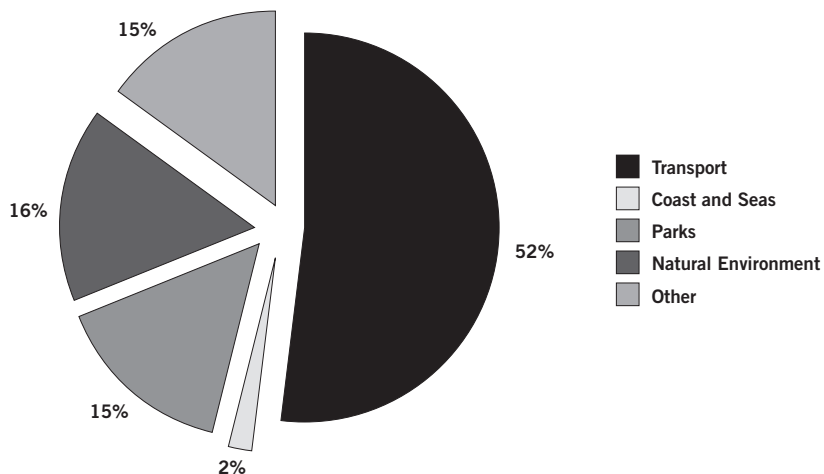
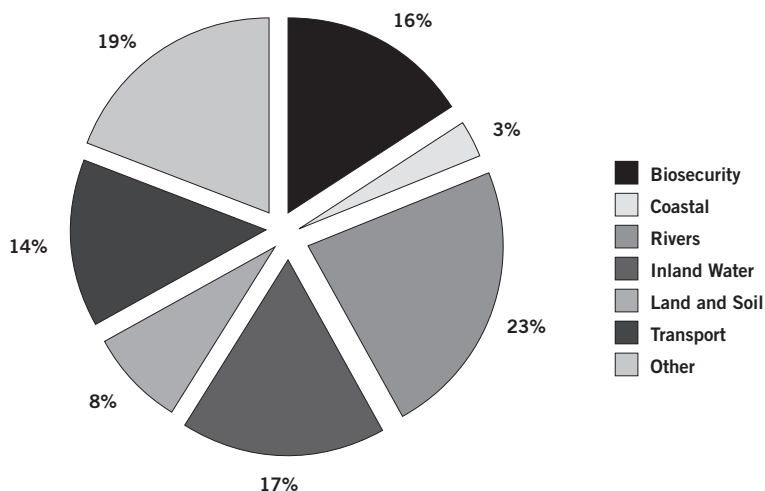


Figure 5.3: EW operating expenditure for year ended 30 June 2006 Source: EW, 2006c



PART TWO

CASE STUDIES



six

Case Study 1: Planning for the Hauraki Gulf and Kaipara Harbour Coastal Marine Areas

Spatial challenges in a fluid environment

We've realised that it's all very well having a good environmental quality plan, but you also do need to [strategically] plan as well. (respondent 13)

It's certainly harder to create zones around the sea than it is on land. (respondent 24)

The policy framework

Three main types of environmental planning are currently being applied to the coastal marine area (CMA):

- *Regional coastal planning* under the RMA
- *Fisheries planning* under the Fisheries Act 1996
- *Marine protection planning*, which is non-statutory, but which is being undertaken within the framework of the Marine Protected Areas Policy Statement and Implementation Plan (DoC & MFish, 2005).

The RMA provides a comprehensive legislative framework for marine planning by local authorities. The Act requires regional councils to prepare regional coastal plans, which apply to the coastal marine area extending from mean high water springs to the outer extent of the territorial sea (twelve nautical miles). The purpose of the plans is very broad, and is to assist the regional council along with DoC, to achieve sustainable management in relation to the CMA (section 63(2)).

The RMA is not very precise as to the content of the plans, or the planning approach which should be taken when preparing them. The plans must simply state the objectives for the CMA, the policies to implement the objectives and any rules to implement the policies (section 67(2)).

Further guidance as to the preparation of the plans is provided in the New Zealand Coastal Policy Statement (NZCPS), which contains policies addressing a wide range of issues in the coastal environment. They have a focus on the protection of the coast's natural character. Some policies specifically address coastal waters and urge regional councils to develop rules to enhance water quality in the coastal marine (policy 5.1.1).

The Fisheries Act 1996 provides for the preparation of fisheries plans, but is also vague as to what they are and how they should be developed. They may relate to fish stocks, fishing years, or areas. Amongst other things, they may include fisheries management objectives, strategies to achieve the objectives, and/or

performance criteria to measure the achievement of the objectives (section 11A). They are prepared within the overriding purpose of the Act which is 'to provide for the utilisation of fisheries resources while ensuring sustainability' (section 8(1)). Although there is a requirement in the Act (section 12) that, before approving a fisheries plan, the Minister must consult with 'such persons or organisations as the Minister considers are representative of those classes of persons having an interest in the stock or the effects of fishing on the aquatic environment in the area concerned' and provide for the input and participation of tangata whenua, there is no statutory submission or appeal process.

A *Fisheries Plan Framework* was issued by MFish in 2005. This identifies fisheries plans as being the mechanism to deliver objectives-based fisheries management. They are also identified as an important mechanism to improve tangata whenua and stakeholder involvement in fisheries management. They are to be developed by consensus. Stakeholders can lead the preparation of fisheries plans themselves, but the *Framework* document envisages that the Ministry will be the facilitator of the plans in most cases. The Ministry, however, is not to determine their content:

... the Ministry will not determine the objectives of the fishery or the fisheries covered by the plan. Instead, to make the most of the opportunity to maximise overall value from the fishery, the Ministry will endeavour to facilitate tangata whenua and stakeholder agreement on objectives and other plan components. Where consensus cannot be achieved, the Ministry will convey tangata whenua and stakeholder views to the Minister, and provide advice on the implications of different options. (MFish, 2005b, 3)

The *Framework* document identifies four key elements of the plans; an information brief, a set of objectives, a risk assessment and an operational plan. Preparation of the plans is in its initial stages, with the Ministry in the process of developing three 'proof of concept' plans. One of these plans, the Coromandel Scallop Fisheries Plan, affects the Hauraki Gulf. None applies to the Kaipara Harbour. To date, all of these plans have been 'stock-focused', that is focused on individual fish stocks within a quota management area, rather than 'area-focused' and addressing all fisheries management issues within a spatial area. Future plans may include more than one stock in the same quota management area.

There is no legislative basis for the preparation of Marine Protection Plans. The Marine Reserves Act 1971 does provide for the Minister of Conservation to approve statements of general policy for the implementation of the Act (section 6) and also requires every conservation management strategy to establish objectives for the integrated management of marine reserves (section

7). The Conservation Management Strategy for Auckland 1995-2005 (DoC, 1995, 313) identifies three sites within the southern Kaipara Harbour, and 23 sites within the Hauraki Gulf, warranting investigation for marine protected areas status. These have been identified solely on ecological grounds rather than through a more broadly-based planning exercise. Of those identified in 1995 as warranting investigation, only one, Te Matuku Bay, has since been designated as a marine reserve.

The *Marine Protected Areas Policy and Implementation Plan*, released by DoC and MFish in 2005 (DoC & MFish 2005) provides a policy framework for the plans. The objective of the policy is to 'protect marine biodiversity by establishing a network of marine protected areas (MPAs) that is comprehensive and representative of New Zealand's marine habitats and ecosystems.' The Implementation Plan identifies a four-stage process to be undertaken to establish the network. Stage one includes preparatory work, such as developing a classification approach and mapping existing management tools. Stage two involves providing an overview of the current situation through developing an inventory of existing MPAs, identifying gaps in the MPA network and prioritizing new MPAs. Stage three deals with the development of an MPA network and an integrated regional approach. In nearshore areas this is to involve the establishment of a marine protection planning forum which will consider the classification and inventory information, compile information on existing uses and interests in the area, identify sites and potential tools for marine protection and seek to establish consensus on areas to be set aside as MPAs.

So how have coastal managers approached planning for the coastal marine area within this legislative and policy framework, and how successful has it been? Information from the interviews, which provides some answers to these questions, is presented in the following sections.

Regional coastal plans

Planning for the coastal marine area under the RMA presents different challenges from those presented when planning on land. This is because the marine area is far less well understood, obtaining information is much more difficult and expensive, and the resources which are being planned for are largely in public ownership. In addition, the regional councils in the study areas had very different resource bases to utilise for management (see Figure 6.1).

Figure 6.1: Regional council statistics Source: www.localcouncils.govt.nz and www.stats.govt.nz

Regional council	Operating expenditure (year ended 30 June 2005)	Number of staff	Land area (km ²)	Resident population (2006)
Auckland Regional Council	\$213.1 million	519	6,059	1,303,068
Environment Waikato	\$65.2 million	271	24,025	382,716
Northland Regional Council	\$13.6 million	104	13,789	148,470

So how did the respective regional councils approach the preparation of the first regional coastal plans? At the beginning of the planning process, council planners endeavoured to identify important values associated with parts of the coastal marine area that were important to protect:

... it's really based around ... what are the elements of the coastal marine area in particular that are critical to protect? ... Then you define the use and developments which overlay or can impinge on those values. That's the general approach that we took. (respondent 7)

One of the key features of the Hauraki Gulf is it is essentially on the doorstep of the country's greatest population. So there's a lot of competing demands and pressures on the values of the Hauraki Gulf. So in the preparation of the [Auckland regional] coastal plan, for example, we wanted to understand what those values were and articulate those, and then develop strategies so that those values could be protected but at the same time recognise that a lot of people come to the region and to the shores of the Hauraki Gulf to make a living and to benefit from those values. So it was always a balancing act of competing interests and issues ... one of the challenges was trying to find where that balance was between the use of the Gulf and in the protection of its values.

Values of natural character, public access, balance between use and development, the importance of the coast to Māori, all came through predictably in the consultation process that we went through. In a lot of areas we identified that there wasn't a lot of information about what those values were. So there was a lot of information gathering, for example, around the cultural and historic values of the coast or around the ecological values. (respondent 24)

The planners endeavoured to move away from prescriptive planning and adopt the effects-based philosophy of the RMA. This resulted in less reliance on zoning and greater reliance on applying environmental assessment criteria to individual applications:

Northland regional coastal plan

I guess the pivotal direction was in my council, and I'm not sure but I think it was shared by others, that they wanted to get away from the formalised planning regime, 'town and country planning' sort of philosophy ... [They] saw merit in bringing people like myself into that process as people who actually understood the ecological functioning of the coastal marine area.

Zoning was a dirty word. We had marine management areas instead ...but the challenge we still had was to have sufficient resource information to identify areas of distinct character, one from the other. We were able to draw lines around port areas which is our Marine 5. Marine Management Area 4 was the moorings and marina areas previously identified by the Northland Harbour Board. But in addition we went through a process of also plotting, as best we could, where moorings were concentrated outside those areas and drew up new ones. These were also Marine 4. Marine 3 was the marine farming zone which was again somewhat simplistic using the existing marine lease and license areas.

At the other end of the spectrum [were] Areas 1 and 2. Marine 1 as it evolved was based on the areas of significant conservation value put together by the Minister of Conservation initially. There was ..., resistance from regional councils to the Minister telling us what to put in our plans, but we in Northland could see value in at least having the areas included.

... which left you with a vast majority of the area as Marine 2 - what we call the Marine 2 (Conservation) management area ... The reason we termed it 'conservation' was that we saw the need to have a cautionary approach to use and development, given that we had limited understanding or resource knowledge of that large area. But it didn't mean that, particularly in light of the RMA, [that] you couldn't still apply for consent. It would just be discretionary.

We had quite a raft of assessment criteria. These are not actually required under the RMA. The Act [refers to] information requirements for consents. There's probably about twelve, if not more, general criteria before you actually

get to the specific ones. It becomes quite a convoluted process for some applications. [But] we're happy with it. (respondent 7)

Waikato regional coastal plan

[The plan] tried to take the spirit of the RMA rhetoric at the time about not being prescriptive – let people do things as long as [they were] based on the merits and there was criteria against which it could be judged. So it's not very specific in terms of planning. It's a good environmental quality type of plan. [But] it doesn't go into areas and say 'what do we want from this piece of marine area and what's our plan for it?' (respondent 13)

Auckland regional plan: coastal

In the initial stages of the RMA there was very much a philosophy of an effects-based planning approach. It's fair to say that it shifted quite significantly from the days of the Town and Country Planning Act. There was a tension in the early days to try and make things as effects-based as possible. But it became apparent that in some circumstances we needed to effectively manage space in the coastal marine area, we needed to apply some zones and some areas to allow development not to occur where it may otherwise have occurred on an effects-based analysis.

Often from a transaction cost point of view, going down the effects-based line was likely to generate large transaction costs for communities, regulators and developers. So going down a zoning line in some circumstances was going to be a more efficient way of determining what activities should take place in certain places, such as the development of mooring management areas, port management areas and coastal protection areas for ecological areas.

I think it certainly is harder to create zones around the sea than it is on the land. One of the principle reasons behind that is the lack of knowledge that we still have about coastal processes and about the inter-relationship between the physical, biological and chemical processes that are operating ... (respondent 24)

The regional coastal plans largely adopted zones for existing activities such as ports, wharves, marinas and moorings. They identified areas of conservation significance guided by information provided by DoC. But they left the bulk of the marine area unzoned, or in a common zone, where activities were largely discretionary and assessed against a set of environmental criteria.

The ARC, which drew from a strong history of marine planning through the former Auckland Harbour Board, adopted the most complex zonings in its coastal plan. The NRC, which had taken over the jurisdiction of the former Northland Harbour Board, also adopted a range, albeit smaller, of zonings. Environment Waikato, whose history was more closely aligned with water and soil management, included very few explicit zones in its plan (see Figure 6.2).

Figure 6.2: Regional coastal plan zonings

Auckland regional plan: coastal	Northland regional coastal plan	Waikato regional coastal plan
Coastal Protection Areas 1 and 2	Marine 1 (Protection) – areas of important conservation value	(Areas of significant conservation value noted but not specifically identified as a zone)
Aquaculture Management Areas	Marine 3 (Marine Farming) – existing leases and farms	Wilson's Bay Marine Farming Zone
Moorings	Marine 4 (Moorings) – existing mooring areas	Zoned Mooring Areas Prohibited Mooring Areas
Marina – existing marinas		Tairua Marina Zone I & II (implementing court decision)
Ports 1A, B & C, 2, 3, 4A, B & C, 5	Marine 5 (Port Facilities)	
Other Port Facilities	Marine 6 (Wharves) – small commercial wharves	
Airport		
Defence		
Special Activity Areas – special organised events		
Tangata Whenua – 2 management areas in the Manukau Harbour		
General – everything not included in other zones	Marine 2 (Conservation) – everything not included in other zones	

The plans were largely reactive in approach – endeavouring to protect current values and minimise the environmental effects of new activities through applying an EIA regime, rather than project a forward-looking vision. None of the regional councils adopted specific objectives, policies or zonings for the Kaipara Harbour or Hauraki Gulf.

So how effective did interviewees think the planning approach adopted was? Although the coastal planners interviewed were generally positive about the performance of the first generation regional coastal plans, their responses also indicated that coastal planning under the RMA was facing several key challenges: the extent to which the marine area needs to be zoned in order to manage it effectively; how the divide at mean high water springs can be effectively bridged; and whether marine planning should adopt a finer-scaled approach through the development of sub-regional plans.

To zone or not to zone

The paucity of zoning in the coastal marine area, although promoting flexibility, can ultimately prove problematic due to the lack of certainty provided to applicants. This is particularly the case where there are strong and competing demands to use the marine area, and competing values. A planner from EW describes the dilemma:

The positive thing is that it's really flexible – you don't need to change it [the coastal plan] a lot as new things come up – they can be accommodated. If it's too prescriptive you need to change the plan.

We've realised that it's all very well having a good environmental quality plan, but you also do need to plan as well. Because we have had problems with pointing out what is appropriate use and development and what is not. This is being decided by the courts and not in a planning process by communities. The court decisions don't consider an overall plan for the area.

Also there's the lack of certainty – people are not able to look at the plan and say 'does my proposal fit in with the way the council has decided this area is going to grow and develop or not'? They look at the plan and think they can do it as is a discretionary activity.

It has not worked well with marinas. At Tairua and Whangamata there have been long running battles over whether it is an appropriate use of the harbour. Part of the philosophy of the plan, when it was written, was that it should be up to the developer to prove that it is appropriate, rather than the council. The general ratepayer should not be paying to say it's an appropriate use. The philosophy was 'why should we spend lots of money designating areas for private enterprise to come along and do things?' But you have to balance that against our duty to make sure things are well planned. There's only a finite space and there are lots of things people want to do in that space. So

surely you need to plan out what could or should happen – you can't do it through ad hoc decisions.

How do you determine what is appropriate in an area – do you do it through individual resource consent applications or do you do it through a plan where everyone has a say and you get some sort of agreement on it? I can see arguments on both sides. (respondent 13)

Difficulties which can occur when attempts are made to zone the marine area, in the context of poor understanding of the marine environment, are illustrated in *Case Study 7: Kaipara Harbour Aquaculture*.

Integrating across mean high water springs

Of the three regional coastal plans studied, the one prepared by the ARC was the only one to attempt to cross the land-water boundary, through incorporating policies (but not rules) for the entire coastal environment rather than only for the coastal marine area:

One of the things about the Auckland region's coastal plan is that it was developed as a coastal environment plan. It's one of only four regional plans in the country that took the opportunity to recognise the arbitrariness of the mean high water springs boundary and has made an attempt to try and cross that boundary. But I don't think we've gone as far as we could or should do and that's in many respects a function of how regional councils work with their territorial authorities.

One of the challenges for the next generation of plans, I think, is to work more closely and in a more integrated fashion with the TAs in the development of policy objectives and rules in that area which spans the high water springs. (respondent 24)

The NRC did not attempt such integration in the first round of planning, because it would have practically and politically complicated an already difficult task. But, as the plans become operative, council planners have more time to evaluate the effectiveness of what they have undertaken and to focus on more effective ways to achieve integration:

[There were] two main reasons [why we split the coastal plan from a possible wider coastal management plan]. One, dealing with the coastal marine area was all we were required to do and that was a big enough chunk to bite off

on its own. Secondly, there was a political environment at the time that we wouldn't impinge on land too much. The district councils would plan for the land and we would stick to the marine area. There was some angst at a political level about traversing the boundary. It was cleaner and simpler to deal with just the things we were required to do and let district councils do their thing.

... now, having finished our plans, [we are] just talking about whether that was a good or bad idea in terms of integrated coastal management. This issue was largely the driver for that sub-regional kind of approach but also has driven things like establishing a system for NRC to be more closely involved in district plan appeals and resource consent applications, to make sure that things are integrated ... we now spend time specifically looking at those cross-boundary issues.

The case in point is the management of moorings and marinas and also aquaculture ... the zone (marine management area) changes for these will be within the coastal plan, but there will be land-based impacts of this, e.g. parking pressure in terms of access, in terms of waste disposal, water, toilets and water supply. We've had to work out ways of dealing with that conflict. This includes developing management plans for specific mooring areas, based on priorities where the pressure was and including the district councils in that process. Through this we (NRC, district councils and mooring owners) sit down and try to agree, okay what is this area going to look like, how are we going to manage the associated cross-boundary issues long term?' (respondent 7)

There has also been community demand for a more integrated approach, as artificial jurisdictional delineations do not necessarily make sense to coastal residents and users:

The focus during the past five years had been to deal with appeals on the plan and get them out of the way. We have also been grappling with the aquaculture thing. Now the plan's operative, there has been an ability to take a bit of time to look at where to go next, what needs to be done. The community is definitely saying that we should be taking a stronger role. At public meetings there is an expectation that [the Regional Council] should be doing more. Some people don't seem to realise that the district council has a major role in coastal land decisions. Some of it is the word 'coastal' being used in [Regional]Coastal Plans. People think it is us that control coastal land use. Some councils, like the ARC, do have a coastal environment plan which

means the plan influences land development, whereas we have quite a hard demarcation [at MHWS]. (respondent 13)

Regional versus sub-regional planning

Bridging the divide between land and water requires a more forward-looking and strategic approach to planning, and a more finely-scaled approach, than was adopted in the first generation of plans. This has led to council planners considering the benefits of developing sub-regional plans:

In the early days of the RMA, the city and district councils had to get on and get their city and district plans prepared, and the regional councils had to get on and get their regional coastal plans and others prepared. Probably in the last five or six years there's been recognition that those plans are through, now we need to get down to a finer resolution in terms of planning and to do that effectively we need to work together ... It's a matter of biting off geographical chunks of the coast and working with the TA and the community and figuring out what is sensible for that section of coast.

There is resistance to doing this. But the alternatives are piecemeal development or lack of development, high transaction costs, [and] paying lawyers rather than doing things on the ground or not doing things on the ground. Sometimes it's a challenge, but again that's where organisations such as the Hauraki Gulf Forum provide opportunities, both at a political and at an officer level, for those groups of people who are involved in those activities to get together and discuss their issues and discuss their concerns. (respondent 24)

For the NRC, undertaking multiple streams of work to further develop their coastal plan has resulted in community disquiet and a demand for more coordinated sub-regional planning:

We've also got potential new Marine 1 areas we're looking at and these marine management area adjustments are all kind of related but separate. Changes for moorings and marinas, aquaculture and mangroves are all happening separately but at the same time ... what it's thrown up, through the consultation around the plan change processes, is people saying to us 'well can't you just talk to us once about all these things? Can we not see the full vision of where aquaculture goes, where the moorings go? etc' ... I think there's an element of logic to that and it's therefore pretty compelling from my point of view to get

the community involved and to get a more integrated approach to planning for harbour areas including the adjacent land. (respondent 7)

But sub-regional planning may consume more resources than councils are willing to devote to marine planning in the short term:

In my experience with the coastal plan, having written it and having seen it through to being made operative, is that I think it's better for the council ultimately and for the community as well that we actually develop some sort of framework which gives us a place [to] have a bit more meaningful interaction, and as a result, a better chance of achieving integration - not just within our council for the functions we undertake - but also with district councils, DOC and MFish and other agencies. But that type of framework does have significant resource implications, which makes this council a wee bit wary of it.

It may just require a different model of governance – the same resources being allocated in a different way. Or it may require additional resources going into it. But is it ultimately a better resource management model. That's probably the key question. (respondent 7)

Both the ARC and EW have embarked on sub-regional planning processes. These are discussed more fully in *Case Study 10: Approaches to integrated management of the Hauraki Gulf*. The ARC has prepared a series of localised coastal compartment plans focusing on the coastal edge. These have had some success in increasing integration between regional council and territorial authority planning for coastal land, but limited success in strengthening marine planning. Environment Waikato is approaching the issue from a more strategic viewpoint, working with the Thames-Coromandel District Council to develop a 'Blueprint' for the entire Coromandel Peninsula. Both the bottom-up and top-down approaches create benefits and problems:

We're going down quite similar approaches but [Environment] Waikato decided to start out by doing a more strategic overview of where they're looking at their whole coastline and prioritising sites to do localised management plans, whereas we started with the localised management plans hoping to use that information to then go up a level and do the regional context next.

We had some really good discussions at those early stages when we were both making those decisions and both seeing the advantage of the other's

approach. Because, taking the top-down approach, you end up with communities who wait another year or two years. From their point of view it looks like you've done absolutely nothing at all, but you've just written a strategy to do localised strategies. But from our point of view [focusing on] hot spots leaves us with patchy information, and different consultants writing slightly different plan structures, and slightly different technical investigations which then can't quite be compared ... ideally I'd rather take a more strategic overview first rather than jumping in.' (respondent 25)

After considering adopting a sub-regional planning approach, the NRC has decided not to proceed in that direction.

Fisheries Plans

Fisheries planning is an attempt to move towards a more values-based and objectives-led approach to fisheries management. Current fisheries management is primarily based on regular reviews of the TAC and TACC aimed at achieving the overall management goal of MSY. Underlying this management approach is often a multitude of historical regulations on where, when and how fishing can take place. Fisheries planning is an endeavour to establish more detailed management objectives and processes to guide year-by-year decision making. It also potentially provides a mechanism through which the sometimes complex regulatory framework can be rationalised:

'It's a very simple concept to have objective-based management but, it's been a little bit difficult to figure out internally how we actually make that come about ... I guess the original idea was that internally we would do our own planning which would be about defining the minimum Crown standards and, once that was made public, the stakeholders, if they wanted to achieve more from the fishery, then they could come up with their own plan and have further proposals. But I guess what became clear, was that that was a double up of effort.

... a lot of effort was originally [spent] just getting stock into the quota management system, having that basic management level in place. So there's been recognition for a while that it works well enough, but it is *ad hoc* compared to actually setting out more specific goals for a fishery and then

making sure that whatever management measures are applying can actually work towards that goal.

The other side of it is, because we moved from a system that had all sorts of controls about what sorts of methods could be used for fishing, even days of the week that you can fish and all sorts of effort controls, we've moved to a system where the most important control is the total level of catch. But you've still got all these other funny little controls in place. The idea was that if you had a clear goal as to what you wanted to achieve from the fishery, [it] would soon become clear that all these other things sitting around the edge may not be helping you achieve that goal, or they may be hindering that goal because it made it simply inefficient.

[In the] Coromandel scallops [fishery] ... they can't fish on Fridays or Saturdays. They can only fish five days a week, which is a bit of an anomaly when you control the total level of catch anyway from that fishery. To us we no longer care what days of the week they fish. You've sort of got that left-over thing ... they probably could be more efficient if they did fish or had the freedom to fish seven days a week. That's a regulation that's in place that could be hindering them achieving certain goals. (respondent 21)

The 'proof of concept' fisheries plans prepared to date are very different to regional coastal plans in that they do not generally have a spatial component in the form of delineations or zonings on maps (although some maps are used to present existing information). Fisheries plans also do not include environmental assessment criteria and are framed around applying risk assessment concepts rather than environmental assessment methodology. For example, the Draft 'Proof of Concept' Fisheries Plan for Coromandel Scallops (MFish, 2006a) sets out a range of goals structured around four categories:

- *Sustainability* – which is not used in its broader commonly-accepted sense but is narrowed to focus on the biomass of the scallop stock in relation to maximum sustainable yield
- *Environment* – which unusually is identified as a separate category from sustainability and which focuses on the environmental effects of scallop harvesting
- *Use* – which focuses on how harvesting is managed in the commercial, recreational and customary sectors, thereby potentially overlapping with the sustainability category

- *Credible management* – which deals with stakeholder and public perceptions of management

These goals are further fleshed out into operational management objectives. A broad assessment is then undertaken of the risk of current management not meeting the objective, and actions to address the risk identified.

There are a number of challenges in planning for fisheries management including the dynamic nature of fisheries, lack of information on the marine environment and overcoming high levels of conflict.

Dynamic nature of fisheries

The life cycle of marine life can encompass a large spatial area and be affected by a dynamic marine environment. This makes it difficult to formulate concrete plans for the management of fisheries:

You can plan up to a point, and then the fish are always going to do what they want to anyway, so it's going to be less detailed in a lot of ways [than other agencies' plans]. The goals are still going to be quite high level and quite broad, and obviously the intention is that it will flow through to specific actions or that there will be an operational plan sitting on the end of it, but I think it's probably still going to be less clear. It would be nice if we had an operational plan that says what we will be doing every year for the next five years, but I personally don't really see that happening. I expect in the councils, for example, they have more ability to do that. (respondent 21)

Lack of environmental information

The marine environment is a complex system and relatively little is understood about it when compared to the terrestrial environment. A large proportion of the research which has been carried out within fisheries management has focused on stock management, and in particular on estimating the size and harvest potential of commercially-significant fish stocks. This has resulted in a paucity of information on the ecosystem effects of fishing activity as well as on environmental factors such as sedimentation which may have a significant effect on the health of fish stocks. This lack of information can make it difficult to effectively address environmental issues within the fish-planning process.

Under Part XIV of the Fisheries Act 1996, the costs of undertaking research relating to the effects of fishing on the aquatic environment can be recovered from quota holders and commercial fishers. There can therefore be a significant

disincentive for commercial fishers to support environmental research, as they may not only have to pay for it, but the findings of the research may result in further restrictions being placed on their fishing activities, thereby resulting in additional costs and/or reduced revenues.

Overcoming high levels of conflict

Fisheries plans are to be developed by the consensus of stakeholders, who can include representatives of commercial fisheries, customary fishers, amateur fishers and environmentalists. Previous attempts at developing consensus-based fisheries plans have broken down due to the inability of key stakeholders to agree (*see Case Study 5: Hauraki Gulf amateur fishers*).

Some of this conflict is long-standing and embedded in the competing interests of these groups in shared resources. An example of this is the ongoing tension over to what extent fisheries planning and management should be devolved to quota holders. An environmentalist describes the dynamics of the issue:

An underlying tension in any management regime in the last 15 years is that the industry has been really pushing for the Ministry to get out of management and hand it over to the industry. They got knocked back on that whole devolution process [when it] became politically unpalatable when Labour came to power, and it sort of slipped off the back burner. You could say that current processes and the current measures are endeavours to sort of move around that ... Broader management plans never suited the industry. They don't want the Ministry and anybody else very involved in management. They would dearly love to have management basically handed over to them as the people who know what they're doing and everybody else can stand on the sidelines and tick a few boxes. (respondent 15)

A representative of the fishing industry puts the argument for devolution:

'To enable quota holders to engage in collective action to manage sustainability and environment concerns, government must externalise the fisheries management planning process ... it also needs to recognise the need to devolve more responsibility and authority of fisheries management to resource users. Such a framework would need to consider the devolution of regulatory decisions. (Soboil & Craig, 2006: 5)

Overcoming these deep divides between stakeholders is likely to be a significant challenge in successfully developing and implementing fisheries plans.

Hauraki Gulf Marine Protection Plan

The first marine reserve in New Zealand was created in 1975 in the Hauraki Gulf at Cape Rodney – Okakari Point, Leigh. Since then, four other marine reserves have been created in the Gulf (see Figure 6.3).

Figure 6.3: Marine Reserves in the Hauraki Gulf

Marine reserve	Year established	Size (km ²)
Cape Rodney – Okakari Point	1975	5.2
Whanganui A Hei – Cathedral Cove	1992	8.4
Long Bay - Okura	1995	9.8
Motu Manawa – Pollen Island	1995	5.0
Te Matuku Bay	2005	6.9
TOTAL		35.3

In addition, a proposal to create a large marine reserve of 50.1 square kilometres off Aotea - Great Barrier Island was notified by DoC in 2004 and subsequently approved by the Minister of Conservation. It is awaiting a concurrence decision by the Ministers of Fisheries and Transport before it can be confirmed.

So why is DoC embarking on a new form of marine planning in the form of marine protection plans? In the Hauraki Gulf, the DoC Auckland conservancy office encountered significant levels of conflict in seeking to establish marine reserves on a case-by-case basis and realised that a radically different approach was required:

As a consequence of the Great Barrier Island project, it became obvious that *ad hoc* marine reserve applications were not going to work, because of the intense competition between different interests: Māori wishing to protect their customary interests, recreational fishing, commercial, AMAs. It was effectively setting up conflict. So I proposed the Hauraki Gulf marine protection project. It is an attempt to take a more holistic approach and to have a multi-stakeholder process. (respondent 14)

In addition, the *ad hoc* approach made it difficult to design an effective science programme to demonstrate the rationale for marine reserves:

Our approach to marine reserves has been *ad hoc*. We've implemented marine reserves generally on the impetus of a community group or the public.

We have done very little baseline work before the reserve is established. We hurriedly try to put together a basic sort of fauna survey to see what is actually there. Each reserve seems to happen in isolation. Sure, there's something interesting and significant about a location, but it is fairly *ad hoc*.

There's no consistency in our monitoring programmes and they are pretty poorly resourced ... We don't actually do sufficient work to validate why we have taken these areas and locked them up. And as a result we will continue to run into a lot of criticism. (respondent 17)

We must set up things so we can test them in a scientific manner. At the moment our monitoring programmes are independent. Even just putting the results of different monitoring programmes on one page, helps to see patterns. (respondent 19)

The new more systematic planning approach drew on developments overseas, and the growing realisation that the current approach did not only create conflict, but was not scientifically defensible:

Our vision is to have protection of some type of a representative range of marine habitats across the Gulf, and probably for them to be replicated, so there's not just one of each habitat, taking into account existing uses. The important thing is to work on it now, as Auckland's population is growing ... If you set these areas aside sooner rather than later, then hopefully there will be less impact and less competition for the space which is left. If we wait until there's two and a half million it will be much harder to find any available space. (respondent 20)

We're starting this now, as globally there's been a push for a more systematic approach to reserves. The reserves in the past have been established in an *ad hoc* way according to particular interests. Where our overall goal is conservation of biodiversity, we need to look at it more strategically, and make sure we are just not protecting habitats, but that we are actually getting a good cross section of biodiversity and a system that will actually work towards conservation, rather than just creating little museums around the place.

We also need to do it in a way that's a bit more open and transparent and credible, and involves the community, and considers other uses, and the fact that there may be lots of possible reserve systems you could come up with to meet the number of goals, and that there are lots of different objectives.

So I think the shift came about because of the development of guidelines and approaches in other countries, and the level of contention here, that it would make sense to go about it in a more orderly way. Our tools for getting information are better and there is a bit more cooperation between marine scientists. (respondent 19).

It is intended that a multi-stakeholder steering group, the Hauraki Gulf Marine Protection Forum, will play a central role in developing the marine protection plan, informed by a technical advisory group and a range of ecological, economic, social and cultural information:

We will get various people around a table identifying what marine environments we have [and] what user interests there are in various places. [This will involve] sitting down with a bunch of maps and saying, from a marine protection point of view, [these are] the best places to achieve a network of protected areas. But recognising you also needed to have, as an outcome, identification of areas for customary fishing and recreational fishing.

The engine of the process is various stakeholder groups [referred to as the Hauraki Gulf Marine Protection Forum]. There is an element of faith that they will come up with the outcome. They need to agree on principles and outcomes so that people participating can have confidence that it will be a fair process and that they can participate in a constructive way. The working group is informed by an advisory group, which provides information but not values. It provides technical, social and economic information. And we also need communications advisory people on how to go about managing the process. (respondent 14)

We are not just looking at biodiversity information. We will also be looking at social, recreational, commercial, fishing and other activities as well as cultural information from iwi. It's not just biodiversity values but it is other values, which you need to balance with your protection. (respondent 19)

The Hauraki Gulf Marine Protection Plan is still in the preparatory stages, so it is not yet possible to determine how successful the approach will be. It will need to overcome some serious challenges, including a lack of scientific understanding of the marine environment, high levels of mistrust generated by earlier processes, and the lack of a clear statutory framework. It does, however, represent a positive step forward in implementing a more strategic and stakeholder-driven approach towards marine protection.

Lack of scientific understanding of the marine environment

Designing a marine protection network requires a more sophisticated understanding of the dynamics of the marine environment than establishing stand-alone marine reserves. Scientific knowledge is not yet sufficient to support network design, but is getting close:

Evidence shows that marine reserves are incredibly effective at a local scale in conserving biodiversity. We are only just starting to look at the broader scale in terms of exporting juvenile larvae or adults to surrounding areas. We know that some animals stay in the reserve and others move away. We need to look at the whole issue of interconnectivity amongst different reserves and neighbouring areas and how we can actually enhance biodiversity values outside reserves and perhaps even fishing values as well. But this is something not taken up by DoC because it is not really their concern and maybe not even by fisheries.

We need to look at how we can design a network that actually contributes to not just preserving animals but helping biodiversity and ecological processes throughout the whole system, as well as the services that depend on it like fishing. It's a bit more of a difficult question. It's hard enough to count animals in the oceans. Even getting the broad baseline data we need for planning is a start, because in many cases we don't even know where the off-shore reef sediment habitats are to start with. What are the connections between them and where are the fish moving? A lot of fishermen know that stuff.

We don't really know how to design a network of marine reserves to enhance the overall marine system. You can model alternative systems. You can model connectivity. I suspect in the Hauraki Gulf we are on the cusp of having sufficient information to understand connections. (respondent 19)

Obtaining stakeholder support and engagement

Obtaining stakeholder support for the process is likely to be a key challenge. This is due both to philosophical differences between stakeholders and the fallout from previous marine reserve applications. This is reflected in the responses from some of the interviewees:

Ministry of Fisheries

The different in cultures in DoC and MFish could be an obstacle, as their main focus is utilisation of the resource whereas ours is conserving it. (respondent 20)

Māori

There was an awful lot of discussion about iwi involvement and how that was going to occur. I think its fair to say that the Department of Conservation took a bit of a caning over that ... because they were favouring one model and the tangata whenua representatives on the [Hauraki Gulf] Forum felt that the functional role of tangata whenua in that group needed to be stronger. There wasn't much tangata whenua [representation] in the steering group and there was some interesting debate on that. (respondent 27)

The intent again is quite good, it's really that thing that I said before, about a set of guidelines and it goes into a process and ... it's a structured and strategic approach to the whole set of issues around marine protection, not just marine reserves. That's got to be better than the way it is at the moment. But being run by DoC and the way it is, it's got its downside ... It's hard to build a co-operative relationship to go forward with somebody who is trying to stab you in the back and rubbish you. It's really difficult. (respondent 22)

Recreational fishers

The jury is out on that as far as the recreational sector is concerned, which is sad because I think there's a will by DoC to do something right. However, if we look at Aotea, the process which DoC conducted with Aotea-Great Barrier Island Marine Reserve, was nothing short of deceitful. There was lies, there was mis-information, there was a whole lot of wriggle, wriggle round the background. They ignored Māori concerns, they ignored locals' concerns, they ignored Auckland's boating fraternity concerns and they ignored the commercial fishing industry's concerns ... We don't trust them at all. (respondent 31)

Lack of a statutory framework

The lack of a statutory framework for the planning process means that there is no certainty that the plan can be implemented, as any outcome will need to be actioned through the processes currently provided in the Marine Reserves Act 1971 and the Fisheries Act 1996, amongst others.

We could go through the whole process and end up in a double process where you have to notify the MPAs and a whole lot of new people pop out of the woodwork who weren't involved in the original process ... You can have an election and a new government with different priorities and it all gets put on the shelf. Because it's just policy it can easily be ignored. (respondent 20)

Prospects for integration

Regional coastal plans, fisheries plans and marine protection plans overlap each other spatially, applying to common marine areas (see Figure 6.5) but very different approaches have been adopted for their preparation. So what are the prospects for integrating them?

There are statutory links between the plans. Section 66(2) of the RMA requires regional councils to have regard to management plans and strategies prepared under other Acts, and also fisheries regulations, when preparing regional coastal plans (as well as other regional plans and the regional policy statement). Section 11(2) of the Fisheries Act 1996 requires the Minister of Fisheries to have regard to the provisions of any RMA regional policy statement and regional plan and any conservation management strategy or plan, when setting or varying a sustainability measure, but not in the preparation of fisheries plans themselves.

Regional coastal plans have, for most of the marine area, established an environmental impact assessment regime which applies when resource consents are sought for new marine activities. The process to prepare the plans is set out in detail in the RMA. It is a council-led process with statutory requirements for public participation. A dispute-resolution mechanism is provided by the Environment Court to which proposed plan provisions can be appealed. Regional coastal plans follow the boundaries of regional councils and therefore two regional coastal plans apply to the Kaipara Harbour and the Hauraki Gulf.

Fisheries plans deal with existing rather than future activities, and seek to provide a more strategic framework within which fisheries management decisions can be made. To date they have been fish stock based rather than regionally based. The plans have a statutory basis in the Fisheries Act, but no legislated process for their preparation. The process to prepare the plans is facilitated by MFish but is intended to be stakeholder-led. There is no formal process for public participation, other than a broad obligation to consult representatives of stakeholder groups, and no plan-based dispute resolution process. At the time of this study, one fisheries plan was being prepared in the Hauraki Gulf and none for the Kaipara Harbour.

Marine protection plans have no statutory basis. They are intended to be prepared by a multi-stakeholder steering group. They seem likely to be strongly spatial, identifying spatial areas suitable for marine protection. Although incorporating a wide range of environmental, social and cultural information in a spatial format, the process will only focus on identifying areas for marine protection rather than for other marine activities. The Hauraki Gulf Marine

Protection Plan is likely to cover the entire Gulf, thereby encompassing the jurisdiction of the two DoC conservancies.

Figure 6.4: Comparison of different types of marine planning

	Regional coastal plans	Fisheries plans	Marine protection plans
Statutory	Yes	Yes	No
Spatial extent	Regional council boundaries (MHWS to 12 nm)	Quota management areas (marine area extending to 200 nm)	Inshore regions (still to be determined)
Spatial zoning	Yes (for at least part of the marine area)	Not usually	Yes (for MPAs)
Statutory linkage	Take into account fisheries plans	No	No
Participation	Council prepares plan; public submission, hearing and appeal rights	MFish prepares plan in association with stakeholder groups; public submission rights	Multi-stakeholder forum prepares plan supported by DoC and MFish; public consultation and submission on draft

Both regional coastal plans and marine protection plans incorporate spatial zoning. The jurisdictional boundaries of regional councils and DoC conservancies coincide, and the Minister of Conservation has a statutory role to approve all regional coastal plans. As a result there seems likely to be some integration of the plans, at least in the longer term. In the shorter term, the preparation of the second generation of regional coastal plans in the Hauraki Gulf may occur in advance of the marine protection plan being completed.

There is also an institutional link between MFish and DoC which are jointly charged with implementing the marine protection areas policy and implementation plan, which should provide a linkage between marine protection plans and those of fisheries.

There is a less obvious interface between fisheries plans and regional coastal ones. During the preparation of the Coromandel Scallops ‘proof of concept’ plan, no information was gathered on the provisions of regional coastal plans which might impact on the scallop fishery. In addition, although the impact of sedimentation and pollution on scallop beds was raised as a potentially significant issue, no information was gathered, or research proposed, on the extent of the problem or how regional councils were currently managing the issue. The Ministry of Fisheries stated in the plan that ‘This is not currently a

core focus for Ministry research, since the Ministry is not able to manage land use' (MFish, 2006a, 14). If research is not undertaken to identify whether or not this is a serious issue, it is hard to see how regional planning can properly take it into account. It is also difficult to see why regional councils would use land-based ratepayers' money to research an issue that is related to fisheries management, an area outside their management focus.

The lack of fit between fisheries and regional plans needs to be urgently addressed. The Hauraki Gulf Forum could potentially play a role in achieving a better interface between these two kinds of planning within the Hauraki Gulf.



seven

Case Study 2: Managing Coastal Development in the Hauraki Gulf and Kaipara Harbour

Struggling to keep ahead of the game

We tend to want to get into moving deck chairs around first before we actually have a look at the deck. (respondent 43)

The [RMA] time frame involved in staying ahead of the issues and making timely changes are ... out of all proportion to what happens in the real world. (respondent 42)

The policy framework

As with marine planning, the RMA provides a comprehensive legislative framework for managing coastal development, through the preparation of district plans focusing on land use, and regional plans focusing on broader catchment management issues such as sedimentation. District plans, prepared by territorial authorities, primarily control the use and subdivision of land. These are the focus of this case study. The purpose of the plans is to assist in the integrated management of the effects of the use, development and protection of land (sections 31(1)(a) and 72). They must contain objectives for the district, policies to implement the objectives and any rules to implement the policies (Section 75(1)). They are prepared within the framework of the New Zealand Coastal Policy Statement and regional policy statements which provide generally broad statements of policy.

Unlike regional coastal plans, district plans have in practice contained detailed zoning provisions and rules, particularly for urban areas. They have built on a long history of land-use planning encapsulated in various former Town and Country Planning Acts.

As well as preparing district plans, territorial authorities along with regional councils, are required to develop Long Term Council Community Plans (LTCCPs) under the Local Government Act 2002. These plans, which have a time frame of at least ten years, are intended to provide a longer-term planning framework for councils as well as to promote better alignment between council spending and the achievement of community outcomes. They encompass the social, economic, environmental and cultural aspects of community well-being, thereby embracing a broader sustainable development paradigm than does the RMA (section 10).

There is no direct legislative linkage between the RMA and plans prepared under the Local Government Act, other than the general provision in the RMA

requiring councils to have regard to management plans and strategies prepared under other Acts when preparing a district plan (section 74(2)(b)(i)).

Planning for land use on the Coromandel Peninsula, the Hauraki Gulf Islands, the Hibiscus Coast and within the catchments of the Kaipara Harbour has faced different challenges and has been approached in a variety of ways as described in the sections below.

Hauraki Gulf: Coromandel Peninsula

The Coromandel Peninsula, situated on the south-eastern side of the Hauraki Gulf, incorporates approximately 680 kilometres of coastline. The eastern side of the Peninsula has a high wave-energy coastline and many attractive beaches. It has long been sought after as a holiday location for people living in Auckland, Hamilton and surrounding towns. Growth pressures on the peninsula are intense. Over seventy per cent of eastern beaches and dunes have houses on them and ongoing subdivision pressures threaten most of the remaining undeveloped beaches (Beca Carter Hollings & Ferner Limited, 2005, 31).

All of the Coromandel Peninsula is within the jurisdiction of the Thames-Coromandel District Council. The main strategy behind the development of the Council's District Plan was to contain future development within seven urban settlements. The council sought to achieve this primarily through imposing a large minimum lot size in order to restrict subdivision in rural and coastal areas outside the settlements:

One of the main things the council is trying to achieve is to contain development within serviced settlements. These are the places we want to concentrate growth ... Generally outside of these areas you have to have a minimum of forty hectares to subdivide (in a rural or coastal zone). That is one of the rules that we really rely upon in terms of minimizing wholesale land fragmentation and sprawl. [There is] a lot of pressure on the rural and coastal areas and we try and maintain that line as much as possible. It's one of the key planks of our plan in terms of trying to achieve our settlement strategy. (respondent 41)

The seven urban settlements strategy is, however, being undermined by several factors. These include the ability to build houses 'as of right' on previously subdivided but undeveloped rural land, the subdivision of rural land using the conservation lot provisions in the district plan, provisions for subdivision and development in some areas as a result of negotiated settlements over the plan provisions, and the lack of adequate infrastructure in identified growth nodes:

Existing use rights

There's still a lot of big parcels out there, that people can [use to] actually meet the twenty hectare minimum lot rule. There's a lot of focus as well within existing lots. If you have an existing vacant lot, you can put a house on it basically ... There's also a lot of redevelopment happening within existing use rights and there are existing titles that people are now putting a house on, or redeveloping.

Conservation lots

You can come in and apply to do a conservation house lot. If you are able to covenant a minimum of five hectares you may be entitled to an extra lot. In the past it seems that people have received a development right without a lot of the costs, and in a lot of cases there's not a lot of obligation on them to maintain these areas ... You get a lot more trees but you also have a lot more houses, you have a lot more cars, driveways, a lot more pets, you have a lot more potential issues around septic tanks, landscape impacts and things like that. Following a recent study, we are currently thinking about reviewing our conservation house lot provisions.

Negotiated settlements

A lot of development provided for around the Peninsula ... was negotiated out of consent orders as a result of appeals on the Proposed Plan. In many cases the community does not seem to have been fully aware of the submissions and downstream processes ... they often seem surprised at development rights that were provided through these processes.

Lack of infrastructure

We've had some pretty big issues with servicing some of our settlements. There's talk of our waste water systems not being up to scratch ... Basically waste water is a big issue for this district at the moment. Most of our main settlements have waste water issues and the costs involved in updating these are significant. Just for the plants on the east coast we're looking at fifty million dollars, with a low ratepayer base ... and then there's the cost of all the other necessary infrastructure – water, stormwater, community facilities and services – the costs are really significant for a small district.

It would be great if we had the infrastructure in place. There's no doubt that we could make our towns really interesting and nice places to live, but without the infrastructure you can't provide for increased development. There's spill over effects ... into the more natural and undeveloped areas ... Basically the

settlement strategy has worked, but if you don't have the infrastructure in place it won't continue to work into the future.

I think that infrastructure provision is something that we should do better in terms of strategic land-use planning. I think it should be the district plan sitting alongside infrastructure provision. But at the moment the district plan seems to be the main tool to manage growth. (respondent 41)

Despite these difficulties, the percentage of houses within settlement areas has been maintained at a fairly constant proportion, although this means that a third of dwellings are now located outside these main urban areas:

In the last ... say ten years, basically from 1991 to 2001, there'd been sixty-nine per cent of dwellings in the main settlements since 1991 and by 2001 it was seventy per cent of all dwellings and pretty much the same number of people/permanent residents lived in those settlements. So on the face of it, it seems that that strategy has been successful. (respondent 41)

The council has been developing a more proactive approach to planning outside the RMA, through the preparation of community plans and the development of a longer-term vision and strategic approach. The community plans have enabled significant local input into the development of the council's LTCCP. However, they have been less successful in influencing the provisions of the district plan, because of the complex statutory process that RMA plan changes are required to undergo, and therefore have yet to effectively impact on development control. Challenges faced in developing the community plans have included a high turnover in members of the community boards and residents, and raised community expectations which are not always able to be met by the council:

The beginnings of community planning

It started off with an issue with mangroves. What happened is Environment Waikato got involved and the council got involved and ... they had a big community plan day. It was decided, because there were so many different views about mangroves, it would be a good idea if the community came together and tried to reach some sort of compromise over it. Once they started talking about mangroves they started talking about water quality and density, a whole range of things, basically all the issues that were in the community. So from there grew the idea of developing a community plan so that the community as a whole designed what it wanted to see in that community into

the future. So that sort of concept was born from there and then it's just been transported across the whole district. (respondent 42)

How it developed

Whangamata was the first one, and I'd probably say it was probably the least successful, although it is a very difficult community ... compared with all the others ... I did a survey in Whangamata when the community plan was finished and went to all different groups and tried to get their views on the community plan ... and all the groups accept there were things in the plans that they would change, that they didn't totally support, but overall they all said they could live with it. So it was a compromise and the whole community did buy into it at the end of the day.

The one at Pauanui was really successful ... Pauanui was very much a community - and community board - led process and I think it's just a reflection of the personalities involved, just some very committed people. They forged ahead and developed a community plan to have a good level of buy-in, as well as a number of detractors who don't agree with bits of it. That worked very well. We're just in the middle of the Tairua one which has worked very well.

A number of the smaller communities went and did their own ones off their own bat because they realized ... when we were developing the LTCCP we tried to pull out the main things out of the community plans and make sure that the main actions were recognised in the LTCCP and had a budget and all the rest of it. So some of the smaller communities cottoned on to this pretty early and they said if we want something, we'd better make sure that we've got a community plan so they can take it into account. (respondent 42)

Effectiveness

I'd say seventy per cent of what was in the plans that relates to council ... has probably found its way into the LTCCP in one way or another ... (respondent 42)

Challenges

The other thing that we have to think about is that, anecdotally, a house changes hands every four years in this area, so you've got a big turnover of people in the community ... People come in and they don't know anything about previous community planning exercises and expectations.

I think the bigger question is though, ultimately, where the community plans sit in the overall framework of the district plan. For example, it came through really strongly in the community plan that they wanted to see infill subdivision stopped or tightened up. Our district plan still hasn't changed to reflect that because we think if the infill stops then there will be pressure for quite intense apartment style development and we haven't yet got really good design and open space provisions in our Plan.

Community plans are interesting, they all want to manage growth and in some cases stop it. We don't see how you can really stop it, you can manage it but you can't stop it. I think the real challenge will be how do we manage the community expectations through the community plans with the district plan and also our infrastructure? (respondent 41)

The council has more recently put effort into strengthening strategic planning guided by a broad vision of the future for the Peninsula. This shift has been driven by a growing realisation that planning ahead enables the council to exert more control and address issues more effectively. It has also been facilitated by the council gradually getting on top of short-term burning issues, associated with failing infrastructure and disputes over the district plan, and having the time and space to look further ahead:

So where we got to with the council in recent parts is there is a lot of talk about protecting what we've got - protecting the unique features of the Coromandel. We all come here to see those unique features, but there's a recognition that all the people coming to see and visit the experiences is part of the problem. There's a recognition that growth is, they use the word, inevitable. It's more that growth is going to happen and growth is not necessarily bad ... But it's managing the growth. We've been grappling with the term 'appropriate development', and trying to define what appropriate development is. Of course it means different things to different people depending on where you're coming from ... I think, for the first time, the current council had a bit of an environmental focus to their vision for the future. I think in the past it's been pretty much development and all of council's response to that development had been the traditional engineering type of responses of build more of and bigger of, etc.

I think it's a recognition that there were a number of functions under the Resource Management Act and Local Government Act that perhaps we weren't addressing as well as we could have. I think it was a recognition

also that we kept coming across the same problems time and time again, successive councils, and we hadn't sort of learnt from what happened last time. That tended to be because we were just reacting to everything that came along. I think there was a general recognition, certainly at management level but also into the governance side, that if you have a plan for some of these things they're a lot easier to manage. We can actually determine the outcome that you want by planning for it, rather than just reacting to everything that came along. So I think there was just a better recognition that planning ahead does pay dividends in the long run.

Perhaps also catching up with a lot of the outstanding issues that have been around, particularly around infrastructure, things like wastewater plants and water plants, and suddenly being able to take a bit of a breather and go 'right what's next' and there's been an opportunity to go forward. The creation of the policy and plan group occurred about five years ago and that was really, I think, the change when we started getting into this future stuff. (respondent 42)

The council is planning to commence a review of its district plan during 2007.

Hauraki Gulf Islands

Many of the islands in the Hauraki Gulf are within the jurisdiction of Auckland City Council, which also manages the country's largest city. The islands have been modified to a varying extent, but retain high natural character values. Great Barrier Island is by far the largest island in the Gulf with a land area of 28,500 hectares. It has a rugged landscape rich in natural features and wildlife. The island has a declining resident population, currently consisting of around 800 people, and around seventy per cent of the island is owned by DoC. Waiheke Island is the second largest island in the Gulf and comprises 9,324 hectares. It is the most heavily settled, with a population of around 8000 people. Most of the population is settled in a series of villages on the western end of the island, which is the part of the island closest to the Auckland central business district (Auckland City Council, 2006, sections 3.2.1 and 3.3.1).

The Auckland City Council is well-resourced, with the largest annual budget of all the management agencies included in this study. It has long experience of urban development and planning on which to draw. The council adopted a more ecologically-based approach to planning for the Gulf islands than applied by any of the other territorial authorities interviewed. This planning approach began with the premise that it was important to protect the natural environment of the Gulf Islands:

... we identified that the Gulf ... had been modified by man, but not to the extent that the other two environments, the isthmus or the CBD, had. So I guess what the plan is trying to do is reflect that [it is] touched by man but not necessarily destroyed in total by man. It's not an urban environment, it's very much the natural environment ...

The first plan probably historically came from the fact that you had Waiheke and Great Barrier as two different planning authorities ... So when Auckland City inherited those two islands ... it set about trying to bring them together. I guess, in a local government sense, to create a community where they had their own identities, trying to retain that identity but at the same time recognise that they were part of a bigger city now.

The common element that they set on was the land unit, which was basically identifying and linking the ability for a piece of land to be developed or used to its underlying topography, to the elements that it made up [although not all land units were linked to topography]. So if it was a wetland it went into the wetland land unit, if it was a sand dune, it went into the sand dune coastal cliffs or another type of land unit, recognising that the topology, topography and the land form was the underlying basis. So the plan basically follows the contour of the land, or the type of soil, or the type of natural environment that you're dealing with. So it didn't recognise property boundaries, it looked at the natural landform as its boundary. (respondent 49)

But this approach, although sounding good in theory, was problematic in its application. It failed to take into account the very different local circumstances on the two islands. On Great Barrier Island this approach was too complicated and costly to effectively administer:

... what tended to happen was you had a single site and one ownership that might have ten land units on it, because [on the] Barrier particularly, the size of the sites were huge ... So you had a member of the public, with not much planning knowledge behind them, trying to manage planning approaches for different parts of their site. Even then it wasn't that simple because it wasn't as if you said, okay here's your site and this applies to that part and that would apply for that ... The land units followed the land and, so this land unit might have cut back into there, because that might have been a wetland and you had this land unit running around through there, maybe another one that popped in through there because that was following that contour or whatever. Someone said 'okay traditionally this is where the access was and we want to

put a building platform on there'. They had three land units then to contend with ... The standards are different so ... [they ask] 'how do I work that out over my whole site because I've been operating this as a farm, how do I work it out over the whole?'

The community out there doesn't have engineers, surveyors, ecologists to call on, on a regular basis. To get them over there was a cost. Most of the folk were land-rich and basically cash-poor. We've got a declining permanent population but an increasing holiday population, so we're getting more and more people buying land and putting a holiday home on it. And the people who were left as a resident population were suffering because, let's face it, if you can afford to pay some of the prices on parts of our coastal environment in the Gulf, you can probably afford to fly an architect, an engineer and an ecologist across. If you're living on Barrier, it's probably not as reasonable from your perspective to bring those people across to be able to say whether or not you can put your farm shed on the site. You've always had a farm shed on there, you just have to shift it from here to there because that one got lost in a fire or flooded out or whatever. The council's rules are a bit more difficult there to work with.

So what we're trying to do is say, 'okay, we don't necessarily want to walk away from that but we need to find a smoother way through the process'. So I think that's one of the things that we'll probably try and achieve, hopefully people will be a bit more satisfied. (respondent 49)

The issues on Waiheke Island were very different, due to the much heavier development pressure experienced, which required more sophisticated urban controls. The initial focus on ecological considerations failed to deal sufficiently with urban issues:

The current operative plan focuses on the impact on the natural environment, but doesn't consider what the human element brings to that natural environment. So, for example, we had controls around vegetation clearance and earthworks, which we will continue to have, but they don't look at the fact that that site that you created your vineyard on [and] then ended up with a restaurant tourist accommodation and attracts a hundred people every Saturday morning to a wine tasting.

We're introducing parking standards or limitations around parking, trying to manage traffic, trying to understand that, when you have an environment

that doesn't have a reticulated sewerage system, that doesn't have the ability to collect its rain water off the streets, that you can't have unlimited capacity on the site so you cover it with concrete. But that you actually have to put controls on how much concrete you can put on a site, which will be new for the plan. Currently there aren't any controls. You can basically cover your residential land unit in Oneroa or wherever with concrete if it's already had its vegetation cleared ... We're putting limitations around those kind of things because we're getting run-off basically hitting the road, scouring out the roads and the loose metal that's still on the drains or whatever and taking that down your local stream and out into the coast. (respondent 49)

The council has found that the plan's approach has proved reasonably successful in protecting and promoting natural vegetation. It has not, however, provided community certainty as to the form that development will take, or addressed the social impacts of the form of development evolving, which can generate conflict:

Areas that had previously been grazed are now absolutely planted out. I think they've been successful, the plan and the ongoing approach has been successful; on that level. There will always be elements of it that haven't been successful in terms of the level of development that may have occurred at Oneroa village, that people might not have envisaged, and so forth. But I think on the whole what it tried to do, as its first step from the Town and Country Planning Act up to an RMA destination, it's been very successful in that sense.

I guess the hard part is the plan as it's set up allows you the ability to do anything if you can prove that you can meet your development controls or alternatively, if you don't, that you're not going to have an adverse environmental effect. So ... some of the community have said it hasn't performed - if you've got the money to throw at litigation, you can get activities or development that the general public might not have anticipated, or a scale of development that the people might not have anticipated.

It's not so much the physical changes that might have happened on Waiheke, it's some of the more social elements. There's a huge community on Waiheke with a lot of money and there's a huge community that don't have so much money. When you think about it, in the seventies and sixties Waiheke was where folk could buy a good size section and a bach for \$20,000. So I think that's part of it. That there's a lot of people that went there in the early days

and now there's this new wave of folk who've got money who can afford to litigate. I think they think the plans are a little more permissive than they would like them back then, because [of] that sort of 'have and have not' environment that's developed on Waiheke.' (respondent 49)

In September 2006, the Council notified a new Proposed District Plan – Hauraki Gulf Islands Section, which includes both Great Barrier and Waiheke Islands. This is now going through the statutory submission process.

Hauraki Gulf: the Hibiscus Coast

The Hibiscus coast incorporates the northern part of the Hauraki Gulf and is managed by Rodney District Council. Because it is a very attractive part of the coast with many white sand beaches and rocky headlands, and is located in close vicinity to the Auckland metropolis, it has experienced some of the strongest coastal development pressures of anywhere in the country. The growth in Rodney District as a whole, which also includes the southern part of the Kaipara Harbour, was 14.8 per cent between 2001 and 2006 compared to a national average of 7.8 per cent (Statistics New Zealand, 2006). So how has the Council endeavoured to manage these pressures and how successful has it been?

The overall strategy of the Council is set out in a document titled *Vision Rodney* which identifies as key priorities keeping 'our country look and feel' and not letting 'our towns and villages sprawl' (Rodney District Council, 2006, 3). The approach to achieving this within the coastal environment included identifying and providing greater protection for special-character areas through the use of conservation zoning, providing for country-side living in zones delineated for that purpose and only permitting subdivision in rural areas where environmental enhancements could be achieved. This was within the framework of applying outcomes-based planning under the RMA:

Overall strategy

I guess the latest version of Vision Rodney, those key intents - we will keep our country look and feel, we will not let our towns and villages sprawl ... Prior to that we've had our Rodney District Development Strategy of 1994 that looked at the whole district and said 'where we should grow' ... [In] places like Snell's Beach, for example, we don't want it to be spreading all the way along the Mahurangi Peninsula. There's some edges to that. So if we're going to do anything it's to try and make the things more dense, not more dense but more deep, I suppose, away from this coast here as opposed to it spreading along. We don't want ribbon development all the way along the coast.

Outcomes-based planning

We introduced [change 55] in 1995 and that was our first attempt to change the rural rules to align more with the Resource Management Act. It lessened control and it tried to express more clearly the outcomes that we wanted to achieve rather than simply saying here is a list of activities you can do, here's some subdivision you can do, but tried to express [that] this is the anticipated environmental results we want or environmental outcomes and ... there was a range of activities that you might allow and the type of subdivision that you might allow.

Conservation zoning

... The pressure for us from people to subdivide and ... to use rural coastal land has come in the east coast, so it's all been over that time that generally we've had some form of protection zoning on the east coast. Over the various plans we've refined them but that's what we're doing, essentially. Running from our district boundary in the north along the east coast all the way down to the Hibiscus Coast, basically, there has been some form of conservation type zone. It's been based on quite a narrow coastal strip that extends inland for several kilometres inland and it's based on some landscape features.

The proposed district plan 2000 - in that we refined ... the extent of the zone because each time we were challenged through submissions and occasionally through to the Environment Court on why we had chosen a conservation-type zone for someone's land. The conservation zone contains more stringent controls on vegetation removal and earthworks and less opportunity for subdivision in each of the conservation zones. Every time we change the plan we probably refine those more.

Countryside living zones

Countryside living ... was where we accepted the notion that, because we're under a lot of pressure from the periphery of the city to subdivide down into what are essentially lifestyle blocks, ... we should allow for that to try and soak up the pressure from everywhere else so we didn't have sporadic lifestyle blocks all over the place interfering with farming ... So we said let's try and concentrate on areas where there's already that character and if you added more it's not going to cause too much of an adverse effect. So we've introduced those countryside living zones. (respondent 52)

So how well has it worked? The rural conservation zones have provided greater control over the form of development. The provisions based on environmental

enhancement create greater environmental benefits but also more subdivision opportunities throughout the rural environment – so more trees, but also more houses, infrastructure, people and pets. And, given the permissive nature of the RMA, there is the ongoing prospect of major developments which do not comply with the structure of the district plan getting consent under the RMA and fundamentally changing the development pattern on the coast:

Here under the new plan, because our subdivision rules are based on environmental outcomes or environmental benefit if you like ... subdivision has given us extra reserve land. You get some conservation lots and you kind of allow that type of subdivision to occur in these areas as well. But generally buildings are a controlled activity, on building form and location in particular, and the range of activities is slightly different.

Generally the [provisions] have worked in a sense that people now know that they are in a rural conservation type zone and that the council [will scrutinise] the style and type of house they're going to put up there and where they're going to put it and that there are some key things. You don't want people building on the high ridge lines and painting their houses red. They've got to blend in and be part of the country and still give the impression of rural country and achieve the outcomes that we want for that area, protecting its special character.

At the moment we've got a series of proposals on the books that could influence patterns on the eastern coast. We've got, for example, the Te Uri o Hau settlement where they originally proposed a 2,000 household settlement. They've dropped it back to 1,400 and we've told them to go away and rethink. We've got a major submission from a group called Omaha Park Limited, to our district plan, I think its 700 hectares, just off the top of my head, they own in there and they want to basically double the size of Omaha north and south as it is and add on to it and then have a whole series of coastline living and rural residential type lots over that 700 hectares. So that would double the size of the existing Omaha at least and more. Then we've got other proposals in the wind ... and we get continual requests for things like that.

(respondent 52)

The Council issued decisions on submissions on its proposed district plan (which was notified in 2000) during late 2006 and the appeal process has now commenced.

Kaipara Harbour

Land use in the northern Kaipara Harbour is largely managed by Kaipara District Council and in the southern harbour by Rodney District Council. They have taken very different approaches to planning, resulting in very different district plan provisions governing land development. Both councils also manage areas of the east coast where development pressure has been much stronger, and much of their coastal development management focus has been away from the Kaipara Harbour.

Although the Kaipara Harbour has not in the past experienced the kind of development pressures occurring on the east coast, as property prices on the east coast have risen dramatically, interest in residential subdivision on the harbour has increased. This is especially in areas of the harbour accessible to Auckland, such as the South Head and along Oneriri Road on the Pahi Peninsula. An example is the Takahoa Bay subdivision on the Pahi Peninsula, which incorporates 48 lots with prices ranging from \$220,000 to \$595,000:

What's happening in Kaipara is that a lot of the larger more productive farms are amalgamating. So you get large land holdings on one side of the coin. The other side of the coin is you get a lot of lifestyle development i.e. people wanting their little blocks of land with rural amenity ... The pressure is on for lifestyle development and there's also pressure in terms of some of the coastal settlements for more urban development ... [in] places like Tinopai, Whakapirau, Bayleys Beach, Pahi, and currently those settlements are not reticulated with sewerage. (respondent 46)

Kaipara District Plan

The development of the Kaipara District Plan was not based on a strong strategic direction. The orientation of the plan was to encourage development and at the same time to broadly reinforce existing settlements. The coastal edge of the Kaipara Harbour was not initially seen as requiring special recognition or protection. A policy overlay, rather than a full coastal protection zone, was subsequently introduced as a result of an appeal to the proposed plan by DoC. This provided a lower level of protection than for other coastal areas in the district, based on a lower rating of the natural character of the harbour.

The provisions of the overlay place extra controls on earthworks and the clearance of indigenous vegetation, and require additional information to be provided when a resource consent application is lodged. They do not, however, place any additional restrictions on subdivision or building development. Subdivision down to a minimum lot size of four hectares is provided for as a controlled activity throughout the entire rural zone bordering the harbour

(rule 11.5.1(1)(i)). This can be compared to a minimum lot size of 20 hectares as a controlled activity for subdivision in the coastal zone within the district which applies to coastal land outside the harbour along the west coast (rule 11.5.3(1)(i)):

[The district plan] was probably one of the first ones in the country to be made operative ... it was probably done on an extremely small budget. It was also partly within a time when people were probably [not] fully aware of the implications around the RMA. Consequently it was a bit of a collation of the district scheme at the time, in terms of RMA speak.

... initially there was no special zone or differentiation for the Kaipara Harbour as opposed to the rural zone ... [A change] was largely driven by the Department of Conservation. To resolve that appeal, Kaipara District Council agreed to do a plan change to ... better recognise and provide for the Kaipara harbour in planning terms ... First of all the Fringes of the Kaipara Harbour Planning Study was undertaken ... (respondent 46)

The Study considered that the harbour as a whole has varied natural character values and that high value landscapes were confined to a relatively small number of more discrete features or units. It recommended a policy overlay over the existing rural zoning. This was based on the opinion that the natural character of the Kaipara Harbour was at the lower end and that the working environment should be retained with a comparatively permissive settlement regime. [In my opinion] the assessment of low to moderate natural character was based on a Eurocentric aesthetic preference for the high energy oceanic seascape which the Kaipara didn't fit into. (respondent 16)

[The Study] resulted in Plan Change Number 10 [introduced in 2001]. What that did was essentially [put] an overlay zone or overlay environment on top of the rural zone. [It] provided for ecological landscape provisions and additional assessment criteria and the objectives and policies. That was subsequently appealed by the Guardians of the Kaipara, but got resolved in a consent order in 2005. (respondent 46)

The Kaipara District Council is currently reviewing its strategy and district plan in the context of developing its LTCCP. The new plan will be more strongly guided by a growth strategy. Due to a lack of resources to carry out detailed planning studies, the plan is likely to place the onus on developers to undertake a more rigorous assessment of the effects of development in rural areas:

'Council went through a visioning exercise for its long-term plan and they talked about at a higher level what they want Kaipara to look like in sort of 20 years time. And [they] talked about focusing residential growth in the Mangawhai area, talked about maintaining the rural heartland and talked about protecting the Kaipara Harbour and protecting the west coast as the wilderness area, promoting Dargaville as a service centre. So at a very high level there is direction.

... Typically a lot of people in the Kaipara community, when the original district plan was developed, were saying 'let's encourage more development' and the people who submitted are saying 'let's encourage more development, we don't want to make it any more restrictive'. ... What we probably will be doing is looking at a comprehensive development type approach, i.e. really encouraging, if people are looking at intensifying settlements or intensifying development, directing to a discretionary planning approach or a private plan change approach and so saying effectively you'll have to consider a whole heap of things because simply we don't have resources to do those base studies and ideally that's what you'd be doing. (respondent 46)

Rodney District Plan

In the southern part of the harbour, which is under the jurisdiction of the Rodney District Council, the generic rural provisions in the Rodney District Plan apply. These only allow subdivision (as a restricted discretionary activity) if an environmental benefit is provided, such as covenanting areas of bush. Where such benefit is not provided, subdivision is a non-complying activity. However, unlike for the east coast, the Rodney District Plan does not specifically recognise the Kaipara Harbour as a coastal environment or provide any protective zoning for the land surrounding it. This is because the level of development pressure surrounding the harbour has been much less than on the Hibiscus coast.

It hasn't had the same pressure on it, from our point of view, for settlement that the east coast has had. When you look at the population numbers and the density of population along here, it's still pretty sparse by comparison ... the density of settlement is much less, the activities are more rural and we don't have much pressure. (respondent 52)

However, appeals have been filed against the proposed plan seeking *inter alia* the creation of a Kaipara Harbour coastal zone. These have yet to be determined.

The planning challenges

The interviews identified a range of challenges facing planners in managing coastal development in a climate of strong development pressure. These include difficulties associated with the RMA itself, difficulties internal to the councils, and external pressures.

RMA challenges

Challenges which councils have faced relating to the RMA itself include the lack of a strategic focus to planning under the Act, the site-specific focus which the Environment Court has adopted in applying it, and the cumbersome process which councils are required to go through to change their plans to address issues as they arise.

Lack of a strategic perspective

The RMA, which is very much focused on problem identification and the assessment of effects of individual applications, has not promoted strategic thinking about the long-term development trajectory of a district. This has made it difficult to manage coastal development other than in an *ad hoc* manner. In addition, councils can struggle developing and implementing a long-term vision, particularly when they are endeavouring to address immediate issues and achieve positive change for their communities in the short term:

The Act says that you have a plan to address the resource management issues. If the thing is sitting there and it's in a fine state you don't have an issue ... even when we put [the district plan] out in 2000 it was the result of thinking that came out of the Ministry for the Environment in the 1990s, late 1990s ... They had a particular view and they published a lot of papers about what district plans could and couldn't address and to say that you've got to address the resource management issues and, if your environment is sitting there and it's fine at this point in time that you're making your rules, then you don't have an issue in our view. That was the view we took at the time. So if you look at that definition of what an issue is ... you need some sort of conflict or problem for it to be an issue. (respondent 52)

Part of the question is why does that frustration with statutory planning persist? I think sometimes it's because of a lack of a strong strategic view. Sometimes it's a result of how we write plans. Too often plans are not clear in the outcome and that is why, for instance, there are things called compartment plans

or whatever because [of] what people are interested in. Well, what is the practical implementation of a statutory plan, what are the things that might be appropriate, what might not be appropriate? ... We tend to want to get into moving deck chairs around first before we actually have a look at the deck. (respondent 43)

Up until probably two years ago ... I don't think the council had a vision of what it wanted to achieve. It had a district plan that said a few things ... The management team had been away a couple of times together on retreats to try and define what their vision for the Coromandel [Peninsula] is. I think it would be fair to say that the elected members had a great deal of difficulty defining what it is. I guess that's a general statement I'd make about people who are not in the planning or policy field, is that most people have a great deal of difficulty looking to the future. They can sort of think to the next Friday's rugby game or whatever but, in actually planning ahead, I think is a skill that a lot of people have difficulty with.

Until three years ago when ... the sort of forward-planning council resourcing was beefed up ... everything was geared towards reacting to the huge growth that was coming and [there] wasn't a lot of effort put into future planning. The council has recognised that, which is good. They've put quite a bit of extra resource into forward planning. But it's always a bit of a balance. I'm very conscious that the policy and planning group tends to be seen as a bunch of bureaucrats sitting upstairs producing screeds of paper and ... that's the reality with the local government environment. There's certainly an expectation by this community that there are things which are much more important than staring at the stars and thinking about what a wonderful future we should all have ... [It is] not helped by the demographic of most of the district tending to 65 plus, [they] tend not to have much of a long-term horizon. That's a bit of a generalisation, but in general terms [they] are worried about this year, and this year's rates and what are you doing now, rather than are you trying to make the place better for the future. (respondent 42)

Site-specific focus

In implementing an environmental outcomes approach in its district plan, Rodney District Council sought to describe and protect the special character of areas. However, this approach has not found favour with the Environment Court, and the council has been forced to use much more generic descriptions to avoid challenge:

We had a couple of court cases where the court said ‘you’ve listed the special features and the special character that exists there but someone went out on to one particular site and said ‘but I can’t see any of these features existing on this particular site’ and you’re saying ‘it’s still part of the catchment that provides that’. But they took it in a literal meaning that every site that was in that zone would have to have those features that we had listed as part of the special character. You’re not going to get that on every site but they took it that way.

You could argue that we weren’t clear enough in the way we expressed the outcomes. But we found that the Environment Court took it to mean that every site had to have those same features that you mention as the definer of special character. Clearly that’s never going to be the case. Some sites will, but ... you have a site and it drops down into a hollow and it goes up into a hill again. The stuff on the hill might have that special character, but you’re not going to have a hole [in the special character zone] where the hollow is because you can’t see the coast from there or you can’t be seen from the coast for example. You’re not going to have another zone there. You have to include it in the same zone, so what we’ve done is pull back a little bit and been a little more circumspect about ‘special’, being more generic. (respondent 52)

Cumbersome plan change process

The lengthy and expensive process which councils are required to undergo to make changes to their plans is a major obstacle to providing a timely planning response:

One of the realities of the Resource Management Act is that the time frame involved in staying ahead of the issues and making timely changes are just ... they’re out of all proportion to what happens in the real world ... We’ve got some issues with infill housing in Whangamata where there’s a perception that people are getting around the rules and things like that. We can do a very easy change to the district plan that we think will sort that out, but we know it will just be in the Environment Court for years while various interests argue their points of view. I’ve sort of come to the cynical view, I guess, that the best way to control land use behaviour is not through the district plan but through the provision or non-provision of infrastructure. If you’ve got an environmental problem brewing in Pauanui because of the wastewater treatment plant, don’t try and address it through the RMA district plan. Put a moratorium on and you’ve stopped [it] just like that, whereas the district plan process will take years and years and years. (respondent 42)

Council internal challenges

Internal problems identified by interviewees included a lack of planning capacity, difficulty in recruiting and retaining staff, a lack of robust information to inform decision making and poor internal links between different council functions, including RMA planning and service provision. The territorial authorities studied each have very different resources available to them. In 2005 Auckland City Council had the largest budget of \$425.6 million which is around 16 times the annual budget of Kaipara District Council which was only \$26.5 million (see Figure 7.1).

Figure 7.1: Territorial authority statistics Source: www.localcouncils.govt.nz

Territorial authority	Operating expenditure (year ended 30 June 2005)	Number of staff	Land Area (km²)
Auckland City Council	\$425.6 million	1,666	633
Rodney District Council	\$94.1 million	374	2,427
Thames-Coromandel District Council	\$41.6 million	176	2,297
Kaipara District Council	\$26.5 million	43	3,117

Lack of planning capacity

Smaller councils struggle to develop and retain sufficient planning expertise to undertake their strategic planning functions and institutional memory can be lost as long-serving staff leave:

We're going to review the district plan, that's the first thing, but ... at the moment there's myself, a senior planner and a policy planner who's basically straight out of university. We also have an administrator. I think there's real big challenges in terms of attracting and retaining staff in the smaller districts. To compensate we've got quite a large consultancy budget - \$450,000 this year, \$600,000 next year, with the intention of starting to review our plan. I think continuity of staffing and retention of corporate knowledge is crucial.

(respondent 41)

Lack of information to support decision making

Councils can lack basic information to inform decisions on how to manage development. This can be due to poor information management systems but also a lack of monitoring, so the effectiveness of policies and environmental changes can be identified:

Even basic stuff like how many houses there are in our settlements wasn't here three years ago ... I think we've gone ... pretty much from zero to probably about 60 per cent of what we need in three years, but there's still information that needs to be obtained - on the number of houses in the coastal zone, for example, or area covered by conservation covenants in the district. All simple baseline information that we are only just getting. (respondent 41)

I guess monitoring throughout the council hasn't been a strong suit. Internally we monitor process and stuff but, in terms of what effect we're having out there in the community, it's not as good as it should be ... I think there's just been a view generally that all the effort goes into getting the thing done and finished and you move on to the next thing and you don't ever go back [and ask] 'is it actually making a difference or not?' (respondent 42)

Poor links between land-use planning and infrastructure provision

In the past land-use decisions have been made without fully considering the infrastructure and service provision implications, due to a poor linkage between these functions within council:

I think in the past the relationship between the district plan and infrastructure planning has been extremely poor ... it makes pretty good sense that if you're zoning land you should be talking to engineers about if it's going to be serviced and vice versa. If they're planning to service land and it's not zoned, what's the point? I think, since the Local Government Act 2002, things are definitely moving in the right way, but there's still a long way to go. (respondent 41)

External factors

Each council is managing very different pressures for coastal development (see Figure 7.2). Rodney District Council is managing a high rate of increase in residential population (14.9 per cent between 2001 and 2006). Thames-Coromandel District Council is managing a low increase in population (2.9 per cent) but a high increase in the number of dwellings, driven by a high demand for holiday homes. Kaipara District Council also has a low population growth (3.6 per cent) from a low population base. However, the rate of growth has significantly increased over the previous five years (1996 to 2001) where it was only 0.5 per cent with only 93 extra people living in the district, compared to 647 additional people during the following five years. The district is also starting to experience demand for holiday homes. Auckland City Council is also managing a large population increase (9.1%) although much of this is on

the isthmus rather than the Hauraki Gulf Islands. The population of Great Barrier Island showed a sixteen per cent decrease in population between the two censuses, whereas Waiheke Island experienced a 7.7 per cent increase. These population increases can be compared to the national per cent increase of 8.4 per cent in the usually resident population (Statistics New Zealand, 2006).

Figure 7.2 District population growth 2001 to 2006 Source: Statistics New Zealand, 2006

Territorial authority	Total population 2001	Total population 2006	Change 2001 - 2006	% Change 2001 - 2006
Rodney District Council	76,185	89,562	13,377	14.94
Auckland City Council	367,737	404,658	36,921	9.1
Kaipara District Council	17,456	18,103	647	3.6
Thames-Coromandel District Council	25,176	25,938	762	2.9

Where there is heavy development pressure there can be strong challenges to the district plan rules, and a whole industry developing around beating the rules. As one planner observed:

So, as far as the east coast goes ... we're seeing more and more pressure as time goes on and the value of sites has risen accordingly. So people are prepared to challenge your rules a little bit more ... there's been a whole series of them where people are challenging our limitations on the subdivision because the land value is so high that you can get a million dollars for your site, spending \$100,000 in the Environment Court.

What we've seen is the rise of rural development ... Previously in the rural areas we'd get Joe Blogs whose father owned the land and he'd have a go and he wanted to subdivide two or three lots off and he'd do that. Now that's changed and what you're getting is people who are set up as developers. They've never been farmers. They come in and there are partnerships and all sorts of things. Urban business, come out here, buy rural sites on the basis that they're going to subdivide them and make money. That's something that's happened over the last five years; we're getting more of that. (respondent 52)

Prospects for integration

Each council has adopted a very different approach to managing coastal development in different parts of the Hauraki Gulf and the Kaipara Harbour, although within a broad strategy, applied with differing levels of vigour, to concentrate coastal development in existing urbanised areas (see Figures 7.3 and 7.4).

In all these areas, subdivision provisions were no more restrictive on the coast than in inland rural areas, even though coastal areas are often more sensitive environments. However, additional controls have commonly been placed on the location and design of buildings and earthworks and on vegetation clearance through different forms of coastal zonings.

In the Gulf, approaches ranged from relying on a twenty hectare minimum subdivision limit in the rural areas of the Coromandel Peninsula, to no permitted subdivision in the Hibiscus Coast without the provision of an environmental benefit. For the Gulf Islands a sophisticated landform-based approach was used to guide development but proved complicated to implement. Controls on development are tighter in the Hauraki Gulf, where development pressures had been much stronger, and where management agencies have placed more effort, than in the Kaipara Harbour.

In the Kaipara Harbour, controls range from subdivision down to four hectares as a controlled activity in the north, to subdivision without environmental benefits being a non-complying activity in the south. The provisions for conservation lots also differ significantly with, for example, a conservation lot provided for 4000m² of protected area in the north compared to two hectares in the south. This marked difference in subdivision rights in the harbour has been the source of community concern. It could lead to a concentration of poorly-designed subdivisions in areas of the harbour within the Kaipara District that have good accessibility to Auckland.

The approach taken by the Kaipara District Council is the most permissive, reflecting the historically low levels of development in the northern part of the Kaipara Harbour's coastal environment, the small size of the council and consequent lack of resources to undertake more detailed planning exercises, and the council's emphasis on promoting economic development. As development pressure starts spilling over from the east coast to this west coast harbour, the district plan is not well equipped to ensure positive environmental outcomes.

The current review of the plan provides an opportunity to better address these issues. A revised plan could provide stronger protection for the coastal

values of the harbour as well as better integrating with other planning documents affecting it. It is not yet clear to what extent this will be achieved.

Although there has been relatively little development pressure in the southern Kaipara Harbour when compared to the east coast, the Rodney District Plan has strong controls on rural subdivision, with a large loophole in the form of environmental enhancement projects. There has been no planning exercise specifically directed at identifying what may be most appropriate for the harbour, and the proposed Rodney District Plan does not propose any specific zoning for the harbour environs. In the absence of such a planning exercise, and the identification of the coastal environment of the harbour as a special character area, it is difficult to see how the largely district-wide provisions of the Rodney District Plan that affect the harbour could be integrated with the provisions of the Kaipara District Plan, most of which are also currently district-wide.

Ten territorial authorities manage the coastal environment of the Hauraki Gulf and this study focused on just three (Rodney District Council, Thames-Coromandel District Council and Auckland City Council). Between them they manage a large proportion of the coastal environment. Rodney and Thames-Coromandel District Councils have taken a similar approach in creating conservation zonings along the rural coastline to provide greater control over development. They both have conservation lot provisions which enable a greater degree of coastal subdivision. Auckland City Council undertook a much more detailed planning exercise for the Hauraki Gulf Islands, mapping different landforms and developing provisions tailored to each.

The differences between the approaches taken in preparing the district plans are much greater than the differences between the regional coastal plans applying to the case study areas, which were generally similar (see *Case Study 1: Planning for the coastal marine area of the Hauraki Gulf and the Kaipara Harbour*). This suggests that, for the catchments of the Hauraki Gulf and the Kaipara Harbour, integrating land use management by territorial authorities will be much harder than integrating marine planning.

The councils are all at different stages with their district planning which is likely to make any attempts at integration even more problematic. Rodney District Council has recently released decisions on submissions on its proposed plan and appeals have been lodged. Auckland City Council has recently notified a proposed section of its district plan for the Hauraki Gulf Islands and the Kaipara and Thames-Coromandel District Councils are just beginning processes to review their district plans. The development of these new plans does provide an opportunity to better recognise the need to improve the management of development in the coastal environment.

Figure 7.3: Territorial authority approaches to coastal planning in the Hauraki Gulf

Territorial authority	Separate delineation	Effect of delineation	Rural subdivision rules
Rodney District Council <i>Proposed district plan</i>	East Coast Rural Zone; Landscape Protection Rural Zone	Additional control over location of buildings, earthworks and vegetation clearance	Environmental enhancement – 1 lot for 2ha protected; 2 lots for 12 ha protected etc. (restricted discretionary); Otherwise non-complying
Auckland City Council <i>Proposed district plan</i>	Great Barrier Strategic Management Area; Waiheke Strategic Management Area	Landform based zoning; additional control over location and design of buildings in rural areas; additional control over development in sensitive landforms (e.g. coastal cliffs, dune systems and wetlands)	25ha min lot size in Rural 2 (discretionary)
Thames-Coromandel District Council <i>Proposed district plan</i>	Coastal zone	Additional control over location and design of buildings	20ha min average lot size (discretionary); conservation lots – 1 lot for minimum of 5ha protected (controlled in rural, discretionary in coastal zone)

Figure 7.4 Territorial authority approaches to coastal planning in the Kaipara Harbour

Territorial authority	Separate delineation	Effect of delineation	Rural subdivision rules
Rodney District Council <i>Proposed district plan</i>	No	Nil	Environmental enhancement - 1 lot for 2ha protected; 2 lots for 12 ha protected, etc. (restricted discretionary); otherwise non-complying
Kaipara District Council <i>Operative district plan</i>	Kaipara Harbour Environment Area	Additional control on earthworks and vegetation clearance (but not buildings)	4ha minimum lot size (controlled); conservation lots – 1 lot for 4000m ² min protected lot size with 4ha average in subdivision and no more than 6 in subdivision



eight

Case Study 3: Reducing Sedimentation in the Hauraki Gulf and Kaipara Harbour

The Sleeping Giant

Accelerated sedimentation is one of the most pervasive issues facing the Gulf. (Hauraki Gulf Forum, 2005, 67)

Little has been done about it, partly because the farming community is a little bit sacrosanct. (respondent 2)

It's twenty years down the track rather than ten till we get to a certain point. [But] how much of an achievement is that? (respondent 3)

The policy framework

The RMA provides the framework for the control of activities which result in land-based sediment entering the marine area. Regional councils are charged with the responsibility of controlling the use of land, for the purpose of maintaining coastal water quality and coastal marine ecosystems amongst other things (section 30(1)(c)). This includes managing the negative effects of sedimentation. Regional councils are able to prepare regional plans to address the issue.

The New Zealand Coastal Policy Statement, somewhat surprisingly, does not directly address the issue of sedimentation although it can be implied from provisions relating to the protection of significant habitats of indigenous fauna (policy 1.1.2) and the maintenance and enhancement of water quality (section 5.1).

The management issue

Sedimentation is a sleeping giant that relates both to farm management and subdivision and other forms of development ... Little has been done about it, partly because the farming community is a little bit sacrosanct. 'Farming does no harm to the marine environment'. That's been part of it. [It] hasn't been on anyone's radar ... I think the issue arises from getting back to our colonial history of deforestation, farming down to the waters' edge and land reclamation ... now it's uncontrolled sediment from earthworks related to house site roadways and driveways. (respondent 2)

The negative impacts of accelerated levels of sediment entering marine areas can be serious and extensive. These include smothering marine organisms living on the sea floor, stressing filter feeders such as shellfish, and reducing seawater clarity and the ability of light-dependent species such as seagrass to photosynthesise. Negative impacts are more pronounced in sheltered areas, such as estuaries, harbours and enclosed embayments, where there is a reduction in wave action and off-shore currents to disperse the sediment (Peart, 2005, 81-83).

Although land erosion and the resultant sedimentation are a natural process, human activities on the land such as the removal of forest and vegetation cover, grazing and earthworks can greatly increase the rate of erosion. Sedimentation of the marine area can be a particular problem where there are steep short catchments and erodible soils (Hauraki Gulf Forum, 2005, 68).

Sedimentation is a significant problem in the Hauraki Gulf and the Kaipara Harbour:

Accelerated sedimentation is one of the most pervasive issues facing the Gulf. Sediment affects the quality of the Gulf's waters, its biodiversity values, its natural character, and people's potential use and enjoyment of the Gulf. (Hauraki Gulf Forum, 2005, 67)

In some of the areas [in the northern Kaipara] we were just pulling up cockle beds that were probably quite extensive, but they were all dead now, and it's just basically the fine sediment that comes off the land ... Some areas aren't fenced off; in others the fences go right down to the water's edge, and I believe that some of the fences that go to the edge are not in existence because of the erosion ... If you're going to farm right down to the water's edge you're going to have a few problems. (respondent 33)

Examples of accelerated rates of sedimentation on the Coromandel Peninsula are shown in Figure 8.1 below. The increase is particularly marked in the Wharekawa and Whangamata Estuaries where production forestry has been established in recent times.

The extent of sedimentation can also be indicated by the spread of mangroves in harbours, although this can also be affected by nutrient run-off. Several harbours in the Coromandel Peninsula have experienced high rates of spread of mangroves in recent years (see Figure 8.2)

Figure 8.1: Sedimentation rates in Coromandel Peninsula Estuaries Source: Mead and Moore, 2004

	Pre Polynesian mm/year	Polynesian Settlement mm/year	Post-European Settlement mm/year	Recent mm/year
Firth of Thames				
Site 1	0.09	0.13	0.5	
Site 2	0.18	0.05	0.05	
Site 3	0.10	0.38	1.3 – 1.5	
Coromandel Harbour				
Site 1	0.02	0.07	0.82	
Site 2	0.07	0.05	0.31	
Site 3	0.94	0.39 – 0.57	1.01	11.7
Whangapoua Harbour				
Site 1	0.03	0.13	1.3	0.89
Site 2	0.08	0.12	1.5	1.33
Wharekawa Estuary	0.09 – 0.12		3.6 – 7.2	5.0 – 8.0
Whangamata Estuary	0.1	0.3	6 - 11	

Figure 8.2 Increase in mangrove coverage in Coromandel Peninsula estuaries

Source: Mead and Moore, 2004

	Time period	Percentage increase in coverage
Te Kouma Harbour	1971 to 1995 (24 years)	66.5%
Manaia Harbour	1971 to 1995 (24 years)	194.6%
Tairua Harbour	1983 to 1996 (13 years)	214.9%
Otahu River	1983 to 1996 (13 years)	214.9%

So, how have coastal managers approached the problem and how successful do they think they have been in getting on top of it?

The management approach

Regional councils have adopted very different management approaches to the main potential sources of sedimentation – earthworks, forestry and farming – with earthworks receiving the greatest control and farming activities the least.

Earthworks

The three regional councils studied have all exerted some control over larger earthworks in their regional plans (see Figure 8.3).

In the Auckland region, measures to reduce the sedimentation caused by earthworks were first introduced during 1979, through the distribution of an Urban Earthworks Guide. This voluntary approach was only partially effective and in 1988 the then Auckland Regional Authority introduced regulatory controls. These were subsequently replaced by the Auckland Regional Plan: Sediment Control which became operative in 2001 (Auckland Regional Council, 2001, 2). Of the regional councils included in this study, the ARC was the only council to have a separate plan for sediment control.

Figure 8.3: Regional council control over earthworks as permitted or controlled activities

Regional council	Permitted activity (subject to compliance with environmental standards)	Controlled activity (subject to compliance with environmental standards)
Northland Regional Council	<5000m ³ volume of soil disturbed over 12 months where not erosion-prone (outside riparian management zone); <1000m ³ volume and 0.1ha surface area of soil disturbed over 12 months where erosion prone (outside riparian management zone)	>5000m ³ volume of soil disturbed over 12 months where not erosion-prone (outside riparian management zone)
Auckland Regional Council	<1ha area of soil disturbed with slope < 15 degrees or <0.25ha area of soil disturbed with slope = or > 15 degrees (outside sediment control area); <0.25ha of soil disturbed (within sediment control protection areas)	1 to 5ha of soil disturbed with slope < 15 degrees (outside sediment control area)
Environment Waikato	No control outside high risk erosion areas; Within high risk erosion areas <250m ³ volume of soil disturbed and <0.2ha of surface area disturbed	Within high risk erosion areas 250 to 1,000m ³ volume of soil disturbed or 0.2 to 2ha surface area disturbed

Research commissioned by the ARC in 1994 indicated that earthworks had a huge potential to generate sediment, which was far greater than other potential sources, when based on the same land area (see Figure 8.4). As a result, most effort was focused on managing sedimentation generated from earthworks.

Figure 8.4: Annual soil loss in the Auckland Region Source: Auckland Regional Council 2001, 11

Land-use	Predicted average annual soil loss (over 20 years) (Tonnes/km ² /year)
Developed Urban – Residential	24
Pasture	46
Market Gardening	52
Developed Urban – Industrial	100
Earthworks	16,800

Through a multi-pronged approach, the council has devoted considerable resources to managing the sedimentation generated by large earthworks. Because the earthworks industry has a relatively small number of large operators, it has been possible for the council to achieve high levels of compliance through the development of positive relationships:

The integrated co-ordinated programme that we have has four components to it. We have a regulation side ... [which adopts a] best practical options approach, using erosion controls ... We have a very strong educational component as well. We have annual workshops, we run industry education courses, we have plan preparers, we had the annual workshop at Long Bay this year. So that's an annual event. We see education as a very strong component of getting change within the industry ... We are also doing research in terms of improving efficiencies of our own quality controls ... We have a very strong compliance programme and we have a whole lot of out-sourced consultants who will check them at least once in a week. (respondent 51)

We have good relationships with the industry, they know what's expected of them. The threshold is quite ... low in terms of enforcement, so they know where the boundaries are. If they step over them, they know what the consequences are. We've worked with that industry for quite some time, so we have quite a good relationship, a good approach to how we deal with it, and therefore there's a lot more focus in that area around compliance. (respondent 44)

Although the ARC has been successful in reducing sediment derived from large earthworks, there are some notable weaknesses in the current approach. Sediment ponds are used as a major tool in reducing the sediment load of run-off, but they only remove up to seventy-five per cent of the waterborne sediment, still allowing twenty-five per cent to escape. This is because the clays

in Auckland soils will not settle out, irrespective of the size of the sediment pond. The Council is currently researching the use of chemical treatments which could increase the efficiency of the ponds up to ninety-five per cent, but they would also significantly increase the cost of undertaking the earthworks (respondent 51).

There is also a major gap with the management of small earthworks, which cumulatively can have a major impact.

So what happens is you have a developer who comes in who develops a big area of land. He has all the sediment controls in place and has consents. He has to go through a quite rigorous process to actually get the consent and carry out the activity appropriately. Then what happens is he sells it, and then comes a whole lot of builders who rip the sites to pieces with no sediment controls at all. Effectively the whole site is opened up and you've got cumulative effects with a whole lot of different activities. Now that's but one example and that's one that we've been talking about for some time. It's at the forefront of a lot of people's thinking, but it's often fallen into the too hard basket. (respondent 44)

Forestry

The harvesting of trees as part of production forestry activities can expose the soil and result in the accelerated sedimentation of marine areas. For example, in the Mahurangi Harbour, where sedimentation has been identified as having a major negative impact on marine communities, forestry may be contributing over half of the recent sediment settled in the harbour mud flats (Gibbs, 2004, iv). In a recent study of the source of sediment in the Whangapoua Harbour it was found that, in the three eastern arms of the harbour, soil from recent exotic pine forest logging contributed between fifty-four and seventy-five per cent of the sediments analysed (Gibbs, 2006).

Controls over accelerated sedimentation resulting from the harvesting of production forests are not generally strong and vary across the study area. Controls over harvesting are strongest on the Coromandel Peninsula and weakest in the Auckland region (covering the northern part of the Hauraki Gulf and southern part of the Kaipara Harbour). In the Auckland region in terms of sediment control, vegetation removal is a permitted activity (as long as it complies with standards in the plan). In the Northland region there is greater control over harvesting in riparian areas. The harvesting of trees planted after the plan became operative, within five metres of a water body, is a discretionary activity. In the Waikato region, large vegetation clearance in most catchments

draining into Coromandel Peninsula estuaries is a controlled activity. There are also more general controls over replanting in riparian areas (see Figure 8.5).

Figure 8.5: Regional council control over vegetation clearance for plantation forestry

Regional council	Type of regulation
Northland Regional Council	Permitted activity (if complies with environmental standards in plan); Discretionary activity where trees planted after the plan became operative (28 August 2004) are within 5 metres of a water body or the coastal marine area
Auckland Regional Council	Permitted activity (if complies with environmental standards in plan)
Environment Waikato	Permitted activity (subject to conditions including no replanting in riparian areas - which on the Coromandel Peninsula are within 5 metres of a water body between 20 and 50 hectares and within 10 metres of a water body greater than 50 hectares); Controlled activity where vegetation clearance exceeds 50 hectares per annum on slopes greater than 20 degrees within specified catchments (those draining into Coromandel Peninsula Estuaries)

In the Auckland region, where forestry is a permitted activity and does not require a resource consent, enforcing compliance with the standards in the plan has been a challenge:

Forestry is a permitted activity that we've often struggled with, because in many cases it's been isolated, and geographically it's not across the entire region. So we still do forestry compliance, and we still try and keep those people on track. Forestry is one we really struggle with through our policy development ... historically everybody planted right down to the stream edge. What we're seeing now is quite fine margins for people in terms of forestry operation, so they're starting to cut corners, and cutting those corners then has consequences in terms of compliance and enforcement. (respondent 44)

In the Waikato region, despite a non-regulatory focus in plan preparation, a more cautious approach was taken with forestry on the Coromandel Peninsula where soil erosion and sedimentation are a significant issue:

The overall philosophy the council told us to write the [regional] plan with was to remove regulation to the maximum extent possible whilst ensuring that the objectives were achieved. So we were really focused on getting rid of, as much as possible, restrictive presumptions in the Act and previous plans. They wanted that philosophy extended to land use, which had quite restrictive

requirements in the transitional plan, where you couldn't disturb more than a hectare of vegetation without getting a resource consent. The direction was to define common activities where we processed resource consents with little variation in conditions, and where we could hence define a permitted activity with conditions which were adequate to protect the environment. Then you define areas of high risk where you need to provide for site-specific controls if standard conditions were not sufficient.

So we ended up with a plan which is permissive except in areas that meet a list of criteria [High Risk Erosion Areas]. But we identified explicitly that the Coromandel Peninsula was to be treated differently, because sedimentation is in the higher end and forestry is a major contributor to the issue. We therefore cannot enable forestry to the same extent as the rest of the region. The Council accepted that, but made it a controlled activity with non-notification presumptions included in the rule. Appeals against the rule have recently been settled by consent. (respondent 12)

Agriculture

The biggest impact to that [Kaipara] harbour has been farming.
(respondent 33)

Although poor farming practices such as overgrazing and allowing stock access to river and coastal margins can result in significant sedimentation, only EW currently regulates grazing and stock access to water bodies and these regulations only came into force recently (subsequent to the interviews with council staff being undertaken) as a result of a consent order regarding a submission on the proposed regional plan (see Figure 8.6).

Figure 8.6: Regional council control over sedimentation generated through grazing and stock access to waterways	
Regional council	Type of regulation
Northland Regional Council	Permitted activity (provided environmental standards are met)
Auckland Regional Council	No applicable rules
Environment Waikato	Discretionary activity within stock exclusion water bodies; Permitted activity outside stock exclusion water bodies (provided environmental standards are met)

The three regional councils have to date largely relied on voluntary initiatives and incentive programmes to reduce the impacts of farming activities on water bodies. These include providing partial funding for the cost of fencing off and replanting riparian areas. When the regional plans were first developed, it was politically difficult to address non-point source farming impacts through regulation. However, as more information on the negative impacts of farming on water bodies has become available, regulatory options are being explored. Environment Waikato has been the first of the three regional councils to adopt regulation to exclude stock from identified water bodies. Those identified include fourteen streams on the Coromandel Peninsula (section 4.3.5.3, Proposed Waikato Regional Plan). The ARC is also considering future regulation. The Northland Regional Council, on the other hand, is still firmly focused on the use of financial incentives to change behaviour.

Auckland Regional Council

The ARC currently has no controls over grazing and stock access to riparian areas or water bodies, but is considering their future use:

I think politically we weren't at a stage where we were really ready to consider that, because it actually had quite big implications for farmers. Dairying and farm practices in Auckland, it's not as extensive as compared with the Waikato or other areas of New Zealand. So, rather than taking a hard line approach, I think our council at the time probably took a softer approach and decided that it wasn't appropriate to have rules around those things at that time. But as I said it's changing.

[We are] now starting to look at the fencing provisions of the Air Land Water Plan going forward, because at the moment we don't have any rules in the plan specifically around fencing of water courses. But that's something that will certainly develop and we'll move on to.

Stock exclusion is a big one. Again there's no specific rules for stock exclusion. In Auckland it's hard to, because you've got to remember we've got such a multitude of small stream networks that it's not easy for farmers to just fence off. If they were to fence off all the stream networks it would be quite expensive. But in saying that we've got all the information about riparian planting, what needs to be done to protect the environment; the substance is all there. The challenge now is to turn that into policy that works and to actually implement it to the point that you actually then start to see some of those benefits. (respondent 44)

Environment Waikato

At the time the interviews were undertaken, EW was moving towards some control over stock access to streams:

Grazing is the other activity that has an effect on sedimentation and that is a permitted activity without controls everywhere in the region. This was because, at the time the council prepared the plan, we didn't have definitive information that would allow us to categorically point the finger at farming in terms of these effects. The information on nutrients in places like Taupo has only come available since 1998. So that creates a vacuum for policy makers to say, "Well we haven't got sufficient information to intervene yet. We'll focus on a mix of non-regulatory measures with a baseline for permitted activities".

One change that we have recently negotiated in the plan [which has subsequently been sealed by a consent order] is the introduction of a livestock exclusion layer into the plan which says livestock must be excluded from streams for the first two kilometres of streams upstream from the CMA, incorporating a large number of streams on the Coromandel, and goes all the way to the salt water. It was justified on habitat protection but also reduces bank erosion and sedimentation. (respondent 12)

However, the main focus has been on better monitoring and the development of non-regulatory tools:

There is quite a lot you can do to respond to agriculture now without changing the policy but simply implementing it more effectively. This year we are going to beef up the amount of monitoring of the compliance of permitted activities for farm animal effluent disposal, soil disturbance, vegetation clearance and stock and water bodies. We are also starting to develop a capacity to develop proper integrated catchment management and not just landcare stuff. We have pulled together a sustainable agriculture strategy which is a comprehensive package of eight or nine work programmes, only one of which is about new policy development. (respondent 12)

Northland Regional Council

The NRC has not focused on reducing sedimentation from farming activities, although it does have a fund which contributes to the fencing of water bodies.

I think throughout Northland we have been steadily changing the behaviour of farmers in relation to effluent from the cowshed and that's been the focus so far and we're probably down to the last, I don't know, 150 odd reluctant dairy farmers throughout Northland of which a percentage would be flowing into the Kaipara Harbour ... I've got to say that ten years ago my councillors weren't happy when we were prosecuting farmers and now the last couple of years they've got through that and said 'well if you prosecuted farmers in the past for this you'd better sort the rest of these out'. And even farmers, most farmers will say 'if you make me comply, you'd better make sure the others do'. So I think we're over the hill in terms of farm effluent stuff.

What we're not so good at anywhere in Northland I think is just the ordinary non-point run-off. So I've got the councillors to increase our environmental fund, which gives about \$400,000, which is sort of a partnership with farmers where we pay the fencing and they do the other half in the riparian management areas ... and of course we signed the accord, the Fonterra one, and we're trying to do quite a lot of promotion with some of that. We just supported the other day the Sharemilker of the Year and the farm environment competition so I'm trying to do sort of carrot and the stick in relation to improvements. (respondent 6)

Catchment-based management approaches

Councils are piloting a range of catchment-based management approaches to address the impacts of sedimentation and nutrient discharges into rivers and the marine area. At least three of these projects, the Mahurangi Action Plan and the Okura Catchment Study undertaken by the ARC and the Muddy Feet II Project undertaken by EW, have linked activities on the land specifically to sedimentation impacts in specific sensitive marine areas.

Mahurangi Harbour Action Plan

Environmental monitoring was commenced in the Mahurangi Harbour during the early 1990s with the intention of providing a pre-urbanised baseline from which future changes could be identified:

The Mahurangi was identified as an area which was deserving of investigation back in the 1980s, really. There was quite a lot of work that was done to say

what's the current status of this area and the idea was really that it was a reasonably pristine harbour and catchment that was yet to be developed, but it was close enough to Auckland, so there was quite a strong potential in the future for that to be developed or for pressure to be brought to bear on it.

In the case of the Mahurangi, there was opportunity to get in well before the receiving environment was heavily impacted, therefore your monitoring programmes had the opportunity to give an early warning as opposed to saying 'yeah, you've got problems already'. So, following on from the investigations that were carried out in the catchment and in the receiving environment, in 1994 a benthic ecology monitoring programme was established and has continued to operate up until now. (respondent 54)



Figure 8.7: Mahurangi Harbour

The results of the monitoring over a ten year period showed a quite unexpected picture – the rural catchment, which was thought to be having little impact on the marine environment, was resulting in the decimation of the biodiversity of the harbour:

What we found was that there was a lot of abundance of predominantly sediment-intolerant species ... so [there was] a lot of effect on shellfish and things like that disappearing from the harbour. You started off with quite a range of community types and physical and biological nature, so habitat and

communities were quite diverse. By the time we got to 2000 or 2001, the results started to show that, in addition to losing the more sediment-sensitive species, we were also finding that the habitats were increasingly showing community structure that was similar. In other words the diversity was being squashed and the habitat was shifting more and more so that they were supporting communities that were similar as opposed to being quite diverse.

(respondent 54)

Once the council had identified the problem, it set about determining what the cause was and what the solutions might be:

The state of the environment monitoring identified a trend, an ecological decline or increasing, in this case, stress on the system. We were able to then say 'alright what's causing it and what should we do about it?' We carried out a desktop evaluation using all of the work that had been done in places like Okura, looking at tentative levels of sediment, habitat preferences and so on. Compiling all of this together pretty much led to the conclusion that sediment was causing the problem, that the harbour was sediment-stressed.

We then followed it up with a desktop evaluation and some studies and things to find out where the sediment might be coming from and then using that as a package of information, because we didn't want to go out and start telling people you've got problems, that's it. We wanted to say 'there's a problem in this environment, this is what's causing it, we think this is where it's coming from and this is what we think we need to do to solve it. (respondent 54)

Subsequent research indicated that exotic pine forest activity may have contributed over half (fifty to fifty-four per cent) of the recent mud flat sediment, with pastoral activities contributing an additional thirty-two to forty-four per cent. (Gibbs, 2004, iv). As previously noted, there are no regional rules which apply to stock grazing and access to waterways within the Auckland region including the Mahurangi Harbour, and plantation forestry harvesting is a permitted activity if it meets the environmental standards in the plan. The regional plan was therefore not addressing the causes of the harbour sedimentation

A response was developed and in 2004 the Mahurangi Harbour Action Plan was launched. The focus of the Action Plan was to reduce sedimentation occurring within the Mahurangi catchment, primarily through fencing off and replanting riparian areas rather than through regulatory means. This was the first time the ARC had launched a rural catchment-based initiative. A key plank

of the plan is a Grants Fund launched in April 2005 which provides between fifty and seventy-five per cent of the cost of projects to reduce sedimentation, up to a maximum of \$20,000.

The initiative has had some notable success with forty-one kilometres of riparian margins being fenced, more than 50,000 native trees and shrubs planted, and 300 hectares of erosion-prone land protected from stock (ARC & Rodney District Council, 2006). However, the plan appears more focused on reducing sedimentation generated by farming rather than by forestry activities which are the major contributor.

The Mahurangi Action Plan very much focuses on achieving results on the ground in terms of fencing and replanting of riparian areas, rather than on taking a strategic view of how future development of the catchment could be managed to achieve better outcomes. This has resulted in dissatisfaction among some community members who see the opportunity of also addressing future development of the area:

There was some general disquiet but the news, that the likes of cockle populations in areas that were being studied or that have been studied were under such stress that they were no longer reproducing, was a complete shock. So yes, we knew the harbour was changing but hadn't realised that it was happening at such a rate that it was seriously impacting on the communities until the ARC work.

In some ways the management agencies are wildly exceeding expectations. They identified the problem. In the bad old days we would somehow have become aware of the problem and then been charged with the task of proving that there was a problem and then convincing the authorities that the problem was real and that they should be concerned. So we've had that handed to us on a platter.

The Mahurangi Action Plan is a plan to a certain extent in name only. It's a budget and it's a determination to address the riparian fencing post haste, and that's all good, but Friends of the Mahurangi feel that it has to look at a longer term strategy and look at a lot of other areas beyond riparian fencing including the contribution that subdivision plays. A lot of the subdivisional focus in the Mahurangi relates to the two hectare bush block and there's almost certainly room to have that bush lot provision, that rule, result in better outcomes in respect to the sedimentation issues. (respondent 29)

But this has not found favour with the Rodney District Council which is keen to see resources devoted to action on the ground rather than localised long-term strategic planning:

That's something that the Friends of the Mahurangi have raised on a regular basis and there's certainly a view that there's [a need for a] wider vision. On resource management grounds it's very desirable. But I have to tell you I went to public meetings arranged by Friends of the Mahurangi and one message I got was 'get on with it. We want plants in the ground and fences and these things done'. At the moment there's a lot of focus of money resources going to actually doing. I think it's very desirable to do a vision for the Mahurangi but I don't want it to be at the cost of on-the-ground work at the moment.

(respondent 30)

Okura Catchment Study

The Okura catchment drains into the Okura estuary which is part of the Long Bay/Okura marine reserve. In 1997 the Environment Court determined that the catchment should not be included in the metropolitan urban limits. This was on the basis that:

Urbanisation of the subject land in the catchment would necessarily result in a significant deterioration of the quality of the waters, to the extent that their life-supporting capacity would be imperilled. Although development controls, settling ponds and other measures would mitigate adverse effects of urbanisation on the waters of the estuary, they would not avoid them. (North

Shore City Council v Auckland Regional Council [1997] NZRMA 59, 92)

The ARC and territorial authorities (North Shore City Council and Rodney District Council) were then faced with determining what level of non-urban development would be appropriate for the catchment. In 1999 they embarked on the Okura Catchment Study which resulted in the development of a sophisticated model to determine the appropriate level of urban development within the catchment.

The modeling was undertaken by NIWA and it assessed the risks to ecology of the estuary as a result of earthworks associated with varying degrees of intensification. The study initially assessed three scenarios: scenario one, which was rural intensification allowed under existing development rights; scenario two, which was rural intensification with a two hectare minimum lot size; and

scenario three, which was rural intensification with a one hectare minimum lot size. Further variants on these scenarios were also assessed.



Figure 8.8: The Okura Estuary

The question then came up about, well, what form of rural development was appropriate in the Okura catchment part of the North Shore City? But you couldn't look at that issue without considering the catchment as a whole. So Rodney was approached to become part of a study looking at rural development within the Okura catchment. Certainly, at an officer level, they became a close part of that study and all of the work that was done by NIWA on behalf of the three councils.

It was jointly funded, although the contribution from North Shore and the ARC tended to be a bit bigger. The three councils reached a joint position of which scenario they felt was most appropriate for the catchment. The NIWA model looked at a number of scenarios ... It was determined that, if you actually gave people a subdivision right, then you're in a much better position to secure the longer-term remediation that you want for the catchment. (respondent 43)

But, in a cross-council initiative, it was not always easy to keep the politicians involved and supportive:

It was left up to each of the local authorities to keep their respective political masters informed and how well that played out at the Rodney is probably a question you're best to ask [them] but certainly my impression at times was

that this was a study process that went over several years and I think there was at least one election in there. When you get a change of politicians, it's quite a hard task to actually bring in new politicians up to speed on an issue which some of them may well regard to be on the periphery of development. So I think there is an issue about keeping alive interest. What I observed was that some members were very interested in it and some hadn't really turned their minds to it.

North Shore, I think, maintained a closer interest in it because the area of the North Shore is relatively small, it's right on their door step, it did very much relate. Development in Okura actually had an adverse effect on ... the adjoining coastline at Long Bay, which is one of North Shore's premier sites. They had a much more immediate need to show an interest. Their electorate is right on its doorstep and that makes a huge difference and, in that sense, they're probably quite similar to the Hauraki Gulf islands where a marine environment is right on its door step. Therefore the political interest increases significantly.'(respondent 43)

The challenge was then to get provisions that aligned, between the two territorial authorities, across the whole catchment:

We went through the respective statutory processes for the North Shore plan and the Rodney plan. The North Shore plan went through an extensive Environment Court hearing and over the course of a year there were three different hearings. The council's preferred position about doing a subdivision that's desirable to protect the marine environment wasn't really challenged in the Environment Court process ... So what we've got is a catchment which is largely, but not all of it, in alignment. There are some parts of the Rodney catchment which still lie outside of what we call the Okura policy. I think it was a success story. (respondent 43)

Muddy Feet Phase II: The Firth of Thames Ramsar Site Project

The Muddy Feet Phase II project was initiated by EW in July 2005 to identify the relative environmental risks to the Ramsar site in the southern part of the Firth of Thames. The project was jointly funded by two regional councils, three territorial authorities and DoC, and co-ordinated by a multi-agency steering committee (EW, 2006b).

The project developed and applied a relative-risk model to identify, describe and assess potential risks to the site. The model incorporated information on

the activities which produce risks, the location of ecological values, and the impact of the activities on values which management is seeking to protect (Elmetri & Felsing, 2006).

Representatives from participating agencies (which included the two regional councils, four district councils with catchments draining into the southern Firth of Thames, DoC and MFish) took part in workshops where the ecological values of the site, and threats to those values, were identified. A workshop was also held for members of the community to identify their values and concerns associated with the site (EW, 2006b).

Initial results from the project indicate that agricultural land use contributes by far the largest risk to the site. Sediment and nutrients from land run-off were found to be the most important stressors of the site (EW, 2006b).

The project is moving into the next phase where priority actions will be identified to reduce the key risks. Recommended actions will be provided to relevant agencies for inclusion in their work plans (EW, 2006b).

It has taken longer as we are getting everyone to buy in as we go along, but hopefully we will get a lot of political support. So hopefully when the recommendations come out as to what EW should do and others, it will be easier to achieve. It might take longer, [be] more involved, take more money and more research to work out what we need to know. But it is hoped that it will be quite easy to get community buy-in when we implement. (respondent 10)

Management challenges

The interviewees identified a range of challenges involved in effectively managing sedimentation. These include difficulties in maintaining a catchment-level focus; lack of information, on the carrying capacity of the marine environment and the links between sedimentation and fisheries health, to drive catchment management; the shortcomings of the plan-making process; and insufficient tools to stem progressive degradation of marine areas.

Maintaining a catchment-level focus

It can be hard to persuade decision makers at a local level to focus on the big catchment-level picture:

The biggest challenge at a regional level is actually convincing any decision maker, whether it's the Environment Court or a local authority. It's a lot easier to deal with it when you're talking about cumulative effects than when you're talking about individual schemes. That's the level that we're able to most effectively articulate a perspective. In those contexts, what happens is that the developer keeps trying to focus the decision maker's attention on the property and we're trying to focus on the bigger picture. Whether you're successful or not depends on where the decision maker is focusing their attention and what their background is.

It becomes quite challenging when you get down to the level of a development on 10 or 20 acres, like we're facing with the Mahurangi. We've got an appeal on at the moment ... and the question there is, is it appropriate to have further urban development on land that's [draining into] the Mahurangi Harbour which is in a stress situation. It's quite hard to make those science-based arguments on a small scale. You can make those science-based arguments on a catchment basis. (respondent 43)

Identifying the carrying capacity of the marine area

The carrying capacity of the marine area to absorb sediment without an adverse effect on its marine ecology is often unknown. It is therefore difficult to identify when the threshold may be crossed by an individual resource consent application. This would require much more sophisticated catchment modelling than has been undertaken to date, other than in a few areas such as for the Okura and Whitford catchments:

That's the game we're into, minimisation. Its really difficult to say, from a statutory sense, 'no' to development ... Someone comes in with an individual consent and we've got no mechanism to say, well potentially they're going to discharge X amount of sediment, how does it fit into the overall sediment yield of the catchment. We just don't have it. We don't have the science behind it. That's something that we've sort of talked about in terms of catchment sediment approach. Effectively you'd look at the catchment as a whole and in terms of what would be hypothetically the sediment yield that you could put into it before you had any adverse effects. So therefore development A could only discharge X amount of sediment, development B or C and if D came along [and exceeded] the catchment quota then you couldn't do that. What we're trying to do with our chemical treatment is use a policy where, if the

cumulative effects are pretty large, [we would have an] internal policy saying each site will have to be chemically treated. (respondent 51)

Linking sedimentation and health of fish stocks

The Ministry of Fisheries, which manages fish stocks, does not actively become involved in land-based issues such as sedimentation which may impact on fish stocks, particularly snapper, flounder and shell fish:

Traditionally [sedimentation] has not been recognised by the Ministry as an area which it really has resources to get greatly into, nor necessarily one that the Ministry has a resource role [in]. Look at the Act and in particular environmental principles. They're not there to act as an advocate for fishing or for management of fisheries in terms of external impacts on fisheries. So unlike DoC, for example, which has an advocacy role to argue for any impact on a species, we do not have a particular role in terms of dealing with third-party aspects on fishing. (respondent 36)

Achieving significant change through the regional plan-making process

It's hard to deal with contentious issues, and make significant changes, through the regional plan-making process:

I also think the process of plan-making is a political process and you can't really do something that turns on its head a set of presumed or real property rights without considerable consultation. You can't say you are operating under a set of presumed property rights, and because you are having this effect on the public good, we are going to regulate immediately. The consultation we did on the plan was quite general, as it is a region-wide document, and doesn't get to the level you need if you are going to change presumed property rights ... particularly in a spatial way. It allows you to deal with big cross-regional issues where there's no single defined community of interest, e.g. spray drift and the management of significant regional infrastructure. Once you are dealing with a bunch of people in a specific catchment, and fundamentally affecting their way of life, you need to adopt a different process and that's where the sub-regional process kicks in. But once you get into proper consultation it's a lot of time and effort and money. (respondent 12)

Slow and resource-intensive plan-making process

Regional plans become out-of-date before they become operative, and energy is focused on getting them operative rather than addressing new challenges as they arise:

The air, land and water plan ... it's been quite a number of years in the making and it's certainly leaps and bounds ahead of where we were. I don't want to be too critical about it, because I know there's a lot of problems with it, but ... I guess over the time it's developed, some of our thinking has moved on. We've started to recognise now, once we get more into the implementation phase, that some of our rules are probably not as good as what they could have been.

If you think about it, our air, land and water plan is already out of date. It's taken us so long to get it to where it is, and things have moved on from there. My personal view is that we should be looking at our second generation plan, and working backwards. But the problem is we're locked into a statutory process now, and we're committed to see that through, warts and all. For all its problems, we just need to get that through and then move on from there.

... the problem is when you are head down, bum up and trying just to get the thing through ... It's incredibly difficult. There's some quite protracted battles with industry representatives over certain rules, and those are things that have to be worked through. So it's not easy, but for us it's still a very powerful tool in terms of getting things done on the ground. It's just recognising you can't implement change really quick. It does actually take some years to start to see that effect on the ground.' (respondent 44)

Stopping degradation rather than only slowing it down

Although councils have, in some cases, succeeded in slowing down the rate of sedimentation from major sources, they have not succeeded in reducing it to a level which will halt the progressive degradation of the marine environment:

I guess largely the best which has been achieved is the slowing up of adverse impact certainly in water quality. [With] most controls all you're doing is you're catching a higher and higher proportion of sediment discharging. But development is carrying on so what you are [only] doing is slowing it up. So you wonder how much of a success it is. [It's] certainly a success – [we're]

getting better at it. It's twenty years down the track rather than ten till we get to a certain point. [But] how much of an achievement is that? (respondent 3)

When we look at the objectives for Clean Streams, I was very keen for us to have a water-quality objective that says 'we'll make streams swimmable'. The scientists who were providing advice said 'we can't, we have no certainty that this work will achieve that, so we shouldn't have it as an objective'. But what they did say and the reason for that was that farming is intensifying all the time so you could stop the increase and it wouldn't get any better. (respondent 11)

With forestry companies one bone of contention is the monitoring programmes they are required to put into place in the harbours to prove they're not having adverse effects. I think they show they are having an effect, but it is probably much less than in the past, so we are being effective at reducing their impact. (respondent 12)

Prospects for integration

Councils are grappling with addressing the many sources of sedimentation impacting on the marine environment, including earthworks, forestry and farming activities. All the regional councils studied require large earthworks to obtain resource consent which enables the councils to exert oversight over sedimentation-control measures. However, even the rigorous use of sedimentation ponds will only reduce sediment release by up to seventy-five per cent. Therefore, such earthworks still potentially contribute to marine degradation.

Control over sedimentation caused by forestry and farming activities is less rigorous. Variable and relatively weak controls have been placed over forestry harvesting, and stock access to water bodies is largely unregulated. The tools under the RMA have proved more effective at addressing significant urban activities (earthworks and rezoning) than multiple non-point source impacts from rural activities. Councils have been particularly reluctant to regulate farming activities and are experimenting with various incentive and educational approaches. The challenge is how to upscale localised farm-based interventions to achieve significant catchment-wide improvements.

Sediment-control measures have been largely based on a broad acknowledgement that sediment has a negative impact on marine communities

and should be mitigated rather than on detailed knowledge of the sediment-carrying capacity of specific marine areas sensitive to sediment, such as estuaries and harbours.

Both the ARC and EW have been experimenting with pilot projects which more explicitly link the health of specific marine environments to catchment-management efforts. These projects can be expensive and time consuming and seem unlikely to be broadly applied in the near term. However, they provide useful insights, into the relationship between land use and marine health and the relative importance of contributing activities, which should be able to be applied elsewhere.

The case study has illustrated the importance of long-term monitoring of the health of the marine environment which can serve as an early warning system for serious degradation. Both the ARC and EW are implementing monitoring programmes to detect changes to benthic habitats, but the coverage of these programmes is still limited.

The ARC has recently initiated a review of sediment management within the region which may result in future plan changes.



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Case Study 4: Marine Biosecurity in the Hauraki Gulf and Kaipara Harbour

The Missing Link?

[It's about] being able to find a needle in a haystack. (respondent 32)

Regional councils have had no mandate, they've had no expertise, they've no experience with the marine issues apart from the planning aspects. (respondent 38)

The policy framework

Marine biosecurity is governed by the Biosecurity Act 1993. This provides a framework for the prevention of organisms entering the country and for their eradication or control once they are here. The Act provides for the preparation of both national and regional pest management strategies. The management objectives for the Hauraki Gulf, set out in sections 7 and 8 of the Hauraki Gulf Marine Park Act, must be paid particular regard in the development of pest-management strategies under the Biosecurity Act (section 13 Hauraki Gulf Marine Park Act).

Although the RMA does not specifically address biosecurity issues, regional councils do have a management function under that Act associated with maintaining indigenous biological diversity (section 30(1)(ga)), which is a key reason for undertaking marine biosecurity activities.

Biosecurity management efforts have traditionally focused on terrestrial organisms, and a pro-active biosecurity programme for the marine environment has only recently been established. Biosecurity New Zealand was established in November 2004, as a division of the Ministry of Agriculture and Forestry, to undertake a lead role in carrying out the government's biosecurity functions. Regional councils undertake terrestrial biosecurity functions for the management or eradication of pests once they have become established in New Zealand. However, apart from a few exceptions, they have not become involved in managing marine incursions.

The management problem

New marine organisms are constantly entering New Zealand, hitching a ride on the hulls of vessels or in their ballast water. Once established in New Zealand, they can modify marine ecosystems and out-compete indigenous species.

Problem species in the Hauraki Gulf include the Pacific oyster which displaces the native rock oyster, the Asian date mussel which smothers tidal and subtidal flats, Japanese kelp (*Undaria*) which displaces native seaweed communities, and the sea squirt *Styela*. The Kaipara Harbour is less susceptible to direct invasions from vessels as it does not have an operational port.

The management approach

Management efforts are being focused in three areas: controlling the main vectors through which marine organisms arrive in New Zealand, monitoring the marine environment so incursions can be quickly identified, and then responding to incursions to eradicate them or manage their spread.

Preventing marine incursions

The main vectors for marine incursions are through the exchange of ballast waters within New Zealand's coastal waters and through the transport of organisms attached to the external fittings and hulls of vessels. For ballast water, an Import Health Standard has been in place in New Zealand since 1997 which requires all ships to exchange their ballast water with mid-ocean water, prior to entering New Zealand. This is within the framework of the International Convention for the Control and Management of Ships' Ballast Water and Sediments, which provides for internationally consistent controls. The convention was adopted in 2004 but is yet to come into force.

Ballast water was the first cab off the rank and it came to the fore as the result of an aquatic pest invasion in the Great Lakes in North America around the mid 1980s. That led the Canadians to raise the issue of transfer in invasive species in ballast water with the UN's International Maritime Organisation, which kicked off international action on the issue. New Zealand has been an active supporter of international action on the issue. We were doing research at least five years ago on the risks associated with ballast water, so we have a reasonable amount of information to contribute to the debate on the issue. In addition, we were one of the first countries to impose controls on the discharge of ballast water in its ports. (respondent 32)

However, it is difficult to ensure that no ballast water is discharged within New Zealand's waters and the country is largely reliant on global-level developments in technology and ship-management practices:

The issue has to be dealt with at the global level to effect the changes to ship-board systems, ship design, as well as the attitudes and behaviours of crew and officers needed to resolve the issue. [The international ballast water management convention is] already having some impact (without yet being in force) in the sense that the major industrial organisations that service the shipping industry are working away on treatments for ballast water and coming up with systems that can be put into new vessels or retro-fitted into existing vessels. The mandatory use of such systems required by the Convention will eventually reduce or remove the biosecurity risk of ballast water. These systems are likely to use a combination of filtration systems (to remove sediment) and something like chemicals or ultra violet rays to kill the organisms in the ballast water. Systems using chemicals will undergo a stringent approval process to ensure that their discharges are not an environmental hazard.

I don't have information on the level of compliance and you'd have to be suspicious if it's not part of the standard behaviour of crew and officers to actually carry out ballast water management. (respondent 32)

In contrast, hull fouling is not currently controlled internationally or nationally as there has not been a crisis similar to that which occurred with the Great Lakes, to put it on the international agenda:

Action to address hull fouling has lagged behind ballast water. There is a growing level of interest, and New Zealand will be working closely with like-minded countries - Australia, Canada and the US, to get this particular issue onto the international agenda. It's a long process to get the level of understanding around an issue where nations are prepared to club together and act.

Hull fouling is not currently controlled but we'd like it to be. An import control under the Biosecurity Act would need to be put in place, and to do that we would have to have good grounds, supported by scientifically-based information. In order to obtain the latter, Biosecurity NZ has embarked on a large research project to survey all types of vessels coming into New Zealand, be they international or domestically-owned, commercial or private, yachts, barges or oil platforms. We expect the survey part of this research to be finished in 2007. The research will then be analysed, before the results are put into a formal biosecurity risk assessment that would provide the basis for imposing controls.

In addressing biofouling as a biosecurity risk, a number of issues will be encountered. The first is about raising awareness of the risk, getting wider understanding of it, getting vessel owners to buy into the need for action on their part to clean their vessels and carry out maintenance. We will need measures to manage larger vessels that arrive in NZ with fouling that poses a risk, as we don't have the facilities in New Zealand to send an oil tanker off for cleaning, for example. Biosecurity New Zealand's biofouling research is likely to highlight a need to focus on slow-moving vessels like barges, that spend a lot of time in an overseas port and come here heavily laden with an ecosystem. We would have to have practical means at our disposal to allow vessels of that type to be cleaned in a manner that does not release the fouling back into the marine environment. (respondent 32)

There are a range of potential management tools which could be implemented:

I think there has to be annual inspections of fishing and other at-risk sector boat hulls. The technologies are improving, they're getting now to where you can actually photograph under the water without having to dive. There has to be better quarantine around things like piling barges. They seem to be about the worst thing for spreading *Undaria* and *Styela*. So there's a number of things that can be done, like codes of practice for marina operators, for barge operators, for dry docks, in much the same way as anti-fouling is applied. (respondent 38)

Identifying marine incursions

A monitoring programme focused on harbours is being undertaken to identify marine incursions. However, it has proved difficult to locate organisms in time to take effective action:

Biosecurity New Zealand is monitoring marine pests in the Waitemata Harbour, and has carried out baseline surveys of both the Waitemata Harbour and Kaipara Harbour. There are inherent difficulties in finding pest organisms using traditional survey methods, given the patchy distribution of organisms in the marine environment. For example, Biosecurity New Zealand had done a number of surveillance surveys in the Waitemata Harbour without detecting *Styela* during the period when it is likely to have been present there. I think it was first found in the Viaduct Basin - an area that wasn't covered by surveys. So one of the major challenges facing biosecurity efforts in the

marine environment is about being able to “find a needle in a haystack”.
(respondent 32)

Responding to marine incursions

Eradicating marine organisms once they have arrived is extremely difficult as few effective control mechanisms have been developed:

... there aren't really any species-specific control mechanisms for marine pests. If there are, they're fairly expensive. Things like wrapping wharf piles in the Waitemata Harbour might be one way of dealing with *Styela* but that would be a very expensive option. More non-specific tools such as poisons might also have an impact on other parts of the ecosystem, which is to be avoided. Therefore, once we know a pest is here usually there isn't a lot we can about it.

Some of the tools we've got under the Biosecurity Act for managing pest incursions aren't effective in the marine environment. Take for instance the power to control movement into and out of an “infected” area. That was one of the mechanisms that were looked at for managing *Styela* in the Waitemata Harbour. On balance, there is such a large boat population in the Auckland area that it was never going to be feasible to actually manage and control the movement of boats around the Waitemata Harbour. Instead, Biosecurity New Zealand has invested heavily in a public awareness programme to encourage boat owners to take more responsibility for regular cleaning and maintenance on their vessels. If this message is taken on board, it will go some way towards slowing the spread of *Styela* and other pest organisms. (respondent 32)

But this low-level approach has been criticised at a regional level as not being appropriate for the level of threat:

Our Council is of the view that Biosecurity New Zealand needs to show a lot more leadership ... our regional council has not been entirely happy with the amount of resources that Biosecurity New Zealand has put into dealing with marine incursions. They tend to have a policy of ‘here it is, we'll publicise it but we won't do much else about it’. BNZ appeared to take quite a long time to make up its mind about whose responsibility it is and then normally the thing is found to be much more widespread than is possible to eradicate. So they tend to redouble their efforts to publicise, but without giving any clear

steer to stakeholders including regional government or marine users about whose responsibility it is.

Regarding Styela, BNZ did some good work talking to marine users, for instance yachties, boaties [and] fishing people. The ARC was involved with discussions around yachting people and the marina operators. That was fine and I think BNZ did a good job of getting that publicity and materials out. Unfortunately there's a lot more to managing incursions than just talking about them, putting out publicity.

When Undaria was first found in the Auckland Harbour, BNZ did a fairly limited delimitation survey, i.e. only in half the harbour. So the ARC was left with the task of either ignoring it, or trying to find out how much was there. So we conducted a delimitation survey of our own and unfortunately found some Undaria at Kawau. Because BNZ had made up its mind that it's not going to fund Undaria control anywhere ... [the] ARC did fund the removal of Undaria from Kawau. It was a very small job and we achieved that in February. (respondent 38)

Prospects for integration

It is currently unclear what role regional councils will ultimately play in marine biosecurity and this may help determine the extent to which marine biosecurity can be integrated into other marine management functions. Since being established in November 2004, Biosecurity New Zealand has endeavoured to more clearly identify central government and regional councils' roles in managing biosecurity issues. Biosecurity New Zealand is a Wellington-based organisation with no regional offices and it envisages a strong role for regional councils in detecting and managing local incursions, and in integrating biosecurity issues into council planning and resource consent functions:

If for example, there was a regional reserve or a site containing marine features of regional importance, there could well be measures that could be taken at the regional level to provide for biosecurity protection. This could take the form of surveillance (to detect the presence of invasive pests), or maybe there are controls that a regional council could put in place to manage the means by which marine pests would get to that particular site.

I think there is not only a case for an active role by regional councils in managing pests. They could also take into consideration marine biosecurity in their other planning and resource consents activities, in relation to activities in the coastal marine area or on land, close to the marine area. For example, if there are inadequate safeguards to retain and dispose of fouling that's cleaned off vessels during maintenance, then it will simply be washed back into the marine environment where it may survive. As we have vessels from overseas being cleaned in our boat yards, this is one way that marine pests can be introduced into our marine environment.

So there is a legitimate role for regional councils, when putting controls and consent conditions on boat-maintenance activities, to consider marine biosecurity as one of the possible adverse effects of that activity. Consideration is presently given to the run-off of chemicals or paint residues into adjacent seawater, but they don't look at the possible biological "contamination". An exception is the Auckland Regional Council, which has considered this matter in its consent guidelines for boat maintenance facilities. (respondent 32)

Regional councils have been reluctant to take on a role in marine biosecurity:

Regional councils are actively engaged in managing pest species on land, but they are quite restrained in their approach to managing marine pests. There are a number of reasons for this. They have issues with capability (they've not done it before, and don't have the knowledge and expertise). They also have an issue of mandate (around whether councillors or ratepayers want their council to be engaged in managing marine pests, or consider it's their role to do so). Last but not least, there is a funding issue (around who will pay for managing marine pests, especially as there isn't a constituency who could be rated for that purpose). (respondent 32)

Due to a perceived lack of appropriate powers:

Marine biosecurity is difficult because of ownership and I guess it's often impossible to fully control the vectors ... the law of the sea effectively means you can't regulate many of those vectors offshore, or it's difficult to regulate them. In any case it's fundamentally a BNZ issue not regional government's. If it's a terrestrial issue in our back yard we can regulate for it. It would be impossible for ARC to make an effective regional pest-management strategy for marine organisms because we just simply don't have the power to regulate

... the people, the shipping people, the yachting people, boating people. We simply don't have those powers, outside of the territorial limits.

Regional councils have absolutely no ability to prevent the passage of marine pest organisms into the country or into the region from outside the country or into the region from another region or out of the region into another region. They can attempt to manage marine pests in their own territory but they have no power at all to prevent the ingress or the egress of marine pests. Biosecurity NZ has all the powers in the world to do that. I'm not saying it wouldn't be possible to manage some of these issues, but there's only one organisation that has the power to do that and that's BNZ. (respondent 38)

And a lack of mandate and expertise:

Regional councils have had no mandate, they have no expertise, they've no experience with marine issues apart from a few planning aspects. The whole programme was managed by Ministry of Fisheries, albeit in a fairly low-level way but they were managed, and some reasonably well, by Ministry of Fisheries. BNZ made an approach to Government to have all biosecurity functions, powers and resources (including staff) transferred from DoC, MFish and MoH [Ministry of Health], and this was successful. Now these functions and staff have been transferred to BNZ so BNZ, although they are short-staffed, they're capable. Those people have the expertise and the experience and they have the mandate. Regional government does not. (respondent 38)

And a concern that councils will end up carrying the funding responsibility:

They've attempted to come up with a working model for shared responsibility for marine issues ... The primary thing wrong with it was [that] it looked like all the tough stuff was going to be done by the regions and the soft tasks, like liaison or provision of training resources, would be provided by BNZ. I think if you want a proper model it's got to be equally funded and managed.

One possible model that the BNZ has put up now is sharing funding for surveillance. I have concerns about that because I think it's a 'thin edge of the wedge' scenario. One of the proposals that BNZ put to regional government was 'we can't do surveillance but if we find you the money could you do it? Could you manage it on our behalf?' I'm very wary of that because you might find that regional contracts disappear and after five years or so the public have got very used to leaning on the regional council to do the work and

it's not being funded ... another case of regional government being handed central government responsibilities. (respondent 38)

One way of integrating marine biosecurity more effectively with other regional council functions is to incorporate guidance into the New Zealand Coastal Policy Statement. Biosecurity New Zealand is exploring this option. The Hauraki Gulf Forum could also potentially play a role in integrating biosecurity management, between regional councils and Biosecurity New Zealand, within the Gulf.



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Case Study 5: Hauraki Gulf Amateur Fishing

A sector under siege?

When I was probably in the order of 20 years old, the most memorable occasion was sailing at six or seven knots for an hour through a single school of kahawai on the other side of Moehau ... today occasionally you'll see a little school of kawahai apologising virtually that they're there. (respondent 48)

The Gulf is the most intensively fished recreational fishing region in New Zealand. (Hauraki Gulf Forum, 2005, 104)

... fishes don't scream of course. You pull them up on a string and they flap around. But if they yelled and moaned and screamed like a bird or a mammal would, there'd be hardly anyone going fishing for fun. (respondent 55)

The policy framework

Three pieces of legislation provide the regulatory and policy framework within which amateur fishing is managed. The predominant legislation is the Fisheries Act 1996 which enables the Minister of Fisheries to restrict recreational fishing activity through setting sustainability measures (section 11). These include limiting the number of fish which each fisher can catch (bag limits), imposing size limits for fish caught, restricting fishing methods and closing areas to fishing. There is no licensing of amateur fishers and anyone can legally fish for non-commercial purposes so long as he or she abides by the applicable regulations.

Management activities under the Fisheries Act also help determine how many fish are available to be caught by amateur fishers, that is, the size and composition of the fish stock. This is through the Minister setting the TAC for a fish stock within a quota management area. The Act requires the Minister to set a TAC that maintains the stock at or above a level that can produce the 'maximum sustainable yield' (MSY) (section 13). This is defined as:

the greatest yield that can be achieved over time while maintaining the stock's productive capacity, having regard to population dynamics of the stock and any environmental factors that influence the stock. (section 2).

Where a fish stock is below B_{MSY} the Minister is required to set a TAC which will enable the stock to increase to B_{MSY} . The time period within which the stock is able to rebuild to B_{MSY} is undefined, but should be a 'period appropriate

to the stock and its biological characteristics' having regard to relevant 'social, cultural and economic factors' (section 13).

Where fish stocks are shared between commercial, amateur and/or customary fishers, the Act is unclear as to what proportion of the TAC should be allocated to each sector. The Minister is required to set a total allowable commercial catch (TACC) which specifies how much of the stock can be harvested by quota holders (commercial fishers) (section 20). However there is no guidance as to what proportion of the TAC, the TACC should consist of, other than that the Minister must 'allow for' Māori customary non-commercial fishing interests and recreational interests before setting the TACC. Both voluntary and regulatory spatial restrictions have been placed on commercial fishing in some areas to reduce conflicts with amateur fishers. There is provision, under the Fisheries Act, for marine areas of traditional importance to iwi or hapū to be managed locally as a taiapure-local fisheries. Provision is also made, under the Kaimoana (Customary Fishing) Regulations 1998, for mataitai reserves which can exclude commercial fishing.

The Marine Reserves Act 1971 also impacts on amateur fishing, negatively in excluding areas from fishing activity, but also positively in terms of spill-over effects whereby fish stocks are increased in neighbouring areas. Under the Act, marine reserves can be established and amateur fishing can be excluded from them. There are currently five marine reserves in the Hauraki Gulf, covering 3.5 square kilometres, all of which exclude amateur fishing.

The Resource Management Act 1991 impacts on amateur fishing through the control regional councils can exert, through regional coastal plans and the resource consent process, over the location of facilities in the coastal marine area such as marinas, moorings, jetties and launching ramps. Pollution from boats can also be controlled under the Act. In addition, the RMA makes provision for the imposition of coastal occupation charges on activities which occupy part of the coastal marine area, such as marina berths and boat moorings (section 64A).

Under the RMA, councils can potentially control the impact of fishing activity on marine ecosystems and habitats, although they have been reluctant to do so. Under section 30(1)(iia) of the Act, a regional council has the function of controlling the use of land (defined in section 2 as including land covered by water) for the purpose of 'the maintenance and enhancement of ecosystems in water bodies and coastal water'. In the coastal marine area regional councils can also control 'activities in relation to the surface of the water' (section 30(1)(d)(vii)), although they must not do so 'to control the harvesting or enhancement of aquatic organisms for the purpose of conserving,

using, enhancing, or developing any fisheries resources controlled under the Fisheries Act 1996' (section 30(3)).

The management issue

The Hauraki Gulf has the largest number of amateur fishers in the country, with several hundred thousand people thought to fish in the Gulf each year. The main target species are snapper and kahawai (Hauraki Gulf Forum, 2005, 104). Amateur fishers and boaties probably comprise the largest user group of the Gulf's marine area outside the narrow land/sea interface. They have a strong interest in healthy fish stocks, as well as public access to the marine areas, so that they can pursue their activities successfully. They also, collectively, hold a large amount of information about the state of the marine area over time. They therefore potentially provide a strong constituency to support marine management efforts:

I have been on the water all my life and I'm very passionate about the environment in which we live and that's the marine environment. So we have probably raised more concerns on issues with the environment than any environment group has ever thought of because they're not out there all the time. We are out there all the time, we see what's happening. (respondent 31)

The interests of the amateur fishing sector are promoted by a number of organisations including the New Zealand Recreational Fishing Council and option4. Representatives from these organisations who were interviewed painted a picture of a sector under siege, caught at the intersection between commercially-orientated fisheries management on the one hand and conservation efforts which seek to exclude all fishing from areas of the Gulf on the other.

As soon as we turn our backs, somebody pops up another marine reserve, so we're in reaction to it. As soon as we turn our back, somebody wants to close this part of the coastline. As soon as we turn our backs, commercial fishermen want more, they want to increase the TACC in this area ... so we're reacting to crises all the time. (respondent 31)

Issues of particular concern to the sector are the shortage and small size of fish to be caught, the lack of a legally-enshrined right to fish, the lack of a defined right where fish stocks are shared with commercial fishers, exclusion from parts of the

coast due to the establishment of marine reserves and the general degradation of the marine environment by land-sourced sediment and pollutants.

Instead of fulfilling a positive stewardship role in the Gulf, the sector has become defensive and combative. In addition, there appears to be a rising new generation of boat owners who are both more individualistic and have greater disposable wealth which can be mobilized to support their interests:

We've got a new generation of people coming in, getting involved. You've only got to have a look at these big gin palaces, the game fishing ones - million dollars game fishing boat and the big reels ... This is younger people in their thirties and forties. They can afford it. They're making big money. But they don't like being stuffed about and those are the people who can just reach in their pocket. You and I, it was five dollar donation, for these people it's \$5,000 and they don't even bat an eyelid. (respondent 31)

Unresolved issues in one area, such as fisheries management, have spilled over into other areas such as the establishment of marine reserves, making it difficult to move forward on marine management issues.

III-defined right to fish

Some people assume it goes right back to the Magna Carta, but there has been a long-held public perception that, unlike England where some fishing was often owned by local landlords, there is intrinsic value in the ability of any New Zealander to go and toss a line in the sea and catch fish and that they don't need a licence or permission to do so. (respondent 15)

The amateur 'right' to fish has never been defined in legislation. The High Court has recently confirmed that there is, however, a long-standing common-law right (which is limited by legislation) to fish for one's needs. This can be compared to commercial fishers who have better-defined rights in legislation to harvest shared fish stocks. The ownership of an individual transferable quota provides a perpetual right to catch a fixed share of the TACC of a fish species within a defined quota management area.

Customary fishing activity has legislative recognition under the Treaty of Waitangi (Fisheries Claims) Settlement Act 1992, the legislation being passed to settle Māori claims over fisheries. The Act provides that the Minister of Fisheries shall recommend the making of regulations 'to recognise and provide

for customary food gathering by Māori' (section 10(c)). Although there is no legal restriction to prevent the Minister reducing the customary catch, it has generally not been the Ministry's policy to do so. More recently, however, there have been some reductions in the allowance made for customary fishing in the kahawai and snapper fisheries.

The conflict between sectors is particularly heightened in the Hauraki Gulf, because of the large number of amateur fishers, and also because snapper in the Gulf is both the preferred target for amateur fishers and the basis of one of the most important commercial inshore fisheries (Hauraki Gulf Forum 2005, 104).

The need to provide a more settled basis for amateur fishing, in the context of the adoption of a quota management system for commercial fisheries, has long been recognised. In a national policy for recreational fishing published by the government in 1989, the Minister of Fisheries stated that non-commercial fishing would get preference over commercial fishing where the stock was not sufficient to support both. However this policy intent has never been translated into law (Hersoug 2002, 89). As a result, there is no defined process to increase or adjust the recreational share of the catch and weak protection for it. Not surprisingly, recreational fishers feel pressured and threatened:

Because the rights of commercial and customary are enshrined in law now, if there's any blowouts, if there's any over-fishing, if there's any illegal theft, if there's any blow out at all of the TAC, the total allowable catch, it is removed off the public share because it's the only share that the Ministry can remove it off without paying compensation. Unless they can demonstrate that it's a sustainability issue, they can't take it off commercial and they definitely can't take it off customary. So we are continuing, particularly in the last five years and some, to see our fishing rights and opportunities eroded because of other impacts in the fishery, predetermined rights and the illegal catch.

(respondent 31)

Conversely, the lack of definition of the amateur share, and the application of management tools to ensure that the amateur sector stays within that allocation, also provides uncertainty for the commercial fishing sector:

The [amateur sector] is only managed by the daily bag limit. That's not an effective quantitative control that ensures that they stay within their overall qualitative allocation. Until you can put in place measures that ensure that the recreational sector stays within the allocation of the fishery, there's no incentive for them to participate in robust management. (respondent 57)

The fears of the amateur sector, that their interests are being sacrificed in favour of commercial fishing, have only been heightened by recent speeches by the new Minister of Fisheries, emphasising the economic development aspect of fisheries:

MFish has much more than a regulatory role in protecting the fishery resource. It also has an economic development role. Its job is to help maximise the value of the resource as well as to protect it. (Anderton 2006)

During 2000, MFish produced a document titled *Soundings* which identified three possible options for the management of amateur fishing: the status quo, where the Minister used his or her discretion to make allowance for the recreational sector; a proportional share based on the allocation of a specific share of the TAC; and recreational management where recognised recreational management groups would have a greater role in day-to-day management. Over 62,000 submissions were lodged responding to the proposals (Hersoug, 2002, 92).

This effort to better define recreational fishing rights also resulted in the establishment of the recreational fishing lobby organisation known as 'option4'. This group, as well as strongly opposing the government's proposals and seeking to preserve recreational fishing rights, has strongly opposed marine reserve proposals such as that for Great Barrier Island.

In November 2006, the Ministry produced a successor to the *Soundings* document titled '*Shared Fisheries: Proposals for managing New Zealand's shared fisheries: A public discussion paper*' (MFish, 2006b). This provides more comprehensive policy proposals to deal with shared fisheries. These include a guaranteed minimum tonnage for the amateur sector; the re-setting of allocations through independent assessment, valuation studies or negotiation; and adjustments between sectors to be made proportionally or based on an assessment of the marginal value of fish to each sector.

In an attempt to obtain a legal definition of the amateur fishing right, the New Zealand Recreational Fishing Council and the New Zealand Big Game Fishing Council lodged judicial review proceedings against the Minister of Fisheries' decisions on the setting of the TAC and TACC for the Kahawai 1 stock (also present in the Hauraki Gulf):

One of the legal points we are challenging in the Kahawai legal challenge is that the Minister has chosen to reduce the sectors in a proportional manner. He doesn't have the statutory right to do that. He 'shall allow for our non-commercial fishing interests' as per section 21. So we want to know what is

allowed for and what does 'interests' mean. What are our interests? That's been interpreted by the Ministry to mean some wide estimate of what we've caught recently. We're not catching many kahawai because there aren't many kahawai to catch ... My interest is abundance and a good chance of a reasonable catch. My interests are not what you and your mates might have caught and my interests are certainly not managing fisheries at below B_{MSY} (respondent 28)

The High Court delivered its decision in March 2007 and provided considerable clarity on the current position of amateur fishers under the Fisheries Act (The NZ Recreational Fishing Council, 2007). This decision is currently being appealed to the Court of Appeal. The High Court stated that, when setting the TACC, the Minister must first make allowance for non-commercial fishers. That means that recreational and customary fishing must be provided for, where they exist. However, the same does not apply for commercial interests and there the Minister could conceivably set the TACC at zero. Presumably this could occur where adequately providing for non-commercial fishers required the entire TAC. This means that, if the decision of the High Court is upheld, under current legislation amateur fishers probably have priority over commercial fishers, at least to the extent that their wellbeing is adequately provided for. This legal clarification should provide more certainty for amateur fishers in future MFish allocation decision making depending on the outcome of the pending appeal.

The Minister had based the allowance made for recreational fishers on past-catch history and had then reduced the commercial and recreational share in the same proportions (as a result of reducing the TAC). The High Court determined that what the Minister should have done was to weigh up the recreational fishers' right to provide for their social, economic and cultural wellbeing by fishing against the extent to which commercial fishing benefited the wider general public. This was to be a subjective evaluation which could not be only determined through means such as financial modelling (The NZ Recreational Fishing Council, 2007, paras 72-73).

The decision also helped clarify the significance of the Hauraki Gulf Marine Park Act in decisions to establish a TAC for a fish stocks within the Hauraki Gulf. The Court found that the Minister had erred in failing to have regard to the provisions of the Act when making his decision. In particular he failed to pay particular regard to the social, economic, recreational and cultural wellbeing of the people of the Hauraki Gulf and to maintain and enhance its physical resources in the form of the kahawai stock (para 81).

Perceived shortage of fish

Snapper is the species of fish of the most importance to both amateur and commercial fishing in the Gulf. They are dominant in northern inshore waters and may live up to more than sixty years of age. They feed on a wide range of sea life found on the sea floor, including molluscs, crustaceans, crabs and worms. The Hauraki Gulf is the largest and most important spawning ground for snapper on the northeast coast.

There has long been a snapper fishery in the Gulf. Commercial fishing effort increased significantly, however, when a new export market in Japan opened up for frozen snapper:

In the seventies, markets opened up in Japan for frozen snapper. What had happened is the Japanese had over exploited their own fisheries and collapsed their snapper fisheries ... Because it was frozen it was capable of being transferred in big quantities. International catching technology was scoured. 'Well how do you catch the most you can in the shortest amount of effort and get it to shore and get it frozen and get it to Japan and get some money?' So there was a great drive for this frozen Jap pack market in the 70s which saw just an explosion in fishing. The whole Danish seining method of catching which involves surrounding a school of fish with a net and ropes and bringing them in ... [and] towing a net between two boats. The size of boats went up. The Government encouraged this interest and there were some pretty dodgy suspensory loans ... but what it led to was just this explosion. Now as the seventies went on and ran into the eighties, it just became more and more extreme ... Building big boats and using them in the Hauraki Gulf, catching, dumping, ripping gear, wasting. An unbelievable amount of fish came out ...' (respondent 58)

Snapper landings in the Hauraki Gulf peaked at 8,600 tonnes in 1971 and then declined. They rose again between 1975 and 1979 to around 6,000 tonnes, due to the use of pair trawlers which caught larger quantities further reducing stocks. But this was shortlived as the population was severely reduced. Landings of snapper in the Gulf between 1999 and 2003 were just under 2,000 tonnes a year. (Hauraki Gulf Forum 2005, 106)

When snapper was brought into the quota management system in 1986, the Hauraki Gulf was included in a large quota management area extending from North Cape to Cape Runaway. Total commercial landings from this area totalled 5,876 tonnes during the 1985-86 fishing year just prior to the quota

management system being introduced. A TACC of 4,710 was initially set for snapper within this area. However, appeals by fishers, challenging the amount of quota they received, increased the total TACC to 6,010 tonnes in 1991. This was subsequently reduced, reaching 4,938 tonnes in 1996 (Hauraki Gulf Forum 2005, 106).

The introduction of the quota management system probably saved the snapper fishery from economic collapse. The biomass of the Hauraki Gulf and Bay of Plenty snapper stock is thought to have been slowly recovering since the late 1980s and may be at a level approximating its B_{MSY} .

Management of this highly-sought-after and contested fish stock has not been easy, and efforts during the 1990s to develop a common strategic plan between the various stakeholders ultimately failed. There was deep suspicion on both sides. When asked for their viewpoints on the plan process, an interviewee from the environmental sector expressed the view that the large commercial fishers were not interested in developing an inter-sectoral plan until the rights of amateur fisheries have been better defined:

I don't think we're really moving anywhere with snapper management ... Snapper 1 which is Hauraki Gulf is probably in a little bit better state than it was ten or twelve years ago, but nowhere near as good as it should be ... There is no plan for snapper. The snapper strategic plan process fell over because basically industrial fishers didn't want a plan. They wanted to get the recreational right and catch defined and quatarised before they would engage in the whole plan. So in that sense we're absolutely no further ahead with snapper than we were then.

There were concerns about the sustainability of Snapper 1. Doug Kidd, who was the Minister at that time [1992], said that he wanted to see a rebuild of the snapper fisheries and he directed all the parties to work together to develop a strategic plan to rebuild the snapper fisheries. A very lengthy process was engaged in over a space of about nine months, with endless day long meetings which got absolutely nowhere for nine months, because the industry basically weren't interested ... Every other party, including the small commercial fishers represented by the Federation of Commercial Fishermen, acknowledged that their time was being wasted. So the Ministry of Fisheries, environmental, recreational and Federation of Commercial Fishers, went away in secret and wrote a plan ourselves without the industrial fishers. (respondent 15)

An interviewee from a large commercial fishing company also emphasized the need for better definition of the recreational right, but also expressed the view that the recreational fishers did not want a snapper plan, as they gained greater traction from political lobbying:

The essence of [fisheries plans] is to endeavour to have stakeholders work together ... Having a process where stakeholders engage is more likely to be achieved by having ... some comparable sets of rights ... we've got the commercial sector robustly managed but you've got ineffectual control of the recreational sector where it's very light-handed management, if any management at all.

Why the recreational sector walked away from the process is because they don't get any political [mileage]. They've always considered that more can be gained by persuading the Minister to reduce the interests of commercial fishing progressively and therefore they agree to nothing that might bring about some form of permanency to the arrangement. (respondent 57)

In 1995, after the major stakeholders in the snapper fishery failed to reach agreement on management, the Minister announced a forty per cent reduction in the TACC for Snapper 1, from 4,928 to 3,000 tonnes. The commercial fishing industry challenged the decision in court and successfully obtained interim relief while the case was pending, allowing them to fish their former quota. The Minister set the same regulation the following season and once again the industry took him to court and sought interim relief. The core of the industry's argument was that, because there was no effective control on recreational fishers, the commercial sector was taking the whole brunt of the need to reduce the overall catch (Hersoug, 2002, 83-4). The Minister won the case in the High Court, but was overturned in the Court of Appeal on the basis that he had failed to consider the possibility of reducing catch through the Crown acquisition of quota. Shortly after the case was finalised, a new Minister of Fisheries reduced the TACC only slightly to 4,500 tonnes and it has remained at that level since.

More progress has, however, been made in reducing direct conflict between commercial and amateur fishers in the Gulf and increasing amateur fishing opportunities through spatially separating the two sectors. A closure was largely achieved through agreement between representatives of the amateur and commercial fishing sectors within the context of the Minister seeking substantial TACC reductions in snapper:

The introduction [in 1995] of the 'inner Hauraki season', the Hauraki Gulf closures from 1 October to 31 March has had a huge impact on improving the recreational fishing opportunity for local boaties fishing inside Tiri, Rangitoto Channel, Motuihi, Waiheke Passage, Tamaki Strait because that area is closed to commercial fishing for six months of the year. So you're not competing with bulk fishing methods of long liners and set nets and stuff like that. So the fishing opportunity has improved immensely to the level where the good guys, that know what they're doing when they're fishing, can go out there and catch their bag limits and be home by midday. The not so good guys can go out there and fish all day and come home with a feed.
(respondent 31)

The most recent stock assessment by MFish concluded that the stock located in the Hauraki Gulf and Bay of Plenty is currently below, but close to, its estimated B_{MSY} which is less than a quarter (around twenty-three per cent) of the estimated original size of the stock (MFish, 2006c, 666). This indicates that management of the stock has been reasonably successful in achieving the narrow goal of B_{MSY} .

However, it is the adoption of the MSY management goal by government which underlies some of the conflict between the amateur and commercial sectors, and between the amateur sector and MFish. Managing stocks to the level where they produce their MSY, in theory, lets fishers catch the greatest amount of fish year after year (all other things being equal). It results in fish stocks being reduced to a small proportion of their original size, and significantly reduces the average size of fish. The rationale behind MSY management is that it is desirable to remove large numbers of older fish which grow slowly if at all, thereby creating the opportunity for more young fish to thrive. And, as young fish can grow very rapidly, they quickly build the biomass thus increasing the amount available to be regularly harvested.

The MSY management goal ultimately benefits commercial fishers who are interested in maximising the total biomass of fish that can be caught. Amateur fishers, on the other hand, have an interest in having a good opportunity to catch fish and the ability to catch bigger fish. This requires fish stocks to be larger (above B_{MSY}) and for a greater number of larger fish to be left in the sea by commercial fishers. A depleted fish stock also disproportionately benefits commercial fishers who, with bulk fishing methods such as trawling are more likely to get a larger share of the resource than amateur fishers with a small number of lines or nets. An amateur fisher expresses this viewpoint:

Our mantra is 'more fish in the water' not more fish in our bag limit. What we want is a rapid rebuild of the fisheries ... What we need to do is move

the management of the fishery so that it is managed above B_{MSY} which puts more fish in the water, which improves the fishing opportunity for non-commercial fishers, and gives you a better-quality fish. What would you rather catch, three small fish like this where you get six small fillets like this or one big fish like that where you get two big fillets which is a substantial feed? So, yes there is a problem of the fishing opportunity for recreation fishers who fish primarily for food or fun. (respondent 28)

Establishment of marine reserves

The establishment of marine reserves in the Gulf can also be perceived as threatening amateur fishers' ill-defined rights to fish. This threat was first clearly perceived by a recreational sector interviewee during the establishment of the marine reserve at Te Matuku Bay off the southern end of Waiheke Island. The reserve was established in 2005 and was the first to be established in the Gulf for ten years. The proposal to establish a much larger marine reserve off the north-eastern coast of Great Barrier Island was to receive the backlash of both the perceived poor process at Te Matuku Bay, and the mobilisation of amateur fishers through option4 as a result of MFish's efforts to better define amateur fishing rights. The large size of the Great Barrier proposal of 50,100 hectares can be compared with the total area covered by marine reserves in the Gulf to date of only 3,538 hectares.

Te Matuku Bay marine reserve



Figure 10.1: Te Matuku Bay, Waiheke Island

Some amateur fishers felt that they had been shafted by the establishment of the Te Matuku Bay marine reserve. Their side of the story as told by an amateur fishing representative is as follows:

Forest and Bird were given a bit of land right up in the head of the bay to ... create this little bird sanctuary ... for wading birds and everything else like that. They wanted to put a little marine reserve around the foreshore. We looked at it. It even went out to discussion ... and I thought I'm going to lose one of my best flounder spots ... [but] yeah, okay, we'll support that because these wading birds come out. Okay we'll forgo our little recreational fishing spot, no problems at all.

Next thing we hear, they want to take it out to the bar ... So notification [was] coming up round the club, we talked about it at the club - you'd still be able to go in there and anchor and do a few things but you just wouldn't be able to fish in there.

Then when it was finally gazetted it came right across the Ponui channel and the big area outside including the Sisters. Now the Sisters is the yachties' area for catching kawhai. It's the one area where everybody catches kahawai. Kahawai are a pelagic species, they swim in and out, they don't honour marine reserves ... Yachties ... manage to snag one of these things and they take it ... ashore and they barbecue it or they smoke it or anything else and its all part of their activity of being a yachtie and that's lost now, that's gone.

The marine reserve should never have come out beyond there. It was deceitful the way it was done. Because there are two steps when a marine reserve is gazetted, everybody said, oh yeah won't bother looking at that because it's inside the bay and yeah we've talked about it, we know where we're at ... Suddenly ... it's way outside. This wasn't part of the rules. Too late. We could have fired up and I thought, no, there's bigger issues coming. Aotea [Great Barrier Island] had hit the horizon by that stage of the game and that was going to be a bigger issue. So you keep your limited resources and everything else.

So Forest and Bird got away with that one and they need to understand they did get away with it. But they'll never get away with another one because the next one they put up, we're just going to stop it, end of story. We've got Māori support now. We're just going to stop it. It doesn't matter what the reason,

doesn't matter even if it's a good idea, we're just going to stop it and say talk to us and because you're not talking to us, stopped. (respondent 31)

Great Barrier Island Marine Reserve Proposal

Opposition by amateur fishers, in a much more organised manner, was evident when the Great Barrier Island Marine Reserve Proposal was progressed in the early 2000s.

It started I think in the late eighties when a steering group made up of island residents, some iwi, scientists and DoC did a bit of a survey of coastal areas and found it was of high value. At about the same time, [as part of] the Auckland Conservation Management Strategy, there was a report done identifying a number of areas across the region as potential reserves. A community steering group got together and decided that there should be a marine reserve there and put their preferred boundaries. DoC prepared a draft application with slightly different boundaries. It was about to be notified in around 1994 and the plug was pulled because government priorities changed.

Then about 2001, when the biodiversity strategy came out, the government was giving conservation priority and marine [issues were] up there again. We were directed to look at it again so we did some more investigations, and discovered some other important data, and decided to go for a pretty big reserve. DoC genuinely thought that not many people would be affected because it's not a very populated area, has very exposed to weather conditions and it takes a long time to get there. (respondent 20)

The reserve proposal had been based predominantly on scientific information, as 'scientific study' was the purpose for marine reserves under the Marine Reserves Act 1971. At the outset it had not seriously taken into account the socio-economic context in which it would be considered:

Great Barrier Island was a unique and small community with people deeply suspicious of government, feeling under siege by a shrinking population and rapidly changing social and economic environment on the Island. Then we came up with a seriously big 'kick arse' marine reserve right at their back door.

While we spent a lot of time sitting down talking with them, they immediately launched into an adversarial mode. They felt powerless –the recreational fishers have for a long time felt powerless – they are a huge body of people but not well organised and there is little active management of their interests by government. Their perception was that nothing was being done to acknowledge that they have a legitimate interest which needs to be protected. What they were saying is that we don't like your approach. (respondent 14)

Initially the boundaries were drawn on a scientific basis as the only purpose in the Act is from a scientific point of view. The standard practice for the department up until the MPA strategy was to decide on an area of interest (based on science), then put out a public discussion document showing the area of interest and asking questions about its use, hoping that that would provide the information we need to make decisions on how people are affected, and then modify the application to mitigate some of that concern. However the questionnaire we used was not actually very useful in getting that information.

Later on, after the application was notified, we commissioned a socio-economic impact assessment report to help us provide advice to the Minister. But we are really reliant on what people tell us. We also did boat surveys over the busy summer period to gauge how many people were boating in the area.

It was pretty hard work managing the proposal ... The volume of response from the recreational fishing sector was huge. I was getting thousands of emails a day. There was quite a lot of abuse. It was only the second time that a big stakeholder group had really organised themselves to encourage public comment and make submissions. Option 4 had a link so people could just put their name to a form and email it off. We got 3,000 responses to the original proposal and about 4,000 to the actual application. (respondent 20)

Despite the level of opposition, the Minister of Conservation decided to approve the marine reserve and it is now awaiting a concurrence decision by the Minister of Fisheries:

At the end we supported the reserve on the basis of biological information and felt the wider public benefit outweighed the local ... We reached the conclusion that the impact of creating a reserve there was not so great that

it warranted not having it. But it was at a cost of alienating sections of the community. (respondent 14)

It may be difficult to move forward on the marine protection and other marine issues in the context of such a high level of mistrust by such a major user group of the Gulf:

So a marine reserve is imposed upon a community. The community basically says in defeat, 'if that's the way you want it, your call, catch me if you can'. That's one attitude - don't expect me to participate in the community. (respondent 28)

Prospects for integration

The alienation of amateur fishers and boaties has meant that the main users of the Gulf are no longer engaged in a positive stewardship or collaborative role, but are more focused on adopting a patch-protection strategy:

The boating community is no longer a champion for the Hauraki Gulf environment because the boating community has been battered too long ... Why the boating community now is starting to step back and not get involved is because we don't get acknowledgement but we continue to get hammered, we continue to get blamed. One of the biggest sacrifices is Island Care. That's when the boating community turns out one day, one weekend, every year. All the charter boats give their boats for nothing and ferries would ship people from Auckland free of charge down on to Rangi wharf and down to the other wharves. You had all these beach clean ups, all round the islands ... It used to be a very good project. It's failed now. They still do some limited areas where they pick some ... but they don't get the boating community on side any more.

The boating community is always being blamed for all the woes in the environment. Recreational fishers are the worst thing for the recreational fishery as far as the eco-ites are concerned ... But hold on ... you're alienating these people. You're not going to stop them going boating but you're alienating them. If you've got a problem in the recreational boating community, the best way to resolve that problem is to embrace the people who are causing the problem. (respondent 31)

This has also made it difficult to move forward on other marine issues such as coastal occupation charges:

In Auckland anyway we only got as far as saying, 'here are some ideas' during [the] consultation period mostly to yacht clubs and boat clubs and organisations like that, those who might be most affected. 'Here's some ideas and we're consulting on whether we should or should not have coastal occupation charging'. My understanding is that a lot of those who might have to pay that, they're now saying 'no we don't want that'. (respondent 54)

Local concerns about fisheries were also raised during consultation on the Auckland regional coastal plan. But with local government no longer having a role in the management of fisheries by virtue of section 30(3) of the RMA, and MFish largely engaging with sector group representatives at a national or broad regional level on fisheries management issues, there was no forum for these concerns to be discussed or resolved:

It was quite interesting in the first round of discussion [on the Auckland regional coastal plan] ... one of the very strong things that came through was [that] the management of fisheries was seen by the community as a very important issue. But regional councils were specifically excluded from having a role in the management of fisheries. So that was quite an interesting dilemma, that the people were saying 'fisheries are an important resource and of value to us' and 'regional councils you don't have a role'. (respondent 24)

The Hauraki Gulf Forum potentially provides a forum for addressing some of these concerns. And in fact, in the opinion of some, the Forum's origins stem from fisheries issues. But this potential has not yet been fulfilled. The Forum technical officers committee is dominated by RMA planners and this group has therefore generated a RMA-focused agenda for the Forum. Although the Forum has become involved in shellfish management through the commissioning of a report on shellfish depletion (Grant & Hay, 2003), it has not become involved in fisheries management more strategically.



eleven

Case Study 6: Kaipara Harbour Commercial Fishing

No room for local management

There's a long history of communities spending a lot of effort trying to get changes from fisheries and getting nowhere. (respondent 15)

The quota management system ...is an economist's nightmare imposed upon a complex ecosystem. (respondent 58)

The policy framework

Commercial fishing is managed under the Fisheries Act 1996 through the allocation of individual transferable quotas. Such quotas provide a right in perpetuity to harvest a share of the TACC for a stock in a quota management area (section 27). Quota management areas are generally very large, and theoretically broadly coincide with the spatial area in which a fish 'stock' is located. Once a quota management area is established, it can only be changed with the agreement of the owners of at least seventy-five per cent of the affected quota (section 25), or if the Minister is satisfied that it is necessary to ensure sustainability, amongst other things (section 25B).

The Minister of Fisheries establishes the TACC and may adjust it from time to time. The Minister may also establish sustainability measures to control the effects of fishing on a fish stock or on the aquatic environment more generally. These measures may include restrictions on fishing methods, the size of fish taken and where and when fishing may be undertaken (section 11). When adjusting the TACC, the Minister is required to allow for Māori customary and recreational interests and must consult with people and organisations representative of Māori, environmental, commercial and recreational interests.

There is no statutory provision for a formal submission or hearing process, or for an independent review of the merits of the Minister's decision in setting the TACC or establishing sustainability measures. In practice, when a decision is required, MFish issues a draft advice paper to the Minister, stakeholders are given the opportunity to make written submissions on the advice paper, and MFish then finalises the advice paper taking into account the submissions made. Finally, the Minister makes a decision based on that advice. Because the advice paper prepared by MFish is the prime technical source of information provided to the Minister, it can be difficult for him or her to reach a decision contrary to the advice provided by the Ministry.

There are dispute-resolution provisions within the Fisheries Act (Part VII) which apply to disputes about the effects of fishing on other people with a

current fishing interest. The Act enables the Minister to appoint a Fisheries Dispute Commissioner to resolve any dispute. In practice the provisions are rarely invoked.

The management issue

The main commercial fish species harvested within the Kaipara Harbour are rig (a small shark species endemic to New Zealand and the Kermadec and Norfolk Islands), flatfish (most commonly yellow-belly and sand flounder) and grey mullet. They are very different species, with rig capable of living more than twenty years and bearing a small number of live young, flounder only living for three to five years and having localized stocks, and grey mullet capable of living up to seventeen years. The fish are predominantly caught by setting nets from small launches or trailered boats.

Little scientific information is available about the size of these fish stocks and it is not known whether they are at a sustainable level. The Ministry of Fisheries undertakes a rough assessment of the sustainability of current fishing effort, primarily through analysing catch per unit effort data (CPUE). The assumption behind this measure is that, if fishers are able to catch at least the same amount of fish over a time period using the same effort, the stock size is likely to be stable. The problem with this measure is that it does not take into account changes in fishing methods or equipment. If fishing becomes more efficient, the stock could be in decline, but show a stable CPUE.

In the case of rig, although the CPUE for the whole quota management area appears stable, there have been shifts in the Kaipara Harbour but with no overall trend. Catch per unit effort on the harbour declined between the 1989-90 and 1991-92 fishing years, increased to the 1996-97 year and declined to the 2003-04 year. Most catches are made when pregnant females enter the harbour to pup (MFish, 2006c, 474). Rig bear a only small number of live young and therefore are slow to recover from overfishing. Targeting of pregnant fish is not currently controlled.

For flatfish, catch rates in the harbour have been decreasing, while at the same time fishing effort has been increasing. This suggests that the fish stocks in the harbour are in decline. Reports prepared by MFish indicate that the causes of this are more likely to be environmental factors (such as siltation) rather than fishing effort, although there is no firm evidence. The grey mullet fishery is also experiencing a decline in the CPUE in some sub-areas within the fishery including the Kaipara Harbour (MFish, 2006c, 156 & 223).

Much concern has been expressed by local fishers, tangata whenua and residents over perceived overharvesting causing localised depletion within the harbour. The lack of fish has, in turn, resulted in ongoing conflict between fishers themselves and between fishers and other harbour users.

When the three species were brought into the new fisheries quota management system, the Kaipara Harbour was included as part of a very large quota management area encompassing the entire top half of the North Island including the Hauraki Gulf. The commercial catch levels (TACC) for rig and flatfish were set at a very high level as a result of unfortunate circumstances. The TACC for rig was based on an over-estimation of historical catches and catches since have been below this. The TACC for flatfish was based on the year which had the highest catch levels on record and has never been caught. The TACC for grey mullet has also never been caught.

The years that were used as a basis for setting the TACC were very productive years and there was also a very productive fishery in the Firth of Thames. That fishery has essentially died, but that TACC was simply used elsewhere.

(respondent 26)

This combination of a large quota management area, high TACCs and a partially-mobile fishing fleet has proved particularly unfortunate for the harbour. It has meant that, for example, when fishing has declined in east coast grounds, fishers with access to quota can trail their boats across to the west coast and concentrate their fishing effort on the Kaipara Harbour. The same boats can move off elsewhere when fishing declines in the harbour, leaving local launch-based fishers, tangata whenua and residents facing a depleted fishery. This acts as a strong disincentive for any local efforts to nurture the health of the harbour and its fisheries, as abundant fish may serve to only attract more 'outsiders' to 'plunder' the resource.

There's a lot of estuaries and embayments in the Kaipara and people living in some of those areas were seeing localised depletion of mullet, in terms of what they could just go out and catch from in front of their place. That at Tinopai resulted in a situation where some shots were fired and there was quite a lot of ill feeling between commercial and the Māori community in those places. (respondent 15)

A lot of Māori were coming back from Auckland and other cities to live at the ancestral land – the marae. They were relying, because there was no

employment, on subsistence fishing and there were some commercial fishers coming in and fishing in front of their marae. (respondent 26)

Māori fishermen were excluded when the QMS [Quota management System] came in as they were part-time fishers. They had to sit on beaches and watch people from Auckland and elsewhere fish their fishing grounds. (respondent 15)

The locals predominantly used launches and are the only fishery in New Zealand that still remains a launch-based fishery. That means they can only fish in the Kaipara because they can't trailer these launches. [It's] a particular type of fishery, a very traditional one and the families have fished for three generations, hundreds of years there.

Because the Kaipara Harbour is accessible, it's sheltered, it's near markets, it was getting an increasing proportion of the TACC ... The Kaipara was one of the few new entry areas too, so it meant that [if] somebody wanted to get in there they just had to buy a very cheap aluminium dinghy, some nets and lease quota. Now a lot of the quota was being bought up by the larger companies and then during the season was simply leased out at very low rates so the companies could get it harvested. So rather than the larger companies or established fishers deciding that it was worth while fishing this quota, you had new entrants with the hope of getting into the industry, who were taking the quota and fishing very high to try and catch what was a dwindling resource.

Towards the latter part of the time we were even getting conflict between local commercial fishers as the pressure really came on, an attempted murder in one place and a number of court cases which tried to not hit the media. (respondent 26)

All the marae in the Kaipara were saying that they weren't getting the mullet coming up past their marae like they used to ... There was a commercial fisherman who ... was saying, because they're ring netting now at the mouth of the Kaipara, just inside the mouth, he's been forced to do it as well because everybody else is doing it ... and the mullet are just not getting up into the arms of the harbour at all. Those sorts of things are known to local fishermen and they could modify their behaviour and organise that amongst themselves if there was control over who was actually fishing in the harbour. But there's nothing to stop a Thames fisherman, if fishing is good in the Kaipara and if

they're having a bad time, well they just trailer their boats over to the Kaipara. When you've got that situation happening, it makes a mockery of any attempt by locals to organise things in a way that keeps everybody in the community happy.' (respondent 15)

There were also concerns about poor fishing practices, such as the catching of juvenile fish and wasting fish through leaving them in nets too long.

The management approach

In the late 1990s MFish was faced with a declining fishery and increasing conflict between fishers. It was under considerable pressure to act. Its initial approach was to bring the stakeholders together in order to negotiate a solution, although what form that solution might take was not clear at the outset.

It wasn't clear what the Ministry was trying to achieve there in terms of the outcome. Certainly it was under quite a bit of pressure because there was armed conflict occurring and the police and local authority politicians had made it clear that they saw the Ministry of Fisheries having some responsibility for trying to resolve that conflict, because they saw that Ministry policies, if not creating, were exacerbating that situation.

We had a real challenge ... I was trying to support a local process where we had all sector groups involved - commercial fishers, tangata whenua, recreational fishers and environmental groups. It started from the situation [where] there was difficult dynamics. It was real animosity. It took a number of meetings, but eventually they developed a relationship and a rapport which allowed them to move forward in order to spend almost three years developing an agreed strategy among them.

When we set up this process to develop a strategy for the Kaipara Harbour, I got the Mayor of Kaipara. We invited, through him, the Auckland Inshore Fishing Company to nominate its representatives and gave them the opportunity of nominating all the commercial representatives on it. They went through an election process and did that, and then strategically decided they didn't want to be a part of this process, because they're not going to run it or control it. So they backed out, and to their credit, the people who were elected said 'stuff you' and went and became part of that group as well.

We held a series of three meetings at Matakoho. I got the Mayor to chair it ... We gained agreement that we would form a group and develop a strategy for the whole of the harbour. I put a report down to the Minister or a draft letter which he sent back to everybody saying 'I'll endorse this process, I'm looking forward to the outcome, I want you to sort out the community process and all the rest of it'. So, armed with this letter from the Minister, I worked for two and half years to produce an agreed strategy, with lots of consultation.

We went through a process of doing a draft that we then circulated, and had a bunch of public meetings and special meetings with the industry, and then [prepared] a final draft of that plan after more consultation. Then [we] invited the Minister to come, which was Pete Hodgson, and delivered it to him with great ceremony and all the rest of it. (respondent 26)

The final report, *Fishing for the Future*, identified a range of options to improve management of the Kaipara fishery. A key proposal was to establish a separate management regime for the harbour and the options presented to achieve this were a licensing regime or the establishment of a separate quota management area. Other proposals included adopting codes of practice through a mix of regulation and voluntary measures to deal with issues such as minimum fish and mesh size, net length, soakage time, net marking and the stalling of nets. The report also proposed initiating a temporary closure of scallop beds which were severely depleted (Kaipara Harbour Sustainable Fisheries Management Study Group, 2003).

A representative of a large fishing company did not accept that there was a stock sustainability issue which needed to be addressed in the Kaipara Harbour, but saw the issue as being about conflict between locals and outsiders as to who should take the fish:

The assessment report doesn't support a notion that there's a stock sustainability issue. What it is, is groups of mobile fishermen who don't come from those communities being seen by those communities as coming and taking their fish locally ... it's an issue of how they, fishers in that community, are relating to each other. (respondent 57)

The Ministry of Fisheries review of the proposals developed by the Sustainable Fisheries Management Study Group was not favourable, with concern expressed about the potential to undermine the quota management system framework, the establishment of an unhelpful precedent for managing local issues and the lack of support for the proposals by some elements of the fishing industry.

Its recommendations placed emphasis on negotiating voluntary mechanisms rather than the use of regulation (Chatterton, undated):

From the Ministry standpoint the group therefore wasn't representative because it didn't have the Auckland Inshore Company on it or, more particularly, it wasn't the Auckland Inshore Company that was going to run that process and so that was the technical reason why the Ministry wasn't supportive ... The second reason was that the philosophy of the Ministry says that they are not sympathetic to local solutions, they really wanted to rely on the quota management system and that would mean a quota area-wide solution. (respondent 26)

The key proposal of creating a separate quota management area for the harbour was considered problematic by the Ministry. There was a considerable legal impediment, as such an area can only be altered under the Fisheries Act if it is supported by quota owners who hold seventy-five per cent or more of quota shares for the affected stocks, or if the Minister deems it necessary to ensure sustainability (sections 25A and B). There was no evidence of support from a sufficient percentage of quota owners and the information on sustainability was equivocal. In addition, it was not seen as biologically supportable as the Kaipara Harbour was not a separate breeding population, and smaller areas would be more costly to manage (Chatterton, undated):

Certainly it makes sense that some locals will be husbanding their harbours and some of the outsiders won't be, and yeah there is again, especially with the Kaipara suggestion to make it a smaller quota management area. And it's an attractive option. But there's a huge technical difficulty which is that we would then have to somehow or other scientifically assess stocks in the Kaipara of things like mullet that don't necessarily stay there. It's enough trying to get information on our major stocks at the moment and part of the reason that they're large [QMA's] is supposedly the biological range. (respondent 18)

Of all the proposals in *Fishing for the Future*, the only one ultimately adopted by the Ministry was the temporary closure of the scallop beds. This was a non-commercial fishery which was not part of the quota management system and therefore did not raise the same internal conflicts as the other proposals. The Ministry did not, however, follow up this step with monitoring of the impact of the closure on shellfish stocks.

Although not adopting a local solution to the harbour fisheries management issue, in response to the report the Ministry did undertake a review of the status

of the three main commercial fish stocks. This, however, also failed to result in any substantive action being taken.

... basically you've got ... quite a few different people there. You've got some commercial fishers, you've got some community groups and also marae around that area, pretty much agreeing that they're not happy with the level of commercial fishing in that harbour. I guess, on the other side, you've got other commercial fishers and more broadly stakeholder organisations for those larger fishery areas who disagree with that. [They] don't really think there's a problem with the level of fishing that happens in the harbour. It's basically come down to disagreement about whether there's maybe just a local problem there or whether maybe those fish are over-fished on a broader scale.

We reviewed the catch limit for flounder and rig and mullet which are the three species most commonly caught in that harbour. It was quite an unusual process. We don't have particularly good science information for any of those stocks even though we'd commissioned some extra research on them. But it was one of those things where it was pretty much you could interpret things different ways, I guess, depending on your starting point. Ultimately the decision was that probably the catch limits were okay for the entire stock, but maybe they weren't very good for within that harbour.

Some of [the conflict] has been to do with the method of fishing, because it's mostly set nets and ...it's a large harbour but it's quite a tidal harbour. So it's kind of about spatial control as well, because set nets, they're fixed and they often are left for quite a long time. It's very visible who's fishing and where they are. So it's quite hard to separate out the problem of having a net there from the problem of what impact that net is having on the fish stocks. There are concerns as well about catching juvenile fish and that sort of thing, but that again is quite hard to untangle. Is that necessarily a concern at a larger scale over that whole stock or is that more to do with how people are interacting with each other there?

The community group came up with quite a few detailed regulations that they'd like to see, to do with different mesh sizes and quite fine-scale controls. Whether you want to do all those changes or whether you actually want to focus your resources on getting enough information to get the overall catch limit right. At the same time I think it's still really important to try and work with those groups and we certainly try to ... Ultimately no changes were

made, but it was actually quite a lot of staff time to do that review of the three main species. That was quite a major thing for us to do. (respondent 21)

Although MFish failed to take substantive action to resolve the conflict, the Northern Inshore Fisheries Company did subsequently develop an Industry Code of Practice for Responsible Fishing in the Kaipara Harbour. This dealt with a range of issues such as minimum mesh sizes, net length and soakage time for nets. There was also an effort by that Company to encourage 'outside' fishers to reduce their effort in the Kaipara Harbour:

Now if you let that conflict go on, I don't argue that that's satisfactory. My experience of it elsewhere is if you let that go on without engaging in it as constructively as you can, then ultimately the parties simply become entrenched and fester with each other. You're best to step in and say to some of the entities, 'well it would be a good idea if ... you didn't need to put your effort into that harbour, then don't. Go and select somewhere else' and the level of tension will diminish. (respondent 57)

Management challenges

The management of commercial fisheries in the Kaipara Harbour has proved particularly challenging and this is for a number of reasons including the 'lack of fit' of the quota management system with local management issues, the lack of an effective conflict-resolution process and lack of a clear process to address non-fishing-related impacts on fish stocks.

Reconciling the philosophy underpinning the quota management system with local management issues

There are several underlying conflicts between the philosophy underpinning the quota management system and local management efforts. The idea that the quota management system would conserve stocks through encouraging spatially-dispersed fishing effort does not necessarily work out in practice in situations such as the Kaipara Harbour.

There was a theory that the fishermen would move around and that it wasn't going to be in their economic interest to fish one area too much, just over-fish it. So they would have a system that they would work out for themselves where

they'd sort of move around. But on the west coast harbours my perception is that that theory hasn't worked. And it's no great surprise because one fisherman doesn't ring up the other one and say, 'well I fished there yesterday don't come' ... It's simply because people know the good stocks and they keep coming back to them and there's nothing to stop them doing that.

(respondent 8)

The quota management system was based on the idea of moving away from regulation to a rights-based system whereas local management may result in more regulation:

My understanding is that part of getting them [commercial fishers] buying into the quota management system was they were told, because no-one knew any different, that they would not have to worry any more about those irksome regulations and restrictions and area closures and method controls and all of that sort of thing, because the quota management system was an output as opposed to an input control. (respondent 18)

Ministry policies at the time were to rely solely on [the] quota management system and that's seen in a national and regional context of providing output controls and tradeable rights and enabling fishers to fish unimpeded, but according to those rights, which is a right to a certain amount of fish. Any local controls were seen as an anathema because it was impacting on the markets of tradeable rights and they saw it as fettering the rights of quota holders. And therefore, even though there has always been and will continue to be, in my view, a regulatory framework of managing fisheries, at that point the Ministry was trying to back away from that regulatory framework and were certainly not looking at enhancing that in terms of local controls.

(respondent 26)

A rights-based system, based on individual transferable quota over large spatial areas, is essentially incompatible with localised management solutions:

The difficulty is how we do respond to those local issues. They're not going to go away. In some ways they are a reaction to people's belief that it hasn't provided for them at a local area in terms of the small-scale traditional fisherman. They haven't seen the benefits of the system, of say the deep-water fishery. What they're trying to find is a way to take over control of their own local fisheries. (respondent 36)

On the other hand, localised management requires more management effort and administrative costs:

I don't think you can necessarily deal with localised abundance in every area by trying to impose sub area limits. As soon as you start doing that you're into micro-management with little wee things being parcelled out, up and down the coastline. Because that's virtually what it would come to. Every local area would want its own particular fisheries with ability to control the level of catch that would come out of their own area. So you're talking about a whole management area broken down into lots of micro-areas which would become quite costly and administratively kind of impractical for a fisherman to have to apply in each one of those areas if they wanted to go into it. (respondent 36)

Addressing local issues may require both a philosophic departure from a pure application of the quota management system and also legislative change. It could require a layered approach where local management plans nest under a group of broader fisheries management plans:

I think those local management issues are going to be the really crunch ones that the Ministry has got to deal with and I think they're going to have to make changes with the Fisheries Act to do it in order to create a mechanism that recognises the fact that communities do want a say in their fisheries. (respondent 15)

Lack of an effective conflict-resolution process

There is currently no effective conflict-resolution process in place between parties jointly accessing a public resource. Although there are dispute-resolution provisions in the Fisheries Act, these are rarely used and were not utilised to resolve the Kaipara fishing conflict. This may have been because Ministry staff did not wish to intervene in what was perceived as an intra-commercial fishing disputes:

In the Kaipara it seems to be partly within the sector, the commercial quota owners. Within-sector issues are the harder ones to deal with. Why should government get in and deal with issues within a sector. It's established a quota and then people want us to get in and further intervene and start establishing rules between groups of commercial fishermen. I'm not sure that that would be a forward step in the way of managing fisheries at all.

Obviously we have to find some way of not letting issues get to the stage where they are firing at each other or shooting at each other but there is no easy answer. I don't think we'd want to be putting in place another whole set of rules and regulations to try and manage an issue within a sector. Yes, there may be issues between the recreational fishing and the commercial fishing and there's rules within the Fisheries Act that are designed to address those sorts of concerns, but the Fisheries Act is not really structured in a way to deal with the intra-sector conflicts. (respondent 36)

The Ministry of Fisheries is rolling out a programme to develop fisheries plans. They are to be developed by consensus and are potentially seen by the Ministry as a way of resolving ongoing conflict over fisheries management. The Ministry may facilitate the development of a plan for the northern west coast harbours, many of which are suffering similar problems. However, there is no clear process if agreement cannot be reached by the parties to the plan:

... at least it will provide a mechanism for those views to come to the table, whereas at the moment there's no incentive for people to talk to each other. There are different groups with different held positions and there's no point of discussion around that. A fishery plan may provide a vehicle where those issues can be tabled and agreements brokered, but it doesn't seem to have occurred to date. There doesn't seem to be sufficient commonality to get agreement on anything. So what that means is they fall back on government to say 'can you intervene for us because we have this problem and we can't fix it ourselves.' [It's a] tricky situation for government to be responding at that scale. (respondent 36)

It may be difficult for the development of another plan to overcome the inherent problems already encountered, being the conflict of interest between harbour fisheries and those owning wider quota (with the current system favouring the wider quota owners), and the fundamental conflict between the quota management system and localized marine management.

Addressing non-fishing-related impacts on fish stocks

The impacts of sedimentation on flatfish stocks, highlighted as a potential major issue in the stock review, remained unexplored and unaddressed both by the Kaipara Harbour Sustainable Management Study Group and MFish. This was because they both saw it important to focus their efforts on managing the impacts of fishing effort:

'When we first set up that group [Kaipara Harbour Sustainable Fisheries Management Group], there was a lot of interest and the Northland and Conservation Boards with financial support from the Auckland Regional Council and Northland Regional Council sponsored, it might have been, two trips around the harbour ... There must have been fifty or sixty people with all the chiefs as well as the Indians ... It was quite clear that the regional councils and DoC wanted to broaden out this process, because by that stage we'd got quite good representation and were starting to get some traction in terms of at least relationships going. [They] wanted to address other issues, catchment-related issues ... We thought about it long and hard but decided that if we broadened it out, simply the group was going to get too big, the focus was going to be lost and we wouldn't be able to achieve any outcomes. (respondent 26)

At the moment the Ministry of Fisheries, we manage fishing. We don't manage fish as such. What our mandate is to do, is to control fishing. Anything to do with fish that isn't to do with fishing, at the moment isn't our job. So whether you make [managing sedimentation] our job or whether you explicitly make it someone else's job, I don't know, but I think that would probably be the first step to take away that ambiguity about whose responsibility it is. (respondent 21)

Traditionally [sedimentation] has not been recognised by the Ministry as an area which it really has the resources to get greatly into, nor is it necessarily one that the Ministry has a resource role in. Look at the Act and in particular environmental principles. They're not there to act as an advocate for fishing or for management of fisheries in terms of external impacts on fisheries. So unlike DOC, for example, which has an advocacy role to argue for any impact on a species, we do not have a particular role in terms of dealing with third party aspects on fishing. So that hasn't had a lot of attention directed to it.

Again, it's having the capacity to engage with regional councils on those issues and I think we see that probably as being the role of fishery stakeholders to also take those issues and engage with regional councils on decisions that impact on their fisheries. In terms of the priority, I think that would rank lower in terms of the issues that we are trying to address, unless it was the predominant cause of the sustainability state of that fishery. The most emphasis that we're looking at is the level of fishing activity, first and foremost, rather than looking at third party effects. (respondent 36)

A more recent initiative to develop an integrated management plan for the Kaipara Harbour (see case study 9), provides the potential to highlight the linkage between catchment and fisheries management.

Prospects for integration

Disputes over the Kaipara Harbour fisheries remains unresolved because of the direct conflict between management of the harbour and the philosophy and approach underpinning the quota management system. There is currently no room for local management solutions to develop, if agreement to them cannot be wrested from most quota holders, and it is not in their interests to agree. In the face of such unresolved conflict, the Fisheries Act lacks any effective conflict-resolution processes. As a result, fisheries management appears to have reached a stalemate position, which favours the status quo, risking further decline of fish stocks. Without local management of the fisheries of the harbour, there is no interface through which to deal with other issues such as the impact of catchment management on harbour fisheries. While this conflict continues and absorbs energy, some of the more insidious impacts on fish stocks, such as sedimentation, which would require a more integrated approach to fisheries management, go unrecognised, unresearched and unaddressed.



twelve

Case Study 7: Kaipara Harbour Aquaculture

Spatial planning with poor information

What they were planning to do [was to] put a 140 hectare artificial reef containing an invasive exotic species on this rare significant unique community. (respondent 16)

These will be the safest oysters in the world, inter-tidal good quality ones ... the world's best oysters. (respondent 56)

We're caught between a rock and a hard place, not pleasing the community, not pleasing the industry, but that's just the way the law is. It's the best we can do. (respondent 7)

Policy framework

Part 7A of the RMA provides the framework for the management of aquaculture. In order to establish a marine farm, it must firstly be located within an aquaculture management area (AMA) and, secondly, a coastal permit must be obtained (section 12A).

There are three main procedures by which AMAs can be created. The first is for a regional council to prepare a change to its regional coastal plan, delineating AMAs. The second is through the invited privately-initiated plan change procedure. In this case the council can identify areas in the coastal marine area which are 'excluded areas'. It may then invite private plan changes to establish an AMA outside those areas. The applicant for a successful private plan change is guaranteed an authorisation to apply for a coastal permit covering 80 per cent of the area (with the balance being allocated to iwi). Any person may also apply for a private plan change through the normal RMA procedures, but in this case authorisations to apply for a coastal permit are allocated by way of public tender. These procedures came into force on 1 January 2005.

In 2002, prior to the new regime for marine farms coming into force, a moratorium was placed on the processing of resource consents for marine farms. This was lifted at the end of 2004. Provision was made for the early lifting of the moratorium in specific cases, but this could only be done through an Order in Council made on the recommendation of the Minister of Conservation, and only after a request had been made by the regional council (section 150C). In order to recommend an early uplift, the Minister of Conservation needed to be satisfied, based on the information supplied by the regional council, that the size and location of the affected coastal marine area would avoid, remedy

or mitigate the adverse effects of aquaculture activities on the environment, amongst other things.

The management problem

There are currently several small-scale oyster farms in the northern Kaipara Harbour and none in the southern part of the harbour. There has been a growing demand to establish new marine farms within the harbour, particularly within the southern area, due to its high water quality.

With the northern Kaipara ... the long thin arms have a problem in that the tide goes in and out in the same place and the water never changes ... [there's] very little mixing ... we get into a problem when things start spawning and the spawn stays there and, what's more, it gets spread onto the other oysters or the other barnacles and that's usually one of the triggers to make more and more spawn so you get massive over-catch problems with barnacles and oysters and other sort of inter-tidal organisms in those upper reaches ... [In the] southern Kaipara ... it's water that's totally unaffected by any human or animal run-off. It has a huge, huge water exchange, two cubic kilometres each tide, each way going through the entrance. (respondent 56)

Neither the NRC nor the ARC had identified areas for new marine farms within the Kaipara Harbour in their proposed regional coastal plans. Once the moratorium had been placed on the processing of resource consents for marine farms to allow AMAs to be put in the place, both councils separately initiated processes to identify AMAs within their respective portions of the harbour. They did so independently and in the context of a paucity of ecological information on the harbour, particularly in respect of benthic habitat.

The management approach

When the process to establish AMAs was commenced, neither council had developed a strategic framework which was specifically tailored to the management of the Kaipara Harbour. The regional coastal plans provided for new marine farms throughout both regions as a discretionary activity and to be considered on a case-by-case basis:

The [Northland] regional plan just has broad policies pretty much chunked down by the protection issues in section 6 and by coastal development activities such as marinas, sand extraction, structures, those sort of things. So there are broad issue-based policy objectives but nothing specific to the Kaipara Harbour at the moment. (respondent 7)

We don't know yet [what the ARC is trying to achieve in the Kaipara]. I guess in the end the same sort of objectives as we have for the Hauraki Gulf - identify and protect the values in the Kaipara ... It's not a priority ... Its location is less used, less known, less valued by a large part of population. [There was an] assumption it was looking after itself. But that's starting to change as more pressure is coming on. (respondent 3)

[In] the Auckland region we produced our regional coastal plan in February '95 and at that time there was limited interest shown in marine farming in the Auckland region. Council made aquaculture a discretionary activity over much of the region. The only exclusions were Coastal Protection Area 1 - they were marine reserves, bird sites and important geological sites, principally, so quite small and generally located hard up against the edge of the coastal marine area. (respondent 27)

Both regional councils sought to be proactive in providing opportunities for aquaculture in their regions. They adopted a traditional constraints-mapping approach, supplemented by industry information on desired locations for marine farms, to drive their planning exercise. In the northern part of the harbour, after the identification of potential AMAs but prior to any plan change being notified, the exercise ground to a halt as a result of public opposition:

It began with aquaculture being one of the four strands of identified regional economic growth potential ...As a result, when the aquaculture law reforms were first mooted, the NRC as a council wanted to be proactive in responding to this initiative rather than wait for the completed law reforms to come through.

[We] instituted a process of, aligned again to what Auckland was doing, [what] we called constraints mapping. This is pretty much plotting on a harbour by harbour basis what we know about the harbour - where the navigation lanes are, where the sewerage discharges are, where the areas of some conservation value are, etc. After plotting this information onto maps, we took these out to

the public, saying 'we know about this information, this is what we have been able to plot. What would you like to add to this, what local information do you know about that isn't yet recorded but which should be?' Obviously there's a bit of sensitivity around fishing spots and things like that, but we managed to work around that to come up with this pictorial representation of existing uses of the coast - special areas included.

At around that same time representatives of the aquaculture industry formed the Northland Aquaculture Collective, a group of industry players who had an interest in getting access to suitable aquaculture space, if not now, in the future. ... We as a council had no in-house expertise in aquaculture management and determining what areas were suitable for aquaculture. They [the Northland Aquaculture Collective] were the ones who obviously knew where they wanted to go ... So what they came up with for us was a 'wish list', they would say a 'needs list', of 125 potential aquaculture management areas.

We then went out a second time to the community and said 'okay here's the constraints maps overlayed with the 125 areas that the marine industry has come up with. What do you think about that? If they are too large, could you live with them if they were to be reduced in size?' ... [We] boiled the list down through that process to eighteen potential AMAs that we felt comfortable with ... However, [the process] also raised increasing concern from the public that we were meeting the needs of an individual industry ... They were increasingly saying 'we don't like what you are doing but you won't listen to us, you're still trying to identify areas where aquaculture can go. (respondent 7)

The process in the southern part of the harbour took a similar approach, but initially proceeded much more quickly, with a variation to the regional coastal plan (Variation 3) being notified by the ARC in October 2002.

The approach we did for the AMAs was a relatively standard sort of constraints analysis – what are the impacts of marine farming, what are the values people want to protect, what constraints does it put on things. Tends to quickly narrow down large chunks of the marine area where you wouldn't want it. (respondent 3)

'The constraints mechanism was something which really came about through aquaculture planning. Environment Waikato did it originally for Wilson's Bay and most other regional councils have picked up on it. Effectively, you create

a series of overlays of values and resources an area has ... It's part of your consultation really, saying 'if you were to put marine farming within a particular area, how would you then choose which place to have it?'

So you know that it conflicts with boating and you know that ... you avoid areas of high ecological diversity and so forth. So if you find all that information, then you map it, you might end up with a box that says 'if you're wishing to avoid all of these issues, this is where you would put it'. The constraints mapping is very useful in that it gives you visual presentation on a whole lot of issues which you wouldn't normally be able to see in that way. We didn't use it to apply weightings, we just said 'it's here or it isn't here' and we took it on face value. (respondent 27)

The adoption of this constraints-mapping approach was criticised by one interviewee as being an unsophisticated approach:

ARC and NRC engage in constraints mapping. From a GIS [geographic information system] perspective it was flawed. Little more than getting layers, changing colours and then saying what's left. This is just a data dump. It just overlays shapes with no interrogation of [the] underlying database and whether it is sufficient. It does not take a systems view and does not address missing information (which shows up as a gap). You can analyse GIS databases quite skilfully and pose questions such as which is the best place to locate AMAs given considerations of the catchments, its permeability, number of building platforms, underlying geology, propensity for sedimentation and faecal coliform counts. (respondent 16)

In the absence of an overall strategic framework for managing the harbour, the constraints-mapping approach has also been criticised as pre-determining the outcome:

As we get well-heeled economic drivers setting the agenda, then science is used for the agenda set by private sector commercial interests. In the absence of a public mandate which stands on its own, you get all these approaches on an ad hoc basis from the commercial sector. The ARC says we are going to have to arbitrate between conflicting economic sectors, better rally the troops, get resources together and see how we can manage economic contenders. You are actually orientated towards an economic outcome rather than a wider-public-good outcome and, in the case of the AMAs, I think that was a classic example. The constraints mapping was done to identify where

AMAs could go, not whether they were appropriate or not. Given that that was the objective, the science was always going to be orientated towards those goals and you end up with the outcome you are looking for. (respondent 2)

Around 1,300 submissions were lodged on the variation. The progression of it was complicated by the fact that there were two major resource consents being processed for marine farms in areas delineated as AMAs in the variation. Biomarine Limited had lodged two applications before the aquaculture moratorium was announced. The first was for a mussel farm which was accepted and considered under the old rules prior to the aquaculture reforms coming into effect. In the case of the second application for an oyster farm, the council considered that it lacked sufficient information, and the application was subsequently caught by the moratorium but Biomarine Limited successfully sought an early uplift of the moratorium, which also enabled it to be processed under the old rules.

Biomarine Limited mussel farm application

The mussel farm application covered thirty hectares some 100 metres or more out from the shoreline of South Head between Te Kawau Point and Omokoiti Bay. The applicant indicated that the farm would generate \$20,000 revenue per hectare per annum and would directly employ four people. When reaching a decision on the application, the council hearing body was split, but a majority decision approved the application (South Kaipara Harbour Environment Trust, 2006).

The grant of resource consent was appealed to the Environment Court by the South Kaipara Harbour Environment Trust, a local resident Britta Heitz, the Royal Forest and Bird Protection Society of New Zealand Incorporated, Te Runanga o Ngati Whatua and the Kakanui Marae Trust Board. The appeals were heard during September 2005 and February 2006 with the decision being released in April 2006. The parties agreed that the effects of the farm on marine ecology were not likely to be significant and the hearing focused on the impacts of the farm on natural character and landscape.

The Environment Court upheld the appeal, overturned the council's decision and declined the application. The Court concluded that the application area and adjacent shoreline had a high level of natural character. It found that the marine farm would be 'jarring' in the context of the availability of panoramic views around the harbour and its precincts and its sense of remoteness, and would significantly degrade its natural character.

Biomarine Limited appealed the Environment Court's decision to the High Court. The case was heard in October 2006 and a decision issued in November.

The ARC changed its position and supported the decline of consent. The appeal was declined by the High Court except on one small point. The Environment Court has now been directed to reconsider the application without taking into account the potential increase of walkers and the attraction of the regional park on the nearby land (Biomarine Limited, 2006). As this is a small point in the context of the overall decision, it seems likely that the Environment Court will reach the same decision and decline the application.

Biomarine Limited oyster farm application

This application was for a 140 hectare oyster farm located on a sandbank off Orongo Point. Biomarine Limited plans to transfer oysters from nurseries in other areas and to fatten them at the application site which has particularly high quality water. The ability to locate an oyster farm in this exposed location was only due to recent technology developments which enable oysters to be farmed in baskets suspended on wires rather than on fixed racks. In rough weather the racks would be damaged but the baskets swing and the waves go right through them without causing damage. Another benefit of the basket technology is that it prevents shells dropping onto the sea floor.

The most significant environmental issue raised by the application was the potential effect of the farm on the benthic habitat below and surrounding the farm. During the hearing of the application the existence of an ARC-commissioned monitoring report was brought to the attention of the hearing's committee by a local objector.

We had been aware that Judy Hewitt from NIWA had been engaged to do benthic habitat mapping. At the hearing I requested a copy of ... DoC's [scientific] evidence as I couldn't stay to listen to it. When I got the evidence it had in it references to Hewitt 2005. I requested a copy of the report and this forced a stay of proceedings so that the report could be released and given to the commissioners. We never found out why the ARC had not produced the report at the hearing, although we asked the question. (respondent 16)

The report in question had been prepared by NIWA as part of the ARC's monitoring programme. In 2000 the ARC commissioned NIWA to develop a state of the environment monitoring programme for the marine ecology of the region. The design that was ultimately adopted included three tiers focused on monitoring sea floor flora and fauna. Tier I monitoring was undertaken every two to three months at small intertidal sites in important harbours, including the Mahurangi, to detect changes in the benthic ecology. Tier II monitoring was planned to be undertaken every ten to fifteen years over a much bigger spatial

area of intertidal and subtidal areas less than twenty metres deep, in order to describe habitat patterns and ecological communities present, and to detect larger changes. Tier III monitoring was planned for deeper marine areas and would focus on broad-scale habitat mapping (Hewitt & Funnell, 2005, 5).

The Kaipara Harbour was selected for the first Tier II monitoring exercise. This was due to the shortage of ecological information on the harbour and the need to consider the potential location of AMAs within the context of spatial information for the whole of the southern Harbour. The monitoring started in 2003 and the report was completed in 2005.

The monitoring report (Hewitt & Funnell, 2005) indicated that there were rare and unique groupings of marine life at the site of the application:

While many of the taxa and habitats found in the Southern Kaipara Harbour occur elsewhere, some are unique. In particular, a subtidal association of tube-building worms was found in the shallow sub-tidal area [the subject site] of the main harbour comprised of high numbers of *Owenia*, *Macroclymenella*, *Euchone* and *Phoronoids*. Subtidal *Zostera* is also comparatively rare in New Zealand.

AMA D [the Biomarine oyster farm site] and E lie across an area of subtidal *Zostera* and high diversity patches of sponges, suspension-feeding bivalves, filamentous seaweeds and the unique tube-dominated community.

One objector described the proposal as follows:

What they were planning to do is put a 140 hectare artificial reef containing an invasive exotic species on this rare significant unique community of assemblage ... It's a bit like going into Fiordland and saying we can run a really good *Pinus Radiata* operation here and, you know what, it will be inside the kauri. You won't know it's there ... [or] just taking a kiwi fruit orchard and putting it inside what is regarded as a rare and unique area. Isn't there something absurd in that ... before you've even looked anywhere else.

(respondent 16)

The commissioners turned down the application and Biomarine Limited appealed this decision to the Environment Court. Before the appeal was heard the company engaged NIWA to undertake a more detailed study of the benthic habitat of the site.

By the end of 2005, not only did the ARC have decisions declining applications for two of the AMAs proposed in Variation 3, but the monitoring

report prepared by NIWA had also found that four of the five proposed AMAs in Variation 3 were located in the areas of highest taxonomic diversity in the harbour and/or important or unique habitats and communities. The area which was not so unique was the subject of the declined biomarine mussel application and had failed on natural character considerations. The Council withdrew Variation 3 in May 2006, without holding hearings on the submissions which had been lodged.

The Northland Regional Council had experienced considerable community backlash in response to its attempts to identify AMAs and so, during 2006, reached a decision not to notify a proposed plan change:

The council was seen to be compromising its role in [deciding] consents through the RMA process ... We were too close to industry, the potential applicants, according to the community's point of view. Secondly, the community, through rates, was bearing the cost for the process which they also didn't like - hundreds of thousands of dollars you're talking about. Thirdly, [with the] aquaculture law reform options at the time, who was to get access to an AMA would have to be determined by an open tender process. This didn't give any certainty as to who was going ultimately to be able to undertake marine farming. The industry group was unhappy with NRC having to follow that approach saying 'Okay, we've helped you ... [in] identifying potential new areas but now you can't give us priority to them.'

So here we are, as reported in the paper, we're caught between a rock and a hard place, not pleasing the community, not pleasing the industry. But that's just the way the law is. It's the best we can do. (respondent 7).

So far as the Biomarine Limited oyster farm application is concerned, in early 2007 the Environment Court overturned the earlier decision and granted consent subject to conditions, including a reduction in the size of the area to be farmed and the gradual development of the balance of the area through the implementation of an adaptive management regime (Biomarine Limited, 2007).

At the hearing in the Environment Court the main issue in dispute was the impact of the oyster farm on the subtidal seagrass (*Zostera*) which is an ecologically important habitat and quite rare throughout New Zealand. The subtidal *Zostera* beds in the Kaipara Harbour were the only ones known in the entire Auckland region. After considering competing expert evidence, the Court decided that a staged monitoring approach to the development of the

farm would be appropriate so that any adverse effect on the *Zostera* beds could be ascertained and addressed.

Biomarine Limited was finally successful in obtaining a resource consent for the oyster farm, but there was no underlying AMA for the site as a result of the ARC withdrawing Variation 3. Whether or not Biomarine was able to proceed without an AMA was unclear. The Environment Court was not required to determine this point but, after hearing legal argument on the matter, tentatively concluded that the provisions of section 12A of the RMA did apply (Biomarine Limited, 2007, para 74). The section states that no person may occupy the coastal marine area for the purpose of aquaculture except in an AMA in a regional coastal plan.

The Environment Court experienced considerable difficulty in interpreting the legislative amendments resulting from the aquaculture reform process. In seeking to determine what rules applied to an application which was the subject of an early uplift of the moratorium, it stated (Biomarine Limited, 2007, para 64):

It is clear that those situations where a consent was already granted or was processed during the moratorium are treated in a different way to applications which were not considered until the end of the moratorium. Beyond that the provisions of the Act are so obscure that it is difficult to immediately understand their relevant application and treatment.

By early 2007, planning for aquaculture in the Kaipara Harbour had reached a stalemate situation where no-one seemed happy. Neither council had been successful in identifying AMAs in their regional coastal plan. And one marine farm application, which had escaped the moratorium, ultimately obtained a resource consent for the Kaipara Harbour, despite the ARC withdrawing support and its proposed AMA covering the area to be occupied by the farm.

Earlier, in October 2006, the NRC notified Proposed Plan Change 4 to the Regional Coastal Plan. This indicates that the invited private plan change procedure will be the primary mechanism for the establishment of AMAs in Northland. In order to provide some guidance for the location of the AMAs, the council has produced a set of Aquaculture Evaluation Maps, building on the constraints mapping work which had previously been undertaken. The maps classify various uses and values of the marine area as 'critical', 'significant' or 'potential'. Critical areas are those where aquaculture is unlikely to be suitable, significant areas are those where there are some constraints, and potential areas are those where there are likely to be few constraints to aquaculture.

The ARC is now exploring an approach which involves identifying excluded areas which are not suitable for aquaculture and then also embarking on an

invited plan change procedure. The council has also commissioned research into the costs and benefits of aquaculture to the region, which should help answer the question of whether aquaculture is the best use of its marine space.

This shift towards utilising a private plan change procedure rather than council initiated planning to identify suitable areas for aquaculture marks a move back to a more *ad hoc* approach to planning for aquaculture. This is because each private plan change application is considered separately, rather than as part of a broader strategic planning approach.

Management challenges

Planning in the context of poor information

The experience of attempting to plan for aquaculture in the Kaipara Harbour illustrates the difficulties in undertaking coastal planning, in the absence of good quality information:

It's very difficult to deal with applications in the absence of good information and I guess the history of aquaculture and aquaculture management areas in the Kaipara is certainly an illustration of how difficult it is. The original applications for mussel farming in the Kaipara - the recommendations were to decline those applications on the grounds that there was a paucity of information and the future impacts on the natural character of the area.

Now the council chose to grant those consents contrary to the officers' advice. It put the council in a position where ... it could hardly not establish an AMA in an area that it had previously granted the resource consent. So draft AMAs were established in the Kaipara. Now the current council has obviously taken a different stance in respect to marine farming in the Kaipara and has reversed some of those policy directions. So I guess [it is] the classic case, where a lack of information and lack of knowledge of what was happening with that environment has led to a flip flop in decisions. (respondent 24)

This case study also highlights the dilemma of whether or not to plan for activities in the absence of sufficient information. The ARC decided to notify AMAs in the Kaipara Harbour in the absence of sufficient information on benthic ecology on the basis that, as there was so little information and it would

take such a long time to obtain an adequate amount, it was better to move forward cautiously rather than wait:

Ideally you'd have all the information available and would debate your proposed decision. One option was to say, 'we don't want to progress anything until we get all the information ...do you wait for the years that you might need in order to actually get to a point where we are at with the Firth of Thames [and that] is still several years away, quite a lot of years away from where we might be in the position to say it's a good idea or not' ... So the conclusion was, in order to bring this thing up to speed to the point where you had all the information, you'd be waiting for many, many years. Do we wait for that? The decision was made by council to go ahead and start the project. (respondent 54).

What we did after we notified [Variation 3], we knew we didn't have ecological information so we got NIWA to do an ecological assessment of the southern part of the Kaipara Harbour for us. So we had all that in [process] but it took two years or so for that report to come out.' (respondent 27)

The case study also illustrates the problems of making decisions on a case-by-case basis in the absence of a broader information base within which to assess proposals adequately. The assessment of environmental effects undertaken for the Biomarine oyster application failed to identify the importance of the benthic habitat on the site and this may have been because the assessment lacked the context within which to fully interpret the significance of site specific data:

These [benthic habitats] are quite special when you look across the entire Kaipara Harbour, or at least the southern part of the harbour. It is sort of special, unique and it is vulnerable as a consequence of that. It is an area of particular biodiversity ... you don't know that until you start looking across the whole of the Kaipara, which is what our study did. When you're carrying out an environmental impact assessment for an application for a consent, you don't want to have to study the entire Kaipara Harbour ... so you're going to be more restrictive. So the context in which you can understand the results that you get from your focused study aren't necessarily going to be there without that broad base. (respondent 54)

The applicants did some research on the benthic habitats. The impacts were that it did lie on sea grass beds, let's say, and the oyster farming would destroy those sea grass patches. You'd know that this area here was sitting adjacent

to biological communities of worth. You should know that, that should have been picked up in the AEE [assessment of environmental effects] which it wasn't. It all comes back to getting the job done properly first of all. Before you can designate anything as AMA, you want to know exactly what's there and that wasn't done just because of the way the process of the consent went through, to the moratorium. (respondent 33)

The Biomarine oyster farm experience also highlights the dynamic nature of the marine area and the dangers of relying on a snapshot view over a limited time period. Very little *Zostera* was present on the site when the application was lodged, but the abundance of the species had increased significantly by the time the application was ultimately decided five years later:

When this application first came out there was a bit of sea grass over that area of it and there was sea grass in other places. Since that application was made, in two years the sea grass now occupies about eighty per cent of that site. So it's been growing, which went against it as well. Biomarine had done some work, NIWA had done small site specific stuff for them, but it was reasonably light. (respondent 27)

And why didn't the ARC introduce the findings of the NIWA report to the Biomarine hearing? As a regional councillor observed:

It does make you wonder, the fact [that] it was a community group that brought the commissioners' attention to the fact [there was a] report out there. Why the report was not used as part of the AEE or council assessment of effects has got to be a serious question. One department commissioned it and another department was doing the assessment of the marine farm, I guess. But it's a pretty sad indictment on interdepartmental communication. (respondent 2)

Allocating the costs of marine planning

Some of the backlash that councils experienced during their endeavours to delineate AMAs was the public perception that ratepayers were funding a process which ultimately was for the benefit of a small commercial sector of marine farmers:

Now no councils are doing any reviewing of their coastal plans and creating an AMA because the cost is huge. It cost me over three-quarters of a

million to do this [oyster farm application] under the old legislation. Now the costs are a lot more ... If they want to make AMAs, they'd have to ask the ratepayers to stump up a million bucks. Half the ratepayers are yachties and the other half don't think anyone should make a living from the sea and that sort of philosophical problem or whatever, and they're just not going to do it. (respondent 56)

The process now favoured by the councils, of inviting private plan change applications, effectively places a much greater proportion of the costs of planning for aquaculture onto marine farmers, but with no certainty that the person paying will ultimately obtain a right to farm. The current provisions of the RMA do not enable the council to make an early choice between competing private plan changes. This acts as a strong disincentive for prospective marine farmers to invest in such applications

You invite private plan changes ... [and] currently if two, three or four people apply for the same area they have to let all the applications run so everybody has got to do all the consultation, all the scientific reports, everything for their private plan change and then they have to run them all together. They can't pick one ... Under the present thing no-one would apply for a private plan change because you're not going to spend several million to be one of five possible people all spending several million and one of you gets picked. The one who gets picked under the rules now is the one who essentially spends the most money... (respondent 56)

The high costs of the plan change procedure may also have the effect of stifling innovation within the marine farming industry, where experimental changes in farming practices of species may require a full-scale plan change:

What's sad is that small aquaculture projects that could be just used to improve an existing farm or experimental things on a modest scale or whatever can't take place ... because it requires a public plan change, complete plan change just to change the species from one thing to another. (respondent 56)

Finding appropriate locations for aquaculture

The case study also highlights the difficulty in finding appropriate sites for aquaculture which are away from centres of population but also away from ecologically significant habitats:

From an aquaculture perspective, you're wanting things such as clean water, you're wanting to be away from large population areas because you end up with a lot of people being opposed to it, a lot of people seeing it and so they're opposed to it. You want to move away from where existing development is because that has implications on the water quality. Generally in coastal marine planning you concentrate development where development is but you want in this case to move away from those things. If you move away from those things, you tend to move towards areas of high water quality which have low levels of foulment, high levels of natural character and you've got the tension between the two. (respondent 27)

Prospects for integration

The case study indicated that there is currently little integration in planning for aquaculture in the Kaipara Harbour. Both regional councils undertook their aquaculture planning exercises independently, both in terms of gathering information to inform the planning process and when identifying potential sites for AMAs. The small amount of environmental monitoring within the harbour is also undertaken independently.

The public backlash which occurred when councils delineated proposed AMAs highlights the difficulty of undertaking spatial planning in the marine area for one activity (in this case aquaculture) in the absence of a broader strategic framework or spatial planning process which considers a range of other potential activities and future development trajectories. This is because, once marine space is delineated for one activity, parties with an interest in other activities for which a similar planning exercise has not been undertaken can feel threatened and adopt a defensive position. Similar problems have been experienced with the establishment of marine reserves in the Hauraki Gulf (see case study 5: Hauraki Gulf amateur fishers). The lack of high quality information on the ecology of the harbour compounds the difficulty of sustaining planning decisions as being technically sound in the face of such opposition.

In the absence of strategic spatial planning for the harbour, however, decisions will continue to be made on a case-by-case basis. The case study of the Biomarine oyster farm illustrated the pitfalls in this approach, where it can be difficult to evaluate the significance of site-specific information without a broader understanding of the dynamics of the ecological system within which it is located. This highlights the need for a well-designed long-term research and

monitoring programme for the harbour. The ARC has made a positive start on this, but only for the southern portion of the harbour.

The case study of the Biomarine mussel farm also highlights the conflicts which can be encountered when there is no strategic framework to guide decision making. The location of the mussel farm in an area which the ARC proposed to delineate as an AMA was ultimately found to be in conflict with a regional park and walkway which the ARC subsequently established on South Head.



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Case Study 8: Integrating Management of the Hauraki Gulf

Building the platform for collaboration

There's too many different views of how best to achieve integration, just even on the Hauraki Gulf. I think everybody is doing their own little thing which is anti-integration really. They're integrating over there and they're integrating over here. (respondent 34)

Approaches

There have been a number of multi-agency initiatives to integrate the management of the Hauraki Gulf, or parts of it, particularly across the land-water divide. These include the establishment of the Hauraki Gulf Forum, developing local area management strategies and coastal compartment plans and preparing a 'Blueprint' for the Coromandel Peninsula.

The spatial extent of each initiative is quite different. The Hauraki Gulf Forum extends across the entire Hauraki Gulf. Blueprint Coromandel encompasses the entire Coromandel Peninsula and surrounding marine areas. The coastal compartment plans and local management plans, on the other hand, have focused on small localised areas of the coast. Marine protection planning is also seeking to integrate across several management agencies and regimes (see *Case Study 1: Planning for the Hauraki Gulf and Kaipara Harbour coastal marine areas*).

Hauraki Gulf Forum

The beginnings

The Hauraki Gulf Forum had its beginnings back in the late 1960s when a Hauraki Gulf Maritime Park Board was established to manage the newly-created Hauraki Gulf Maritime Park as a result of the Hauraki Gulf Maritime Park Act 1967. The Park included Crown-owned islands from the Poor Knights Islands in the north to Mayor Island in the South. The Board developed a management plan for the Gulf and pioneered work on island pest destruction and habitat restoration (Waitangi Tribunal, 2001, 7) and was proactive in advocating for the protection of the Gulf:

Often Lands and Survey said 'we couldn't get involved in that issue' and the Parks Board said 'go for it'. So I was in boots and all, in all sorts of planning issues. [If] somebody wanted to subdivide or develop, we could quite easily get involved, and we didn't have [to get approval from] the Ministry of Works or our Minister. That often was under-estimated. Where is the government free agent now? The Ministry for the Environment aren't doing it, DOC are hamstrung, they're politically really getting choked in terms of their advocacy role and yet these Parks Boards were in boots and all.' (respondent 26)

The Board was disestablished by the Conservation Law Reform Act 1990 and DoC took over the management of the Park. Management of the Gulf was split between three conservancies: Northland, Auckland and Waikato.

During the 1980s, a former member of the Board, Allan Brewster, lobbied for the establishment of a marine park in the inner gulf where commercial fishing would be excluded. This failed to get political traction, but in 1990 the National Party's spokesperson on conservation announced an intention to promote a complete marine park for the Gulf. In 1991 a technical working party was established to investigate the options for taking the proposal forward (Waitangi Tribunal, 2001, 8).

In 1994, in the hiatus between the Park Board being disestablished and the political machinations to develop a new approach, the Auckland City Council launched a *Vision Hauraki* project. This sought to bring together the different statutory agencies and to provide conservation leadership for the Gulf:

Auckland City Council had its Vision Hauraki Project which was investigating or trying to bring agencies together who had statutory functions in the Hauraki Gulf and it did so at an officer level. It didn't really work particularly well ... It was to get the various agencies at an officer level to talk to each other about what they were doing. Effectively, it was Auckland City Council telling everyone else what they were doing ... The key frailties of the Vision Hauraki was that it was at an officer level and didn't have political buy-in. (respondent 27)

The Auckland City Council initiative was effectively superseded when in 1997 the ARC established a collaborative group of local and central government representatives (but not tangata whenua), which was called the Hauraki Gulf Forum, to coordinate management efforts. Meanwhile work on legislation to establish a Hauraki Gulf Marine Park continued and in 1998 the Hauraki Gulf Marine Park Bill was introduced to Parliament. It was finally passed into law in February 2000.

The Act consists of three main parts. The first part sets out objectives for the management of the Gulf. These are incorporated into decision making under the RMA, the Conservation Act 1987 and the Fisheries Act 1996. The second part establishes the Hauraki Gulf Forum as a coordinating body. The third part establishes the Hauraki Gulf Marine Park. The Park is managed by DoC and there is no direct linkage between the Park and the Forum, as there was between the Hauraki Gulf Maritime Park Board and the Hauraki Gulf Maritime Park which it managed under the previous legislation.

Operation

The Forum includes representatives appointed by the Ministers of Conservation, Fisheries and Māori Affairs, representatives of the twelve local authorities with jurisdiction in the Hauraki Gulf and its catchments, six tangata whenua representatives appointed by the Minister of Conservation, and two further representatives appointed by the ARC. The Forum meets every three months at the ARC offices in Auckland. A technical officers committee of the Forum meets at least every month.

In terms of scope, the Forum has been based on a 'catchment to the sea' approach and includes management agencies with jurisdiction for the marine area as well as for catchments draining into the sea.

The Forum did not have an easy birth, receiving varying levels of political support:

I think with the Hauraki Gulf there was a little bit of baggage in the past. There were all sorts of things going on. The time the Hauraki Gulf Marine Park was under contemplation and [was undergoing] its slow and tortuous route through to being formed, the ARC was against it. It was and it wasn't. Officially the policy position was not antagonistic and there are certainly those of us there who were strongly supportive, both amongst politicians and amongst staff. But the politicians and staff who were in support of it were not in the influential positions. You had a situation where elements of relevant committee chairs and management were antagonistic and Phil Warren was very, very lukewarm about it ... [The Hauraki Gulf Maritime Park Act's] been ignored. Sometimes it's given a bit of lip service. Very, very seldom has it been given the status that it warrants ... We will just carry on business as usual thank you very much. (respondent 1)

The Forum really struggled in the early years. This was because the ARC, while it had responsibilities to support it, didn't believe in it. So it wasn't given the resources it deserved. And a number of other local authorities didn't give

it priority. The other problem was it took iwi some time to become comfortable with it and agree among themselves. They still haven't agreed, but things have improved markedly. (respondent 14)

When the Forum was originally set up as a committee of the ARC, ARC was in support of it. When it got set up under legislation, at about that time our CEO wasn't particularly keen on it ... Her focus was principally on transport issues. (respondent 27)

But a recent political shift in the ARC may give it more traction:

Now, as I say all those players have gone. Most recently we've got [] starting as General Manager of Policy and Planning and he identified the lack of proper accounting for the Hauraki Gulf Marine Park Act as being a key issue that he as a manager wanted to address. And I said 'you've got the politicians right behind you'. (respondent 1)

Since its inception, the Forum has produced a range of documents including a strategic issues document, a state of environment report, an inventory of activities and programmes undertaken in the Gulf and a shellfish-depletion study (Willis, 2004, 8). In addition, the Forum has supported community-based shell fish monitoring programmes, commenced a scoping exercise to determine Māori cultural indicators and commenced a project to establish a set of templates for effects-based cultural heritage management (Hauraki Gulf Forum, 2006, 13).

Evaluation

So how well has the Forum been seen as performing? On the positive side, it has helped to improve communication and the exchange of expertise:

Maybe it's a talkfest, but one of those things that it was supposed to be solving was the lack of communication between agencies. The thing that does really work is the officers group. That works really well as far as communication goes. We know what each of us is doing and at that level it's working pretty well, whereas in the past we in the Ministry [of Fisheries] wouldn't have had a clue what Environment Waikato were doing. (respondent 18)

It's kind of an interesting thing to be involved in from a professional point of view, because we talk about a lot of issues between councils and that does bring out what is going on and new ideas. (respondent 13)

I think there's a lot of unrealised potential there. It can't be underestimated how important it is that the councils are talking to each other, and doing detailed constituent party reports, so that the councils know what each other is doing. The Forum is not about abrogating responsibilities of the different parties. It's about understanding how those responsibilities relate and how various activities relate and looking for opportunities to add value. The Forum has been valuable in terms of networking. We often jointly report on the various things happening in the Gulf. We share information between ourselves at officer level. (respondent 30)

There are a number of reports which go to the Forum which basically inform the Forum and the various members what other agencies are doing and that's quite useful. For example, in the Auckland region we've got a big issue with growth and we have a growth strategy etc. Coromandel has significant issues with growth and there are some similarities between our growth strategy and what's happening in Coromandel. We've had the growth strategy stuff presented to the Forum and because of that TCDC [Thames-Coromandel District Council] are much more aware of that growth information and have created links through the people in our agencies and assisted with some of that stuff.

Some people say the Forum is just a talkfest. Well it is in some ways, and there's nothing wrong with that, because it's transferring information, it's creating ties and links between agencies and so forth. So there are a lot of benefits in just the transfer of information between agencies. (respondent 27)

It has facilitated joint projects between agencies:

Then things like the aquaculture project. We [ARC] have developed very strong links with Environment Waikato, in part because of the Forum, looking at aquaculture in the Firth of Thames principally, but we looked at the whole of the Hauraki Gulf Marine Park to start with. Through that there's been a number of joint projects, joint-funded projects and joint-led projects with Environment Waikato on aquaculture. A lot of their meetings we go to and they come to our meetings as well and all that, so there's a lot of integration there.

There's other things like the recent work that's presented at this next Forum's meeting on the relative-risk-assessment work in the Firth of Thames. ARC have been strongly involved in that, as has all of the TAs [territorial authorities]

around the Firth of Thames, Department of Conservation and so forth. They've all been linked in together in that particular project. (respondent 27)

And provides a political forum where agencies can present ideas:

We have benefited from it in that we have an opportunity to put issues that are significant to us before other agencies at a political level. It shoves them further up the agenda. (respondent 14)

And can exert moral pressure on members:

It has no teeth but it does have moral authority. (respondent 14)

I encourage those reports and their reporting and the publicising of those things because that's the peer pressure between large organisations and small ones that will eventually see the quality of water, the quality of the environment come up together. (respondent 48)

Option 4 came and they were talking about the Department of Conservation's proposed ban in terms of the Great Barrier Marine Reserve ... There was a lot of debate about the Department's consultation policy through that ... members of the Forum were quite concerned about the level of consultation which the Department had with that particular project. The Department was made very aware of those concerns. So some of that stuff I think is very useful with other agencies there. The agencies are there in front of their peers and there is an unwritten level of performance [to be attained]. In that particular situation the Department's peers said, 'well actually we think your performance of consultation of this matter is not adequate. We think you need to review that'. (respondent 27)

It provides a forum where iwi are resourced to articulate their perspectives:

In the first year of its operation I didn't click on to the Forum's importance or value. I went to one or two things but not much. Then I realised when the strategic issues document was being put together, I had a paradigm shift I suppose, because I'd been so used to receiving documents from public bodies and submitting to them, commenting on them and waiting to see what little if anything, happened. I suddenly realised that this was one that you could work on from the inside, being part author of ... it's been an opportunity to be paid to articulate iwi perspectives to a reasonably receptive but reasonably

influential audience. That's an opportunity that doesn't often happen, with some back-up resources to an extent from the various other constituents, so you can draw on their information sources and other things like that ... There's a whole lot of work that has been done to better inform that Forum on tangata whenua perspectives. Generally, as a group of local body politicians, I would say they're better informed as a result on those issues than most. So that's progress. (respondent 22)

On the negative side there has been no clear vision or purpose. The 2004 Enfocus review of the Forum identified a tension between the potential roles of the Forum as simply a mechanism which passively shares information and knowledge or a more proactive body that undertakes projects and programmes under its own banner (Willis, 2004, 1):

I think it's struggling. I think most people you talk to will probably say that, but not without cause. The difficulty it's struggling to overcome is I don't think it did have a clear objective to start with or vision. It's had to come up with a strategic issues document because it had very general objectives ... it really didn't have any direction about what it was actually supposed to do, so hard from the start with that. (respondent 18)

The Forum I suppose has always struggled a little bit to find out what its real function and role is. Each of its agencies are obviously independent authorities in their own right. They have functions and tools and so forth, and they get on and do what they're required to do legislatively. The Forum has always found itself a little bit uncertain about where its function is in terms of that co-ordinating role, and I think rightly so, because it's actually not an easy place to find yourself. I think what the Forum really needs to develop is some strong political leadership in that area. (respondent 27)

Relationships are weak amongst politicians:

The second difficulty I observe is that because the elected reps only meet four times a year, they can't really build up a working relationship, meeting each other that infrequently. I'm not exactly sure how it rates in the hierarchy of jobs that elected people get to do, but I have a feeling that for some councils it's kind of 'oh I'm sorry but you're going to have to do the Hauraki Gulf rep as well' ... They just don't see each other often enough, so it's very hard to get things going. (respondent 18)

Participant bodies don't necessarily give it priority:

I was involved when I was at a district council. When I started, it was very much the job given to the graduate planner, the job that senior planners knew they had to do but couldn't be bothered getting involved in. I think it's changed a bit now. (respondent 13)

Involvement of different agencies in the Forum is very, very variable ... In the last year it's been really noticeable that Environment Waikato have had very little involvement ... I think probably at the Forum they have been, but technical officers have very little input from there and I think that's a real gap. (respondent 22)

There is still an uneasy relationship with iwi, as iwi are not directly represented on the Forum (rather there are six tangata whenua representatives appointed by the Minister), and there is no formal mechanism for accounting back to iwi:

Prior to the inception of the Hauraki Gulf Marine Park Act, amongst other things, the Hauraki Maori Trust Board ... [was] actively involved in advocating for the integrated management of the Gulf. There was, however, some cynicism as to whether a piece of legislation could make the different agencies act in an integrated way. One of the concerns after the Act came into being, as an iwi resource management practitioner, was the lack of communication between tangata whenua of the Hauraki Gulf and the Hauraki Gulf Forum. That concern exists to this day. For example, iwi do not receive agendas or minutes of the Forum work. From where I am sitting as Environmental Manager working for one of the key iwi authorities of the Hauraki Gulf, I am saying that we have no relationship with the Hauraki Gulf Forum. We hope that will change and we don't know why the information hasn't been getting out to our whanau, hapū and iwi. If you look at the provisions in the Act that recognize tangata whenua relationships with the Gulf, there has got to be some accountability back by the Forum to tangata whenua of the Hauraki Gulf. (respondent 34)

At times it has been unable to rise above parochial interests:

I think the benefit of the Hauraki Gulf Forum is actually having everybody around the table and having some serious and considered discussion on the issues affecting the Hauraki Gulf and associated values, with a focus on integrated management. Yes, I think it's mostly about dialogue. It may not

necessarily be about solutions but it is about putting the onus of solutions squarely on the agencies and groups that are represented on the Forum. Healthy and informed debate of course will lead to solutions. The problem right now is that I don't think the quality of debate is there because everyone is protecting their own patch or own jurisdiction. I think tangata whenua representatives on the Forum are in a prime position to break through that impasse and take the discussion to a different level, but that hasn't happened yet. Taking everybody's agendas off the table, and starting to talk about the real issues is the step the Forum has yet to take. (respondent 34)

It has a low public profile and this makes it difficult to gain political support:

Most people in the Auckland region have never heard of it. It's not really seen to be doing useful things on the ground. It's had a lot of trouble to find things to do that give it a stamp. It still doesn't have its own kind of logo or anything like that, They're talking about that now. It's kind of getting there. (respondent 18)

... different councils have had different priorities it wants to set, basically because they're responsible to their ratepayers to start with and then they move up through the wider good for the general environment and for the Gulf as a whole. I don't think there's a strong enough visibility of the Gulf Forum to those ratepayers. I don't think there'd be many Auckland City ratepayers that would even know that it exists. As a consequence, there's very little dedicated support for it at officer level. At the political level where decisions are made about funding, there's little wide political support for it because it has such a low visibility.' (respondent 49)

And finally, it hasn't directly affected what the management agencies are doing on the ground:

Has it affected what MFish is doing in the Gulf? The short answer is no. I think it's probably the same for all the other agencies. We're doing things ... that the Forum would like, but we're not doing it for that reason. We're doing it for our own management reasons ... I think it's pretty much the same for the others. It's sort of business as usual ... (respondent 18)

The council had never got round to incorporating references to it into the regional policy statement until we got niggly last year and said 'right, we need to have this. Okay, it might be almost a one-liner thing but that was what that

Act required. Again the policy statement sets the scene for district plans. So the whole thing, the whole planning hierarchy, is supposed to mesh together and work in a coherent sort of way but it has still been isolationist in some respects. (respondent 1)

So we've got this Hauraki Gulf Forum made up of all these local authorities who do a whole lot of talking and you get a nice lunch and get paid and they pat each other on the back and they feel good. I have yet to see any substantial evidence of where the marine environment is improving by their efforts, because we are continuing to have urban run-off, which is something they should be able to stop. Sure, they couldn't stop it in a year, but ten years down the track you'd expect that they would have reasonable sort of plans in place to stop the continual blowouts we're still getting. (respondent 31)

The Forum is estimated to cost \$320,000 to run annually. This includes \$98,000 to fund tangata whenua participation and \$60,000 for the ARC to provide administration and servicing (Willis, 2004, 9 & 15). So is it worth the cost and effort? There are a wide range of viewpoints:

There hasn't been a limitation on what's been spent there. There have certainly been questions asked 'are we getting value for money? Is this actually the best way to achieve integrated management of the Hauraki Gulf?' Because the Forum costs bucketloads to run. And would we not be better off ... not cutting back on the budget as such but doing it in smarter ways? Now that's only been asked by the politicians, so it's sort of carried on in a status quo bit of inertia in that respect. (respondent 1)

In terms of whether the benefits are worth the effort it's kind of a fifty/fifty thing, I would say. It's kind of useful, but, yes, it does take a lot of time and I think in here [Ministry of Fisheries] and most other agencies, the higher managers still don't really know what it is, don't have much of an understanding. So I'm not sure how much support their staff get. It's all right. (respondent 18)

I think the Forum is great. It has a regular and formalised group who share ideas, try to financially do things more efficiently and group together for common goals. (respondent 13)

I'm still prepared to put time into it. I think it is valuable but it's limited because it is, on the one hand, only an advisory body and, secondly, it does have to overcome all the sectional interests. Even though it has representatives from

those interests in the room talking to each other and working together, there's a hell of a lot of entrenched positions within the agency. (respondent 22)

Yes, the ARC does think it's getting value back. But I don't think, at a managerial level, that we're getting as much as we should be. I think that's what we're interested in looking at. How can we actually achieve better outcomes through this and the strategic issues document and so forth will help towards that.' (respondent 27)

In March 2007, the Forum resolved to establish a jointly-funded dedicated staff resource to co-ordinate Forum activities and promote the Forum. This may help to enhance the Forum's profile and effectiveness.

Local Area Management Strategies

Local Area Management Strategies (LAMS) were an attempt by EW to achieve integrated management at a local level. One of the strategies focused on the management of mangroves in Whangamata Harbour:

[The Whangamata initiative] began initially as a concern about mangroves spreading in the estuary ... A landcare approach was seen as a way of addressing issues around mangroves ... My role was to facilitate a discussion which would identify community aspirations and solutions. So I worked on that as a kind of harbour-care process for a couple of years. About that time the council decided to try three approaches to integrated management ... Local Area Management Strategies ... So there was one for Whangamata.

We worked with the district council and we did a community involvement process to identify what people wanted and the district council took more and more ownership of that until, finally, the actions that came out of it were primarily district council actions. Some were related to getting consents for removal of mangroves and the potential to review the coastal plan to adjust the way we treat mangroves. The processes have been going on until last year when it was evident that a non-notified consent wasn't going to be achievable from removing mangroves because of the divisions in the community. (respondent 11)

But there were significant difficulties in implementing the concept due to the broad scope adopted:

The plan ended up being much wider than EW's brief and there was a bit of internal stress because we began the consultation process in a totally open way. We just asked people what issues they had. So we got things like street lighting and footpaths as well as environmental concerns. We spent two years coming up with things that everyone could agree on. Water quality was a big issue, so the district council set up a working party to address those issues. That's gone on for four years now ... The overall process took eight years. I count it as a really significant failure, because in that time I would have expected people to find compromises, but I wasn't able to do that.

(respondent 11)

And the concept was ultimately abandoned:

The LAMS were an attempt to integrate resource management in the region. The vision we had was that we would trial the methods, find out what worked and then expand out into the rest of the region. They were scheduled to take two years each. They all went different ways. Taupo identified the lake was in serious trouble and it became the Taupo project. Whaingaroa Harbour drew in Landcare Research and they identified a plan of action and set up an environment centre – basically a community-run process. Whangamata produced a community plan but didn't address the original issue or solve it. What we learnt is that we have grossly underestimated the cost and they dredge up a lot of stuff which we couldn't deal with. So there was no way we could carry on over the region, so we pulled the plug around 2002.

(respondent 11)

Coastal Compartment Plans

Objectives

The ARC has initiated a programme of developing coastal compartment plans. The two key drivers of the initiative were the need to obtain better information to inform coastal planning and the need to achieve better integration and less conflict with territorial authorities:

[We] had a couple of objectives for the coastal compartment plans, to address two different problems. One was lack of information, about the coast and coastal management problems, for reviewing the regional coastal plan. At the moment the coastal plan has got the general management areas and ... [little] guidance for what's appropriate in those areas, the carrying capacities, the structures or activities in different areas.

The other driver really was a lack of integration with TAs ... [There was] quite a series of issues with things like sea walls or coastal protection works coming in as consent applications. The ARC was saying, 'have you looked at the bigger picture, studied coastal erosion on that stretch of coast' or 'if you do these works here, how will it affect the next beach?' Consent applications often take a very location-specific view and [we want to] incorporate more of our policies in their initial planning of their works programme. [We are] trying to be proactive and stop those fights before they came to a consent application, and by developing projects that were fifty per cent funded by the TA and the regional council, having more of an active joint way of solving those problems before they came in. (respondent 25)

Scope

The scope of the plans was kept tight, primarily restricted to the coastal fringe. This enabled a focused approach to be adopted, but meant that the planning lacked direct integration with broader catchment-related causal factors:

[The geographic extent of the plans was] one of the issues that we spent a long time discussing in the very early stages of the scoping and it has continued to be an issue. What we decided to do was focus on the coastal fringe. We haven't defined it explicitly as a set distance inland because of the difficulties. We tried to keep it a fuzzy boundary, relevant to the particular issues in the area. So when we've been explaining the extent to the public, we've said 'the land adjacent to the coast' and generally most of the issues are to do with these sites, it's about parks and sites initially straight on the edge.

The issue comes with dealing with things like water quality and sediment where it's the whole catchment. So mostly the issues are either just [to do with] the initial first ten or twenty metres of land or the whole catchment. But the catchment has been dealt with through integrated catchment-management plans and other processes under the whole stormwater action-plan team's

work. So we didn't want to start duplicating that work or confuse people thinking we were covering those issues as well. So I've tried to integrate with that team and keep them involved and aware of the different plans and the draft plans. [The plans] highlight water quality or sedimentation as an issue but then, as an action, they'll just say 'this area is highlighted in the integrated catchment-management plan where this issue has been dealt with through the controls being promoted by compliance team, better practice on farms' or something. The problem [is] with defining that split and explaining it to people ... [when] we're just dependent on those other processes. (respondent 25)

The plans have also focused on short-term solutions so that communities can see timely improvement:

We don't ignore it [catchment management], because it's the main thing that we try and explain to people when we talk about the issues [e.g. water quality or mangroves]. But just doing something about it, addressing the sedimentation coming into a stream, is going to take decades to make a difference to the mangroves. So most of the actions that we're talking about are to help local communities. They're more short-term - how do they get their boating space back in the next five years? - so that's where the short-term management is which needs to be done irrespective of long-term management. (respondent 25)

Process

The plan preparation process adopted a fairly standard planning approach:

The general approach that we decided to take was consultation on general issues or a vision for the coast, then develop a draft plan, do more consultation on the draft plan and then finalise the draft plan. We've expanded the basic model, with two stages of consultation: one that is fairly open with the issues and options stage and one that's on the draft plan. What we've done, with all of the plans, is to get a consultant on-board at the beginning. So the two councils have developed a joint brief of what they want to achieve for the area and in that we've identified what we know are current issues and concerns, and tendered that out to a consultant that comes on-board and manages the whole process. We then work with the consultant to go out and do the consultation and any technical studies that we decide to include in developing the draft plan. (respondent 25)

Evaluation

The planning approach has not been without its problems. During the preparation of five plans, a number of difficulties were encountered, including lack of information and lack of sufficient budget to undertake new investigations:

Problems identified were things like having insufficient budgets or just identifying that further investigations are needed. So, as soon as you start looking at a section of coast you say, 'well, I can't develop policies in this area until I've done a study of what the flora and fauna is or what the coastal processes are' and the budgets that we had have been about working with existing information, not doing the investigations.

As soon as you start looking at areas, you decide you need more investigations done, then you need a research programme. It's a fundamental problem with this type of work that it can't just be researching forever; but if you're doing them all around the coast, you end up doing huge research projects. Then when you get on to implementing them, [you've used up] the budget just doing the research, but at the same time you've still got problems on the ground. How do you make proper management decisions without adequate information? It's just a continual struggle. (respondent 25)

There was also the difficulty in getting buy-in from staff in participant councils:

We've tried to do that [get buy-in] by having briefings and meetings and getting people to comment. But there's always tight deadlines getting things back to consultants. It's quite hard to actually keep everyone in the council informed of what's going on. (respondent 25)

And community expectations were raised through the community consultation process which the council is consequently unable to meet:

You go out to the community and say 'what do you want with this piece of coastline?' They might give you their vision and then you come down to reality. 'Well we're not doing any more investigations so we can't get to the level of detail that you want, and we don't have a huge budget for doing the work anyway, so we're not going to end up with the vision that you've asked for.' But by going in there with the process, [it] starts raising their expectations that you are planning to do what they say.

We go out and do consultations, get feedback that says eighty per cent of people think mangroves are the problem with the coast and ninety per cent of those people think they should all be removed. Then we come back with a draft plan that says you can remove them in these limited areas that we have prioritised, and they just can't understand it. 'We all said this is what we want, why aren't you doing it?' Then it comes back, 'well we're balancing it: your community desires for the coast to meet the social and economic well-being aspects of the Act, but also the protection of the environment.' They don't see the importance of any scientific information that we come back with, they don't trust the consultants that we use and they think their local knowledge is far more relevant than any science that you come back with.

We end up not quite satisfying anybody, not giving the people the vision and we're not meeting the expectations of the scientists that were involved, because we're coming up with things like recommendations for mangrove removal where ecological groups are inclined to want to keep them.
(respondent 25)

There is also the problem that non-statutory plans do not necessarily carry any weight when statutory decisions, such as granting resource consents, are made:

They're a non-statutory plan and no-one has to take them into account. There has been a few environment court cases with judges looking at non-statutory plans like the Auckland City coastal management strategies and saying 'we didn't really have to take them into account, we can read them, but you don't have to follow them'. So, in some ways it's a huge waste of effort because you can't guarantee that anything will ever be achieved.

I think, for me, that's one driver of why it's been so important to have strong buy-in from the ... District Council of an issue that they know needs to be addressed and that they're willing to put the time and money into. They're going to put actions into the plan that they're actually going to follow through and do it, because their community want follow up, and be concerned that this plan has effect. Otherwise ... the council can just do this process and then ignore it in terms of putting it into their LTCCP [long term council community plan] process or their own consent application or processes on the land.

What we try and do in the compartment plan is identify a whole list of actions [so] that people can see a change will be made and not just high-level strategies of what we're trying to achieve. We do try and have a list of actions

with general or indicative costings and priorities. But whether those things ever happen is just dependent on subsequent processes. So they need to go into an LTCCP or a district plan change or a regional council plan change or council works programme, or a community group might get funding to do a planting. (respondent 25)

The plans have cost between \$40,000 and \$60,000 each. They take the form of action plans which are more linked to LTCCPs and council budgets than statutory planning under the RMA. They also focus more on identifying actions for territorial authorities to take, rather than action by the regional council. But are they achieving the goals the ARC initially set out to achieve in embarking on the plans?

Papakura Council ... they've put about two million dollars into their draft LTCCP for the next few years to implement the plan that we've helped to develop. The ARC haven't put in any budget, and we haven't because the only actions that we [have] ... are things like when we review the coastal plan we'll take into account the list of issues that we're putting through this process. So it will help do a review of the coastal plan or help in other projects that we do here, but it's not actually work by regional parks or anything. I guess that might change if we get occupation charges through as a new funding stream. At the moment we're seeing [coastal compartment plans] more as a planning role.

I think we're achieving more with the integration with the TAs, where we're helping the long term TA planning for coastal management quite well. But we're not doing so well on improving future coastal plan reviews. It's partly because we're just not doing enough investigation of the CMA; we focus more on the land ... At the same time it's taken a lot longer and been more expensive than we thought. (respondent 25)

What about the territorial authorities, how do they see the plans? Rodney District Council had undertaken a coastal management strategy review in 1999 and identified 37 compartments for which more detailed planning was seen as desirable. The officer interviewed at Rodney District Council was positive about the Algies Bay Coastal Compartment Plan and the relationship established with the ARC. This was a plan initiated by the RDC to which the ARC subsequently gave support as part of its coastal compartment plan programme:

We [Rodney District Council] wanted a plan that was going to be easy to read, easy to understand and implement and was going to be suitable for effective

community engagement ... I wanted a departure from a plan that has an enormous amount of descriptive information at the front and, when it came to the policy, it was kind of relegated to the back somewhere or hard to link together with the issues. In our draft you can read the whole document in several pages by looking at the executive summary. All the actions for each issue are together. It sets priorities for actions. ... No plan is ever perfect, but I feel very pleased with the draft as a basis for good consultation ... we've had positive feedback and we've attained valuable insights from the community pre and post the draft plan being released. We're looking very hard at that through the submission process ... I'm looking forward to finalising it and I feel that it's going to be a useful model template for future coastal compartment management plans.

It is essential to work together with the ARC because one of the main issues, for example, is [that] the community is concerned about loss of sand, and wanting to do sand replenishment on the beach. The ARC has to be an integral part of that discussion. A lot of issues and concerns are at least partly in the area of ARC jurisdiction ... So we're releasing a draft plan, that in real terms is joint. . Both the authorities are saying this is the kind of future for Algies Bay that we're agreeing with the community and seeking to achieve in this environment. (respondent 30)

On the other hand, Auckland City Council had declined to participate in the coastal compartment plans, due to concern about the ability to deliver on an integrated plan through different budgetary processes. During the 1980s Auckland City Council had started developing several coastal management plans around the city, but the work-stream was discontinued when the council's coastal planner resigned:

The theory sounded absolutely brilliant of getting them integrated ... [But] you could actually have a piece of work that requires funding from the two agencies, and different groups within those two agencies, where you don't get a co-ordinated budget approach. Partly that's because the annual plan decisions are responsive to the consultation that you do with your communities, and the particular drivers that the council might have for that particular year as a priority. So we might have done a piece of work and said, 'okay, here's the coastal environment from Point Erin to Point Chevalier, and we've got a compartment plan for that and we think these are top priorities for that.' But then that goes into a mix with a whole lot of other projects from around the council and then it goes into the ARC's list.

We wanted to see how the ARC dealt with all that process ... before we committed to it. Because once you commit to those kinds of projects, you have a community expectation, and what we've discovered with the coastal management plans is you go out and you consult and you engage the community and then you can't deliver on those things. It's actually very difficult to work in those environments again, with that community, because there's a level of suspicion. 'You've promised, you've said, you've asked us, we told you what we wanted, but then the funding went somewhere else.'

The coastal compartment plans at the time that the ARC came to us, I think they hadn't even completed their first one, and they hadn't even got into the funding of those. So we wanted to see how those worked, because we'd seen the difficulties that we'd had with our own when we controlled it. We wanted to see how they worked when they had a TA as well as the regional council to work through. (respondent 49)

So, are coastal compartment plans a good idea? They can be helpful in focusing on specific locally-based issues and coordinating an inter-agency response. However, they are unlikely to provide a tool for region-wide coastal planning, because they are too expensive and localised:

It's different to more strategic tools. They have their strengths in doing certain things. [They make it] much more easy to identify issues, and in the implementation sense, it's easier to integrate. I think they are good as long as they have a framework to work within ... [But] it's resource-hungry and I don't know how far you go. It might follow the squeaky wheel. Some politicians are better at getting them, but is it the right place to do it? ... Coastal compartment plans put things into the LTCCP, into budgets ... so pieces of work are done to achieve an outcome, to overcome a coastal erosion problem, add coastal access and improve water quality ... [They are a] driver of the budget rather than regulation. (respondent 3)

I think it's definitely been worth while. Good plans will make a difference. But there are real problems with trying to replicate it around the whole region. From the ARC's point of view, a different approach probably is needed, and I think we'll probably step it up a level and either go to much bigger spatial areas ... or focus on particular issues, for example having a mangrove- focus strategy or a natural character strategy for the whole region. (respondent 25)

The ARC has not made budgetary provision to develop additional coastal compartment plans going forward. This is because the plans require significant resources to be devoted to relatively small areas of the coast which is not necessarily cost-effective.

Blueprint Coromandel

The beginnings

Blueprint Coromandel is an inter-agency project focused on developing a strategic coastal plan for the Coromandel Peninsula, including the land and surrounding marine area. Its inception stems from a common need of EW and Thames-Coromandel District Council to have a stronger strategic basis for statutory planning:

The blueprint project that we're doing with Environment Waikato, it started off as Environment Waikato wanting to do these harbour plans. We [TCDC] said we've done a whole [lot of] community plans, we don't want you to come along and say now we need harbour plans, and the community say but we've done our community plan. After all, everything that happens in the harbour is the result of something that's happened further up in the catchment, so how can we integrate these together?

TCDC was, as a separate exercise, trying to scope out this thing we called the growth strategy which was to try and deal with this whole issue of growth and appropriate development. What we've ended up doing is merging the two projects. TCDC and EW have just started now on what we call the Blueprint strategy. What we're looking for, as an output out of that, is essentially a plan of the whole peninsula. There's likely to be individual local, essentially structure plans, that say here is the appropriate place for sawmills or whatever, these areas are appropriate for residential development or industrial development or they're not appropriate places. (respondent 42)

The genesis of the idea within Thames-Coromandel District Council appears to have been, at least partly, a result of a high-profile resource consent case where the Environment Court turned down an application by Blue Mountain Lumber to establish a sawmill in the catchment behind Whangapoua:

The idea behind that was the Blue Mountain Lumber. Those people had come along to council in Thames and said we want to build a sawmill in Thames-Coromandel somewhere, where can we go? Rather than then having to find a location and going through the whole resource consent process and all the rest we would say, 'as a result of this study, here are some locations that would be appropriate for a sawmill', or 'no, there's nowhere in the Coromandel that's appropriate.' They could then go and either it already had the zoning on it or the Blueprint structure plan would indicate that we've already looked at all the constraints and opportunities around it and, if you applied for resource consent here, a lot of that scoping, that's already been done, you're likely to get a sympathetic response from council in terms of the resource consent application. (respondent 42).

The concept behind this project, from the regional council's perspective, was similar to that for coastal compartment planning, to provide more local detail in the regional coastal plan:

This was based on the idea of local coastal plans. It will be focused on, essentially, adjusting the regional coastal plan for local conditions, so it has a strong statutory planning component. (respondent 11)

Partnership

The success of the project is dependent on a positive working relationship between EW and Thames-Coromandel District Council. This would not have been possible in the past:

We are working much closer with Environment Waikato than we ever have in the past, and I think that is the way things will happen in the future, it seems crazy not to have the two aligned. I suspect the Blueprint project, where it's necessary, will result in joint plan changes to the regional plan and district plan to achieve the same outcomes happening at the same time. It seems to be a no-brainer to me that that's what you would do. That's only been possible because the two organisations are working in closely together. If we were still at loggerheads with each other it would just carry on how it used to ... A lot of it has been personalities ... our Chief Executive used to work for Environment Waikato ... I guess a lot of the issues we've been grappling with are around coastal process, and harbour processes and stuff. They are often in EW's jurisdiction, so we've had to work with them anyway and it's just that, working together, I think we've discovered that we've actually come from the

same planet and stuff like that. We're all trying to achieve the same ends and it's just been that gradual building of relationships and it's working really well now. (respondent 42)

The project has involved tangata whenua from its inception, and included funding for the collection of information both of relevance to tangata whenua and the larger project:

The Waikato Regional Council, the Thames-Coromandel District Council and the local Department of Conservation involved us really early, looking at what does an integrated coastal strategy look like for the peninsula. [They] involved us really early in the scoping of that, so that project is something we were all working together on, it ... has real potential for us because we've agreed on some principles around the need to be integrated, both in terms of the sea and the land, and we've made a commitment to find our way through that in terms of what that will look like. So that's a success for us because we hope that, with a bit of effort and thinking now, that there's going to be some long-term aims and a getting right down to some practical things on the ground.

Attached to that are our own initiatives. We have a heritage mapping initiative ... with the support of the regional council and DoC ... to map out the values that we have within a given area, whether they might be fishing grounds, whether they might be wāhi tapu sites ... We've been doing that on a catchment basis. We've kind of got little projects like this and we're finding everything is starting to align now. (respondent 34)

Scope

The scope of the project is similar to that of coastal compartment plans, with the focus on the coastal edge where most of the activity takes place, rather than the broader marine environment:

Although it won't technically leave anything out to the twelve nautical mile zone, it will definitely focus on the coastal edge. We are using the mountains to the sea approach. The nature of the Coromandel is that people are clustered around the coastal edge and a lot of the upper land is DoC or Māori and fairly stable in terms of nothing really happening, or is in forestry. The focus will be very much on coastal townships, how streams are being managed, near-shore recreation, harbours and open beaches. There is not a lot going on further out except aquaculture, which we are also working on. (respondent 13)

The project is based on EW boundaries which cut across the Firth of Thames, so it does not constitute a fully integrated initiative for the Hauraki Gulf marine area. This creates particular difficulties for iwi who lack the resources to participate meaningfully in multiple initiatives.

We [Hauraki Trust Board] were disappointed that the Auckland Regional Council [and the] Auckland Department of Conservation didn't come on board because their boundary, of course, is right through the Firth of Thames. Now we've spent the last few years working with the Waikato regional council over here to just get a framework right to work together and to achieve this thing that we all call integrated management, whatever it means, and got some good things that we didn't want to have to repeat again for the other side of the Firth of Thames line.

So one of our suggestions was 'come on board over here, Auckland. These people over here have listened to us and recognised what some of our priorities are, the frameworks in which we want to participate. How about you fellows join so that line can get taken away?' But no, because they've got their own priorities about what integrated management means, they don't want to work over here because they've got their own thing going over there. I think the thing going over there is that whole marine protected area stuff that is being driven by Auckland DoC.

So that puts us in the position of still having to deal with two separate processes to achieve integrated management, which for us is just ridiculous, it's the antithesis of integration and all we are saying is we want somehow to work together. And it's at a more practical manageable level than say a forum, because we're talking to the managers and staff that are actually doing the job. That for us is when a real change can happen at a practical level, but it has to be supported strategically. But it's still impossible, even though that catch phrase of integrated management is being bandied about, it's still not being practised.

There's too many different views of how best to achieve integration, just even on the Hauraki Gulf, and I think everybody is doing their own little thing which is anti-integration really. They're integrating over there and they're integrating over here. Here's the coastal environment; here's the catchment; here's the issue. Let's all work on it together. It's still not happening.

All we can do is work over here, try and participate there, because we've limited capacity and skill. We can't effectively interface with both, so we have to make a priority and our priority is, well, they've invited us at the scoping stage, we're best over here, to put our energy over here hoping that the information that develops out of here can inform over there, but we can't do both. I know there will be a perception in Auckland that we're not interested or that we're not a player in integrated management on that side of the line, which is not right, but we're just being really practical and pragmatic about what we can manage at this point in time.

From Auckland's point of view we're kind of the periphery of their region, we're the rural area, the back block, we're not where all the action happens, we're their parkland, or a place of recreation or for its development potential for their constituent members in Auckland. So I think probably their priority is more about their boating, being able to hop in the car and go and visit a nice picture-postcard place outside the ugly subdivisions that are happening there. It's like 'keep the green space for Auckland'. Whereas on this side, we're talking about communities that live and work around here and rely on this area for their wellbeing and livelihood. We've been denied the opportunity to develop our economic potential, our enterprise potential, through historical actions of the Crown and of local government and so who are you Auckland, who are you to say we can't make a living off our resources and look after our own livelihood as a community and do so in a way that is sustainable, just because we're a green space for you, Auckland. That's the dilemma.

(respondent 34)

Outputs

The planned outputs are a set of reference documents to guide council decision making. These include a district strategic framework in the form of an illustrative map and accompanying text, structure plans of major settlements and growth nodes showing both land and sea, and an action plan identifying implementation including projects to incorporate into the LTCCP (Environment Waikato, 2006a).

There's not a great link between land and harbour management and a lot of that is just because there's not the information available, it's not available in a form that can have an impact on decision making ... I certainly haven't seen anything that makes the link between the two. That again is part of what this Blueprint project is looking at, it's trying to pull together all that information

that all the different agencies have, so that we can actually make some informed decisions about, well, we've got a problem down here because of something that's happening up here, so if we stop that thing up here, what's the impact going to be?

The community plans will also feed into that Blueprint project because a lot of the issues we're talking about, particularly about where is growth appropriate or not, had been dealt with through the community plan. Some of the community plans, like the Coromandel one and the Whitianga one, they've gone to the next step and produced draft structure plans that say we want further residential in this area and we want some further light industrial over here, but we don't want light industrial over there ... so the Blueprint project will pick up some of that type of work, and at the end of the day, we hope to have a really very visual representation of where we think it's appropriate to have different types of development. (respondent 42)

Evaluation

The project is in its early stages so can not yet be meaningfully evaluated. It is ambitious and is likely to rely on the continued energy and political support of the participating agencies. It should provide a rich learning environment for how integrated coastal planning might be applied in practice. But it is unclear to what extent the outputs will be able to affect development decision on the ground.

Prospects for integration

The Hauraki Gulf has a long history of initiatives to integrate coastal management extending back to the 1960s. Models which have been tried include a dedicated management agency (the Hauraki Gulf Maritime Parks Board), a co-ordinating body (the Hauraki Gulf Forum) and multi-agency collaborative planning exercises of various scales.

The Hauraki Gulf Marine Park Act 2000 establishes a set of very broad common objectives for the Gulf. But there has yet to be developed a single strategic framework, a plan, a set of targets or a monitoring programme for the Gulf and this is not in prospect. There has also been no 'champion' for the Gulf since the demise of the Hauraki Gulf Maritime Park Board in 1990. Although the Forum could theoretically play a champion role, it is hampered

by the fact that the management bodies charged with managing the Gulf are its members and are, not surprisingly, reluctant to turn a spotlight onto their own management performance in order to champion the broader common good.

Councils are innovating and are trialling different approaches to integration. But this is at a much smaller scale than the whole Gulf. Maybe detailed planning can only be realistically carried out at a localised level. There does appear to be a gradual strengthening of the collaborative relationship between different management agencies within the Gulf, facilitated by the Forum's activities. This may provide a stronger basis for more ambitious integrated management initiatives in the future.

The Forum has provided the opportunity for iwi representatives to regularly meet with management agencies on an equal basis. This has created opportunities for the integration of Māori resource management perspectives into mainstream thinking, as well as providing resources to help strengthen iwi coastal management on the ground. The Blueprint Coromandel project has also achieved a partnership approach with iwi, who have been involved in the project since its inception.



fourteen

Case Study 9: Integrated Management of the Kaipara Harbour

Similar aims but different visions

We are the only management body with a fully-integrated perspective.
(respondent 23a)

We are clear that we are not talking about an integrated catchment plan which is an holistic all-encompassing process of managing everything that happens within the Kaipara catchment. (respondent 4)

The institutional framework

There is a major jurisdictional split through the middle of the Kaipara Harbour, with the northern half being managed by the NRC, Kaipara District Council and the Northland conservancy of DoC and the southern half being managed by the ARC, Rodney District Council and the Auckland conservancy of DoC. The rohe of Te Uri o Hau encompasses the northern part of the harbour with Ngāti Whātua having jurisdiction over the south. The other main management body, MFish, manages most of the major fish stocks of the Kaipara Harbour as part of a large quota management area encompassing the top half of the North Island.

The resources available for the management of the southern part of the harbour are potentially much larger than those in the economically-depressed north. For example, for the year ended June 2005, the ARC had an operating expenditure of \$213 million compared to the NRC's of only \$13.6 million. Similarly, Rodney District Council's operating expenditure for the same year of \$94 million was much larger than the modest \$27 million that was available to Kaipara District Council.

There are significant differences in the way each part of the harbour is managed. For example, in the northern part it is possible to subdivide rural land down to four hectare lots as a controlled activity. These generous subdivision rules are due to the developmental focus of Kaipara District Council which, in a climate of little population or economic growth, is endeavouring to attract development to the district. Between 1996 and 2001, population within the district increased by only 0.5 per cent, although the rate of increase rose to 3.6 per cent between 2001 and 2006 (see Case Study 2: Managing Coastal Development in the Hauraki Gulf and Kaipara Harbour).

In stark contrast, in the southern part of the harbour, rural subdivision is a non-complying activity unless environmental benefits are provided. These much more stringent rules stem from high levels of growth experienced by Rodney

District Council over many years, particularly on the east coast as described in Case Study 2. Population growth rate in the district was a very high fifteen per cent between 2001 and 2006.

The catchments of the harbour are large, encompassing in the north around half of the Northland region. There are differences in the regional council catchment management of the two parts of the harbour. For example in the southern part, forestry harvesting including in riparian areas is a permitted activity, whereas in the northern part there is greater control over riparian harvesting (see Case Study 3: Reducing Sedimentation in the Hauraki Gulf and Kaipara Harbour). This appears to be due to the stronger urban focus of the ARC, which has historically placed more emphasis on managing the effects of urban development such as earthworks, on the marine area than those of rural activities such as farming and forestry.

The split jurisdictions and management approaches have created some difficulties in achieving cohesive management of the harbour:

Guardians of the Kaipara from the Kaipara District side came to us [the ARC] and said would we go and meet with them. They had a lot of concerns about the future environmental quality of the Kaipara. So some of us went up and met with them and they sort of outlined quite a sorry story. They acknowledged that on the Auckland Regional side there is control, in that we have a growth strategy and that we actually control growth and urbanisation to a certain extent around the lower Kaipara.

But that's not happening on the northern and if you go up there you see massive development on the northern side in Kaipara district. There's no decentralised wastewater system, it's all septic tanks and it's all on a hill running down into the Kaipara. One group said to us, 'we used to be oyster farmers but we were closed more often than we were open and that's without this development happening' ... So they've taken up commercial fishing in that they hire themselves out to people that want to go out and fish ... They're saying 'we could lose our livelihood, we've already lost our oyster farm, we managed to sell it. Now we're in fishing, but we see that if there's going to be continuation of aquaculture in here, then something has to be done to control the urbanisation around the Kaipara Harbour.' (respondent 5)

The main issue is that different councils are all doing different projects here, there and everywhere and there's money going into them. The RDC is doing run-off and effluent studies and the NRC and ARC have done constraints in terms of aquaculture. But there is no overall view/picture of the research

being undertaken by the various councils or what they are all spending money on. It's all piecemeal and at the same time we're seeing a rapid pace of development. We've got gated communities all around here. (respondent 23a)

So I would say [the] ARC and Rodney have quite different philosophies about their interests in the Kaipara. Rodney [is] very interested in seeing it as a resource for economic values. [The] ARC [is] more interested in seeing [the] resource, politically, for environmental values. So those two are quite different and [there is a] lot of potential for conflict. (respondent 2)

Approaches

All the statutory management agencies involved in managing the Kaipara Harbour also have jurisdiction over eastern coastal areas which have experienced much greater development pressure. As a result, the management of the harbour has been given relatively low priority on the basis that it was looking after itself. Although Te Uri o Hau has been lobbying for a more integrated management approach towards the harbour for over a decade, the idea only gained traction in 2005.

During 2005 two projects were initiated to focus on integrating the management of the Kaipara Harbour. One project has been led by the ARC and is now a joint ARC and NRC project. The second project has been led by Te Uri o Hau. When the interviews for this project were undertaken, the nature of the ARC-led project was still being determined as was the relationship between that project and the Te Uri o Hau-led project.

The ARC-led project

The ARC-led project was generated at a political rather than a technical level. This appears to be due to increased environmental activism within harbour communities resulting from more people moving to the area for lifestyle reasons, a recently-elected environmentally-active regional councillor who lives in the area and a more environmentally-supportive council being elected:

There are more people living there and they are more articulate. A lot of them are professionals. So they have personal resources and they are seeing what's going on with development pressure ... They provide the credibility to

substantiate me [regional councillor] raising concerns. Otherwise it doesn't get traction. (respondent 2)

A number of concerns have been expressed through submissions to plan changes, and submissions to aquaculture applications that have been processed. They have generated sufficient concerns with some of our councillors who have suggested that we need to look at what the issues are in terms of achieving better integrated management of the Kaipara Harbour. This initiative came out of a resolution of the Environmental Management Committee of the ARC and also a resolution from the Inter-regional Committee which is a joint committee of ARC and Northland Regional Council. That resolution was that a scoping report be prepared to identify the issues in terms of achieving integrated management in the Kaipara Harbour. (respondent 4)

There is also now greater appreciation of the benefits that can be derived from integrated initiatives:

I think there's a huge benefit in terms of saving resource and money and using a joint approach to achieving wider objectives ... You might do a joint project with a district council which achieves both objectives, but you jointly fund it so it doesn't cost X dollars twice.. There's obvious benefits in this approach and also in terms of getting the community engaged and involved as well, because the community can deliver a lot of things through care groups. There are people out there who undertake such activities on their own land but are also really willing to plant up dunes and streams or whatever else all over the place for the wider benefit and public good.

I think it could save everybody money and stop a duplication of process ... all of the agencies are running staff, administrators, financial departments, etc. I don't think you're necessarily giving up your autonomy, you're working in a much more integrated way to achieve better outcomes than you would probably achieve individually at the moment.

You have the ability to share knowledge and information and use other people's knowledge and information collectively to deal with the impacts of development. For example, if we're doing a major research project which is costing a significant amount, then by taking a joint approach there is the possibility that the focus of the research can be extended to address wider issues, which can then be delivered through that research project. (respondent 4)

The project has, at least initially, been designed as a joint ARC and NRC project rather than as a partnership between the territorial authorities, iwi and other government departments involved in managing the harbour:

It is considered that the NRC and the ARC, as the major policy and consent agencies in the Kaipara, need to have a clear understanding of the present growth pressures, the potential impacts, and possible responses to ensure there is integrated management of the harbour. Having determined this the two councils will be in a better position to decide what they individually as well as jointly need to commit or contribute, to give effect to their statutory obligations and actively participate in community processes (such as the Te Uri o Hau process). (ARC & NRC 2006, 1)

The focus of the exercise has been limited to the harbour's coastal environment for which the NRC and ARC have responsibility, particularly under the RMA, rather than including the broader catchments and wider non-RMA issues such as fisheries.

We've been very clear that what we're talking about at the moment is not an integrated catchment management plan which is an holistic all-encompassing process for managing everything that happens within the Kaipara catchment ... Integrated coastal planning in my opinion means looking in an integrated manner about how you're managing activities that happen within the coastal environment, which includes the use of resources, as well as the impact of activities. So for example, you don't put a jetty in an area you have earmarked as being of high natural value, or an area of outstanding landscape, which is a pristine area that you want to protect ... Integrated planning in a coastal environment means, for example, that you decide where your stormwater outfalls are going to be and you locate them so they don't cause coastal erosion on a beautiful recreation beach.

[It's looking at] the wider implications and making decisions about the use, protection and development options and how these would impact on the coastal environment in a comprehensive way.

We've been careful to take this project from a low-key point of view to date, because we don't want to create large-scale community or political expectations that a hugely-extensive integrated catchment-management planning process is going to come out the other end, at this stage of the process. (respondent 4)

The councils have produced a scoping report (ARC & NRC 2006) which proposes that they jointly review existing information, undertake a gap analysis of research information and review statutory policy and planning documents. Additional actions will then be considered once the outcome of this work is available. The scoping report was circulated for public submission and eleven submissions were received. One of the major issues raised by submitters was the limited scope of the project and the need to eventually extend it to include the whole catchment (ARC 2006b).

Te Uri o Hau-led project

During 2005 Te Uri o Hau, with the assistance of Landcare Research, initiated two hui to discuss integrated management of the Kaipara Harbour. Te Uri o Hau and Ngāti Whātua Nga Rima o Kaipara subsequently joined together to form Nga Kaitiaki Taiao o Te Kaipara to lead a project titled 'Integrated Management for a Healthy and Productive Kaipara Harbour Project'. An interim steering group including members from Nga Kaitiaki Taiao o Te Kaipara, the Kaipara Harbour Sustainable Fisheries Management Group, DoC, Landcare Manaaki Whenua, Kaipara District Council, Rodney District Council, the ARC and the NRC, has been established to oversee the project. (ARC & NRC, 2006, 15-16).

The scope of the project is much wider than the ARC-led project, taking a more holistic 'catchment to the sea' approach. It will encompass catchments draining into the harbour and non-RMA activities such as fisheries and economic and cultural issues.

For many years iwi had been seeking integrated management of the Kaipara but gained little traction with the statutory authorities:

We have been calling for some kind of integrated management of the harbour for at least the last 10 years. I think they met once a month in 1996 but nothing came to fruition. (respondent 23a)

They were able to progress the matter themselves, when they were successful in obtaining a small amount of funding through a partnership with Landcare Research, and were able to initiate two hui in July and November 2005. The hui involved people from organisations such as the Sustainable Fisheries Management Group, Kaipara water group, MFish, DoC, territorial authorities, Ngāti Whātua and Te Puni Kokiri.

The feedback we got was that everyone was in agreement that the harbour was in decline, that there was an unsurpassed amount of pressure from the

general weight of subdivision, along with the pressure on fishing. There was a general shared view that something needed to happen quickly. Everyone had a vision of a healthier, productive Kaipara. An integrated management plan would be a vehicle to achieve that. We've now formed an interim steering committee. What also came out is that they would like to see it as an iwi-driven project rather than the usual council-driven project. We have responsibility for the project. (respondent 23a)

Te Uri o Hau sees itself as coordinating the management of the harbour, but currently lacks capacity to fulfil this vision. It needs the support of the statutory management bodies:

We are the only management body which has a fully integrated perspective. We also have the backing of other stakeholders that have come to us and agreed this is the way forward - development and commercial interests and the Kaipara Sustainable Fishing Group.

We see ourselves in a coordination role with all those agencies and various stakeholders. We would like to have the capacity to be able to do that on the ground. But it will take support and capacity and resourcing, perhaps from these larger councils. The support needs to be there from the central government side as well as local. We have the mandate from the group to be the driver of the integrated project. We've got to get the buy-in of the ARC to really be able to continue.

I see it as getting all the stakeholders and agencies on board in some kind of forum, where they're all having input and able to input into decisions. I see it as an iwi-driven project whereas I think the ARC perhaps see the same sort of integration but they would still be driving the project. I think it would be different if the ARC drove it, because if we didn't have the input we are having now, the fisheries wouldn't be included, it wouldn't go wider than the CMA [coastal marine area] and I don't know [about the] inclusion of other stakeholders – so it would be based solely on a RMA perspective and biophysical issues I think. (respondent 23a).

But at that stage the relationship between this iwi-led initiative and the ARC-led project was uncertain:

Then the ARC decided after ten years that they're going to do their own scoping study on the Kaipara looking at what is needed to achieve an integrated plan.

They're working with the NRC. 'Te Uri o Hau has a new memorandum of understanding with both those councils ... It's all about clear and honest communication and things like that ... [but] the ARC study only came to my attention when I was browsing through a council agenda, otherwise I probably still would not have heard about it. There was poor communication on their part. I think they are starting to clear it up and look at how we can do this project without duplicating each other.

I think the ARC project will be useful as well. It was just poorly communicated to us. So what they will look at is how all their statutory documents can be tied together. They will also look at all the research that's been done. We just have to identify who's going to be responsible for undertaking which parts of the research.

We are trying to meet with the ARC and ask which bits they are going to cover because they are uncomfortable about the catchment wide view which we have taken. We have talked about an integrated catchment management plan. That is very daunting to most agencies – the sheer size and complexity. But I think you could look at it in terms of zones. Perhaps have more detailed specific studies in the CMA but then look at the whole catchment which would not be as detailed. That's why we are getting Landcare and these people to help us ... That's the other thing some agencies couldn't get their head around is that we wanted to extend the plan right into fisheries so as to have a holistic scope. That made them stand back about getting on board our committee (respondent 23a)

The Kaipara District Council is supportive of the iwi-led process but is also not well-resourced:

... we see the strength in Te Uri O Hau actually leading ... insofar as they potentially will be able to achieve buy-in from a lot of the community: buy-in, in terms of getting all the stakeholders around the table. They have [memoranda of understanding].....with most of the statutory agencies and also they can probably drive with quite a bit of focus on it. The downside is the capacity and the whole funding of it, but equally the district council is probably in the same position as well ...

We've got a good relationship with Te Uri O Hau. It's really about supporting them. The key driver or key objective for Te Uri o Hau and also Kaipara District recognises that the Kaipara Harbour is the life blood of the Kaipara district.

So we've got much more focus on the Kaipara Harbour and it's possibly because, for the regional councils, it's one of a number of things they've got to manage ... (respondent 46)

So resourcing the iwi-led project is an uphill battle:

I've done an estimate of costs and I don't think anyone is going to give Ngāti Whātua \$2.5 million to do it. I do not think it is likely to happen as there are not enough resources. Something will happen but not on the scale we would hope for, nor will it be directed at the outcomes that would suit us.

(respondent 23b)

There is a proposal to divert sand-mining royalties from sand mined in the harbour, to help fund the development of an integrated management plan, but it is not clear if this will gain any traction.

The ultimate goal – different visions?

There are a range of visions of what integrated management of the Kaipara Harbour might look like and how it might be governed. These range from low-key efforts to achieve better coordination between statutory bodies to a formalised forum and integrated plan to jointly guide development decisions:

Typically, once you are setting up a political structure, you have to service that. Therefore it can tend to drive work programmes. They will want to understand a lot more about what they are looking at and that takes work. So just chewing through more and more budget and more and more people and resources – to look at that and study that and look at approaches and, if you don't put that amount of resource in, it results in frustration. If you set up a forum and don't do much, what is the forum there for? It becomes a talk shop and you quickly lose credibility.

'My feeling [is that the Kaipara] isn't at the level of the Hauraki Gulf. So on that basis, the Hauraki Gulf model wouldn't be appropriate. Maybe something else between what we're doing at the moment and that. Maybe going back to a joint approach with the NRC, three to four times a year, meeting regularly at an officer level to talk about a range of issues ... So maybe that model is enough and you bring in various of the TAs at times. I don't know. But it firstly is back to issue identification – what are the issues coming up, are they different from elsewhere, are they more important, is there a better way we can deal with them? (respondent 3)

For the Kaipara I see it as everyone with an interest in the area working together irrespective of boundaries, putting aside artificial boundaries and acknowledging a single body needs to be dealt with, without a defensive proprietorship. So everyone [is] saying we have a responsibility here and we all share it, so let's get on with it.'

'You would have to identify where [the] differences are and disparities between regulation, get a mutual understanding of [the] threats and what [is] happening on each side of [the] boundary, mutual recognition of those, for people to want to do something about it and then agree [on] what to do about it ...

Nothing like the Hauraki Gulf Forum. They do a lot of talking but nothing happens on the ground. The Hauraki Gulf is under so much pressure and so degraded so, in that context, maybe it's appropriate. I think a very clear set of objectives for the Kaipara, what people want and expect, taking into account environmental and economic values, not just economic. A holistic set of objectives, but ones that recognise the particular environmental conditions of the Kaipara and that are truly integrated and consistent across the catchment ... a consistent administrative or regulatory regime across the catchment which is enforced and delivered at grass-roots level through community partnerships, people on the ground who make a difference. (respondent 2)

We need a governance model which is going to be working, living, ongoing and able to change ... I think there'll be a forum established and I think there will be some kind of database and digital mapping system and perhaps some kind of zone or ideas of where things are. We'll be able to see where things are happening, where the pressure is and maybe have areas for that, rather than a strict plan. I imagine some sort of guidelines or something. (respondent 23a)

I don't think [an integrated plan] would make that much difference to the work that's already been undertaken. It would, however, provide a comprehensive framework for managing the use and development within the harbour and its environment, which the various agencies that have those responsibilities could work under. At present the problem with the Kaipara Harbour and its coastal environment is that the harbour itself is an entity but it's split between at least four statutory agencies, and they each have different approaches to how they manage activities within that one coastal environment. So the work that we're doing in all of those activities is basically issue-driven ... you might

actually have more of a focus if you're working under one policy framework to achieve whatever vision you have determined that you would be working towards.

There are a number of options in terms of what sort of plan you would develop. For example you could develop a regional plan for the Kaipara Harbour and coastal environment, which is a joint regional plan to be delivered by both regional councils and which the district plans had to fit within, so you've got a consistent set of rules that all of those parties will then need to implement. That's one option. There are a whole raft of other options. (respondent 4)

Prospects for integration

It is early days for efforts to better integrate the management of the Kaipara Harbour. Historically, most of the management agencies have not specifically focused on the harbour as a management entity, either in terms of policy and rule making or for information-gathering purposes. Changes to the aquaculture regime, and the resultant consideration of aquaculture management areas for the harbour, more recently have required that regional councils place some focus on the harbour's marine environment for planning purposes (see case study 7).

Integrated management is likely to require significantly more management resources to be devoted to the harbour than is currently the case, as well as additional funds for information gathering and research, to guide management. There is some reluctance by managers to shift such resources to the harbour's management from other areas which are seen as being under greater pressure.

The ARC has narrowed the scope of its harbour project, to include only the NRC as a partnership entity, to (at least initially) only involve information gathering and to only include elements of the coastal environment under the regional councils' jurisdiction. This ensures that the project is manageable, can be more readily funded and can be kept under control, and enables the councils to move ahead cautiously step-by-step. It lacks, however, a fully-integrated approach.

Te Uri o Hau, on the other hand, has been much more ambitious, basing its project on a wide partnership group and encompassing cultural and economic as well as environmental issues throughout the entire catchment of the harbour. This is a much more integrated approach, but is a much larger project which will require significant funding. The project currently lacks any obvious source

of funds to make real progress unless one or more statutory authorities make a major financial commitment. This does not seem to be on the cards, at least in the short term, particularly as the project is occurring in parallel with the regional council initiative which is absorbing ARC funding allocated to the harbour. This highlights the difficulties that poorly-resourced iwi encounter when seeking to undertake their kaitiakitanga role in the coastal environment. It also highlights the difficulties of seeking to adopt a fully-integrated approach.

PART THREE

CONCLUSIONS



fifteen

Conclusions and Recommendations

The coast is extraordinarily important in terms of New Zealand's natural and cultural heritage, quality of life and economy. At the same time, the coast is a high-conflict zone, where there are many competing uses and where sensitive ecological systems are under growing pressure from human impacts. This study has demonstrated that effective management of the coastal environment at a regional level has proved difficult to achieve.

New Zealand's current coastal management system is fragmented and suffers from inconsistent objectives and management approaches. The legislative framework has developed in an *ad hoc* manner and without a clear vision of how integrated coastal management might be delivered at a regional level. There is a wide disparity in the extent to which key elements of a successful integrated coastal management system are being implemented in the various management spheres. The levels of information and resourcing to support effective coastal management are insufficient. There is no legislative framework to support integrative initiatives when they do occur.

There is clearly a need to 'step through' the complexities of the current administration arrangements to develop a more strategic and integrated approach. This does not require a grand national oceans plan to overlay the other multiplicity of planning activity already under way. What it does require is a review and reform of the key pieces of legislation impacting on coastal management, including the RMA, the Fisheries Act, the Marine Reserves Act and the Biosecurity Act, to ensure that a fully-integrated and effective coastal management system can be practically delivered at the regional level.

Much can be also done to support greater integration of New Zealand's coastal management efforts prior to such a legislative reform program. The recommendations below identify some ways in which this could be achieved.

Summary of recommendations

This study has identified some serious shortcomings in the current governance framework and the extent to which it can deliver effective and integrated coastal management at a regional level. It is therefore recommended that:

Recommendation 1

The Minister in charge of oceans policy, as a matter of urgency, initiates a comprehensive review of legislation impacting on coastal management, with the objective of implementing a programme of legislative reform to ensure that effective and integrated coastal management can be delivered at a regional level.

Pending the implementation of the results of such a review, a number of recommendations have been developed which identify ways in which integrated coastal management could be strengthened at a regional level. These recommendations have been organised in accordance with the factors supporting inter-agency collaboration identified in Figure 2.5 and draw on the conclusions in chapter five as to the extent to which these elements are currently in place.

Broad agency alignment

The study identified fundamental conflicts in the way different coastal managers approach coastal management tasks and in the objectives they are seeking to achieve. It is therefore recommended that:

Recommendation 2

Broad alignment between coastal management agencies be encouraged and supported through the identification of core common elements which are required to underpin all coastal management efforts. These could include commonalities in the objectives sought to be achieved, such as integrated ecosystems-based management, as well as the approaches and processes which are used to reach the objectives. The core common elements could be incorporated into a formal memorandum of understanding between the agencies in the short term but would need to be embedded into legislation in the longer term.

Inter-agency dependency

The study identified a particularly weak relationship between regional councils and MFish, with each agency generally undertaking their activities independently of the other. It is therefore recommended that:

Recommendation 3

A clearer statutory relationship between MFish and regional councils be created so that they are jointly charged with managing the ecological health of the marine environment and have defined roles in each other's decision-making processes. Pending any necessary legislative changes, formal and regular consultative engagements should take place.

The study identified a very unequal role between tangata whenua and other coastal management agencies, which was exacerbated by the lack of a clear statutory basis for the role of tangata whenua as coastal managers. It is therefore recommended that:

Recommendation 4

A clearer statutory basis be developed for the role of tangata whenua in coastal management.

Climate of trust

The study identified strong levels of mistrust in some areas of coastal management, including marine protection and fisheries management. The lack of effective participation and dispute-resolution procedures in these areas contributes to the level of conflict. It is therefore recommended that:

Recommendation 5

Strengthened statutory procedural requirements be developed for stakeholder engagement and broader public participation in decisions affecting the coastal environment, which are to be applied to all areas of coastal management as a minimum standard, accompanied by a best-practice guideline.

Recommendation 6

Additional stakeholder capacity-building funding be made available, particularly in relation to stakeholder engagement in fisheries management and marine protection.

Recommendation 7

Effective conflict-resolution procedures, including mediation and an independent review of the merits of the decision, be incorporated into the Fisheries Act 1996 and the Marine Reserves Act 1971.

Compatible scales

The study identified spatial planning as being an important tool to enable different management regimes to interact with each other. Although spatial planning at a regional level is well established under the RMA, it is less well used in other coastal management regimes. It is therefore recommended that:

Recommendation 8

Statutory regional spatial planning be explicitly provided for in marine protection and fisheries management, and that a stronger interface between plans prepared under the different marine management systems be defined.

Decentralised decision making

Decentralised decision making enables a degree of regional and local autonomy which helps facilitate local partnership building and innovation. The level of centralisation of MFish makes it difficult for integrated regional responses to be developed to coastal issues. It is therefore recommended that:

Recommendation 9

Those elements of MFish decision making which are of regional significance to regional offices be decentralized, to foster greater regional diversity in approaches to fisheries management, to enable greater integration with other coastal management agencies, to enable regional and local circumstances to be addressed, and to promote innovation and richer learning opportunities.

Strategic approach

Integrated coastal management requires a strategic focus, so that the myriad activities which impact on a single coastal system can be guided towards the achievement of common long-term visions and goals. There has been a lack of effective strategic planning within all the coastal management agencies studied although recent initiatives, such as coastal management strategies, fisheries plans and marine protection plans, are seeking to strengthen such planning. It is therefore recommended that:

Recommendation 10

The move towards strategic planning for the coast be encouraged through the provision of legislative support. This could include providing for a strategic element in RMA Plans, providing a more detailed strategic framework for fisheries plans in the Fisheries Act and providing a legislative framework for marine protection plans in new marine protection legislation.

Good information base

Information plays an essential role in coastal management, but the study identified large gaps in current knowledge and poor monitoring. It is therefore recommended that:

Recommendation 11

A comprehensive monitoring framework for the coastal environment, which promotes better consistency and integration between existing programmes and helps progressively fill key knowledge gaps, be developed and resourced.

Adequate resourcing

Coastal management is poorly resourced, as coastal users do not currently pay the full cost of their use of public resources and the public is reluctant to pay for coastal management. It is therefore recommended that:

Recommendation 12

Additional funding sources be developed to support coastal management, including an effective and implementable coastal user-charging system, while ensuring that those paying user charges do not directly influence how they are spent.

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New Zealand has a long and spectacular coastline and the coast lies close to the heart of many New Zealanders. But as the population grows, fragile coastal environments are becoming degraded and conflicts over the use of the coast are increasing.

Based on a detailed study of the Hauraki Gulf and Kaipara Harbour, this publication investigates how the coast is currently managed and why. It explores the challenges of integrating coastal management efforts and identifies practical ways in which management can be strengthened in the future.

This publication is essential reading for coastal managers, developers, professionals, academics, activists, students and everyone with an interest in the better management of New Zealand's coasts.

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